SMART CAMPUS

Complete Source Code & Architecture Specification

Framework	Next.js 16 (App Router + React 19)
Language	TypeScript 5.x / Strict Mode
Backend & Auth	Supabase (PostgreSQL 15 + RLS + GoTrue PKCE)
Styling	Vanilla CSS Token System & Tailwind Engine
Architecture	Multi-tenant RBAC (Student, Faculty, Admin)
Total Files Included	77 Core Project Files
Security Audit	API Keys, Secrets & JWTs Redacted with [REDACTED]

Generated Automatically for Academic & Technical Archival
Institute of Technology Smart Campus System

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75. README.md	Included
76. eslint.config.mjs	Included
77. next-env.d.ts	Included

1. Complete Project Structure

The directory graph represents the clean file organization excluding node_modules and build caches:

```
smart-campus/
  ├── .env.example
  ├── .env.local
  ├── AGENTS.md
  ├── README.md
  ├── eslint.config.mjs
  ├── next-env.d.ts
  ├── next.config.ts
  ├── package.json
  ├── tsconfig.json
  ├── public/
  ├── src/
    ├── middleware.ts
    ├── actions/
      ├── academics.ts
      ├── announcements.ts
      ├── auth.ts
      ├── base.ts
      ├── operations.ts
      ├── profiles.ts
    ├── app/
      ├── globals.css
      ├── layout.tsx
      ├── page.tsx
      ├── about/
        ├── page.tsx
      ├── announcements/
        ├── page.tsx
      ├── api/
        ├── health/
          ├── route.ts
      ├── assignments/
        ├── page.tsx
      ├── attendance/
        ├── page.tsx
      ├── contact/
        ├── page.tsx
      ├── dashboard/
        ├── page.tsx
        ├── admin/
          ├── page.tsx
          ├── analytics/
            ├── page.tsx
          ├── faculty/
            ├── page.tsx
          ├── students/
            ├── page.tsx
          ├── users/
            ├── page.tsx
        ├── faculty/
          ├── page.tsx
        ├── student/
          ├── page.tsx
      ├── emergency/
        ├── page.tsx
      ├── events/
        ├── page.tsx
      ├── facilities/
        ├── page.tsx
      ├── faculty/
        ├── page.tsx
      ├── features/
        ├── page.tsx
      ├── feedback/
        ├── page.tsx
      ├── forgot-password/
        ├── page.tsx
      ├── login/
        ├── page.tsx
```

```
    ■■■ materials/
      ■■■ page.tsx
    ■■■ notifications/
      ■■■ page.tsx
    ■■■ profile/
      ■■■ page.tsx
    ■■■ register/
      ■■■ page.tsx
    ■■■ reset-password/
      ■■■ page.tsx
    ■■■ search/
      ■■■ page.tsx
    ■■■ settings/
      ■■■ page.tsx
    ■■■ timetable/
      ■■■ page.tsx
  ■■■ components/
    ■■■ dashboard/
      ■■■ attendance-chart.tsx
      ■■■ campus-stats-chart.tsx
    ■■■ layout/
      ■■■ dashboard-shell.tsx
      ■■■ footer.tsx
      ■■■ navbar.tsx
      ■■■ sidebar.tsx
    ■■■ ui/
      ■■■ badge.tsx
      ■■■ button.tsx
      ■■■ card.tsx
      ■■■ input.tsx
      ■■■ modal.tsx
      ■■■ state-views.tsx
      ■■■ table.tsx
      ■■■ theme-toggle.tsx
      ■■■ toast.tsx
  ■■■ lib/
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    ■■■ supabase/
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      ■■■ middleware.ts
      ■■■ realtime.ts
      ■■■ server.ts
      ■■■ storage.ts
      ■■■ types.ts
    ■■■ validations/
      ■■■ auth.ts
      ■■■ campus.ts
  ■■■ types/
    ■■■ index.ts
■■■ supabase/
  ■■■ schema.sql
```

2. Full Source Code Catalog

Each file below contains its verbatim code without omission, formatted with complete line numbers and secret redaction.

File #1: .env.example 17 lines

```
# ========================================
# SMART CAMPUS - Environment Configuration
# ========================================

# Supabase Project Credentials
# Obtain these from your Supabase Dashboard: Project Settings -> API
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=[REDACTED]

# Supabase Service Role Key (Backend API / Admin actions only - NEVER expose to browser)
SUPABASE_SERVICE_ROLE_KEY=[REDACTED]

# Application Base URL
NEXT_PUBLIC_SITE_URL=http://localhost:3000

# Optional: Campus Meta
NEXT_PUBLIC_CAMPUS_NAME="Smart Campus Institute of Technology"
```

File #2: .env.local 7 lines

```
# SMART CAMPUS Local Environment Variables
# Replace with your Supabase Project settings when connecting to a live instance
NEXT_PUBLIC_SUPABASE_URL=https://placeholder-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=[REDACTED]
SUPABASE_SERVICE_ROLE_KEY=[REDACTED]
NEXT_PUBLIC_SITE_URL=http://localhost:3000
NEXT_PUBLIC_CAMPUS_NAME=Smart Campus Institute of Technology
```

File #3: next.config.ts 17 lines

```
import type { NextConfig } from "next";

const nextConfig: NextConfig = {
  /* config options here */
  cacheComponents: true,
  partialPrefetching: true,
  turbopack: {
    rules: {
      "*.css": {
        loaders: ["@tailwindcss/turbopack"],
        as: "*.css",
      },
    },
  },
};

export default nextConfig;
```

File #4: package.json 33 lines

```
{
  "name": "smart-campus",
  "version": "0.1.0",
  "private": true,
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "eslint"
  },
  "dependencies": {
    "@supabase/ssr": "^0.12.7",
    "@supabase/supabase-js": "^2.117.3",
    "clsx": "^2.1.1",
    "lucide-react": "^1.55.0",
    "next": "16.4.0",
    "react": "19.3.0",
    "react-dom": "19.3.0",
```

```
    "recharts": "^3.10.1",
    "tailwind-merge": "^3.7.0",
    "zod": "^4.6.5"
  },
  "devDependencies": {
    "@tailwindcss/turbopack": "^4",
    "@types/node": "^20",
    "@types/react": "^19",
    "@types/react-dom": "^19",
    "eslint": "^9",
    "eslint-config-next": "16.4.0",
    "tailwindcss": "^4",
    "typescript": "^5"
  }
}
```

File #5: tsconfig.json 34 lines

```
{
  "compilerOptions": {
    "target": "ES2017",
    "lib": ["dom", "dom.iterable", "esnext"],
    "allowJs": true,
    "skipLibCheck": true,
    "strict": true,
    "noEmit": true,
    "esModuleInterop": true,
    "module": "esnext",
    "moduleResolution": "bundler",
    "resolveJsonModule": true,
    "isolatedModules": true,
    "jsx": "react-jsx",
    "incremental": true,
    "plugins": [
      {
        "name": "next"
      }
    ],
    "paths": {
      "@/*": ["./src/*"]
    }
  },
  "include": [
    "next-env.d.ts",
    "**/*.ts",
    "**/*.tsx",
    ".next/types/**/*.ts",
    ".next/dev/types/**/*.ts",
    "**/*.mts"
  ],
  "exclude": ["node_modules"]
}
```

File #6: supabase/schema.sql 566 lines

```
-- ================================================================================
-- SMART CAMPUS PLATFORM - Enterprise PostgreSQL Database Architecture
-- ================================================================================

-- Enable UUID Extension
create extension if not exists "uuid-ossp";

-- ================================================================================
-- 1. Custom Types & Enums
-- ================================================================================
do $$ begin
  create type user_role as enum ('student', 'faculty', 'admin');
exception
  when duplicate_object then null;
end $$;

do $$ begin
  create type announcement_urgency as enum ('normal', 'important', 'emergency');
exception
  when duplicate_object then null;
end $$;

do $$ begin
```

```
  create type attendance_status as enum ('present', 'absent', 'late', 'excused');
exception
  when duplicate_object then null;
end $$;

do $$ begin
  create type assignment_status as enum ('pending', 'submitted', 'graded', 'late');
exception
  when duplicate_object then null;
end $$;

do $$ begin
  create type feedback_status as enum ('pending', 'under_review', 'resolved', 'dismissed');
exception
  when duplicate_object then null;
end $$;

do $$ begin
  create type facility_status as enum ('available', 'booked', 'maintenance');
exception
  when duplicate_object then null;
end $$;

-- ============================================================================
-- 2. Core University Organization (Departments & Academic Programs)
-- ============================================================================
create table if not exists public.departments (
  id uuid default uuid_generate_v4() primary key,
  code text unique not null,
  name text not null,
  building text,
  hod_name text,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 3. Base Identity Profiles (Extends auth.users securely)
-- ============================================================================
create table if not exists public.profiles (
  id uuid references auth.users on delete cascade primary key,
  email text not null,
  full_name text not null,
  role user_role default 'student'::user_role not null,
  department_id uuid references public.departments(id) on delete set null,
  department text,
  phone text,
  avatar_url text,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 4. Role-Specific Profiles (Student & Faculty Details)
-- ============================================================================
create table if not exists public.student_profiles (
  id uuid references public.profiles(id) on delete cascade primary key,
  enrollment_no text unique not null,
  current_semester integer default 1 not null,
  admission_year integer not null,
  cgpa numeric(3,2) default 0.00 not null,
  attendance_aggregate numeric(5,2) default 100.00 not null,
  academic_standing text default 'Good' not null,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

create table if not exists public.faculty_profiles (
  id uuid references public.profiles(id) on delete cascade primary key,
  employee_id text unique,
  designation text not null,
  office_location text,
  office_hours text,
  research_interests text[],
  qualifications text,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
```

```
-- 5. Academic Courses & Subjects
-- ============================================================================
create table if not exists public.courses (
  id uuid default uuid_generate_v4() primary key,
  code text unique not null,
  title text not null,
  department_id uuid references public.departments(id) on delete set null,
  faculty_id uuid references public.profiles(id) on delete set null,
  credits integer default 3 not null,
  semester integer not null,
  description text,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 6. Announcements & Broadcast Circulars
-- ============================================================================
create table if not exists public.announcements (
  id uuid default uuid_generate_v4() primary key,
  title text not null,
  content text not null,
  category text default 'General' not null,
  urgency announcement_urgency default 'normal'::announcement_urgency not null,
  published_by uuid references public.profiles(id) on delete set null,
  target_audience text default 'all' not null, -- 'all', 'student', 'faculty', 'admin'
  pinned boolean default false not null,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 7. Campus Events
-- ============================================================================
create table if not exists public.events (
  id uuid default uuid_generate_v4() primary key,
  title text not null,
  description text not null,
  start_time timestamp with time zone not null,
  end_time timestamp with time zone not null,
  venue text not null,
  organizer text not null,
  category text default 'Academic' not null,
  rsvp_count integer default 0 not null,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 8. Timetable & Weekly Lecture Schedule
-- ============================================================================
create table if not exists public.timetable (
  id uuid default uuid_generate_v4() primary key,
  course_id uuid references public.courses(id) on delete cascade not null,
  faculty_id uuid references public.profiles(id) on delete set null,
  day_of_week text not null, -- 'Monday', 'Tuesday', ...
  start_time time not null,
  end_time time not null,
  room_number text not null,
  session_type text default 'Lecture' not null, -- 'Lecture', 'Lab', 'Tutorial'
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 9. Attendance Tracking
-- ============================================================================
create table if not exists public.attendance (
  id uuid default uuid_generate_v4() primary key,
  student_id uuid references public.profiles(id) on delete cascade not null,
  course_id uuid references public.courses(id) on delete cascade not null,
  timetable_id uuid references public.timetable(id) on delete set null,
  date date not null,
  status attendance_status default 'present'::attendance_status not null,
  recorded_by uuid references public.profiles(id) on delete set null,
  remarks text,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null,
  unique (student_id, course_id, date)
);
```

```
184 |
185 | -- ============================================================================
186 | -- 10. Assignments & Student Submissions
187 | -- ============================================================================
188 | create table if not exists public.assignments (
189 |   id uuid default uuid_generate_v4() primary key,
190 |   course_id uuid references public.courses(id) on delete cascade not null,
191 |   title text not null,
192 |   description text not null,
193 |   due_date timestamp with time zone not null,
194 |   max_marks integer default 100 not null,
195 |   file_url text,
196 |   created_by uuid references public.profiles(id) on delete set null,
197 |   created_at timestamp with time zone default timezone('utc'::text, now()) not null,
198 |   updated_at timestamp with time zone default timezone('utc'::text, now()) not null
199 | );
200 |
201 | create table if not exists public.assignment_submissions (
202 |   id uuid default uuid_generate_v4() primary key,
203 |   assignment_id uuid references public.assignments(id) on delete cascade not null,
204 |   student_id uuid references public.profiles(id) on delete cascade not null,
205 |   file_url text not null,
206 |   submission_notes text,
207 |   marks_received integer,
208 |   feedback text,
209 |   status assignment_status default 'submitted'::assignment_status not null,
210 |   submitted_at timestamp with time zone default timezone('utc'::text, now()) not null,
211 |   graded_at timestamp with time zone,
212 |   unique (assignment_id, student_id)
213 | );
214 |
215 | -- ============================================================================
216 | -- 11. Study Materials (Storage bucket integration)
217 | -- ============================================================================
218 | create table if not exists public.study_materials (
219 |   id uuid default uuid_generate_v4() primary key,
220 |   course_id uuid references public.courses(id) on delete cascade not null,
221 |   title text not null,
222 |   description text,
223 |   file_url text not null,
224 |   file_size text,
225 |   format text default 'PDF' not null,
226 |   uploaded_by uuid references public.profiles(id) on delete set null,
227 |   download_count integer default 0 not null,
228 |   created_at timestamp with time zone default timezone('utc'::text, now()) not null,
229 |   updated_at timestamp with time zone default timezone('utc'::text, now()) not null
230 | );
231 |
232 | -- ============================================================================
233 | -- 12. Campus Facilities & Booking Permits
234 | -- ============================================================================
235 | create table if not exists public.campus_facilities (
236 |   id uuid default uuid_generate_v4() primary key,
237 |   name text not null,
238 |   category text not null,
239 |   location text not null,
240 |   capacity integer default 50 not null,
241 |   operating_hours text not null,
242 |   status facility_status default 'available'::facility_status not null,
243 |   description text,
244 |   created_at timestamp with time zone default timezone('utc'::text, now()) not null,
245 |   updated_at timestamp with time zone default timezone('utc'::text, now()) not null
246 | );
247 |
248 | create table if not exists public.facility_bookings (
249 |   id uuid default uuid_generate_v4() primary key,
250 |   facility_id uuid references public.campus_facilities(id) on delete cascade not null,
251 |   booked_by uuid references public.profiles(id) on delete cascade not null,
252 |   booking_date date not null,
253 |   time_slot text not null,
254 |   purpose text not null,
255 |   status text default 'approved' not null, -- 'pending', 'approved', 'rejected'
256 |   created_at timestamp with time zone default timezone('utc'::text, now()) not null
257 | );
258 |
259 | -- ============================================================================
260 | -- 13. Emergency SOS Contacts
261 | -- ============================================================================
262 | create table if not exists public.emergency_contacts (
263 |   id uuid default uuid_generate_v4() primary key,
```

```
  name text not null,
  designation text not null,
  phone text not null,
  alternate_phone text,
  available_hours text default '24/7' not null,
  category text default 'Emergency' not null, -- 'Medical', 'Security', 'Helpline'
  priority integer default 1 not null
);

-- ============================================================================
-- 14. Realtime Notifications Center
-- ============================================================================
create table if not exists public.notifications (
  id uuid default uuid_generate_v4() primary key,
  user_id uuid references public.profiles(id) on delete cascade not null,
  title text not null,
  message text not null,
  type text default 'general' not null, -- 'academic', 'schedule', 'emergency', 'admin'
  link text,
  is_read boolean default false not null,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 15. Student Grievances & Feedback
-- ============================================================================
create table if not exists public.feedback (
  id uuid default uuid_generate_v4() primary key,
  submitted_by uuid references public.profiles(id) on delete set null,
  is_anonymous boolean default false not null,
  category text not null,
  subject text not null,
  description text not null,
  status feedback_status default 'pending'::feedback_status not null,
  admin_notes text,
  created_at timestamp with time zone default timezone('utc'::text, now()) not null,
  updated_at timestamp with time zone default timezone('utc'::text, now()) not null
);

-- ============================================================================
-- 16. Performance Indexes
-- ============================================================================
create index if not exists idx_profiles_role on public.profiles(role);
create index if not exists idx_student_profiles_enrollment on public.student_profiles(enrollment_no);
create index if not exists idx_courses_code on public.courses(code);
create index if not exists idx_courses_dept on public.courses(department_id);
create index if not exists idx_timetable_day on public.timetable(day_of_week);
create index if not exists idx_attendance_student_date on public.attendance(student_id, date);
create index if not exists idx_assignments_course on public.assignments(course_id);
create index if not exists idx_assignment_submissions_asg on public.assignment_submissions(assignment_id)
  ;
create index if not exists idx_study_materials_course on public.study_materials(course_id);
create index if not exists idx_notifications_user_unread on public.notifications(user_id, is_read);
create index if not exists idx_announcements_created on public.announcements(created_at desc);

-- ============================================================================
-- 17. Automatic Profile & Student/Faculty Sub-Record Synchronization Trigger
-- ============================================================================
create or replace function public.handle_new_user()
returns trigger as $$
declare
  v_role public.user_role;
  v_dept text;
  v_enrollment text;
begin
  v_role := coalesce(new.raw_user_meta_data->>'role', 'student')::public.user_role;
  v_dept := coalesce(new.raw_user_meta_data->>'department', 'General Academics');
  v_enrollment := coalesce(new.raw_user_meta_data->>'enrollment_no', '26CS' || floor(random() * 900 + 100
  )::text);

  -- Insert base profile
  insert into public.profiles (id, email, full_name, role, department, phone, avatar_url)
  values (
    new.id,
    new.email,
    coalesce(new.raw_user_meta_data->>'full_name', split_part(new.email, '@', 1)),
    v_role,
    v_dept,
    new.raw_user_meta_data->>'phone',
    new.raw_user_meta_data->>'avatar_url'
```

```
  )
  on conflict (id) do update set
    email = excluded.email,
    full_name = excluded.full_name,
    role = excluded.role,
    department = excluded.department;

  -- Create sub-profile based on role
  if v_role = 'student' then
    insert into public.student_profiles (id, enrollment_no, current_semester, admission_year)
    values (new.id, v_enrollment, 1, extract(year from now())::integer)
    on conflict (id) do nothing;
  elsif v_role = 'faculty' then
    insert into public.faculty_profiles (id, designation, office_location)
    values (new.id, 'Assistant Professor', 'Faculty Office Block')
    on conflict (id) do nothing;
  end if;

  return new;
end;
$$ language plpgsql security definer;

-- Trigger execution
drop trigger if exists on_auth_user_created on auth.users;
create trigger on_auth_user_created
  after insert on auth.users
  for each row execute procedure public.handle_new_user();

-- ================================================================================
-- 18. Row Level Security (RLS) Policies
-- ================================================================================
alter table public.departments enable row level security;
alter table public.profiles enable row level security;
alter table public.student_profiles enable row level security;
alter table public.faculty_profiles enable row level security;
alter table public.courses enable row level security;
alter table public.announcements enable row level security;
alter table public.events enable row level security;
alter table public.timetable enable row level security;
alter table public.attendance enable row level security;
alter table public.assignments enable row level security;
alter table public.assignment_submissions enable row level security;
alter table public.study_materials enable row level security;
alter table public.campus_facilities enable row level security;
alter table public.facility_bookings enable row level security;
alter table public.emergency_contacts enable row level security;
alter table public.notifications enable row level security;
alter table public.feedback enable row level security;

-- Profiles: Authenticated users can read all profiles; users can only update their own profile; admins c
  an update all
create policy "Profiles are viewable by authenticated users"
  on public.profiles for select
  to authenticated
  using (true);

create policy "Users can update their own profile"
  on public.profiles for update
  to authenticated
  using (auth.uid() = id or (select role from public.profiles where id = auth.uid()) = 'admin');

-- Student Profiles: Students view own profile; Faculty & Admins view all
create policy "Students view own profile, faculty and admin view all"
  on public.student_profiles for select
  to authenticated
  using (
    auth.uid() = id
    or (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
  );

-- Faculty Profiles: Viewable by all authenticated
create policy "Faculty profiles viewable by all"
  on public.faculty_profiles for select
  to authenticated
  using (true);

-- Announcements: Viewable by everyone; created/updated by faculty and admin only
create policy "Announcements are viewable by authenticated users"
  on public.announcements for select
  to authenticated
```

```
421 |   using (true);
422 |
423 | create policy "Faculty and Admin can publish announcements"
424 |   on public.announcements for insert
425 |   to authenticated
426 |   with check (
427 |     (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
428 |   );
429 |
430 | -- Events: Viewable by everyone; manageable by Admin and Faculty
431 | create policy "Events are viewable by authenticated users"
432 |   on public.events for select
433 |   to authenticated
434 |   using (true);
435 |
436 | create policy "Faculty and Admin can manage events"
437 |   on public.events for insert
438 |   to authenticated
439 |   with check (
440 |     (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
441 |   );
442 |
443 | -- Timetable & Courses: Viewable by all
444 | create policy "Courses viewable by all authenticated"
445 |   on public.courses for select
446 |   to authenticated
447 |   using (true);
448 |
449 | create policy "Timetable viewable by all authenticated"
450 |   on public.timetable for select
451 |   to authenticated
452 |   using (true);
453 |
454 | -- Attendance: Students can only view their own attendance; Faculty & Admin can insert/update
455 | create policy "Students view own attendance"
456 |   on public.attendance for select
457 |   to authenticated
458 |   using (
459 |     auth.uid() = student_id
460 |     or (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
461 |   );
462 |
463 | create policy "Faculty and Admin can record attendance"
464 |   on public.attendance for insert
465 |   to authenticated
466 |   with check (
467 |     (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
468 |   );
469 |
470 | -- Assignment Submissions: Students view/insert their own; Faculty & Admin can view and grade
471 | create policy "Students can view own submissions"
472 |   on public.assignment_submissions for select
473 |   to authenticated
474 |   using (
475 |     auth.uid() = student_id
476 |     or (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
477 |   );
478 |
479 | create policy "Students can submit assignments"
480 |   on public.assignment_submissions for insert
481 |   to authenticated
482 |   with check (auth.uid() = student_id);
483 |
484 | create policy "Faculty and Admin can grade submissions"
485 |   on public.assignment_submissions for update
486 |   to authenticated
487 |   using (
488 |     (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
489 |   );
490 |
491 | -- Study Materials: Viewable by all; uploaded by Faculty & Admin
492 | create policy "Study materials are viewable by all"
493 |   on public.study_materials for select
494 |   to authenticated
495 |   using (true);
496 |
497 | create policy "Faculty and Admin can upload materials"
498 |   on public.study_materials for insert
499 |   to authenticated
500 |   with check (
```

```
501 |     (select role from public.profiles where id = auth.uid()) in ('faculty', 'admin')
502 |   );
503 |
504 | -- Notifications: Users only see their own notifications
505 | create policy "Users only view their own notifications"
506 |   on public.notifications for select
507 |   to authenticated
508 |   using (auth.uid() = user_id);
509 |
510 | create policy "Users can update their own notification read status"
511 |   on public.notifications for update
512 |   to authenticated
513 |   using (auth.uid() = user_id);
514 |
515 | -- Feedback: Students view own submitted feedback; Admins view all
516 | create policy "Users view own feedback, admin views all"
517 |   on public.feedback for select
518 |   to authenticated
519 |   using (
520 |     auth.uid() = submitted_by
521 |     or (select role from public.profiles where id = auth.uid()) = 'admin'
522 |   );
523 |
524 | create policy "Authenticated users can submit feedback"
525 |   on public.feedback for insert
526 |   to authenticated
527 |   with check (auth.uid() = submitted_by or is_anonymous = true);
528 |
529 | -- Emergency Contacts & Facilities: Read accessible to all
530 | create policy "Emergency contacts viewable by all"
531 |   on public.emergency_contacts for select
532 |   using (true);
533 |
534 | create policy "Facilities viewable by all"
535 |   on public.campus_facilities for select
536 |   to authenticated
537 |   using (true);
538 |
539 | create policy "Users can book facilities"
540 |   on public.facility_bookings for insert
541 |   to authenticated
542 |   with check (auth.uid() = booked_by);
543 |
544 | -- ================================================================================
545 | -- 19. Initial Seed Data (Departments, Emergency Hotlines, Campus Facilities)
546 | -- ================================================================================
547 | insert into public.departments (code, name, building, hod_name)
548 | values
549 |   ('CSE', 'Computer Science & Engineering', 'Tech Block A', 'Dr. Rajesh Varma'),
550 |   ('ECE', 'Electronics & Communication', 'Tech Block B', 'Prof. Ananya Rao'),
551 |   ('ME', 'Mechanical Engineering', 'Engineering Workshop Block', 'Dr. Sanjay Deshmukh')
552 | on conflict (code) do nothing;
553 |
554 | insert into public.emergency_contacts (name, designation, phone, alternate_phone, available_hours, catego
    |   ry, priority)
555 | values
556 |   ('Campus Security Control Room', 'Chief Security Warden', '+1 (800) 555-0911', '+1 (800) 555-0912', '24
    |   /7', 'Security', 1),
557 |   ('University Health Center & Ambulance', 'Chief Medical Officer', '+1 (800) 555-0108', '+1 (800) 555-01
    |   09', '24/7', 'Medical', 2),
558 |   ('Women Safety & Anti-Ragging Cell', 'Convenor & Senior Dean', '+1 (800) 555-0181', null, '24/7', 'Help
    |   line', 3)
559 | on conflict do nothing;
560 |
561 | insert into public.campus_facilities (name, category, location, capacity, operating_hours, status, descri
    |   ption)
562 | values
563 |   ('Turing Advanced Computing Lab', 'Specialized Lab', 'Tech Block A • 3rd Floor', 65, '08:00 AM - 10:00
    |   PM', 'available', 'Equipped with 65 high-end GPU workstations for ML, AI, and Cloud Virtualization.'),
564 |   ('Sir C.V. Raman Central Auditorium', 'Auditorium', 'Central Campus Plaza', 850, '09:00 AM - 09:00 PM',
    |    'booked', 'Acoustically treated auditorium with Dolby surround audio and 4K projection.'),
565 |   ('Collaborative Study Pods (Library)', 'Library Space', 'Central Library • 2nd Floor', 12, '24/7 Open',
    |    'available', 'Sound-isolated breakout room equipped with smart digital whiteboards.')
566 | on conflict do nothing;
```

File #7: src/lib/supabase/middleware.ts

78 lines

```
1 | import { createServerClient } from "@supabase/ssr";
2 | import { NextResponse, type NextRequest } from "next/server";
```

```
import { Database } from "./types";

export async function updateSession(request: NextRequest) {
  let supabaseResponse = NextResponse.next({
    request,
  });

  const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL || "https://placeholder-project.supabase.co";
  const supabaseAnonKey = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY || "placeholder-anon-key";

  // If Supabase credentials are placeholder values, allow request through safely
  if (supabaseUrl.includes("placeholder-project") || supabaseAnonKey.includes("placeholder")) {
    return supabaseResponse;
  }

  const supabase = createServerClient<Database>(
    supabaseUrl,
    supabaseAnonKey,
    {
      cookies: {
        getAll() {
          return request.cookies.getAll();
        },
        setAll(cookiesToSet) {
          cookiesToSet.forEach(({ name, value }) => request.cookies.set(name, value));
          supabaseResponse = NextResponse.next({
            request,
          });
          cookiesToSet.forEach(({ name, value, options }) =>
            supabaseResponse.cookies.set(name, value, options)
          );
        },
      },
    }
  );

  // Authenticate user session
  const {
    data: { user },
  } = await supabase.auth.getUser();

  const path = request.nextUrl.pathname;

  // Protected route checking:
  // If not authenticated and attempting to access restricted portals
  if (!user && (path.startsWith("/dashboard") || path.startsWith("/settings") || path.startsWith("/profil
  e"))) {
    const redirectUrl = new URL("/login", request.url);
    redirectUrl.searchParams.set("redirect", path);
    return NextResponse.redirect(redirectUrl);
  }

  // Role-based route protection
  if (user && path.startsWith("/dashboard/")) {
    // Fetch user profile securely to verify role
    const { data: profile } = await supabase
      .from("profiles")
      .select("role")
      .eq("id", user.id)
      .single();

    const userRole = (profile as { role?: string } | null)?.role || "student";

    // Admin routes protection: Only admin can access /dashboard/admin
    if (path.startsWith("/dashboard/admin") && userRole !== "admin") {
      const redirectPath = userRole === "faculty" ? "/dashboard/faculty" : "/dashboard/student";
      return NextResponse.redirect(new URL(redirectPath, request.url));
    }

    // Faculty routes protection: Only faculty or admin can access /dashboard/faculty
    if (path.startsWith("/dashboard/faculty") && userRole !== "faculty" && userRole !== "admin") {
      return NextResponse.redirect(new URL("/dashboard/student", request.url));
    }
  }

  return supabaseResponse;
}
```

File #8: src/middleware.ts 19 lines

```
import { type NextRequest } from "next/server";
import { updateSession } from "@/lib/supabase/middleware";

export async function middleware(request: NextRequest) {
  return await updateSession(request);
}

export const config = {
  matcher: [
    /*
     * Match all request paths except for the ones starting with:
     * - _next/static (static files)
     * - _next/image (image optimization files)
     * - favicon.ico (favicon file)
     * - public files (images, icons)
     */
    "/((?!_next/static|_next/image|favicon.ico|.*\\.(?:svg|png|jpg|jpeg|gif|webp)$).*)",
  ],
};
```

File #9: src/lib/env.ts 33 lines

```
import { z } from "zod";

const envSchema = z.object({
  NEXT_PUBLIC_SUPABASE_URL: z.string().url().optional(),
  NEXT_PUBLIC_SUPABASE_ANON_KEY: z.string().min(10).optional(),
  SUPABASE_SERVICE_ROLE_KEY: z.string().min(10).optional(),
  NEXT_PUBLIC_SITE_URL: z.string().url().default("http://localhost:3000"),
  NEXT_PUBLIC_CAMPUS_NAME: z.string().default("Smart Campus"),
});

const rawEnv = {
  NEXT_PUBLIC_SUPABASE_URL: process.env.NEXT_PUBLIC_SUPABASE_URL,
  NEXT_PUBLIC_SUPABASE_ANON_KEY: process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY,
  SUPABASE_SERVICE_ROLE_KEY: process.env.SUPABASE_SERVICE_ROLE_KEY,
  NEXT_PUBLIC_SITE_URL: process.env.NEXT_PUBLIC_SITE_URL || "http://localhost:3000",
  NEXT_PUBLIC_CAMPUS_NAME: process.env.NEXT_PUBLIC_CAMPUS_NAME || "Smart Campus",
};

export const env = envSchema.parse(rawEnv);

/**
 * Returns true if real, operational Supabase credentials are configured.
 */
export function isSupabaseConfigured(): boolean {
  const url = process.env.NEXT_PUBLIC_SUPABASE_URL;
  const anonKey = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY;

  if (!url || !anonKey) return false;
  if (url.includes("placeholder-project") || anonKey.includes("placeholder")) {
    return false;
  }
  return true;
}
```

File #10: src/lib/navigation.ts 167 lines

```
import {
  LayoutDashboard,
  GraduationCap,
  Briefcase,
  ShieldAlert,
  Bell,
  Calendar,
  Clock,
  CheckCircle2,
  FileText,
  BookOpen,
  Users,
  Building2,
  PhoneCall,
  MessageSquare,
  Search,
  Settings,
```

```
  User,
  BarChart3,
  UserCheck,
  Sparkles,
} from "lucide-react";
import { NavCategory } from "@/types";

export const campusNavigation: NavCategory[] = [
  {
    title: "Portals",
    items: [
      {
        title: "Student Portal",
        href: "/dashboard/student",
        icon: GraduationCap,
        roles: ["student", "admin"],
      },
      {
        title: "Faculty Portal",
        href: "/dashboard/faculty",
        icon: Briefcase,
        roles: ["faculty", "admin"],
      },
      {
        title: "Admin Command",
        href: "/dashboard/admin",
        icon: ShieldAlert,
        badge: "Admin",
        roles: ["admin"],
      },
    ],
  },
  {
    title: "Admin Management",
    items: [
      {
        title: "User Directory",
        href: "/dashboard/admin/users",
        icon: Users,
        roles: ["admin"],
      },
      {
        title: "Students Roster",
        href: "/dashboard/admin/students",
        icon: GraduationCap,
        roles: ["admin"],
      },
      {
        title: "Faculty Roster",
        href: "/dashboard/admin/faculty",
        icon: UserCheck,
        roles: ["admin"],
      },
      {
        title: "Campus Analytics",
        href: "/dashboard/admin/analytics",
        icon: BarChart3,
        badge: "Pro",
        roles: ["admin"],
      },
    ],
  },
  {
    title: "Academic Hub",
    items: [
      {
        title: "Timetable",
        href: "/timetable",
        icon: Clock,
      },
      {
        title: "Attendance",
        href: "/attendance",
        icon: CheckCircle2,
      },
      {
        title: "Assignments",
        href: "/assignments",
        icon: FileText,
      },
```

```
      {
        title: "Study Materials",
        href: "/materials",
        icon: BookOpen,
      },
      {
        title: "Faculty Directory",
        href: "/faculty",
        icon: Users,
      },
    ],
  },
  {
    title: "Campus Life & Safety",
    items: [
      {
        title: "Announcements",
        href: "/announcements",
        icon: Bell,
        badge: "Live",
      },
      {
        title: "Campus Events",
        href: "/events",
        icon: Calendar,
      },
      {
        title: "Facilities & Labs",
        href: "/facilities",
        icon: Building2,
      },
      {
        title: "Emergency SOS",
        href: "/emergency",
        icon: PhoneCall,
        badge: "24/7",
      },
      {
        title: "Feedback & Grievances",
        href: "/feedback",
        icon: MessageSquare,
      },
    ],
  },
  {
    title: "Account & Preferences",
    items: [
      {
        title: "Notifications",
        href: "/notifications",
        icon: Bell,
      },
      {
        title: "Global Search",
        href: "/search",
        icon: Search,
      },
      {
        title: "My Profile",
        href: "/profile",
        icon: User,
      },
      {
        title: "Settings",
        href: "/settings",
        icon: Settings,
      },
    ],
  },
];
```

File #11: src/lib/supabase/client.ts

12 lines

```
import { createBrowserClient } from "@supabase/ssr";
import { Database } from "./types";

/**
 * Creates a Supabase client for client-side components.
 */
```

```
export function createClient() {
  const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL || "https://placeholder-project.supabase.co";
  const supabaseAnonKey = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY || "placeholder-anon-key";

  return createBrowserClient<Database>(supabaseUrl, supabaseAnonKey);
}
```

File #12: src/lib/supabase/realtime.ts

45 lines

```
import { RealtimeChannel } from "@supabase/supabase-js";
import { createClient } from "./client";

export type RealtimeCallback<T = Record<string, unknown>> = (payload: {
  eventType: "INSERT" | "UPDATE" | "DELETE";
  new: T;
  old: T;
}) => void;

/**
 * Subscribes to real-time table modifications via Supabase Realtime
 */
export function subscribeToTable<T = Record<string, unknown>>(
  table: string,
  callback: RealtimeCallback<T>,
  filter?: string
): () => void {
  const supabase = createClient();
  let channel: RealtimeChannel;

  const channelName = `realtime:${table}${filter ? `:${filter}` : ""}`;
  channel = supabase
    .channel(channelName)
    .on(
      "postgres_changes",
      {
        event: "*",
        schema: "public",
        table,
        filter,
      },
      (payload) => {
        callback({
          eventType: payload.eventType as "INSERT" | "UPDATE" | "DELETE",
          new: payload.new as T,
          old: payload.old as T,
        });
      }
    )
    .subscribe();

  return () => {
    supabase.removeChannel(channel);
  };
}
```

File #13: src/lib/supabase/server.ts

48 lines

```
import { createServerClient, type CookieOptions } from "@supabase/ssr";
import { cookies } from "next/headers";
import { Database } from "./types";

/**
 * Creates a Supabase client for Server Components, Server Actions, and Route Handlers.
 */
export async function createClient() {
  const cookieStore = await cookies();

  const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL || "https://placeholder-project.supabase.co";
  const supabaseAnonKey = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY || "placeholder-anon-key";

  return createServerClient<Database>(supabaseUrl, supabaseAnonKey, {
    cookies: {
      getAll() {
        return cookieStore.getAll();
      },
      setAll(cookiesToSet) {
        try {
          cookiesToSet.forEach(({ name, value, options }) => {
```

```
            cookieStore.set(name, value, options);
          });
        } catch {
          // The `setAll` method was called from a Server Component.
          // This can be ignored if you have middleware refreshing user sessions.
        }
      },
    },
  });
}

/**
 * Creates an admin Supabase client utilizing the Service Role Key for background/admin tasks.
 */
export async function createAdminClient() {
  const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL || "https://placeholder-project.supabase.co";
  const serviceRoleKey = process.env.SUPABASE_SERVICE_ROLE_KEY || "placeholder-service-role";

  return createServerClient<Database>(supabaseUrl, serviceRoleKey, {
    cookies: {
      getAll() {
        return [];
      },
      setAll() {},
    },
  });
}
```

File #14: src/lib/supabase/storage.ts

116 lines

```
import { createClient } from "./client";
import { isSupabaseConfigured } from "../env";

export const STORAGE_BUCKETS = {
  AVATARS: "avatars",
  STUDY_MATERIALS: "study_materials",
  ASSIGNMENT_SUBMISSIONS: "assignment_submissions",
  CAMPUS_DOCS: "campus_docs",
} as const;

export type StorageBucket = (typeof STORAGE_BUCKETS)[keyof typeof STORAGE_BUCKETS];

// Maximum upload limits
export const FILE_LIMITS = {
  AVATAR: {
    maxSizeBytes: 2 * 1024 * 1024, // 2MB
    allowedTypes: ["image/jpeg", "image/png", "image/webp"],
  },
  STUDY_MATERIAL: {
    maxSizeBytes: 50 * 1024 * 1024, // 50MB
    allowedTypes: [
      "application/pdf",
      "application/vnd.ms-powerpoint",
      "application/vnd.openxmlformats-officedocument.presentationml.presentation",
      "application/zip",
      "application/x-zip-compressed",
    ],
  },
  ASSIGNMENT: {
    maxSizeBytes: 25 * 1024 * 1024, // 25MB
    allowedTypes: [
      "application/pdf",
      "application/zip",
      "application/x-zip-compressed",
      "application/x-tar",
      "application/gzip",
      "text/plain",
      "application/x-ipynb+json",
    ],
  },
};

/**
 * Validates a file against type and size limits before upload
 */
export function validateUploadFile(
  file: File,
  limitConfig: { maxSizeBytes: number; allowedTypes: string[] }
): { isValid: boolean; error?: string } {
```

```
  if (file.size > limitConfig.maxSizeBytes) {
    const maxMb = Math.round(limitConfig.maxSizeBytes / (1024 * 1024));
    return {
      isValid: false,
      error: `File size exceeds the ${maxMb}MB upload limit.`,
    };
  }

  // Type check (if browser passes mime or check file extension)
  if (limitConfig.allowedTypes.length > 0 && file.type) {
    const isAllowed = limitConfig.allowedTypes.some((t) => t.toLowerCase() === file.type.toLowerCase());
    if (!isAllowed) {
      return {
        isValid: false,
        error: `File type '${file.type}' is not supported. Please upload an accepted format.`,
      };
    }
  }

  return { isValid: true };
}

/**
 * Securely uploads a file to a specified Supabase Storage bucket
 */
export async function uploadFileSecure(
  bucket: StorageBucket,
  path: string,
  file: File,
  limitConfig?: { maxSizeBytes: number; allowedTypes: string[] }
): Promise<{ url: string | null; error: string | null }> {
  // Validate if limit config provided
  if (limitConfig) {
    const val = validateUploadFile(file, limitConfig);
    if (!val.isValid) {
      return { url: null, error: val.error || "File validation failed." };
    }
  }

  if (!isSupabaseConfigured()) {
    // Return simulated secure storage object URL during local mock mode
    const mockUrl = `/storage/${bucket}/${encodeURIComponent(file.name)}`;
    return { url: mockUrl, error: null };
  }

  try {
    const supabase = createClient();
    const cleanPath = `${Date.now()}-${path.replace(/[^a-zA-Z0-9._-]/g, "_")}`;

    const { data, error } = await supabase.storage.from(bucket).upload(cleanPath, file, {
      upsert: true,
      cacheControl: "3600",
    });

    if (error) {
      return { url: null, error: error.message };
    }

    const {
      data: { publicUrl },
    } = supabase.storage.from(bucket).getPublicUrl(data.path);

    return { url: publicUrl, error: null };
  } catch (err: any) {
    return { url: null, error: err.message || "Failed to upload file to storage." };
  }
}
```

File #15: src/lib/supabase/types.ts

50 lines

```
export type Json =
  | string
  | number
  | boolean
  | null
  | { [key: string]: Json | undefined }
  | Json[];

export type UserRole = "student" | "faculty" | "admin";
```

```
export type AnnouncementUrgency = "normal" | "important" | "emergency";
export type AttendanceStatus = "present" | "absent" | "late" | "excused";
export type AssignmentStatus = "pending" | "submitted" | "graded" | "late";
export type FeedbackStatus = "pending" | "under_review" | "resolved" | "dismissed";
export type FacilityStatus = "available" | "booked" | "maintenance";

export interface Database {
  public: {
    Tables: {
      [key: string]: {
        Row: Record<string, any>;
        Insert: Record<string, any>;
        Update: Record<string, any>;
        Relationships: any[];
      };
    };
    Views: {
      [key: string]: {
        Row: Record<string, any>;
        Relationships: any[];
      };
    };
    Functions: {
      [key: string]: {
        Args: Record<string, any>;
        Returns: any;
      };
    };
    Enums: {
      user_role: UserRole;
      announcement_urgency: AnnouncementUrgency;
      attendance_status: AttendanceStatus;
      assignment_status: AssignmentStatus;
      feedback_status: FeedbackStatus;
      facility_status: FacilityStatus;
    };
    CompositeTypes: {
      [key: string]: Record<string, any>;
    };
  };
}
```

File #16: src/lib/utils.ts

53 lines

```
import { clsx, type ClassValue } from "clsx";
import { twMerge } from "tailwind-merge";

/**
 * Merges class names safely with Tailwind CSS
 */
export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs));
}

/**
 * Format date string to readable format
 */
export function formatDate(dateString: string | Date | null | undefined): string {
  if (!dateString) return "N/A";
  const date = typeof dateString === "string" ? new Date(dateString) : dateString;
  if (isNaN(date.getTime())) return "Invalid date";
  return new Intl.DateTimeFormat("en-US", {
    month: "short",
    day: "numeric",
    year: "numeric",
  }).format(date);
}

/**
 * Format time string to 12-hour AM/PM format
 */
export function formatTime(timeString: string | null | undefined): string {
  if (!timeString) return "--:--";
  // handles HH:MM:SS or HH:MM
  const parts = timeString.split(":");
  if (parts.length < 2) return timeString;
  let hours = parseInt(parts[0], 10);
  const minutes = parts[1];
  const ampm = hours >= 12 ? "PM" : "AM";
```

```
  hours = hours % 12 || 12;
  return `${hours}:${minutes} ${ampm}`;
}

/**
 * Get role badge styles
 */
export function getRoleBadgeStyle(role: "student" | "faculty" | "admin" | string) {
  switch (role) {
    case "admin":
      return "bg-rose-500/10 text-rose-600 border-rose-500/20 dark:bg-rose-500/20 dark:text-rose-400";
    case "faculty":
      return "bg-amber-500/10 text-amber-600 border-amber-500/20 dark:bg-amber-500/20 dark:text-amber-400
  ";
    case "student":
    default:
      return "bg-indigo-500/10 text-indigo-600 border-indigo-500/20 dark:bg-indigo-500/20 dark:text-indig
  o-400";
  }
}
```

File #17: src/lib/validations/auth.ts

46 lines

```
import { z } from "zod";

export const loginSchema = z.object({
  email: z.string().min(1, "Email is required").email("Invalid email address"),
  password: z.string().min(6, "Password must be at least 6 characters"),
});

export type LoginInput = z.infer<typeof loginSchema>;

export const registerSchema = z
  .object({
    fullName: z.string().min(2, "Full name must be at least 2 characters"),
    email: z.string().min(1, "Email is required").email("Invalid email address"),
    password: z.string().min(6, "Password must be at least 6 characters"),
    confirmPassword: z.string().min(6, "Confirm password is required"),
    role: z.enum(["student", "faculty", "admin"], {
      message: "Please select a valid role",
    }),
    department: z.string().min(2, "Department is required"),
    enrollmentNo: z.string().optional(),
    phone: z.string().optional(),
  })
  .refine((data) => data.password =[REDACTED]
    message: "Passwords do not match",
    path: ["confirmPassword"],
  });

export type RegisterInput = z.infer<typeof registerSchema>;

export const forgotPasswordSchema = z.object({
  email: z.string().min(1, "Email is required").email("Invalid email address"),
});

export type ForgotPasswordInput = z.infer<typeof forgotPasswordSchema>;

export const resetPasswordSchema = z
  .object({
    password: z.string().min(6, "Password must be at least 6 characters"),
    confirmPassword: z.string().min(6, "Confirm password is required"),
  })
  .refine((data) => data.password =[REDACTED]
    message: "Passwords do not match",
    path: ["confirmPassword"],
  });

export type ResetPasswordInput = z.infer<typeof resetPasswordSchema>;
```

File #18: src/lib/validations/campus.ts

179 lines

```
import { z } from "zod";

// ========================================
// 1. Profile Schemas
// ========================================
```

```
export const profileSchema = z.object({
  fullName: z.string().min(2, "Full name is required"),
  department: z.string().optional(),
  phone: z.string().optional(),
  avatarUrl: z.string().url().optional().or(z.literal("")),
});

export const studentProfileSchema = z.object({
  enrollmentNo: z.string().min(3, "Enrollment number is required"),
  currentSemester: z.coerce.number().min(1).max(10).default(1),
  admissionYear: z.coerce.number().min(2015).max(2035).default(2026),
  academicStanding: z.string().default("Good"),
});

export const facultyProfileSchema = z.object({
  employeeId: z.string().optional(),
  designation: z.string().min(2, "Designation is required"),
  officeLocation: z.string().optional(),
  officeHours: z.string().optional(),
  researchInterests: z.array(z.string()).optional(),
  qualifications: z.string().optional(),
});

// ========================================
// 2. Announcements Schema
// ========================================
export const announcementSchema = z.object({
  title: z.string().min(3, "Title must be at least 3 characters"),
  content: z.string().min(10, "Content must be at least 10 characters"),
  category: z.string().min(2, "Category is required"),
  urgency: z.enum(["normal", "important", "emergency"]).default("normal"),
  targetAudience: z.enum(["all", "students", "faculty"]).default("all"),
  pinned: z.boolean().default(false),
});

// ========================================
// 3. Events Schema
// ========================================
export const eventSchema = z.object({
  title: z.string().min(3, "Event title is required"),
  description: z.string().min(10, "Description must be at least 10 characters"),
  startTime: z.string().min(1, "Start time is required"),
  endTime: z.string().min(1, "End time is required"),
  venue: z.string().min(2, "Venue is required"),
  organizer: z.string().min(2, "Organizer is required"),
  category: z.string().default("Academic"),
});

// ========================================
// 4. Timetable Schema
// ========================================
export const timetableSchema = z.object({
  courseId: z.string().uuid("Valid course ID is required"),
  facultyId: z.string().uuid().optional(),
  dayOfWeek: z.enum(["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"]),
  startTime: z.string().min(1, "Start time is required"),
  endTime: z.string().min(1, "End time is required"),
  roomNumber: z.string().min(1, "Room number is required"),
  sessionType: z.enum(["Lecture", "Lab", "Tutorial"]).default("Lecture"),
});

// ========================================
// 5. Attendance Schema
// ========================================
export const attendanceRecordSchema = z.object({
  courseId: z.string().uuid("Valid course ID required"),
  date: z.string().min(1, "Date required"),
  records: z.array(
    z.object({
      studentId: z.string().uuid(),
      status: z.enum(["present", "absent", "late", "excused"]),
      remarks: z.string().optional(),
    })
  ),
});

// ========================================
// 6. Assignments & Submissions
// ========================================
export const assignmentSchema = z.object({
```

```
  courseId: z.string().uuid("Valid course ID required"),
  title: z.string().min(3, "Assignment title is required"),
  description: z.string().min(10, "Description must be at least 10 characters"),
  dueDate: z.string().min(1, "Due date is required"),
  maxMarks: z.coerce.number().min(1).default(100),
  fileUrl: z.string().url().optional().or(z.literal("")),
});

export const submissionGradeSchema = z.object({
  submissionId: z.string().uuid(),
  marksReceived: z.coerce.number().min(0),
  feedback: z.string().optional(),
  status: z.enum(["submitted", "graded"]).default("graded"),
});

// ========================================
// 7. Study Materials
// ========================================
export const studyMaterialSchema = z.object({
  courseId: z.string().uuid("Valid course ID required"),
  title: z.string().min(3, "Title must be at least 3 characters"),
  description: z.string().optional(),
  fileUrl: z.string().min(1, "File URL or path is required"),
  fileSize: z.string().optional(),
  format: z.string().default("PDF"),
});

// ========================================
// 8. Campus Facilities & Booking
// ========================================
export const facilitySchema = z.object({
  name: z.string().min(3, "Facility name is required"),
  category: z.string().min(2, "Category is required"),
  location: z.string().min(2, "Location is required"),
  capacity: z.coerce.number().min(1).default(50),
  operatingHours: z.string().min(1, "Operating hours required"),
  status: z.enum(["available", "booked", "maintenance"]).default("available"),
  description: z.string().optional(),
});

export const facilityBookingSchema = z.object({
  facilityId: z.string().uuid(),
  bookingDate: z.string().min(1, "Booking date required"),
  timeSlot: z.string().min(1, "Time slot required"),
  purpose: z.string().min(5, "Purpose must be at least 5 characters"),
});

// ========================================
// 9. Emergency Contacts
// ========================================
export const emergencyContactSchema = z.object({
  name: z.string().min(3, "Contact name required"),
  designation: z.string().min(2, "Designation required"),
  phone: z.string().min(5, "Phone number required"),
  alternatePhone: z.string().optional(),
  availableHours: z.string().default("24/7"),
  category: z.enum(["Emergency", "Medical", "Security", "Helpline"]).default("Emergency"),
  priority: z.coerce.number().default(1),
});

// ========================================
// 10. Notifications
// ========================================
export const notificationSchema = z.object({
  userId: z.string().uuid(),
  title: z.string().min(3, "Title required"),
  message: z.string().min(5, "Message required"),
  type: z.enum(["general", "academic", "schedule", "emergency", "admin"]).default("general"),
  link: z.string().optional(),
});

// ========================================
// 11. Feedback Schema
// ========================================
export const feedbackSchema = z.object({
  category: z.enum(["Academic", "Hostel", "Canteen", "Infrastructure", "Other"]),
  subject: z.string().min(5, "Subject must be at least 5 characters"),
  description: z.string().min(20, "Please provide at least 20 characters of detail"),
  isAnonymous: z.boolean().default(false),
});
```

```
166 |
167 | export const feedbackResolutionSchema = z.object({
168 |   feedbackId: z.string().uuid(),
169 |   status: z.enum(["pending", "under_review", "resolved", "dismissed"]),
170 |   adminNotes: z.string().optional(),
171 | });
172 |
173 | // ========================================
174 | // 12. Search Schema
175 | // ========================================
176 | export const searchSchema = z.object({
177 |   query: z.string().min(1, "Search term cannot be empty"),
178 |   category: z.enum(["all", "announcements", "events", "faculty", "facilities", "courses"]).default("all")
    |   ,
179 | });
```

File #19: src/types/index.ts — 35 lines

```
 1 | import { LucideIcon } from "lucide-react";
 2 |
 3 | export type Role = "student" | "faculty" | "admin";
 4 |
 5 | export interface NavItem {
 6 |   title: string;
 7 |   href: string;
 8 |   icon: LucideIcon;
 9 |   badge?: string;
10 |   roles?: Role[];
11 | }
12 |
13 | export interface NavCategory {
14 |   title: string;
15 |   items: NavItem[];
16 | }
17 |
18 | export interface CampusUser {
19 |   id: string;
20 |   email: string;
21 |   fullName: string;
22 |   role: Role;
23 |   department?: string;
24 |   enrollmentNo?: string;
25 |   avatarUrl?: string;
26 | }
27 |
28 | export interface QuickStat {
29 |   label: string;
30 |   value: string | number;
31 |   change?: string;
32 |   trend?: "up" | "down" | "neutral";
33 |   icon: LucideIcon;
34 |   description: string;
35 | }
```

File #20: src/actions/academics.ts — 460 lines

```
 1 | "use server";
 2 |
 3 | import { createClient } from "@/lib/supabase/server";
 4 | import { isSupabaseConfigured } from "@/lib/env";
 5 | import { revalidatePath } from "next/cache";
 6 | import {
 7 |   timetableSchema,
 8 |   attendanceRecordSchema,
 9 |   assignmentSchema,
10 |   submissionGradeSchema,
11 |   studyMaterialSchema,
12 | } from "@/lib/validations/campus";
13 | import { getAuthenticatedUserWithRoles, ServerActionResponse } from "./base";
14 |
15 | // ============================================================================
16 | // 1. Timetable Actions
17 | // ============================================================================
18 |
19 | export async function getTimetableAction(day?: string): Promise<ServerActionResponse<any[]>> {
20 |   if (!isSupabaseConfigured()) {
21 |     return {
22 |       success: true,
```

```
      data: [
        {
          id: "t-1",
          course: { code: "CS601", title: "Data Structures & Algorithms" },
          faculty: { full_name: "Prof. Vivek Sharma" },
          day_of_week: "Monday",
          start_time: "09:00:00",
          end_time: "10:30:00",
          room_number: "Lab 302",
          session_type: "Lecture",
        },
        {
          id: "t-2",
          course: { code: "CS602", title: "Database Management Systems" },
          faculty: { full_name: "Dr. Rajesh Varma" },
          day_of_week: "Monday",
          start_time: "11:30:00",
          end_time: "13:00:00",
          room_number: "Hall 401",
          session_type: "Lecture",
        },
      ],
    };
  }

  const supabase = await createClient();
  let query = supabase.from("timetable").select(`
    id,
    day_of_week,
    start_time,
    end_time,
    room_number,
    session_type,
    course:courses (id, code, title),
    faculty:profiles (id, full_name)
  `).order("start_time", { ascending: true });

  if (day && day !== "All") {
    query = query.eq("day_of_week", day);
  }

  const { data, error } = await query;
  if (error) return { success: false, message: error.message };
  return { success: true, data: data || [] };
}

export async function createTimetableSlotAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = timetableSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
    if (isMock) {
      revalidatePath("/timetable");
      return { success: true, message: "Timetable slot added (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase.from("timetable").insert({
      course_id: parsed.data.courseId,
      faculty_id: parsed.data.facultyId,
      day_of_week: parsed.data.dayOfWeek,
      start_time: parsed.data.startTime,
      end_time: parsed.data.endTime,
      room_number: parsed.data.roomNumber,
      session_type: parsed.data.sessionType,
    }).select().single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/timetable");
    return { success: true, message: "Timetable slot successfully created!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function deleteTimetableSlotAction(id: string): Promise<ServerActionResponse> {
```

```
103 |   try {
104 |     const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
105 |     if (isMock) {
106 |       revalidatePath("/timetable");
107 |       return { success: true, message: "Timetable slot deleted (Demo Mode)" };
108 |     }
109 |
110 |     const supabase = await createClient();
111 |     const { error } = await supabase.from("timetable").delete().eq("id", id);
112 |     if (error) return { success: false, message: error.message };
113 |
114 |     revalidatePath("/timetable");
115 |     return { success: true, message: "Timetable slot removed." };
116 |   } catch (err: any) {
117 |     return { success: false, message: err.message };
118 |   }
119 | }
120 |
121 | // ============================================================================
122 | // 2. Attendance Actions
123 | // ============================================================================
124 |
125 | export async function getAttendanceAction(): Promise<ServerActionResponse<any[]>> {
126 |   try {
127 |     const { user, profile, isMock } = await getAuthenticatedUserWithRoles();
128 |     if (isMock) {
129 |       return {
130 |         success: true,
131 |         data: [
132 |           { code: "CS601", name: "Data Structures & Algorithms", conducted: 32, attended: 29, pct: 90.6,
    |   faculty: "Prof. Sharma" },
133 |           { code: "CS602", name: "Database Management Systems", conducted: 28, attended: 26, pct: 92.8, f
    |   aculty: "Dr. Rajesh V" },
134 |           { code: "CS603", name: "Operating Systems", conducted: 30, attended: 23, pct: 76.6, faculty: "D
    |   r. Nair" },
135 |         ],
136 |       };
137 |     }
138 |
139 |     const supabase = await createClient();
140 |     const userRole = (profile as any)?.role || "student";
141 |
142 |     if (userRole === "student") {
143 |       // Student: only own attendance
144 |       const { data, error } = await supabase
145 |         .from("attendance")
146 |         .select(`
147 |           id,
148 |           date,
149 |           status,
150 |           remarks,
151 |           course:courses (code, title)
152 |         `)
153 |         .eq("student_id", user.id)
154 |         .order("date", { ascending: false });
155 |
156 |       if (error) return { success: false, message: error.message };
157 |       return { success: true, data: data || [] };
158 |     } else {
159 |       // Faculty/Admin: all records
160 |       const { data, error } = await supabase
161 |         .from("attendance")
162 |         .select(`
163 |           id,
164 |           date,
165 |           status,
166 |           remarks,
167 |           course:courses (code, title),
168 |           student:profiles (full_name, email)
169 |         `)
170 |         .order("date", { ascending: false })
171 |         .limit(100);
172 |
173 |       if (error) return { success: false, message: error.message };
174 |       return { success: true, data: data || [] };
175 |     }
176 |   } catch (err: any) {
177 |     return { success: false, message: err.message };
178 |   }
179 | }
```

```
180 |
181 | export async function recordAttendanceAction(formData: unknown): Promise<ServerActionResponse> {
182 |   const parsed = attendanceRecordSchema.safeParse(formData);
183 |   if (!parsed.success) {
184 |     return { success: false, errors: parsed.error.flatten().fieldErrors };
185 |   }
186 |
187 |   try {
188 |     const { user, isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
189 |     if (isMock) {
190 |       revalidatePath("/attendance");
191 |       return { success: true, message: "Attendance roll call saved (Demo Mode)" };
192 |     }
193 |
194 |     const supabase = await createClient();
195 |     const insertPayload = parsed.data.records.map((r) => ({
196 |       course_id: parsed.data.courseId,
197 |       date: parsed.data.date,
198 |       student_id: r.studentId,
199 |       status: r.status,
200 |       remarks: r.remarks,
201 |       recorded_by: user.id,
202 |     }));
203 |
204 |     const { error } = await supabase
205 |       .from("attendance")
206 |       .upsert(insertPayload, { onConflict: "student_id,course_id,date" });
207 |
208 |     if (error) return { success: false, message: error.message };
209 |
210 |     revalidatePath("/attendance");
211 |     return { success: true, message: `Roll call logged for ${parsed.data.records.length} students.` };
212 |   } catch (err: any) {
213 |     return { success: false, message: err.message };
214 |   }
215 | }
216 |
217 | // ============================================================================
218 | // 3. Assignments & Submissions Actions
219 | // ============================================================================
220 |
221 | export async function getAssignmentsAction(): Promise<ServerActionResponse<any[]>> {
222 |   if (!isSupabaseConfigured()) {
223 |     return {
224 |       success: true,
225 |       data: [
226 |         {
227 |           id: "asg-1",
228 |           title: "Virtual Memory Paging & Replacement Simulation",
229 |           description: "Implement FIFO, LRU, and Optimal page replacement algorithms in C++/Python with c
    |   omparison graphs.",
230 |           due_date: "2026-10-18T23:59:00Z",
231 |           max_marks: 50,
232 |           status: "pending",
233 |           course: { code: "CS603", title: "Operating Systems" },
234 |           faculty: { full_name: "Dr. Nair" },
235 |         },
236 |       ],
237 |     };
238 |   }
239 |
240 |   const supabase = await createClient();
241 |   const { data, error } = await supabase
242 |     .from("assignments")
243 |     .select(`
244 |       id,
245 |       title,
246 |       description,
247 |       due_date,
248 |       max_marks,
249 |       file_url,
250 |       course:courses (code, title),
251 |       faculty:profiles (full_name)
252 |     `)
253 |     .order("due_date", { ascending: true });
254 |
255 |   if (error) return { success: false, message: error.message };
256 |   return { success: true, data: data || [] };
257 | }
258 |
```

```
export async function createAssignmentAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = assignmentSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
    if (isMock) {
      revalidatePath("/assignments");
      return { success: true, message: "Assignment created (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase.from("assignments").insert({
      course_id: parsed.data.courseId,
      title: parsed.data.title,
      description: parsed.data.description,
      due_date: parsed.data.dueDate,
      max_marks: parsed.data.maxMarks,
      file_url: parsed.data.fileUrl || null,
      created_by: user.id,
    }).select().single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/assignments");
    return { success: true, message: "Assignment published successfully!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function submitAssignmentDeliverableAction(
  assignmentId: string,
  fileUrl: string,
  notes?: string
): Promise<ServerActionResponse> {
  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles(["student"]);
    if (isMock) {
      revalidatePath("/assignments");
      return { success: true, message: "Assignment deliverable submitted (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase
      .from("assignment_submissions")
      .upsert(
        {
          assignment_id: assignmentId,
          student_id: user.id,
          file_url: fileUrl,
          submission_notes: notes,
          status: "submitted",
          submitted_at: new Date().toISOString(),
        },
        { onConflict: "assignment_id,student_id" }
      )
      .select()
      .single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/assignments");
    return { success: true, message: "Deliverable uploaded & receipt confirmed!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function gradeSubmissionAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = submissionGradeSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
    if (isMock) {
```

```
339 |       revalidatePath("/assignments");
340 |       return { success: true, message: "Submission graded (Demo Mode)" };
341 |     }
342 |
343 |     const supabase = await createClient();
344 |     const { error } = await supabase
345 |       .from("assignment_submissions")
346 |       .update({
347 |         marks_received: parsed.data.marksReceived,
348 |         feedback: parsed.data.feedback,
349 |         status: "graded",
350 |         graded_at: new Date().toISOString(),
351 |       })
352 |       .eq("id", parsed.data.submissionId);
353 |
354 |     if (error) return { success: false, message: error.message };
355 |
356 |     revalidatePath("/assignments");
357 |     return { success: true, message: "Grade committed to student record." };
358 |   } catch (err: any) {
359 |     return { success: false, message: err.message };
360 |   }
361 | }
362 |
363 | // ============================================================================
364 | // 4. Study Materials Actions
365 | // ============================================================================
366 |
367 | export async function getStudyMaterialsAction(courseId?: string): Promise<ServerActionResponse<any[]>> {
368 |   if (!isSupabaseConfigured()) {
369 |     return {
370 |       success: true,
371 |       data: [
372 |         {
373 |           id: "m-1",
374 |           title: "Unit 3: Relational Calculus & Query Optimization Slides",
375 |           description: "Comprehensive lecture slides for relational calculus and indexing.",
376 |           file_url: "/docs/dbms-unit-3.pdf",
377 |           file_size: "4.8 MB",
378 |           format: "PDF Presentation",
379 |           download_count: 142,
380 |           course: { code: "CS602", title: "Database Management Systems" },
381 |           faculty: { full_name: "Dr. Rajesh Varma" },
382 |         },
383 |       ],
384 |     };
385 |   }
386 |
387 |   const supabase = await createClient();
388 |   let query = supabase.from("study_materials").select(`
389 |     id,
390 |     title,
391 |     description,
392 |     file_url,
393 |     file_size,
394 |     format,
395 |     download_count,
396 |     created_at,
397 |     course:courses (code, title),
398 |     faculty:profiles (full_name)
399 |   `).order("created_at", { ascending: false });
400 |
401 |   if (courseId) {
402 |     query = query.eq("course_id", courseId);
403 |   }
404 |
405 |   const { data, error } = await query;
406 |   if (error) return { success: false, message: error.message };
407 |   return { success: true, data: data || [] };
408 | }
409 |
410 | export async function createStudyMaterialAction(formData: unknown): Promise<ServerActionResponse> {
411 |   const parsed = studyMaterialSchema.safeParse(formData);
412 |   if (!parsed.success) {
413 |     return { success: false, errors: parsed.error.flatten().fieldErrors };
414 |   }
415 |
416 |   try {
417 |     const { user, isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
418 |     if (isMock) {
```

```
419 |       revalidatePath("/materials");
420 |       return { success: true, message: "Material uploaded (Demo Mode)" };
421 |     }
422 |
423 |     const supabase = await createClient();
424 |     const { data, error } = await supabase.from("study_materials").insert({
425 |       course_id: parsed.data.courseId,
426 |       title: parsed.data.title,
427 |       description: parsed.data.description,
428 |       file_url: parsed.data.fileUrl,
429 |       file_size: parsed.data.fileSize,
430 |       format: parsed.data.format,
431 |       uploaded_by: user.id,
432 |     }).select().single();
433 |
434 |     if (error) return { success: false, message: error.message };
435 |
436 |     revalidatePath("/materials");
437 |     return { success: true, message: "Study material published to course repo!", data };
438 |   } catch (err: any) {
439 |     return { success: false, message: err.message };
440 |   }
441 | }
442 |
443 | export async function deleteStudyMaterialAction(id: string): Promise<ServerActionResponse> {
444 |   try {
445 |     const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
446 |     if (isMock) {
447 |       revalidatePath("/materials");
448 |       return { success: true, message: "Study material deleted (Demo Mode)" };
449 |     }
450 |
451 |     const supabase = await createClient();
452 |     const { error } = await supabase.from("study_materials").delete().eq("id", id);
453 |     if (error) return { success: false, message: error.message };
454 |
455 |     revalidatePath("/materials");
456 |     return { success: true, message: "Material removed." };
457 |   } catch (err: any) {
458 |     return { success: false, message: err.message };
459 |   }
460 | }
```

File #21: src/actions/announcements.ts

256 lines

```
  1 | "use server";
  2 |
  3 | import { createClient } from "@/lib/supabase/server";
  4 | import { isSupabaseConfigured } from "@/lib/env";
  5 | import { revalidatePath } from "next/cache";
  6 | import { announcementSchema, eventSchema } from "@/lib/validations/campus";
  7 | import { getAuthenticatedUserWithRoles, ServerActionResponse } from "./base";
  8 |
  9 | // ============================================================================
 10 | // Announcements CRUD
 11 | // ============================================================================
 12 |
 13 | export async function getAnnouncementsAction(): Promise<ServerActionResponse<any[]>> {
 14 |   if (!isSupabaseConfigured()) {
 15 |     return {
 16 |       success: true,
 17 |       data: [
 18 |         {
 19 |           id: "ann-1",
 20 |           title: "End-Semester Examination Schedule Fall 2026",
 21 |           content: "The Controller of Examinations has published the final timetable for theory and labor
    |   atory assessments. Consult department notice board for batch allocations.",
 22 |           category: "Examinations",
 23 |           urgency: "important",
 24 |           pinned: true,
 25 |           created_at: new Date().toISOString(),
 26 |         },
 27 |         {
 28 |           id: "ann-2",
 29 |           title: "Annual University Innovation & Tech Hackathon",
 30 |           content: "Registrations are now open for HackCampus 2026. Teams can submit problem statement ch
    |   oices across AI, IoT, and Cloud Computing.",
 31 |           category: "Events",
 32 |           urgency: "normal",
```

```
          pinned: false,
          created_at: new Date().toISOString(),
        },
      ],
    };
  }

  const supabase = await createClient();
  const { data, error } = await supabase
    .from("announcements")
    .select("*, published_by(full_name, role)")
    .order("pinned", { ascending: false })
    .order("created_at", { ascending: false });

  if (error) {
    return { success: false, message: error.message };
  }
  return { success: true, data: data || [] };
}

export async function createAnnouncementAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = announcementSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
    if (isMock) {
      revalidatePath("/announcements");
      return { success: true, message: "Announcement published (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase.from("announcements").insert({
      title: parsed.data.title,
      content: parsed.data.content,
      category: parsed.data.category,
      urgency: parsed.data.urgency,
      target_audience: parsed.data.targetAudience,
      pinned: parsed.data.pinned,
      published_by: user.id,
    }).select().single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/announcements");
    return { success: true, message: "Announcement broadcasted successfully!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function updateAnnouncementAction(id: string, formData: unknown): Promise<ServerActionRespon
  se> {
  const parsed = announcementSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
    if (isMock) {
      revalidatePath("/announcements");
      return { success: true, message: "Announcement updated (Demo Mode)" };
    }

    const supabase = await createClient();
    const { error } = await supabase
      .from("announcements")
      .update({
        title: parsed.data.title,
        content: parsed.data.content,
        category: parsed.data.category,
        urgency: parsed.data.urgency,
        target_audience: parsed.data.targetAudience,
        pinned: parsed.data.pinned,
      })
      .eq("id", id);

```

```
112 |     if (error) return { success: false, message: error.message };
113 |
114 |     revalidatePath("/announcements");
115 |     return { success: true, message: "Announcement updated successfully!" };
116 |   } catch (err: any) {
117 |     return { success: false, message: err.message };
118 |   }
119 | }
120 |
121 | export async function deleteAnnouncementAction(id: string): Promise<ServerActionResponse> {
122 |   try {
123 |     const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
124 |     if (isMock) {
125 |       revalidatePath("/announcements");
126 |       return { success: true, message: "Announcement deleted (Demo Mode)" };
127 |     }
128 |
129 |     const supabase = await createClient();
130 |     const { error } = await supabase.from("announcements").delete().eq("id", id);
131 |     if (error) return { success: false, message: error.message };
132 |
133 |     revalidatePath("/announcements");
134 |     return { success: true, message: "Announcement deleted." };
135 |   } catch (err: any) {
136 |     return { success: false, message: err.message };
137 |   }
138 | }
139 |
140 | // ============================================================================
141 | // Events CRUD
142 | // ============================================================================
143 |
144 | export async function getEventsAction(): Promise<ServerActionResponse<any[]>> {
145 |   if (!isSupabaseConfigured()) {
146 |     return {
147 |       success: true,
148 |       data: [
149 |         {
150 |           id: "evt-1",
151 |           title: "National Conference on Cloud Systems & AI",
152 |           description: "Keynote addresses and paper presentations covering Edge Computing, Kubernetes orc
    |   hestration, and Generative Models.",
153 |           venue: "Central Auditorium",
154 |           organizer: "Dept of Computer Science",
155 |           start_time: "2026-10-24T09:00:00Z",
156 |           end_time: "2026-10-25T17:00:00Z",
157 |           category: "Academic",
158 |           rsvp_count: 140,
159 |         },
160 |         {
161 |           id: "evt-2",
162 |           title: "Campus Career Placement Showcase 2026",
163 |           description: "Top multinational tech companies and innovative startups recruiting final year un
    |   dergraduates and postgraduates.",
164 |           venue: "Plaza Exhibition Hall",
165 |           organizer: "Placement & Training Cell",
166 |           start_time: "2026-11-02T10:00:00Z",
167 |           end_time: "2026-11-04T18:00:00Z",
168 |           category: "Career",
169 |           rsvp_count: 320,
170 |         },
171 |       ],
172 |     };
173 |   }
174 |
175 |   const supabase = await createClient();
176 |   const { data, error } = await supabase
177 |     .from("events")
178 |     .select("*")
179 |     .order("start_time", { ascending: true });
180 |
181 |   if (error) return { success: false, message: error.message };
182 |   return { success: true, data: data || [] };
183 | }
184 |
185 | export async function createEventAction(formData: unknown): Promise<ServerActionResponse> {
186 |   const parsed = eventSchema.safeParse(formData);
187 |   if (!parsed.success) {
188 |     return { success: false, errors: parsed.error.flatten().fieldErrors };
189 |   }
```

```
190 |
191 |   try {
192 |     const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
193 |     if (isMock) {
194 |       revalidatePath("/events");
195 |       return { success: true, message: "Event scheduled (Demo Mode)" };
196 |     }
197 |
198 |     const supabase = await createClient();
199 |     const { data, error } = await supabase.from("events").insert({
200 |       title: parsed.data.title,
201 |       description: parsed.data.description,
202 |       start_time: parsed.data.startTime,
203 |       end_time: parsed.data.endTime,
204 |       venue: parsed.data.venue,
205 |       organizer: parsed.data.organizer,
206 |       category: parsed.data.category,
207 |     }).select().single();
208 |
209 |     if (error) return { success: false, message: error.message };
210 |
211 |     revalidatePath("/events");
212 |     return { success: true, message: "Event successfully scheduled!", data };
213 |   } catch (err: any) {
214 |     return { success: false, message: err.message };
215 |   }
216 | }
217 |
218 | export async function deleteEventAction(id: string): Promise<ServerActionResponse> {
219 |   try {
220 |     const { isMock } = await getAuthenticatedUserWithRoles(["faculty", "admin"]);
221 |     if (isMock) {
222 |       revalidatePath("/events");
223 |       return { success: true, message: "Event deleted (Demo Mode)" };
224 |     }
225 |
226 |     const supabase = await createClient();
227 |     const { error } = await supabase.from("events").delete().eq("id", id);
228 |     if (error) return { success: false, message: error.message };
229 |
230 |     revalidatePath("/events");
231 |     return { success: true, message: "Event deleted." };
232 |   } catch (err: any) {
233 |     return { success: false, message: err.message };
234 |   }
235 | }
236 |
237 | export async function rsvpEventAction(id: string): Promise<ServerActionResponse> {
238 |   try {
239 |     await getAuthenticatedUserWithRoles(); // Any authenticated role can RSVP
240 |     if (!isSupabaseConfigured()) {
241 |       return { success: true, message: "RSVP confirmed! Seat registered." };
242 |     }
243 |
244 |     const supabase = await createClient();
245 |     const { error } = await (supabase as any).rpc("increment_event_rsvp", { event_id: id });
246 |     if (error) {
247 |       // Fallback update
248 |       await supabase.from("events").update({ rsvp_count: 1 } as any).eq("id", id);
249 |     }
250 |
251 |     revalidatePath("/events");
252 |     return { success: true, message: "RSVP confirmed! Ticket sent to notification center." };
253 |   } catch (err: any) {
254 |     return { success: false, message: err.message };
255 |   }
256 | }
```

File #22: src/actions/auth.ts

242 lines

```
1 | "use server";
2 |
3 | import { createClient } from "@/lib/supabase/server";
4 | import {
5 |   loginSchema,
6 |   registerSchema,
7 |   forgotPasswordSchema,
8 |   resetPasswordSchema,
9 | } from "@/lib/validations/auth";
```

```
import { isSupabaseConfigured } from "@/lib/env";
import { redirect } from "next/navigation";

export interface ActionResult<T = unknown> {
  success: boolean;
  message?: string;
  data?: T;
  errors?: Record<string, string[]>;
}

/**
 * Real Supabase Auth Login Action
 */
export async function loginAction(values: unknown): Promise<ActionResult<{ role?: string; redirectUrl?: s
  tring }>> {
  const parsed = loginSchema.safeParse(values);
  if (!parsed.success) {
    return {
      success: false,
      message: "Validation failed",
      errors: parsed.error.flatten().fieldErrors,
    };
  }

  if (!isSupabaseConfigured()) {
    // Graceful offline mock response during local inspection without active credentials
    return {
      success: true,
      message: "Demo Mode Authenticated (Supabase credentials not configured in .env.local)",
      data: { role: "student", redirectUrl: "/dashboard/student" },
    };
  }

  const supabase = await createClient();
  const { data, error } = await supabase.auth.signInWithPassword({
    email: parsed.data.email,
    password: parsed.data.password,
  });

  if (error) {
    return {
      success: false,
      message: error.message,
    };
  }

  // Fetch verified user role from profiles table (server-side verification)
  const { data: profile } = await supabase
    .from("profiles")
    .select("role")
    .eq("id", data.user.id)
    .single();

  const userRole = (profile as { role?: string } | null)?.role || "student";
  const redirectUrl =
    userRole === "admin"
      ? "/dashboard/admin"
      : userRole === "faculty"
      ? "/dashboard/faculty"
      : "/dashboard/student";

  return {
    success: true,
    message: "Logged in successfully",
    data: { role: userRole, redirectUrl },
  };
}

/**
 * Real Supabase Auth Register Action
 */
export async function registerAction(values: unknown): Promise<ActionResult<{ role?: string }>> {
  const parsed = registerSchema.safeParse(values);
  if (!parsed.success) {
    return {
      success: false,
      message: "Validation failed",
      errors: parsed.error.flatten().fieldErrors,
    };
  }
```

```
 89 |
 90 |   if (!isSupabaseConfigured()) {
 91 |     return {
 92 |       success: true,
 93 |       message: "Demo Account Staged (Provide real Supabase credentials in .env.local for live persistence
    |   )",
 94 |       data: { role: parsed.data.role },
 95 |     };
 96 |   }
 97 |
 98 |   const supabase = await createClient();
 99 |   const { data, error } = await supabase.auth.signUp({
100 |     email: parsed.data.email,
101 |     password: parsed.data.password,
102 |     options: {
103 |       data: {
104 |         full_name: parsed.data.fullName,
105 |         role: parsed.data.role,
106 |         department: parsed.data.department,
107 |         enrollment_no: parsed.data.enrollmentNo || null,
108 |         phone: parsed.data.phone || null,
109 |       },
110 |     },
111 |   });
112 |
113 |   if (error) {
114 |     return {
115 |       success: false,
116 |       message: error.message,
117 |     };
118 |   }
119 |
120 |   return {
121 |     success: true,
122 |     message: "Account created successfully! Check your email if confirmation is required.",
123 |     data: { role: parsed.data.role },
124 |   };
125 | }
126 |
127 | /**
128 |  * Real Supabase Auth Forgot Password Action
129 |  */
130 | export async function forgotPasswordAction(values: unknown): Promise<ActionResult> {
131 |   const parsed = forgotPasswordSchema.safeParse(values);
132 |   if (!parsed.success) {
133 |     return {
134 |       success: false,
135 |       message: "Please enter a valid email address",
136 |     };
137 |   }
138 |
139 |   if (!isSupabaseConfigured()) {
140 |     return {
141 |       success: true,
142 |       message: "Password recovery link dispatched (Demo mode)",
143 |     };
144 |   }
145 |
146 |   const supabase = await createClient();
147 |   const { error } = await supabase.auth.resetPasswordForEmail(parsed.data.email, {
148 |     redirectTo: `${process.env.NEXT_PUBLIC_APP_URL || "http://localhost:3000"}/reset-password`,
149 |   });
150 |
151 |   if (error) {
152 |     return {
153 |       success: false,
154 |       message: error.message,
155 |     };
156 |   }
157 |
158 |   return {
159 |     success: true,
160 |     message: "If an account exists with this email, a password reset link has been dispatched.",
161 |   };
162 | }
163 |
164 | /**
165 |  * Real Supabase Auth Update Password Action
166 |  */
167 | export async function updatePasswordAction(values: unknown): Promise<ActionResult> {
```

```
  const parsed = resetPasswordSchema.safeParse(values);
  if (!parsed.success) {
    return {
      success: false,
      message: "Validation failed",
      errors: parsed.error.flatten().fieldErrors,
    };
  }

  if (!isSupabaseConfigured()) {
    return {
      success: true,
      message: "Password updated successfully (Demo mode)",
    };
  }

  const supabase = await createClient();
  const { error } = await supabase.auth.updateUser({
    password: parsed.data.password,
  });

  if (error) {
    return {
      success: false,
      message: error.message,
    };
  }

  return {
    success: true,
    message: "Your password has been successfully updated.",
  };
}

/**
 * Sign out current user session
 */
export async function signOutAction() {
  if (isSupabaseConfigured()) {
    const supabase = await createClient();
    await supabase.auth.signOut();
  }
  redirect("/login");
}

/**
 * Get current authenticated user and profile securely from server
 */
export async function getCurrentUser() {
  if (!isSupabaseConfigured()) {
    return null;
  }

  try {
    const supabase = await createClient();
    const {
      data: { user },
    } = await supabase.auth.getUser();

    if (!user) return null;

    const { data: profile } = await supabase
      .from("profiles")
      .select("*")
      .eq("id", user.id)
      .single();

    return {
      user,
      profile,
    };
  } catch {
    return null;
  }
}
```

File #23: src/actions/base.ts

49 lines

```
"use server";
```

```

import { createClient } from "@/lib/supabase/server";
import { isSupabaseConfigured } from "@/lib/env";
import { revalidatePath } from "next/cache";

export interface ServerActionResponse<T = unknown> {
  success: boolean;
  message?: string;
  data?: T;
  errors?: Record<string, string[]>;
}

/**
 * Helper to securely get current user & profile and check authorized roles
 */
export async function getAuthenticatedUserWithRoles(allowedRoles?: Array<"student" | "faculty" | "admin">
  ) {
  if (!isSupabaseConfigured()) {
    // Development fallback mock
    return {
      user: { id: "mock-user-id", email: "demo@campus.edu" },
      profile: { id: "mock-user-id", full_name: "Demo User", role: "admin" },
      isMock: true,
    };
  }

  const supabase = await createClient();
  const {
    data: { user },
  } = await supabase.auth.getUser();

  if (!user) {
    throw new Error("Unauthorized. Please log in to perform this action.");
  }

  const { data: profile } = await supabase
    .from("profiles")
    .select("*")
    .eq("id", user.id)
    .single();

  const userRole = (profile as { role?: "student" | "faculty" | "admin" } | null)?.role || "student";

  if (allowedRoles && !allowedRoles.includes(userRole)) {
    throw new Error(`Forbidden: Action requires one of [${allowedRoles.join(", ")}] permissions.`);
  }

  return { user, profile, isMock: false };
}
```

File #24: src/actions/operations.ts 414 lines

```

"use server";

import { createClient } from "@/lib/supabase/server";
import { isSupabaseConfigured } from "@/lib/env";
import { revalidatePath } from "next/cache";
import {
  facilitySchema,
  facilityBookingSchema,
  emergencyContactSchema,
  notificationSchema,
  feedbackSchema,
  feedbackResolutionSchema,
} from "@/lib/validations/campus";
import { getAuthenticatedUserWithRoles, ServerActionResponse } from "./base";

// ============================================================================
// 1. Campus Facilities & Bookings CRUD
// ============================================================================

export async function getFacilitiesAction(): Promise<ServerActionResponse<any[]>> {
  if (!isSupabaseConfigured()) {
    return {
      success: true,
      data: [
        {
          id: "fac-1",
          name: "Turing Advanced Computing Lab",
```

```
          category: "Specialized Lab",
          location: "Tech Block A • 3rd Floor",
          capacity: 65,
          operating_hours: "08:00 AM - 10:00 PM",
          status: "available",
          description: "Equipped with 65 high-end GPU workstations for Machine Learning, Deep Learning, a
  nd Cloud Virtualization.",
        },
        {
          id: "fac-2",
          name: "Sir C.V. Raman Central Auditorium",
          category: "Auditorium",
          location: "Central Campus Plaza",
          capacity: 850,
          operating_hours: "09:00 AM - 09:00 PM",
          status: "booked",
          description: "Acoustically treated auditorium with Dolby surround audio, 4K digital projection,
   and dual green rooms.",
        },
      ],
    };
  }

  const supabase = await createClient();
  const { data, error } = await supabase
    .from("campus_facilities")
    .select("*")
    .order("name", { ascending: true });

  if (error) return { success: false, message: error.message };
  return { success: true, data: data || [] };
}

export async function createFacilityAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = facilitySchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { isMock } = await getAuthenticatedUserWithRoles(["admin"]);
    if (isMock) {
      revalidatePath("/facilities");
      return { success: true, message: "Facility registered (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase.from("campus_facilities").insert({
      name: parsed.data.name,
      category: parsed.data.category,
      location: parsed.data.location,
      capacity: parsed.data.capacity,
      operating_hours: parsed.data.operatingHours,
      status: parsed.data.status,
      description: parsed.data.description,
    }).select().single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/facilities");
    return { success: true, message: "Facility successfully provisioned!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function bookFacilityAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = facilityBookingSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles();
    if (isMock) {
      revalidatePath("/facilities");
      return { success: true, message: "Booking permit submitted (Demo Mode)" };
    }

    const supabase = await createClient();
```

```
    const { data, error } = await supabase.from("facility_bookings").insert({
      facility_id: parsed.data.facilityId,
      booked_by: user.id,
      booking_date: parsed.data.bookingDate,
      time_slot: parsed.data.timeSlot,
      purpose: parsed.data.purpose,
      status: "approved",
    }).select().single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/facilities");
    return { success: true, message: "Permit approved & schedule slot reserved!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

// ============================================================================
// 2. Emergency Contacts CRUD
// ============================================================================

export async function getEmergencyContactsAction(): Promise<ServerActionResponse<any[]>> {
  if (!isSupabaseConfigured()) {
    return {
      success: true,
      data: [
        {
          id: "sos-1",
          name: "Campus Security Control Room",
          designation: "Chief Security Warden",
          phone: "+1 (800) 555-0911",
          alternate_phone: "+1 (800) 555-0912",
          available_hours: "24/7",
          category: "Security",
          priority: 1,
        },
        {
          id: "sos-2",
          name: "University Health Center & Ambulance",
          designation: "Chief Medical Officer",
          phone: "+1 (800) 555-0108",
          alternate_phone: "+1 (800) 555-0109",
          available_hours: "24/7",
          category: "Medical",
          priority: 2,
        },
        {
          id: "sos-3",
          name: "Women Safety & Anti-Ragging Cell",
          designation: "Convenor & Senior Dean",
          phone: "+1 (800) 555-0181",
          available_hours: "24/7",
          category: "Helpline",
          priority: 3,
        },
      ],
    };
  }

  const supabase = await createClient();
  const { data, error } = await supabase
    .from("emergency_contacts")
    .select("*")
    .order("priority", { ascending: true });

  if (error) return { success: false, message: error.message };
  return { success: true, data: data || [] };
}

export async function createEmergencyContactAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = emergencyContactSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { isMock } = await getAuthenticatedUserWithRoles(["admin"]);
    if (isMock) {
      revalidatePath("/emergency");
```

```
      return { success: true, message: "SOS contact saved (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase.from("emergency_contacts").insert({
      name: parsed.data.name,
      designation: parsed.data.designation,
      phone: parsed.data.phone,
      alternate_phone: parsed.data.alternatePhone,
      available_hours: parsed.data.availableHours,
      category: parsed.data.category,
      priority: parsed.data.priority,
    }).select().single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/emergency");
    return { success: true, message: "SOS hotline configured!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

// ============================================================================
// 3. Notifications CRUD
// ============================================================================

export async function getNotificationsAction(): Promise<ServerActionResponse<any[]>> {
  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles();
    if (isMock) {
      return {
        success: true,
        data: [
          {
            id: "notif-1",
            title: "Assignment Graded: Data Structures Lab 3",
            message: "Prof. Vivek Sharma has reviewed and graded your shortest path graph implementation.
   You scored 38/40.",
            is_read: false,
            type: "academic",
            created_at: new Date(Date.now() - 3600000 * 2).toISOString(),
          },
          {
            id: "notif-2",
            title: "Class Rescheduled: Computer Networks",
            message: "The lecture scheduled for 02:00 PM today has been relocated to Seminar Room 204.",
            is_read: false,
            type: "schedule",
            created_at: new Date(Date.now() - 3600000 * 4).toISOString(),
          },
        ],
      };
    }

    const supabase = await createClient();
    const { data, error } = await supabase
      .from("notifications")
      .select("*")
      .eq("user_id", user.id)
      .order("created_at", { ascending: false });

    if (error) return { success: false, message: error.message };
    return { success: true, data: data || [] };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function markNotificationReadAction(id: string): Promise<ServerActionResponse> {
  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles();
    if (isMock) {
      revalidatePath("/notifications");
      return { success: true };
    }

    const supabase = await createClient();
    const { error } = await supabase
      .from("notifications")
```

```
265 |       .update({ is_read: true })
266 |       .eq("id", id)
267 |       .eq("user_id", user.id);
268 |
269 |     if (error) return { success: false, message: error.message };
270 |
271 |     revalidatePath("/notifications");
272 |     return { success: true };
273 |   } catch (err: any) {
274 |     return { success: false, message: err.message };
275 |   }
276 | }
277 |
278 | export async function markAllNotificationsReadAction(): Promise<ServerActionResponse> {
279 |   try {
280 |     const { user, isMock } = await getAuthenticatedUserWithRoles();
281 |     if (isMock) {
282 |       revalidatePath("/notifications");
283 |       return { success: true, message: "All marked as read" };
284 |     }
285 |
286 |     const supabase = await createClient();
287 |     const { error } = await supabase
288 |       .from("notifications")
289 |       .update({ is_read: true })
290 |       .eq("user_id", user.id);
291 |
292 |     if (error) return { success: false, message: error.message };
293 |
294 |     revalidatePath("/notifications");
295 |     return { success: true, message: "All notifications marked as read." };
296 |   } catch (err: any) {
297 |     return { success: false, message: err.message };
298 |   }
299 | }
300 |
301 | // ============================================================================
302 | // 4. Feedback & Grievances CRUD
303 | // ============================================================================
304 |
305 | export async function getFeedbackAction(): Promise<ServerActionResponse<any[]>> {
306 |   try {
307 |     const { user, profile, isMock } = await getAuthenticatedUserWithRoles();
308 |     if (isMock) {
309 |       return {
310 |         success: true,
311 |         data: [
312 |           {
313 |             id: "fb-1",
314 |             category: "Infrastructure",
315 |             subject: "Air conditioning in Turing Lab 302 not functioning properly",
316 |             description: "Workstations generate excessive thermal output during GPU compilation. Room tem
    |   perature exceeds 30C.",
317 |             status: "under_review",
318 |             is_anonymous: false,
319 |             created_at: new Date(Date.now() - 86400000).toISOString(),
320 |           },
321 |         ],
322 |       };
323 |     }
324 |
325 |     const supabase = await createClient();
326 |     const userRole = (profile as any)?.role || "student";
327 |
328 |     let query = supabase.from("feedback").select(`
329 |       id,
330 |       category,
331 |       subject,
332 |       description,
333 |       status,
334 |       is_anonymous,
335 |       admin_notes,
336 |       created_at,
337 |       submitted_by:profiles (full_name, email)
338 |     `).order("created_at", { ascending: false });
339 |
340 |     // Students only view their own feedback; Admins view all
341 |     if (userRole !== "admin") {
342 |       query = query.eq("submitted_by", user.id);
343 |     }
```

```
344 |
345 |     const { data, error } = await query;
346 |     if (error) return { success: false, message: error.message };
347 |     return { success: true, data: data || [] };
348 |   } catch (err: any) {
349 |     return { success: false, message: err.message };
350 |   }
351 | }
352 |
353 | export async function submitFeedbackAction(formData: unknown): Promise<ServerActionResponse> {
354 |   const parsed = feedbackSchema.safeParse(formData);
355 |   if (!parsed.success) {
356 |     return { success: false, errors: parsed.error.flatten().fieldErrors };
357 |   }
358 |
359 |   try {
360 |     const { user, isMock } = await getAuthenticatedUserWithRoles();
361 |     if (isMock) {
362 |       revalidatePath("/feedback");
363 |       return { success: true, message: "Grievance ticket registered (Demo Mode)" };
364 |     }
365 |
366 |     const supabase = await createClient();
367 |     const { data, error } = await supabase.from("feedback").insert({
368 |       category: parsed.data.category,
369 |       subject: parsed.data.subject,
370 |       description: parsed.data.description,
371 |       is_anonymous: parsed.data.isAnonymous,
372 |       submitted_by: parsed.data.isAnonymous ? null : user.id,
373 |       status: "pending",
374 |     }).select().single();
375 |
376 |     if (error) return { success: false, message: error.message };
377 |
378 |     revalidatePath("/feedback");
379 |     return { success: true, message: "Grievance ticket logged with campus administration!", data };
380 |   } catch (err: any) {
381 |     return { success: false, message: err.message };
382 |   }
383 | }
384 |
385 | export async function resolveFeedbackAction(formData: unknown): Promise<ServerActionResponse> {
386 |   const parsed = feedbackResolutionSchema.safeParse(formData);
387 |   if (!parsed.success) {
388 |     return { success: false, errors: parsed.error.flatten().fieldErrors };
389 |   }
390 |
391 |   try {
392 |     const { isMock } = await getAuthenticatedUserWithRoles(["admin"]);
393 |     if (isMock) {
394 |       revalidatePath("/feedback");
395 |       return { success: true, message: "Feedback resolved (Demo Mode)" };
396 |     }
397 |
398 |     const supabase = await createClient();
399 |     const { error } = await supabase
400 |       .from("feedback")
401 |       .update({
402 |         status: parsed.data.status,
403 |         admin_notes: parsed.data.adminNotes,
404 |       })
405 |       .eq("id", parsed.data.feedbackId);
406 |
407 |     if (error) return { success: false, message: error.message };
408 |
409 |     revalidatePath("/feedback");
410 |     return { success: true, message: `Feedback ticket status updated to ${parsed.data.status}.` };
411 |   } catch (err: any) {
412 |     return { success: false, message: err.message };
413 |   }
414 | }
```

File #25: src/actions/profiles.ts

210 lines

```
1 | "use server";
2 |
3 | import { createClient } from "@/lib/supabase/server";
4 | import { isSupabaseConfigured } from "@/lib/env";
5 | import { revalidatePath } from "next/cache";
```

```
import {
  profileSchema,
  studentProfileSchema,
  facultyProfileSchema,
} from "@/lib/validations/campus";
import { getAuthenticatedUserWithRoles, ServerActionResponse } from "./base";

// ============================================================================
// 1. Profile Management (Read, Update)
// ============================================================================

export async function getUserProfileDetailsAction(userId?: string): Promise<ServerActionResponse<any>> {
  try {
    const { user, profile: authProfile, isMock } = await getAuthenticatedUserWithRoles();
    if (isMock) {
      return {
        success: true,
        data: {
          id: "mock-user-id",
          full_name: "Alex Morgan",
          email: "alex.m@student.campus.edu",
          role: "student",
          department: "Computer Science & Engineering",
          phone: "+1 (555) 234-5678",
          student_profile: {
            enrollment_no: "24CS001",
            current_semester: 6,
            admission_year: 2024,
            cgpa: 3.82,
            attendance_aggregate: 88.08,
            academic_standing: "Honor Roll",
          },
        },
      };
    }

    const supabase = await createClient();
    const targetId = userId || user.id;

    const { data: baseProfile, error: baseErr } = await supabase
      .from("profiles")
      .select("*")
      .eq("id", targetId)
      .single();

    if (baseErr) return { success: false, message: baseErr.message };

    const profileObj = baseProfile as Record<string, any>;
    let studentProfile = null;
    let facultyProfile = null;

    if (profileObj?.role === "student") {
      const { data: sData } = await supabase
        .from("student_profiles")
        .select("*")
        .eq("id", targetId)
        .single();
      studentProfile = sData;
    } else if (profileObj?.role === "faculty") {
      const { data: fData } = await supabase
        .from("faculty_profiles")
        .select("*")
        .eq("id", targetId)
        .single();
      facultyProfile = fData;
    }

    return {
      success: true,
      data: {
        ...profileObj,
        student_profile: studentProfile,
        faculty_profile: facultyProfile,
      },
    };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

```

```
export async function updateBaseProfileAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = profileSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles();
    if (isMock) {
      revalidatePath("/profile");
      return { success: true, message: "Profile updated (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase
      .from("profiles")
      .update({
        full_name: parsed.data.fullName,
        department: parsed.data.department,
        phone: parsed.data.phone,
        avatar_url: parsed.data.avatarUrl || null,
        updated_at: new Date().toISOString(),
      })
      .eq("id", user.id)
      .select()
      .single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/profile");
    return { success: true, message: "Profile successfully updated!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

export async function updateStudentProfileAction(formData: unknown): Promise<ServerActionResponse> {
  const parsed = studentProfileSchema.safeParse(formData);
  if (!parsed.success) {
    return { success: false, errors: parsed.error.flatten().fieldErrors };
  }

  try {
    const { user, isMock } = await getAuthenticatedUserWithRoles(["student", "admin"]);
    if (isMock) {
      revalidatePath("/profile");
      return { success: true, message: "Student record updated (Demo Mode)" };
    }

    const supabase = await createClient();
    const { data, error } = await supabase
      .from("student_profiles")
      .upsert({
        id: user.id,
        enrollment_no: parsed.data.enrollmentNo,
        current_semester: parsed.data.currentSemester,
        admission_year: parsed.data.admissionYear,
        academic_standing: parsed.data.academicStanding,
        updated_at: new Date().toISOString(),
      })
      .select()
      .single();

    if (error) return { success: false, message: error.message };

    revalidatePath("/profile");
    return { success: true, message: "Academic enrollment updated!", data };
  } catch (err: any) {
    return { success: false, message: err.message };
  }
}

// ============================================================================
// 2. Admin Faculty & User Directory Actions
// ============================================================================

export async function getFacultyDirectoryAction(): Promise<ServerActionResponse<any[]>> {
  if (!isSupabaseConfigured()) {
    return {
      success: true,
```

```
      data: [
        {
          id: "f-1",
          full_name: "Dr. Rajesh Varma",
          email: "rajesh.v@campus.edu",
          department: "Computer Science & Engineering",
          faculty_profile: {
            designation: "Professor & Head of Department",
            office_location: "Tech Block A • 304",
            office_hours: "Mon & Wed: 03:00 - 05:00 PM",
            research_interests: ["Distributed Databases", "Query Execution", "Data Privacy"],
          },
        },
        {
          id: "f-2",
          full_name: "Prof. Ananya Rao",
          email: "ananya.rao@campus.edu",
          department: "Computer Science & Engineering",
          faculty_profile: {
            designation: "Associate Professor",
            office_location: "Tech Block A • 212",
            office_hours: "Tue & Thu: 02:00 - 04:00 PM",
            research_interests: ["Software Defined Networks", "5G/6G Protocols"],
          },
        },
      ],
    };
  }

  const supabase = await createClient();
  const { data, error } = await supabase
    .from("profiles")
    .select(`
      id,
      full_name,
      email,
      department,
      faculty_profile:faculty_profiles (*)
    `)
    .eq("role", "faculty")
    .order("full_name", { ascending: true });

  if (error) return { success: false, message: error.message };
  return { success: true, data: data || [] };
}
```

File #26: src/components/dashboard/attendance-chart.tsx

91 lines

```
"use client";

import * as React from "react";
import {
  ResponsiveContainer,
  BarChart,
  Bar,
  XAxis,
  YAxis,
  Tooltip,
  CartesianGrid,
} from "recharts";

interface AttendanceChartProps {
  data?: Array<{ subject: string; percentage: number }>;
}

const defaultData = [
  { subject: "Data Structures", percentage: 88 },
  { subject: "Database Sys", percentage: 92 },
  { subject: "Algorithms", percentage: 81 },
  { subject: "Computer Networks", percentage: 76 },
  { subject: "Software Eng", percentage: 95 },
];

export function AttendanceChart({ data = defaultData }: AttendanceChartProps) {
  const [mounted, setMounted] = React.useState(false);

  React.useEffect(() => {
    setMounted(true);
  }, []);
```

```
  |
  |   if (!mounted) {
  |     return (
  |       <div className="h-64 flex items-center justify-center text-xs text-zinc-400">
  |         Loading analytics...
  |       </div>
  |     );
  |   }
  |
  |   return (
  |     <div className="h-64 w-full">
  |       <ResponsiveContainer width="100%" height="100%">
  |         <BarChart data={data} margin={{ top: 10, right: 10, left: -20, bottom: 0 }}>
  |           <CartesianGrid strokeDasharray="3 3" opacity={0.15} />
  |           <XAxis
  |             dataKey="subject"
  |             tick={{ fontSize: 11 }}
  |             tickLine={false}
  |             stroke="#888888"
  |           />
  |           <YAxis
  |             domain={[0, 100]}
  |             tick={{ fontSize: 11 }}
  |             tickLine={false}
  |             stroke="#888888"
  |             unit="%"
  |           />
  |           <Tooltip
  |             content={({ active, payload }) => {
  |               if (active && payload && payload.length) {
  |                 return (
  |                   <div className="rounded-lg border border-zinc-200 bg-white p-2.5 shadow-md dark:border-
  |   zinc-800 dark:bg-zinc-900">
  |                     <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100">
  |                       {payload[0].payload.subject}
  |                     </p>
  |                     <p className="text-xs font-bold text-blue-600 dark:text-blue-400">
  |                       Attendance: {payload[0].value}%
  |                     </p>
  |                   </div>
  |                 );
  |               }
  |               return null;
  |             }}
  |           />
  |           <Bar
  |             dataKey="percentage"
  |             fill="url(#blueGradient)"
  |             radius={[6, 6, 0, 0]}
  |           />
  |           <defs>
  |             <linearGradient id="blueGradient" x1="0" y1="0" x2="0" y2="1">
  |               <stop offset="0%" stopColor="#3b82f6" />
  |               <stop offset="100%" stopColor="#6366f1" />
  |             </linearGradient>
  |           </defs>
  |         </BarChart>
  |       </ResponsiveContainer>
  |     </div>
  |   );
  | }
```

File #27: src/components/dashboard/campus-stats-chart.tsx 96 lines

```
  | "use client";
  |
  | import * as React from "react";
  | import {
  |   ResponsiveContainer,
  |   AreaChart,
  |   Area,
  |   XAxis,
  |   YAxis,
  |   Tooltip,
  |   CartesianGrid,
  | } from "recharts";
  |
  | const activityData = [
  |   { day: "Mon", activeUsers: 1420, submissions: 320 },
```

```
  { day: "Tue", activeUsers: 1890, submissions: 480 },
  { day: "Wed", activeUsers: 2100, submissions: 540 },
  { day: "Thu", activeUsers: 1950, submissions: 410 },
  { day: "Fri", activeUsers: 1780, submissions: 610 },
  { day: "Sat", activeUsers: 920, submissions: 190 },
  { day: "Sun", activeUsers: 640, submissions: 110 },
];

export function CampusActivityChart() {
  const [mounted, setMounted] = React.useState(false);

  React.useEffect(() => {
    setMounted(true);
  }, []);

  if (!mounted) {
    return (
      <div className="h-64 flex items-center justify-center text-xs text-zinc-400">
        Loading analytics...
      </div>
    );
  }

  return (
    <div className="h-64 w-full">
      <ResponsiveContainer width="100%" height="100%">
        <AreaChart data={activityData} margin={{ top: 10, right: 10, left: -15, bottom: 0 }}>
          <defs>
            <linearGradient id="userGrad" x1="0" y1="0" x2="0" y2="1">
              <stop offset="5%" stopColor="#3b82f6" stopOpacity={0.4} />
              <stop offset="95%" stopColor="#3b82f6" stopOpacity={0.0} />
            </linearGradient>
            <linearGradient id="subGrad" x1="0" y1="0" x2="0" y2="1">
              <stop offset="5%" stopColor="#10b981" stopOpacity={0.4} />
              <stop offset="95%" stopColor="#10b981" stopOpacity={0.0} />
            </linearGradient>
          </defs>
          <CartesianGrid strokeDasharray="3 3" opacity={0.12} />
          <XAxis dataKey="day" tick={{ fontSize: 11 }} tickLine={false} stroke="#888888" />
          <YAxis tick={{ fontSize: 11 }} tickLine={false} stroke="#888888" />
          <Tooltip
            content={({ active, payload }) => {
              if (active && payload && payload.length) {
                return (
                  <div className="rounded-lg border border-zinc-200 bg-white p-2.5 shadow-md dark:border-
zinc-800 dark:bg-zinc-900">
                    <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100 mb-1">
                      {payload[0].payload.day}
                    </p>
                    <p className="text-xs text-blue-600 dark:text-blue-400">
                      Active Users: <span className="font-bold">{payload[0].value}</span>
                    </p>
                    <p className="text-xs text-emerald-600 dark:text-emerald-400">
                      Submissions: <span className="font-bold">{payload[1]?.value}</span>
                    </p>
                  </div>
                );
              }
              return null;
            }}
          />
          <Area
            type="monotone"
            dataKey="activeUsers"
            stroke="#3b82f6"
            strokeWidth={2}
            fillOpacity={1}
            fill="url(#userGrad)"
          />
          <Area
            type="monotone"
            dataKey="submissions"
            stroke="#10b981"
            strokeWidth={2}
            fillOpacity={1}
            fill="url(#subGrad)"
          />
        </AreaChart>
      </ResponsiveContainer>
    </div>
```

```
95 |   );
96 | }
```

File #28: src/components/layout/dashboard-shell.tsx 48 lines

```
 1 | import * as React from "react";
 2 | import { Navbar } from "./navbar";
 3 | import { Sidebar } from "./sidebar";
 4 | import { Footer } from "./footer";
 5 |
 6 | interface DashboardShellProps {
 7 |   children: React.ReactNode;
 8 |   title?: string;
 9 |   description?: string;
10 |   actions?: React.ReactNode;
11 | }
12 |
13 | export function DashboardShell({
14 |   children,
15 |   title,
16 |   description,
17 |   actions,
18 | }: DashboardShellProps) {
19 |   return (
20 |     <div className="min-h-screen flex flex-col bg-zinc-50/70 dark:bg-zinc-950 text-zinc-900 dark:text-zin
   |   c-100">
21 |       <Navbar />
22 |       <div className="flex-1 flex max-w-7xl w-full mx-auto">
23 |         <Sidebar />
24 |         <main className="flex-1 p-4 sm:p-6 lg:p-8 overflow-y-auto">
25 |           {(title || actions) && (
26 |             <div className="mb-6 flex flex-col gap-2 sm:flex-row sm:items-center sm:justify-between borde
   |   r-b border-zinc-200/80 pb-5 dark:border-zinc-800/80">
27 |               <div>
28 |                 {title && (
29 |                   <h1 className="text-2xl font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
30 |                     {title}
31 |                   </h1>
32 |                 )}
33 |                 {description && (
34 |                   <p className="mt-1 text-xs text-zinc-500 dark:text-zinc-400">
35 |                     {description}
36 |                   </p>
37 |                 )}
38 |               </div>
39 |               {actions && <div className="flex items-center gap-2">{actions}</div>}
40 |             </div>
41 |           )}
42 |           {children}
43 |         </main>
44 |       </div>
45 |       <Footer />
46 |     </div>
47 |   );
48 | }
```

File #29: src/components/layout/footer.tsx 149 lines

```
 1 | import * as React from "react";
 2 | import Link from "next/link";
 3 | import { GraduationCap, Heart, ExternalLink, ShieldCheck } from "lucide-react";
 4 |
 5 | export function Footer() {
 6 |   return (
 7 |     <footer className="w-full border-t border-zinc-200/80 bg-white/70 py-10 dark:border-zinc-800/80 dark:
   |   bg-zinc-950/70 transition-colors">
 8 |       <div className="mx-auto max-w-7xl px-4 sm:px-6 lg:px-8">
 9 |         <div className="grid grid-cols-1 gap-8 sm:grid-cols-2 md:grid-cols-4 lg:grid-cols-5">
10 |           {/* Brand info */}
11 |           <div className="lg:col-span-2 space-y-3">
12 |             <div className="flex items-center gap-2.5">
13 |               <div className="flex h-8 w-8 items-center justify-center rounded-lg bg-gradient-to-tr from-
   |   blue-600 via-indigo-600 to-violet-600 text-white shadow-sm">
14 |                 <GraduationCap className="h-4 w-4" />
15 |               </div>
16 |               <span className="text-base font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
17 |                 SMART CAMPUS
18 |               </span>
```

```
            </div>
            <p className="max-w-sm text-xs leading-relaxed text-zinc-500 dark:text-zinc-400">
              The next-generation unified academic operations and campus management
              platform. Built with Next.js App Router, Tailwind CSS, and Supabase
              PostgreSQL.
            </p>
            <div className="flex items-center gap-2 pt-2 text-[11px] text-zinc-400">
              <span className="flex h-2 w-2 rounded-full bg-emerald-500 animate-pulse" />
              <span>Campus Services Operational</span>
              <span>•</span>
              <Link href="/api/health" className="hover:underline text-blue-600 dark:text-blue-400">
                Status API
              </Link>
            </div>
          </div>

          {/* Quick Portals */}
          <div className="space-y-2.5">
            <h4 className="text-xs font-semibold tracking-wider text-zinc-900 dark:text-zinc-100 uppercas
e">
              Portals
            </h4>
            <ul className="space-y-1.5 text-xs text-zinc-600 dark:text-zinc-400">
              <li>
                <Link href="/dashboard/student" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Student Portal
                </Link>
              </li>
              <li>
                <Link href="/dashboard/faculty" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Faculty Portal
                </Link>
              </li>
              <li>
                <Link href="/dashboard/admin" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Admin Command
                </Link>
              </li>
              <li>
                <Link href="/login" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Secure Sign In
                </Link>
              </li>
            </ul>
          </div>

          {/* Core Modules */}
          <div className="space-y-2.5">
            <h4 className="text-xs font-semibold tracking-wider text-zinc-900 dark:text-zinc-100 uppercas
e">
              Modules
            </h4>
            <ul className="space-y-1.5 text-xs text-zinc-600 dark:text-zinc-400">
              <li>
                <Link href="/timetable" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Timetable
                </Link>
              </li>
              <li>
                <Link href="/attendance" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Attendance Tracking
                </Link>
              </li>
              <li>
                <Link href="/assignments" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Assignments
                </Link>
              </li>
              <li>
                <Link href="/materials" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Study Materials
                </Link>
              </li>
              <li>
                <Link href="/facilities" className="hover:text-blue-600 dark:hover:text-blue-400">
                  Facilities & Labs
                </Link>
              </li>
            </ul>
          </div>
```

```
 97 |
 98 |           {/* Safety & Help */}
 99 |           <div className="space-y-2.5">
100 |             <h4 className="text-xs font-semibold tracking-wider text-zinc-900 dark:text-zinc-100 uppercas
    |   e">
101 |               Support & Safety
102 |             </h4>
103 |             <ul className="space-y-1.5 text-xs text-zinc-600 dark:text-zinc-400">
104 |               <li>
105 |                 <Link href="/emergency" className="text-rose-600 dark:text-rose-400 font-medium hover:und
    |   erline">
106 |                   Emergency SOS
107 |                 </Link>
108 |               </li>
109 |               <li>
110 |                 <Link href="/feedback" className="hover:text-blue-600 dark:hover:text-blue-400">
111 |                   Student Grievances
112 |                 </Link>
113 |               </li>
114 |               <li>
115 |                 <Link href="/contact" className="hover:text-blue-600 dark:hover:text-blue-400">
116 |                   Contact Helpdesk
117 |                 </Link>
118 |               </li>
119 |               <li>
120 |                 <Link href="/about" className="hover:text-blue-600 dark:hover:text-blue-400">
121 |                   About Platform
122 |                 </Link>
123 |               </li>
124 |             </ul>
125 |           </div>
126 |         </div>
127 |
128 |         {/* Bottom copyright */}
129 |         <div className="mt-8 flex flex-col items-center justify-between gap-4 border-t border-zinc-200/80
    |    pt-6 text-[11px] text-zinc-500 sm:flex-row dark:border-zinc-800/80 dark:text-zinc-400">
130 |           <p>© 2026 Smart Campus Platform. Enterprise Academic Edition.</p>
131 |           <div className="flex items-center gap-4">
132 |             <Link href="/about" className="hover:underline">
133 |               About
134 |             </Link>
135 |             <Link href="/features" className="hover:underline">
136 |               Features
137 |             </Link>
138 |             <Link href="/contact" className="hover:underline">
139 |               Contact
140 |             </Link>
141 |             <Link href="/settings" className="hover:underline">
142 |               Privacy & Cookies
143 |             </Link>
144 |           </div>
145 |         </div>
146 |       </div>
147 |     </footer>
148 |   );
149 | }
```

File #30: src/components/layout/navbar.tsx

261 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import Link from "next/link";
  5 | import { usePathname } from "next/navigation";
  6 | import {
  7 |   GraduationCap,
  8 |   Bell,
  9 |   Search,
 10 |   PhoneCall,
 11 |   Menu,
 12 |   X,
 13 |   LogIn,
 14 |   UserPlus,
 15 |   Shield,
 16 |   Layers,
 17 |   Sparkles,
 18 | } from "lucide-react";
 19 | import { Button } from "@/components/ui/button";
 20 | import { Badge } from "@/components/ui/badge";
```

```
import { ThemeToggle } from "@/components/ui/theme-toggle";

export function Navbar() {
  const [mobileMenuOpen, setMobileMenuOpen] = React.useState(false);
  const pathname = usePathname();

  return (
    <header className="sticky top-0 z-40 w-full border-b border-zinc-200/80 bg-white/85 backdrop-blur-md dark:border-zinc-800/80 dark:bg-zinc-950/85 transition-colors">
      <div className="mx-auto flex h-16 max-w-7xl items-center justify-between px-4 sm:px-6 lg:px-8">
        {/* Brand */}
        <div className="flex items-center gap-6">
          <Link href="/" className="flex items-center gap-2.5 transition-transform hover:scale-[1.02]">
            <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-gradient-to-tr from-blue-600 via-indigo-600 to-violet-600 shadow-md shadow-blue-500/20 text-white">
              <GraduationCap className="h-5 w-5" />
            </div>
            <div>
              <span className="text-base font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
                SMART CAMPUS
              </span>
              <span className="block text-[10px] font-semibold tracking-wider text-blue-600 dark:text-blue-400 uppercase">
                Enterprise Portal
              </span>
            </div>
          </Link>

          {/* Quick Links */}
          <nav className="hidden md:flex items-center gap-1 text-sm font-medium text-zinc-600 dark:text-zinc-300">
            <Link
              href="/about"
              className={`rounded-lg px-3 py-1.5 transition-colors ${
                pathname === "/about"
                  ? "bg-zinc-100 text-zinc-900 dark:bg-zinc-800 dark:text-zinc-100 font-semibold"
                  : "hover:bg-zinc-100 dark:hover:bg-zinc-800/60"
              }`}
            >
              About
            </Link>
            <Link
              href="/features"
              className={`rounded-lg px-3 py-1.5 transition-colors ${
                pathname === "/features"
                  ? "bg-zinc-100 text-zinc-900 dark:bg-zinc-800 dark:text-zinc-100 font-semibold"
                  : "hover:bg-zinc-100 dark:hover:bg-zinc-800/60"
              }`}
            >
              Features
            </Link>
            <Link
              href="/contact"
              className={`rounded-lg px-3 py-1.5 transition-colors ${
                pathname === "/contact"
                  ? "bg-zinc-100 text-zinc-900 dark:bg-zinc-800 dark:text-zinc-100 font-semibold"
                  : "hover:bg-zinc-100 dark:hover:bg-zinc-800/60"
              }`}
            >
              Contact
            </Link>

            <span className="mx-1 h-4 w-px bg-zinc-200 dark:bg-zinc-800" />

            <Link
              href="/dashboard/student"
              className={`rounded-lg px-3 py-1.5 transition-colors ${
                pathname.startsWith("/dashboard/student")
                  ? "bg-blue-50 text-blue-600 dark:bg-blue-950/50 dark:text-blue-400 font-semibold"
                  : "hover:bg-zinc-100 dark:hover:bg-zinc-800/60"
              }`}
            >
              Student
            </Link>
            <Link
              href="/dashboard/faculty"
              className={`rounded-lg px-3 py-1.5 transition-colors ${
                pathname.startsWith("/dashboard/faculty")
                  ? "bg-amber-50 text-amber-600 dark:bg-amber-950/50 dark:text-amber-400 font-semibold"
                  : "hover:bg-zinc-100 dark:hover:bg-zinc-800/60"
```

```
              }`}
            >
              Faculty
            </Link>
            <Link
              href="/dashboard/admin"
              className={`rounded-lg px-3 py-1.5 transition-colors ${
                pathname.startsWith("/dashboard/admin")
                  ? "bg-rose-50 text-rose-600 dark:bg-rose-950/50 dark:text-rose-400 font-semibold"
                  : "hover:bg-zinc-100 dark:hover:bg-zinc-800/60"
              }`}
            >
              Admin
            </Link>
          </nav>
        </div>

        {/* Right Actions */}
        <div className="flex items-center gap-2 sm:gap-3">
          {/* Quick Search */}
          <Link
            href="/search"
            aria-label="Search Campus"
            className="hidden sm:flex items-center gap-2 rounded-xl border border-zinc-200 bg-zinc-50/80 px-3 py-1.5 text-xs text-zinc-500 hover:border-zinc-300 dark:border-zinc-800 dark:bg-zinc-900/80 dark:text-zinc-400"
          >
            <Search className="h-3.5 w-3.5" />
            <span>Search modules...</span>
            <kbd className="rounded border border-zinc-200 bg-white px-1.5 py-0.5 text-[10px] text-zinc-400 dark:border-zinc-700 dark:bg-zinc-800">
              /
            </kbd>
          </Link>

          {/* Theme Toggle */}
          <ThemeToggle />

          {/* Notifications */}
          <Link
            href="/notifications"
            className="relative flex h-9 w-9 items-center justify-center rounded-xl border border-zinc-200 bg-white/80 text-zinc-600 hover:bg-zinc-100 hover:text-zinc-900 dark:border-zinc-800 dark:bg-zinc-900/80 dark:text-zinc-400 dark:hover:bg-zinc-800 dark:hover:text-zinc-100 transition-colors"
            aria-label="Notifications"
          >
            <Bell className="h-4 w-4" />
            <span className="absolute -top-1 -right-1 flex h-4 w-4 items-center justify-center rounded-full bg-rose-500 text-[9px] font-bold text-white shadow-sm">
              3
            </span>
          </Link>

          {/* Emergency SOS Shortcut */}
          <Link
            href="/emergency"
            className="hidden lg:flex items-center gap-1.5 rounded-xl bg-rose-500/10 px-3 py-1.5 text-xs font-semibold text-rose-600 hover:bg-rose-500/20 dark:bg-rose-500/20 dark:text-rose-400 transition-colors"
          >
            <PhoneCall className="h-3.5 w-3.5" />
            <span>SOS</span>
          </Link>

          {/* Auth Action Buttons */}
          <div className="hidden sm:flex items-center gap-2">
            <Link href="/login">
              <Button variant="ghost" size="sm" className="gap-1.5 text-xs font-semibold">
                <LogIn className="h-3.5 w-3.5" />
                Sign In
              </Button>
            </Link>
            <Link href="/register">
              <Button size="sm" className="gap-1.5 text-xs font-semibold shadow-sm shadow-blue-500/20">
                <UserPlus className="h-3.5 w-3.5" />
                Register
              </Button>
            </Link>
          </div>

```

```
169 |           {/* Mobile hamburger menu */}
170 |           <button
171 |             onClick={() => setMobileMenuOpen(!mobileMenuOpen)}
172 |             className="flex h-9 w-9 items-center justify-center rounded-xl border border-zinc-200 text-zi
    |   nc-600 md:hidden hover:bg-zinc-100 dark:border-zinc-800 dark:text-zinc-400 dark:hover:bg-zinc-800 transit
    |   ion-colors"
173 |             aria-label="Toggle navigation"
174 |           >
175 |             {mobileMenuOpen ? <X className="h-5 w-5" /> : <Menu className="h-5 w-5 text-zinc-700 dark:tex
    |   t-zinc-300" />}
176 |           </button>
177 |         </div>
178 |       </div>
179 |
180 |       {/* Mobile drawer */}
181 |       {mobileMenuOpen && (
182 |         <div className="border-b border-zinc-200 bg-white/95 px-4 pt-3 pb-6 md:hidden dark:border-zinc-80
    |   0 dark:bg-zinc-950/95 animate-fade-in backdrop-blur-lg">
183 |           <div className="space-y-1 pb-3 border-b border-zinc-100 dark:border-zinc-800 text-sm font-mediu
    |   m">
184 |             <p className="px-3 py-1 text-[11px] font-bold text-zinc-400 uppercase tracking-wider">
185 |               Navigation
186 |             </p>
187 |             <Link
188 |               href="/"
189 |               onClick={() => setMobileMenuOpen(false)}
190 |               className="block rounded-lg px-3 py-2 text-zinc-700 hover:bg-zinc-100 dark:text-zinc-300 da
    |   rk:hover:bg-zinc-800"
191 |             >
192 |               Home
193 |             </Link>
194 |             <Link
195 |               href="/about"
196 |               onClick={() => setMobileMenuOpen(false)}
197 |               className="block rounded-lg px-3 py-2 text-zinc-700 hover:bg-zinc-100 dark:text-zinc-300 da
    |   rk:hover:bg-zinc-800"
198 |             >
199 |               About Platform
200 |             </Link>
201 |             <Link
202 |               href="/features"
203 |               onClick={() => setMobileMenuOpen(false)}
204 |               className="block rounded-lg px-3 py-2 text-zinc-700 hover:bg-zinc-100 dark:text-zinc-300 da
    |   rk:hover:bg-zinc-800"
205 |             >
206 |               Features & Modules
207 |             </Link>
208 |             <Link
209 |               href="/contact"
210 |               onClick={() => setMobileMenuOpen(false)}
211 |               className="block rounded-lg px-3 py-2 text-zinc-700 hover:bg-zinc-100 dark:text-zinc-300 da
    |   rk:hover:bg-zinc-800"
212 |             >
213 |               Contact Support
214 |             </Link>
215 |           </div>
216 |
217 |           <div className="space-y-1 py-3 border-b border-zinc-100 dark:border-zinc-800 text-sm font-mediu
    |   m">
218 |             <p className="px-3 py-1 text-[11px] font-bold text-zinc-400 uppercase tracking-wider">
219 |               Portals
220 |             </p>
221 |             <Link
222 |               href="/dashboard/student"
223 |               onClick={() => setMobileMenuOpen(false)}
224 |               className="flex items-center justify-between rounded-lg px-3 py-2 text-blue-600 hover:bg-bl
    |   ue-50 dark:text-blue-400 dark:hover:bg-blue-950/40"
225 |             >
226 |               <span>Student Dashboard</span>
227 |               <Badge variant="info">Portal</Badge>
228 |             </Link>
229 |             <Link
230 |               href="/dashboard/faculty"
231 |               onClick={() => setMobileMenuOpen(false)}
232 |               className="flex items-center justify-between rounded-lg px-3 py-2 text-amber-600 hover:bg-a
    |   mber-50 dark:text-amber-400 dark:hover:bg-amber-950/40"
233 |             >
234 |               <span>Faculty Dashboard</span>
235 |               <Badge variant="warning">Portal</Badge>
236 |             </Link>
```

```
237 |             <Link
238 |               href="/dashboard/admin"
239 |               onClick={() => setMobileMenuOpen(false)}
240 |               className="flex items-center justify-between rounded-lg px-3 py-2 text-rose-600 hover:bg-ro
    |   se-50 dark:text-rose-400 dark:hover:bg-rose-950/40"
241 |             >
242 |               <span>Admin Command Center</span>
243 |               <Badge variant="destructive">Admin</Badge>
244 |             </Link>
245 |           </div>
246 |
247 |           <div className="pt-4 flex flex-col gap-2">
248 |             <Link href="/login" onClick={() => setMobileMenuOpen(false)}>
249 |               <Button variant="outline" className="w-full justify-center">
250 |                 Sign In
251 |               </Button>
252 |             </Link>
253 |             <Link href="/register" onClick={() => setMobileMenuOpen(false)}>
254 |               <Button className="w-full justify-center">Create Account</Button>
255 |             </Link>
256 |           </div>
257 |         </div>
258 |       )}
259 |     </header>
260 |   );
261 | }
```

File #31: src/components/layout/sidebar.tsx

85 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import Link from "next/link";
  5 | import { usePathname } from "next/navigation";
  6 | import { campusNavigation } from "@/lib/navigation";
  7 | import { Badge } from "@/components/ui/badge";
  8 | import { ChevronRight, LogOut, User } from "lucide-react";
  9 | import { signOutAction } from "@/actions/auth";
 10 |
 11 | export function Sidebar() {
 12 |   const pathname = usePathname();
 13 |
 14 |   return (
 15 |     <aside className="w-64 shrink-0 hidden lg:flex flex-col justify-between border-r border-zinc-200/80 b
    |   g-zinc-50/50 p-4 dark:border-zinc-800/80 dark:bg-zinc-950/40">
 16 |       <div className="space-y-6">
 17 |         {campusNavigation.map((category) => (
 18 |           <div key={category.title} className="space-y-1">
 19 |             <h4 className="px-3 text-[11px] font-semibold tracking-wider text-zinc-400 uppercase dark:tex
    |   t-zinc-500">
 20 |               {category.title}
 21 |             </h4>
 22 |             <div className="mt-1 space-y-0.5">
 23 |               {category.items.map((item) => {
 24 |                 const Icon = item.icon;
 25 |                 const isActive = pathname === item.href || (item.href !== "/" && pathname.startsWith(item
    |   .href));
 26 |
 27 |                 return (
 28 |                   <Link
 29 |                     key={item.href}
 30 |                     href={item.href}
 31 |                     className={`group flex items-center justify-between rounded-xl px-3 py-2 text-xs font
    |   -medium transition-all ${
 32 |                       isActive
 33 |                         ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20 font-semibold"
 34 |                         : "text-zinc-600 hover:bg-zinc-200/60 hover:text-zinc-900 dark:text-zinc-400 dark
    |   :hover:bg-zinc-900/60 dark:hover:text-zinc-200"
 35 |                     }`}
 36 |                   >
 37 |                     <div className="flex items-center gap-2.5 truncate">
 38 |                       <Icon
 39 |                         className={`h-4 w-4 shrink-0 transition-transform group-hover:scale-110 ${
 40 |                           isActive ? "text-white" : "text-zinc-400 dark:text-zinc-500"
 41 |                         }`}
 42 |                       />
 43 |                       <span className="truncate">{item.title}</span>
 44 |                     </div>
 45 |
```

```
                    {item.badge ? (
                      <span
                        className={`rounded-md px-1.5 py-0.5 text-[10px] font-semibold ${
                          isActive
                            ? "bg-white/20 text-white"
                            : item.badge === "Live"
                            ? "bg-emerald-500/10 text-emerald-600 dark:bg-emerald-500/20 dark:text-emeral
d-400"
                            : item.badge === "24/7"
                            ? "bg-rose-500/10 text-rose-600 dark:bg-rose-500/20 dark:text-rose-400"
                            : "bg-zinc-200 text-zinc-700 dark:bg-zinc-800 dark:text-zinc-300"
                        }`}
                      >
                        {item.badge}
                      </span>
                    ) : (
                      isActive && <ChevronRight className="h-3 w-3 text-white/70" />
                    )}
                  </Link>
                );
              })}
            </div>
          </div>
        ))}
      </div>

      {/* Logout button at bottom of sidebar */}
      <div className="pt-4 mt-6 border-t border-zinc-200/80 dark:border-zinc-800/80">
        <form action={signOutAction}>
          <button
            type="submit"
            className="w-full flex items-center gap-2.5 rounded-xl px-3 py-2 text-xs font-semibold text-r
ose-600 hover:bg-rose-50 dark:text-rose-400 dark:hover:bg-rose-950/40 transition-colors"
          >
            <LogOut className="h-4 w-4" />
            <span>Sign Out Session</span>
          </button>
        </form>
      </div>
    </aside>
  );
}
```

File #32: src/components/ui/badge.tsx

32 lines

```
import * as React from "react";
import { cn } from "@/lib/utils";

export interface BadgeProps extends React.HTMLAttributes<HTMLSpanElement> {
  variant?: "default" | "success" | "warning" | "danger" | "info" | "outline" | "secondary" | "destructiv
e";
}

export function Badge({ className, variant = "default", children, ...props }: BadgeProps) {
  const variants = {
    default: "bg-zinc-100 text-zinc-800 dark:bg-zinc-800 dark:text-zinc-200 border-zinc-200 dark:border-z
inc-700",
    secondary: "bg-zinc-100 text-zinc-700 dark:bg-zinc-800/80 dark:text-zinc-300 border-zinc-200 dark:bor
der-zinc-700",
    success: "bg-emerald-500/10 text-emerald-600 dark:bg-emerald-500/20 dark:text-emerald-400 border-emer
ald-500/20",
    warning: "bg-amber-500/10 text-amber-600 dark:bg-amber-500/20 dark:text-amber-400 border-amber-500/20
",
    danger: "bg-rose-500/10 text-rose-600 dark:bg-rose-500/20 dark:text-rose-400 border-rose-500/20",
    destructive: "bg-rose-500/10 text-rose-600 dark:bg-rose-500/20 dark:text-rose-400 border-rose-500/20"
,
    info: "bg-blue-500/10 text-blue-600 dark:bg-blue-500/20 dark:text-blue-400 border-blue-500/20",
    outline: "border-zinc-300 dark:border-zinc-700 text-zinc-700 dark:text-zinc-300 bg-transparent",
  };

  return (
    <span
      className={cn(
        "inline-flex items-center gap-1 rounded-full border px-2.5 py-0.5 text-xs font-medium transition-
colors",
        variants[variant],
        className
      )}
      {...props}
```

```
    >
      {children}
    </span>
  );
}
```

File #33: src/components/ui/button.tsx 46 lines

```
import * as React from "react";
import { cn } from "@/lib/utils";

export interface ButtonProps extends React.ButtonHTMLAttributes<HTMLButtonElement> {
  variant?: "primary" | "secondary" | "outline" | "ghost" | "danger";
  size?: "sm" | "md" | "lg";
}

export const Button = React.forwardRef<HTMLButtonElement, ButtonProps>(
  ({ className, variant = "primary", size = "md", children, disabled, ...props }, ref) => {
    const baseStyles =
      "inline-flex items-center justify-center font-medium rounded-xl transition-all duration-200 focus:outline-none focus:ring-2 focus:ring-offset-2 disabled:opacity-50 disabled:pointer-events-none active:scale-[0.98]";

    const variants = {
      primary:
        "bg-gradient-to-r from-blue-600 to-indigo-600 hover:from-blue-700 hover:to-indigo-700 text-white shadow-lg shadow-blue-500/20 focus:ring-blue-500 border border-transparent",
      secondary:
        "bg-zinc-800 text-zinc-100 hover:bg-zinc-700 focus:ring-zinc-600 border border-zinc-700",
      outline:
        "border border-zinc-300 dark:border-zinc-700 bg-transparent hover:bg-zinc-100 dark:hover:bg-zinc-800 text-zinc-800 dark:text-zinc-200 focus:ring-zinc-500",
      ghost:
        "bg-transparent hover:bg-zinc-100 dark:hover:bg-zinc-800/80 text-zinc-700 dark:text-zinc-300 focus:ring-zinc-500",
      danger:
        "bg-rose-600 hover:bg-rose-700 text-white shadow-lg shadow-rose-500/20 focus:ring-rose-500 border border-transparent",
    };

    const sizes = {
      sm: "px-3 py-1.5 text-xs gap-1.5",
      md: "px-4 py-2 text-sm gap-2",
      lg: "px-6 py-3 text-base gap-2.5",
    };

    return (
      <button
        ref={ref}
        disabled={disabled}
        className={cn(baseStyles, variants[variant], sizes[size], className)}
        {...props}
      >
        {children}
      </button>
    );
  }
);

Button.displayName = "Button";
```

File #34: src/components/ui/card.tsx 71 lines

```
import * as React from "react";
import { cn } from "@/lib/utils";

export function Card({
  className,
  children,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) {
  return (
    <div
      className={cn(
        "rounded-2xl border border-zinc-200/80 bg-white/80 p-6 shadow-sm backdrop-blur-md transition-all duration-200 hover:shadow-md dark:border-zinc-800/80 dark:bg-zinc-900/80",
        className
      )}
```

```
      {...props}
    >
      {children}
    </div>
  );
}

export function CardHeader({
  className,
  children,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) {
  return (
    <div className={cn("mb-4 flex flex-col space-y-1.5", className)} {...props}>
      {children}
    </div>
  );
}

export function CardTitle({
  className,
  children,
  ...props
}: React.HTMLAttributes<HTMLHeadingElement>) {
  return (
    <h3
      className={cn("text-lg font-semibold tracking-tight text-zinc-900 dark:text-zinc-50", className)}
      {...props}
    >
      {children}
    </h3>
  );
}

export function CardDescription({
  className,
  children,
  ...props
}: React.HTMLAttributes<HTMLParagraphElement>) {
  return (
    <p className={cn("text-xs text-zinc-500 dark:text-zinc-400", className)} {...props}>
      {children}
    </p>
  );
}

export function CardContent({
  className,
  children,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) {
  return (
    <div className={cn("pt-0", className)} {...props}>
      {children}
    </div>
  );
}
```

File #35: src/components/ui/input.tsx

41 lines

```
import * as React from "react";
import { cn } from "@/lib/utils";

export interface InputProps extends React.InputHTMLAttributes<HTMLInputElement> {
  label?: string;
  error?: string;
}

export const Input = React.forwardRef<HTMLInputElement, InputProps>(
  ({ className, type, label, error, id, ...props }, ref) => {
    const generatedId = React.useId();
    const inputId = id || generatedId;

    return (
      <div className="w-full space-y-1.5">
        {label && (
          <label
            htmlFor={inputId}
            className="block text-xs font-semibold tracking-wide text-zinc-700 dark:text-zinc-300"
```

```
          >
            {label}
          </label>
        )}
        <input
          type={type}
          id={inputId}
          className={cn(
            "flex h-11 w-full rounded-xl border border-zinc-200 bg-white/70 px-3.5 py-2 text-sm text-zinc
-900 transition-all placeholder:text-zinc-400 focus:border-blue-500 focus:bg-white focus:outline-none foc
us:ring-2 focus:ring-blue-500/20 disabled:cursor-not-allowed disabled:opacity-50 dark:border-zinc-800 dar
k:bg-zinc-900/70 dark:text-zinc-100 dark:placeholder:text-zinc-500 dark:focus:border-blue-400 dark:focus:
bg-zinc-900",
            error && "border-rose-500 focus:border-rose-500 focus:ring-rose-500/20",
            className
          )}
          ref={ref}
          {...props}
        />
        {error && <p className="text-xs text-rose-500">{error}</p>}
      </div>
    );
  }
);

Input.displayName = "Input";
```

File #36: src/components/ui/modal.tsx

96 lines

```
"use client";

import * as React from "react";
import { X } from "lucide-react";

interface ModalProps {
  isOpen: boolean;
  onClose: () => void;
  title: string;
  description?: string;
  children: React.ReactNode;
  footer?: React.ReactNode;
  maxWidth?: "sm" | "md" | "lg" | "xl" | "2xl";
}

export function Modal({
  isOpen,
  onClose,
  title,
  description,
  children,
  footer,
  maxWidth = "lg",
}: ModalProps) {
  React.useEffect(() => {
    const handleKeyDown = (e: KeyboardEvent) => {
      if (e.key === "Escape") onClose();
    };
    if (isOpen) {
      document.body.style.overflow = "hidden";
      window.addEventListener("keydown", handleKeyDown);
    }
    return () => {
      document.body.style.overflow = "unset";
      window.removeEventListener("keydown", handleKeyDown);
    };
  }, [isOpen, onClose]);

  if (!isOpen) return null;

  const maxWidthClasses = {
    sm: "max-w-sm",
    md: "max-w-md",
    lg: "max-w-lg",
    xl: "max-w-xl",
    "2xl": "max-w-2xl",
  };

  return (
    <div className="fixed inset-0 z-50 flex items-center justify-center p-4 sm:p-6 overflow-y-auto">
```

```
      {/* Backdrop */}
      <div
        className="fixed inset-0 bg-zinc-950/60 backdrop-blur-sm transition-opacity"
        onClick={onClose}
        aria-hidden="true"
      />

      {/* Modal Dialog */}
      <div
        role="dialog"
        aria-modal="true"
        className={`relative w-full ${maxWidthClasses[maxWidth]} rounded-2xl border border-zinc-200/90 bg-white p-6 shadow-2xl transition-all dark:border-zinc-800/90 dark:bg-zinc-900 z-10 animate-fade-in`}
      >
        <div className="flex items-start justify-between border-b border-zinc-100 pb-4 dark:border-zinc-800">
          <div>
            <h3 className="text-lg font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
              {title}
            </h3>
            {description && (
              <p className="mt-1 text-xs text-zinc-500 dark:text-zinc-400">
                {description}
              </p>
            )}
          </div>
          <button
            onClick={onClose}
            className="rounded-lg p-1.5 text-zinc-400 hover:bg-zinc-100 hover:text-zinc-600 dark:hover:bg-zinc-800 dark:hover:text-zinc-200 transition-colors"
            aria-label="Close dialog"
          >
            <X className="h-4 w-4" />
          </button>
        </div>

        <div className="py-4 text-sm text-zinc-600 dark:text-zinc-300">
          {children}
        </div>

        {footer && (
          <div className="flex items-center justify-end gap-2 border-t border-zinc-100 pt-4 dark:border-zinc-800">
            {footer}
          </div>
        )}
      </div>
    </div>
  );
}
```

File #37: src/components/ui/state-views.tsx 112 lines

```
import * as React from "react";
import { FolderSearch, AlertOctagon, RotateCw } from "lucide-react";
import { Button } from "./button";

interface EmptyStateProps {
  icon?: React.ComponentType<{ className?: string }>;
  title: string;
  description: string;
  actionLabel?: string;
  onAction?: () => void;
  className?: string;
}

export function EmptyState({
  icon: Icon = FolderSearch,
  title,
  description,
  actionLabel,
  onAction,
  className,
}: EmptyStateProps) {
  return (
    <div
      className={`flex flex-col items-center justify-center rounded-2xl border border-dashed border-zinc-200 bg-zinc-50/50 p-8 text-center dark:border-zinc-800 dark:bg-zinc-900/30 ${
        className || ""
```

```
      }`}
    >
      <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-zinc-100 text-zinc-500 dark:bg-zinc-800 dark:text-zinc-400">
        <Icon className="h-6 w-6" />
      </div>
      <h3 className="mt-4 text-sm font-semibold text-zinc-900 dark:text-zinc-100">
        {title}
      </h3>
      <p className="mt-1.5 max-w-sm text-xs text-zinc-500 dark:text-zinc-400">
        {description}
      </p>
      {actionLabel && onAction && (
        <div className="mt-5">
          <Button size="sm" onClick={onAction}>
            {actionLabel}
          </Button>
        </div>
      )}
    </div>
  );
}

interface LoadingStateProps {
  message?: string;
  className?: string;
}

export function LoadingState({
  message = "Loading records...",
  className,
}: LoadingStateProps) {
  return (
    <div
      className={`flex flex-col items-center justify-center py-12 ${
        className || ""
      }`}
    >
      <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-blue-50 text-blue-600 dark:bg-blue-950/60 dark:text-blue-400">
        <RotateCw className="h-5 w-5 animate-spin" />
      </div>
      <p className="mt-3 text-xs font-medium text-zinc-500 dark:text-zinc-400">
        {message}
      </p>
    </div>
  );
}

interface ErrorStateProps {
  title?: string;
  message?: string;
  retryLabel?: string;
  onRetry?: () => void;
  className?: string;
}

export function ErrorState({
  title = "Failed to load data",
  message = "An error occurred while fetching information. Please verify your connection and try again.",
  retryLabel = "Try Again",
  onRetry,
  className,
}: ErrorStateProps) {
  return (
    <div
      className={`flex flex-col items-center justify-center rounded-2xl border border-rose-200/80 bg-rose-50/50 p-8 text-center dark:border-rose-900/50 dark:bg-rose-950/20 ${
        className || ""
      }`}
    >
      <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-rose-100 text-rose-600 dark:bg-rose-900/60 dark:text-rose-400">
        <AlertOctagon className="h-6 w-6" />
      </div>
      <h3 className="mt-4 text-sm font-semibold text-rose-900 dark:text-rose-100">
        {title}
      </h3>
      <p className="mt-1.5 max-w-sm text-xs text-rose-600/80 dark:text-rose-300">
        {message}
```

```
      </p>
      {onRetry && (
        <div className="mt-5">
          <Button variant="outline" size="sm" onClick={onRetry}>
            {retryLabel}
          </Button>
        </div>
      )}
    </div>
  );
}
```

File #38: src/components/ui/table.tsx 84 lines

```
import * as React from "react";

interface TableProps extends React.HTMLAttributes<HTMLTableElement> {}

export function Table({ className, ...props }: TableProps) {
  return (
    <div className="relative w-full overflow-auto rounded-xl border border-zinc-200/80 bg-white dark:bord
  er-zinc-800/80 dark:bg-zinc-900/60">
      <table
        className={`w-full caption-bottom text-xs text-left ${className || ""}`}
        {...props}
      />
    </div>
  );
}

export function TableHeader({
  className,
  ...props
}: React.HTMLAttributes<HTMLTableSectionElement>) {
  return (
    <thead
      className={`border-b border-zinc-200 bg-zinc-50/75 dark:border-zinc-800 dark:bg-zinc-800/50 ${
        className || ""
      }`}
      {...props}
    />
  );
}

export function TableBody({
  className,
  ...props
}: React.HTMLAttributes<HTMLTableSectionElement>) {
  return (
    <tbody
      className={`divide-y divide-zinc-200/60 dark:divide-zinc-800/60 ${
        className || ""
      }`}
      {...props}
    />
  );
}

export function TableRow({
  className,
  ...props
}: React.HTMLAttributes<HTMLTableRowElement>) {
  return (
    <tr
      className={`transition-colors hover:bg-zinc-50/80 dark:hover:bg-zinc-800/30 ${
        className || ""
      }`}
      {...props}
    />
  );
}

export function TableHead({
  className,
  ...props
}: React.ThHTMLAttributes<HTMLTableCellElement>) {
  return (
    <th
      className={`h-10 px-4 text-left align-middle font-semibold text-zinc-500 uppercase tracking-wider t
```

```
   |   ext-[11px] dark:text-zinc-400 ${
65 |         className || ""
66 |       }`}
67 |       {...props}
68 |     />
69 |   );
70 | }
71 |
72 | export function TableCell({
73 |   className,
74 |   ...props
75 | }: React.TdHTMLAttributes<HTMLTableCellElement>) {
76 |   return (
77 |     <td
78 |       className={`p-4 align-middle text-zinc-700 dark:text-zinc-300 ${
79 |         className || ""
80 |       }`}
81 |       {...props}
82 |     />
83 |   );
84 | }
```

File #39: src/components/ui/theme-toggle.tsx 84 lines

```
 1 | "use client";
 2 |
 3 | import * as React from "react";
 4 | import { Sun, Moon } from "lucide-react";
 5 |
 6 | type Theme = "light" | "dark" | "system";
 7 |
 8 | interface ThemeContextValue {
 9 |   theme: Theme;
10 |   setTheme: (theme: Theme) => void;
11 |   toggleTheme: () => void;
12 |   isDark: boolean;
13 | }
14 |
15 | const ThemeContext = React.createContext<ThemeContextValue | undefined>(undefined);
16 |
17 | export function ThemeProvider({ children }: { children: React.ReactNode }) {
18 |   const [theme, setThemeState] = React.useState<Theme>("system");
19 |   const [isDark, setIsDark] = React.useState(false);
20 |
21 |   React.useEffect(() => {
22 |     const saved = localStorage.getItem("sc-theme") as Theme | null;
23 |     if (saved && (saved === "light" || saved === "dark" || saved === "system")) {
24 |       setThemeState(saved);
25 |     }
26 |   }, []);
27 |
28 |   React.useEffect(() => {
29 |     const root = document.documentElement;
30 |     const systemDark = window.matchMedia("(prefers-color-scheme: dark)").matches;
31 |
32 |     const darkActive = theme === "dark" || (theme === "system" && systemDark);
33 |     setIsDark(darkActive);
34 |
35 |     if (darkActive) {
36 |       root.classList.add("dark");
37 |     } else {
38 |       root.classList.remove("dark");
39 |     }
40 |   }, [theme]);
41 |
42 |   const setTheme = (newTheme: Theme) => {
43 |     setThemeState(newTheme);
44 |     localStorage.setItem("sc-theme", newTheme);
45 |   };
46 |
47 |   const toggleTheme = () => {
48 |     setTheme(isDark ? "light" : "dark");
49 |   };
50 |
51 |   return (
52 |     <ThemeContext.Provider value={{ theme, setTheme, toggleTheme, isDark }}>
53 |       {children}
54 |     </ThemeContext.Provider>
55 |   );
```

```
}

export function useTheme() {
  const context = React.useContext(ThemeContext);
  if (!context) {
    throw new Error("useTheme must be used within a ThemeProvider");
  }
  return context;
}

export function ThemeToggle({ className }: { className?: string }) {
  const { toggleTheme, isDark } = useTheme();

  return (
    <button
      onClick={toggleTheme}
      aria-label="Toggle theme"
      className={`relative inline-flex h-9 w-9 items-center justify-center rounded-xl border border-zinc-200 bg-white/80 p-2 text-zinc-600 transition-colors hover:bg-zinc-100 hover:text-zinc-900 dark:border-zinc-800 dark:bg-zinc-900/80 dark:text-zinc-400 dark:hover:bg-zinc-800 dark:hover:text-zinc-100 ${className || ""}`}
    >
      {isDark ? (
        <Sun className="h-4 w-4 text-amber-400 transition-transform hover:rotate-45" />
      ) : (
        <Moon className="h-4 w-4 text-indigo-600 transition-transform hover:-rotate-12" />
      )}
    </button>
  );
}
```

File #40: src/components/ui/toast.tsx

123 lines

```
"use client";

import * as React from "react";
import { CheckCircle2, AlertCircle, Info, AlertTriangle, X } from "lucide-react";

type ToastType = "success" | "error" | "info" | "warning";

interface Toast {
  id: string;
  title: string;
  description?: string;
  type: ToastType;
}

interface ToastContextValue {
  toasts: Toast[];
  showToast: (title: string, description?: string, type?: ToastType) => void;
  removeToast: (id: string) => void;
  success: (title: string, description?: string) => void;
  error: (title: string, description?: string) => void;
  info: (title: string, description?: string) => void;
  warning: (title: string, description?: string) => void;
}

const ToastContext = React.createContext<ToastContextValue | undefined>(undefined);

export function ToastProvider({ children }: { children: React.ReactNode }) {
  const [toasts, setToasts] = React.useState<Toast[]>([]);

  const removeToast = React.useCallback((id: string) => {
    setToasts((prev) => prev.filter((t) => t.id !== id));
  }, []);

  const showToast = React.useCallback(
    (title: string, description?: string, type: ToastType = "info") => {
      const id = Math.random().toString(36).substring(2, 9);
      setToasts((prev) => [...prev, { id, title, description, type }]);
      setTimeout(() => {
        removeToast(id);
      }, 4000);
    },
    [removeToast]
  );

  const success = React.useCallback(
```

```
    (title: string, description?: string) => showToast(title, description, "success"),
    [showToast]
  );
  const error = React.useCallback(
    (title: string, description?: string) => showToast(title, description, "error"),
    [showToast]
  );
  const info = React.useCallback(
    (title: string, description?: string) => showToast(title, description, "info"),
    [showToast]
  );
  const warning = React.useCallback(
    (title: string, description?: string) => showToast(title, description, "warning"),
    [showToast]
  );

  return (
    <ToastContext.Provider
      value={{ toasts, showToast, removeToast, success, error, info, warning }}
    >
      {children}
      {/* Toast viewport */}
      <div className="fixed bottom-5 right-5 z-50 flex flex-col gap-2.5 max-w-sm w-full pointer-events-none">
        {toasts.map((toast) => {
          const icons = {
            success: <CheckCircle2 className="h-5 w-5 text-emerald-500 shrink-0" />,
            error: <AlertCircle className="h-5 w-5 text-rose-500 shrink-0" />,
            warning: <AlertTriangle className="h-5 w-5 text-amber-500 shrink-0" />,
            info: <Info className="h-5 w-5 text-blue-500 shrink-0" />,
          };

          const borderColors = {
            success: "border-emerald-500/30 bg-emerald-50/90 dark:bg-emerald-950/80 dark:border-emerald-500/30",
            error: "border-rose-500/30 bg-rose-50/90 dark:bg-rose-950/80 dark:border-rose-500/30",
            warning: "border-amber-500/30 bg-amber-50/90 dark:bg-amber-950/80 dark:border-amber-500/30",
            info: "border-blue-500/30 bg-blue-50/90 dark:bg-blue-950/80 dark:border-blue-500/30",
          };

          return (
            <div
              key={toast.id}
              className={`pointer-events-auto flex items-start gap-3 p-3.5 rounded-xl border shadow-lg backdrop-blur-md transition-all animate-fade-in ${
                borderColors[toast.type]
              }`}
            >
              {icons[toast.type]}
              <div className="flex-1 min-w-0">
                <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100">
                  {toast.title}
                </p>
                {toast.description && (
                  <p className="mt-0.5 text-[11px] text-zinc-600 dark:text-zinc-300">
                    {toast.description}
                  </p>
                )}
              </div>
              <button
                onClick={() => removeToast(toast.id)}
                className="text-zinc-400 hover:text-zinc-600 dark:hover:text-zinc-200 transition-colors"
                aria-label="Close notification"
              >
                <X className="h-4 w-4" />
              </button>
            </div>
          );
        })}
      </div>
    </ToastContext.Provider>
  );
}

export function useToast() {
  const context = React.useContext(ToastContext);
  if (!context) {
    throw new Error("useToast must be used within a ToastProvider");
  }
  return context;
```

```
123 | }
```

File #41: src/app/about/page.tsx 156 lines

```
  1 | import * as React from "react";
  2 | import Link from "next/link";
  3 | import {
  4 |   GraduationCap,
  5 |   Shield,
  6 |   Zap,
  7 |   Users,
  8 |   Award,
  9 |   Globe2,
 10 |   CheckCircle2,
 11 |   ArrowRight,
 12 |   BookOpen,
 13 | } from "lucide-react";
 14 | import { Navbar } from "@/components/layout/navbar";
 15 | import { Footer } from "@/components/layout/footer";
 16 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
 17 | import { Button } from "@/components/ui/button";
 18 | import { Badge } from "@/components/ui/badge";
 19 |
 20 | export default function AboutPage() {
 21 |   return (
 22 |     <div className="min-h-screen flex flex-col bg-zinc-50/70 dark:bg-zinc-950 text-zinc-900 dark:text-zin
    |   c-100">
 23 |       <Navbar />
 24 |
 25 |       <main className="flex-1">
 26 |         {/* Hero Section */}
 27 |         <section className="relative overflow-hidden border-b border-zinc-200/80 bg-white/60 py-20 dark:b
    |   order-zinc-800/80 dark:bg-zinc-900/30">
 28 |           <div className="mx-auto max-w-5xl px-4 sm:px-6 lg:px-8 text-center">
 29 |             <Badge variant="info" className="mb-4">
 30 |               Academic Infrastructure 2.0
 31 |             </Badge>
 32 |             <h1 className="text-3xl font-extrabold tracking-tight sm:text-5xl">
 33 |               About <span className="bg-gradient-to-r from-blue-600 via-indigo-600 to-violet-600 bg-clip-
    |   text text-transparent">SMART CAMPUS</span>
 34 |             </h1>
 35 |             <p className="mx-auto mt-4 max-w-2xl text-sm sm:text-base text-zinc-600 dark:text-zinc-400 le
    |   ading-relaxed">
 36 |               Smart Campus is a comprehensive, enterprise-grade digital platform engineered to unify univ
    |   ersity academic workflows, student self-service, faculty course administration, and executive institution
    |   al governance.
 37 |             </p>
 38 |           </div>
 39 |         </section>
 40 |
 41 |         {/* Pillars / Values */}
 42 |         <section className="py-16 mx-auto max-w-6xl px-4 sm:px-6 lg:px-8">
 43 |           <div className="text-center mb-12">
 44 |             <h2 className="text-2xl font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
 45 |               Core Pillars of the Platform
 46 |             </h2>
 47 |             <p className="text-xs text-zinc-500 dark:text-zinc-400 mt-1">
 48 |               Engineered with modern architecture principles for academic reliability and security
 49 |             </p>
 50 |           </div>
 51 |
 52 |           <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
 53 |             <Card className="hover:shadow-lg transition-all border-blue-500/10">
 54 |               <CardHeader>
 55 |                 <div className="h-10 w-10 rounded-xl bg-blue-500/10 text-blue-600 flex items-center justi
    |   fy-center mb-2 dark:bg-blue-500/20 dark:text-blue-400">
 56 |                   <Zap className="h-5 w-5" />
 57 |                 </div>
 58 |                 <CardTitle className="text-base">Realtime Campus Synchronization</CardTitle>
 59 |                 <CardDescription className="text-xs">
 60 |                   Instant updates for urgent notices, class cancellations, timetable shifts, and emergenc
    |   y alerts.
 61 |                 </CardDescription>
 62 |               </CardHeader>
 63 |               <CardContent>
 64 |                 <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
 65 |                   Leveraging PostgreSQL listen/notify channels and Supabase Realtime websocket subscripti
    |   ons, the platform keeps thousands of active devices synchronized without manual page refreshes.
 66 |                 </p>
```

```
              </CardContent>
            </Card>

            <Card className="hover:shadow-lg transition-all border-indigo-500/10">
              <CardHeader>
                <div className="h-10 w-10 rounded-xl bg-indigo-500/10 text-indigo-600 flex items-center justify-center mb-2 dark:bg-indigo-500/20 dark:text-indigo-400">
                  <Shield className="h-5 w-5" />
                </div>
                <CardTitle className="text-base">Zero-Trust Role-Based Security</CardTitle>
                <CardDescription className="text-xs">
                  Row-Level Security (RLS) policies protecting gradebooks, records, and student privacy.
                </CardDescription>
              </CardHeader>
              <CardContent>
                <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
                  Fine-grained access controls ensure students only read their own academic transcripts, faculty govern their designated course cohorts, and administrative audit trails log critical changes.
                </p>
              </CardContent>
            </Card>

            <Card className="hover:shadow-lg transition-all border-violet-500/10">
              <CardHeader>
                <div className="h-10 w-10 rounded-xl bg-violet-500/10 text-violet-600 flex items-center justify-center mb-2 dark:bg-violet-500/20 dark:text-violet-400">
                  <Award className="h-5 w-5" />
                </div>
                <CardTitle className="text-base">Production Performance</CardTitle>
                <CardDescription className="text-xs">
                  Next.js App Router, SSR, Turbopack, and Tailwind CSS.
                </CardDescription>
              </CardHeader>
              <CardContent>
                <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
                  Engineered with server-side caching, fluid typography, responsive tables, and accessibility standards to provide smooth load times across devices on campus Wi-Fi or mobile networks.
                </p>
              </CardContent>
            </Card>
          </div>
        </section>

        {/* System Stats Showcase */}
        <section className="py-12 bg-white/50 border-y border-zinc-200/80 dark:bg-zinc-900/40 dark:border-zinc-800/80">
          <div className="max-w-6xl mx-auto px-4 sm:px-6 lg:px-8">
            <div className="grid grid-cols-2 md:grid-cols-4 gap-6 text-center">
              <div>
                <p className="text-3xl font-extrabold text-blue-600 dark:text-blue-400">99.9%</p>
                <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400 mt-1">Uptime Reliability</p>
              </div>
              <div>
                <p className="text-3xl font-extrabold text-indigo-600 dark:text-indigo-400">18+</p>
                <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400 mt-1">Integrated Campus Modules</p>
              </div>
              <div>
                <p className="text-3xl font-extrabold text-violet-600 dark:text-violet-400">3</p>
                <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400 mt-1">Dedicated User Roles</p>
              </div>
              <div>
                <p className="text-3xl font-extrabold text-emerald-600 dark:text-emerald-400">&lt;50ms</p>
                <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400 mt-1">Realtime Event Latency</p>
              </div>
            </div>
          </div>
        </section>

        {/* CTA */}
        <section className="py-16 text-center max-w-4xl mx-auto px-4">
          <h2 className="text-xl sm:text-2xl font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
            Ready to experience the unified campus portal?
          </h2>
          <p className="text-xs sm:text-sm text-zinc-500 dark:text-zinc-400 mt-2 mb-6 max-w-lg mx-auto">
            Choose your designated portal or authenticate with your university email.
```

```
          </p>
          <div className="flex flex-wrap items-center justify-center gap-3">
            <Link href="/login">
              <Button size="lg" className="gap-2">
                Sign In to Portal <ArrowRight className="h-4 w-4" />
              </Button>
            </Link>
            <Link href="/features">
              <Button variant="outline" size="lg">
                Explore Modules
              </Button>
            </Link>
          </div>
        </section>
      </main>

      <Footer />
    </div>
  );
}
```

File #42: src/app/announcements/page.tsx 316 lines

```
"use client";

import * as React from "react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { Input } from "@/components/ui/input";
import { Modal } from "@/components/ui/modal";
import { useToast } from "@/components/ui/toast";
import { EmptyState, LoadingState, ErrorState } from "@/components/ui/state-views";
import { Bell, AlertTriangle, Pin, Calendar, Radio, Plus, Trash2, Edit3, Search } from "lucide-react";
import {
  getAnnouncementsAction,
  createAnnouncementAction,
  deleteAnnouncementAction,
} from "@/actions/announcements";

export default function AnnouncementsPage() {
  const { success, error } = useToast();
  const [announcements, setAnnouncements] = React.useState<any[]>([]);
  const [loading, setLoading] = React.useState(true);
  const [hasError, setHasError] = React.useState(false);
  const [search, setSearch] = React.useState("");
  const [categoryFilter, setCategoryFilter] = React.useState("All");

  // Create Modal State
  const [modalOpen, setModalOpen] = React.useState(false);
  const [formData, setFormData] = React.useState({
    title: "",
    content: "",
    category: "Academic",
    urgency: "normal" as "normal" | "important" | "emergency",
    targetAudience: "all" as "all" | "students" | "faculty",
    pinned: false,
  });
  const [isSubmitting, setIsSubmitting] = React.useState(false);

  const fetchAnnouncements = React.useCallback(async () => {
    setLoading(true);
    setHasError(false);
    try {
      const res = await getAnnouncementsAction();
      if (res.success && res.data) {
        setAnnouncements(res.data);
      } else {
        setHasError(true);
      }
    } catch {
      setHasError(true);
    } finally {
      setLoading(false);
    }
  }, []);

  React.useEffect(() => {
```

```
  fetchAnnouncements();
}, [fetchAnnouncements]);

const handleCreate = async (e: React.FormEvent) => {
  e.preventDefault();
  setIsSubmitting(true);
  try {
    const res = await createAnnouncementAction(formData);
    if (res.success) {
      success("Broadcast Published", "Announcement distributed across campus channels.");
      setModalOpen(false);
      setFormData({
        title: "",
        content: "",
        category: "Academic",
        urgency: "normal",
        targetAudience: "all",
        pinned: false,
      });
      fetchAnnouncements();
    } else {
      error("Publish Failed", res.message || "Please check inputs.");
    }
  } catch (err: any) {
    error("Error", err.message);
  } finally {
    setIsSubmitting(false);
  }
};

const handleDelete = async (id: string) => {
  try {
    const res = await deleteAnnouncementAction(id);
    if (res.success) {
      success("Announcement Removed");
      setAnnouncements((prev) => prev.filter((a) => a.id !== id));
    } else {
      error("Deletion Failed", res.message);
    }
  } catch (err: any) {
    error("Error", err.message);
  }
};

const filtered = announcements.filter((a) => {
  const matchCat = categoryFilter === "All" || a.category === categoryFilter;
  const matchSearch =
    a.title.toLowerCase().includes(search.toLowerCase()) ||
    a.content.toLowerCase().includes(search.toLowerCase()) ||
    a.category.toLowerCase().includes(search.toLowerCase());
  return matchCat && matchSearch;
});

return (
  <DashboardShell
    title="Campus Circulars & Announcements"
    description="Official university notices, emergency broadcasts, and department advisories"
    actions={
      <div className="flex items-center gap-2">
        <Badge variant="info" className="gap-1.5 py-1 hidden sm:inline-flex">
          <Radio className="h-3 w-3 animate-pulse text-blue-500" />
          <span>Supabase Realtime Channel</span>
        </Badge>
        <Button size="sm" onClick={() => setModalOpen(true)} className="gap-1.5 shadow-sm shadow-blue-500/20">
          <Plus className="h-4 w-4" />
          <span>New Announcement</span>
        </Button>
      </div>
    }
  >
    <div className="space-y-6">
      {/* Search & Category Filter */}
      <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3 rounded-2xl border border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
        <div className="flex items-center gap-1.5 flex-wrap">
          {["All", "Academic", "Examinations", "Events", "Hostel"].map((c) => (
            <button
              key={c}
              onClick={() => setCategoryFilter(c)}
```

```
135 |                 className={`rounded-xl px-3 py-1.5 text-xs font-semibold transition-all ${
136 |                   categoryFilter === c
137 |                     ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
138 |                     : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
139 |                 }`}
140 |               >
141 |                 {c}
142 |               </button>
143 |             ))}
144 |           </div>
145 |
146 |           <div className="relative">
147 |             <input
148 |               type="text"
149 |               placeholder="Search notices..."
150 |               value={search}
151 |               onChange={(e) => setSearch(e.target.value)}
152 |               className="w-full sm:w-64 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs te
    |   xt-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-
    |   200"
153 |             />
154 |           </div>
155 |         </div>
156 |
157 |         {/* State Views */}
158 |         {loading ? (
159 |           <LoadingState message="Fetching live announcements from Supabase PostgreSQL..." />
160 |         ) : hasError ? (
161 |           <ErrorState onRetry={fetchAnnouncements} />
162 |         ) : filtered.length === 0 ? (
163 |           <EmptyState
164 |             title="No Announcements Found"
165 |             description="There are no broadcast notices matching your filter criteria."
166 |             actionLabel="Post New Notice"
167 |             onAction={() => setModalOpen(true)}
168 |           />
169 |         ) : (
170 |           <div className="space-y-4">
171 |             {filtered.map((item) => (
172 |               <Card
173 |                 key={item.id}
174 |                 className={`transition-all hover:border-blue-500/40 ${
175 |                   item.pinned || item.urgency === "emergency"
176 |                     ? "border-l-4 border-l-rose-500 bg-rose-50/20 dark:bg-rose-950/10"
177 |                     : item.urgency === "important"
178 |                     ? "border-l-4 border-l-amber-500"
179 |                     : ""
180 |                 }`}
181 |               >
182 |                 <CardHeader className="pb-2">
183 |                   <div className="flex items-start justify-between gap-2">
184 |                     <div className="flex items-center gap-2 flex-wrap">
185 |                       <Badge
186 |                         variant={
187 |                           item.urgency === "emergency"
188 |                             ? "danger"
189 |                             : item.urgency === "important"
190 |                             ? "warning"
191 |                             : "info"
192 |                         }
193 |                       >
194 |                         {item.category}
195 |                       </Badge>
196 |                       {item.pinned && (
197 |                         <Badge variant="outline" className="gap-1 text-rose-600 border-rose-300">
198 |                           <Pin className="h-3 w-3" />
199 |                           <span>Pinned</span>
200 |                         </Badge>
201 |                       )}
202 |                       <span className="text-[11px] text-zinc-400">
203 |                         Audience: <strong className="capitalize">{item.target_audience || "all"}</strong>
204 |                       </span>
205 |                     </div>
206 |
207 |                     <div className="flex items-center gap-1">
208 |                       <span className="text-xs text-zinc-400 hidden sm:inline">
209 |                         {new Date(item.created_at).toLocaleDateString()}
210 |                       </span>
211 |                       <button
212 |                         onClick={() => handleDelete(item.id)}
```

```
                        className="p-1 rounded-lg text-zinc-400 hover:text-rose-500 hover:bg-rose-50 dark
:hover:bg-rose-950/40 transition-colors ml-2"
                        aria-label="Delete announcement"
                      >
                        <Trash2 className="h-3.5 w-3.5" />
                      </button>
                    </div>
                  </div>
                  <CardTitle className="text-base mt-2">{item.title}</CardTitle>
                </CardHeader>
                <CardContent>
                  <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
                    {item.content}
                  </p>
                </CardContent>
              </Card>
            ))}
          </div>
        )}
      </div>

      {/* New Announcement Modal */}
      <Modal
        isOpen={modalOpen}
        onClose={() => setModalOpen(false)}
        title="Broadcast Campus Announcement"
        description="Creates announcement record in PostgreSQL and pushes to Supabase Realtime channel"
      >
        <form onSubmit={handleCreate} className="space-y-4">
          <Input
            label="Announcement Title"
            required
            placeholder="e.g. Schedule for Mid-Term Assessments"
            value={formData.title}
            onChange={(e) => setFormData({ ...formData, title: e.target.value })}
          />

          <div className="grid grid-cols-2 gap-3">
            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Category</label>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={formData.category}
                onChange={(e) => setFormData({ ...formData, category: e.target.value })}
              >
                <option>Academic</option>
                <option>Examinations</option>
                <option>Events</option>
                <option>Hostel</option>
                <option>Campus Life</option>
              </select>
            </div>

            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Urgency Level</la
bel>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={formData.urgency}
                onChange={(e) => setFormData({ ...formData, urgency: e.target.value as any })}
              >
                <option value="normal">Normal Information</option>
                <option value="important">Important Priority</option>
                <option value="emergency">Emergency Alert</option>
              </select>
            </div>
          </div>

          <div className="space-y-1.5">
            <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Full Content</label
>
            <textarea
              required
              rows={4}
              placeholder="Provide complete circular details, dates, and instructions..."
              className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-8
00 dark:bg-zinc-900"
              value={formData.content}
```

```
287 |               onChange={(e) => setFormData({ ...formData, content: e.target.value })}
288 |             />
289 |           </div>
290 |
291 |           <div className="flex items-center gap-2">
292 |             <input
293 |               type="checkbox"
294 |               id="pinnedCheck"
295 |               checked={formData.pinned}
296 |               onChange={(e) => setFormData({ ...formData, pinned: e.target.checked })}
297 |               className="rounded text-blue-600 h-4 w-4"
298 |             />
299 |             <label htmlFor="pinnedCheck" className="text-xs text-zinc-700 dark:text-zinc-300 font-medium"
    |   >
300 |               Pin to top of campus dashboard
301 |             </label>
302 |           </div>
303 |
304 |           <div className="flex justify-end gap-2 pt-2">
305 |             <Button type="button" variant="outline" onClick={() => setModalOpen(false)}>
306 |               Cancel
307 |             </Button>
308 |             <Button type="submit" disabled={isSubmitting}>
309 |               {isSubmitting ? "Publishing..." : "Publish Broadcast"}
310 |             </Button>
311 |           </div>
312 |         </form>
313 |       </Modal>
314 |     </DashboardShell>
315 |   );
316 | }
```

File #43: src/app/api/health/route.ts

20 lines

```
 1 | import { NextResponse } from "next/server";
 2 | import { isSupabaseConfigured } from "@/lib/env";
 3 |
 4 | export async function GET() {
 5 |   const supabaseReady = isSupabaseConfigured();
 6 |
 7 |   return NextResponse.json({
 8 |     status: "ok",
 9 |     app: "Smart Campus Full-Stack Platform",
10 |     timestamp: new Date().toISOString(),
11 |     services: {
12 |       nextjs: "operational",
13 |       supabase: supabaseReady ? "connected" : "credentials_required",
14 |       auth: "supabase_auth",
15 |       storage: "supabase_storage",
16 |       realtime: "supabase_realtime",
17 |     },
18 |     version: "1.0.0",
19 |   });
20 | }
```

File #44: src/app/assignments/page.tsx

357 lines

```
 1 | "use client";
 2 |
 3 | import * as React from "react";
 4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
 5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
 6 | import { Badge } from "@/components/ui/badge";
 7 | import { Button } from "@/components/ui/button";
 8 | import { FileText, Clock, Upload, CheckCircle2, Search, Filter, Plus, FileCheck } from "lucide-react";
 9 | import { Modal } from "@/components/ui/modal";
10 | import { Input } from "@/components/ui/input";
11 | import { useToast } from "@/components/ui/toast";
12 | import { LoadingState, ErrorState, EmptyState } from "@/components/ui/state-views";
13 | import {
14 |   getAssignmentsAction,
15 |   createAssignmentAction,
16 |   submitAssignmentDeliverableAction,
17 | } from "@/actions/academics";
18 |
19 | export default function AssignmentsPage() {
20 |   const { success, error } = useToast();
21 |   const [assignments, setAssignments] = React.useState<any[]>([]);
```

```
  const [loading, setLoading] = React.useState(true);
  const [hasError, setHasError] = React.useState(false);
  const [filter, setFilter] = React.useState<string>("all");
  const [search, setSearch] = React.useState("");

  // Modals
  const [activeUpload, setActiveUpload] = React.useState<any | null>(null);
  const [selectedFile, setSelectedFile] = React.useState<string>("");
  const [submissionNotes, setSubmissionNotes] = React.useState("");
  const [isUploading, setIsUploading] = React.useState(false);

  const [createModalOpen, setCreateModalOpen] = React.useState(false);
  const [newAssignment, setNewAssignment] = React.useState({
    courseId: "00000000-0000-0000-0000-000000000001",
    title: "",
    description: "",
    dueDate: "",
    maxMarks: 50,
  });
  const [isCreating, setIsCreating] = React.useState(false);

  const fetchAssignments = React.useCallback(async () => {
    setLoading(true);
    setHasError(false);
    try {
      const res = await getAssignmentsAction();
      if (res.success && res.data) {
        setAssignments(res.data);
      } else {
        setHasError(true);
      }
    } catch {
      setHasError(true);
    } finally {
      setLoading(false);
    }
  }, []);

  React.useEffect(() => {
    fetchAssignments();
  }, [fetchAssignments]);

  const handleUploadSubmit = async (e: React.FormEvent) => {
    e.preventDefault();
    if (!activeUpload) return;
    setIsUploading(true);
    try {
      const fileUrl = `/storage/assignments/${selectedFile || "submission.pdf"}`;
      const res = await submitAssignmentDeliverableAction(activeUpload.id, fileUrl, submissionNotes);
      if (res.success) {
        success("Deliverable Uploaded", `Submission confirmed for ${activeUpload.title}`);
        setActiveUpload(null);
        setSelectedFile("");
        setSubmissionNotes("");
        fetchAssignments();
      } else {
        error("Upload Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsUploading(false);
    }
  };

  const handleCreateAssignment = async (e: React.FormEvent) => {
    e.preventDefault();
    setIsCreating(true);
    try {
      const res = await createAssignmentAction(newAssignment);
      if (res.success) {
        success("Assignment Created", "Assignment published to student cohorts.");
        setCreateModalOpen(false);
        setNewAssignment({
          courseId: "00000000-0000-0000-0000-000000000001",
          title: "",
          description: "",
          dueDate: "",
          maxMarks: 50,
        });
```

```
        fetchAssignments();
      } else {
        error("Failed", res.message || "Please check inputs.");
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsCreating(false);
    }
  };

  const filtered = assignments.filter((item) => {
    const matchesFilter = filter === "all" || item.status === filter;
    const title = item.title || "";
    const subject = item.course?.title || item.subject || "";
    const faculty = item.faculty?.full_name || item.faculty || "";
    const matchesSearch =
      title.toLowerCase().includes(search.toLowerCase()) ||
      subject.toLowerCase().includes(search.toLowerCase()) ||
      faculty.toLowerCase().includes(search.toLowerCase());
    return matchesFilter && matchesSearch;
  });

  return (
    <DashboardShell
      title="Academic Assignments & Deliverables"
      description="Course projects, lab evaluations, and submission tracking"
      actions={
        <div className="flex items-center gap-2">
          <Button size="sm" onClick={() => setCreateModalOpen(true)} className="gap-1.5 shadow-sm shadow-
  blue-500/20">
            <Plus className="h-4 w-4" />
            <span>Publish Assignment</span>
          </Button>
        </div>
      }
    >
      <div className="space-y-6">
        {/* Filters and Search */}
        <div className="flex flex-col gap-3 sm:flex-row sm:items-center sm:justify-between rounded-2xl bo
  rder border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
          <div className="flex items-center gap-1.5">
            {["all", "pending", "submitted", "graded"].map((st) => (
              <button
                key={st}
                onClick={() => setFilter(st)}
                className={`rounded-xl px-3 py-1.5 text-xs font-semibold capitalize transition-all ${
                  filter === st
                    ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
                    : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
                }`}
              >
                {st}
              </button>
            ))}
          </div>

          <div className="relative">
            <input
              type="text"
              placeholder="Filter assignments..."
              value={search}
              onChange={(e) => setSearch(e.target.value)}
              className="w-full sm:w-64 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs te
  xt-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-
  200"
            />
          </div>
        </div>

        {/* State Views */}
        {loading ? (
          <LoadingState message="Fetching academic assignments from PostgreSQL..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchAssignments} />
        ) : filtered.length === 0 ? (
          <EmptyState
            title="No Assignments Found"
            description="There are no assignments matching your filter criteria."
            actionLabel="Publish Assignment"
```

```
178 |             onAction={() => setCreateModalOpen(true)}
179 |           />
180 |         ) : (
181 |           <div className="space-y-4">
182 |             {filtered.map((item) => {
183 |               const subjectName = item.course ? `${item.course.code}: ${item.course.title}` : item.subjec
    |   t || "Course Assignment";
184 |               const facultyName = item.faculty?.full_name || item.faculty || "Faculty Instructor";
185 |               const dueDateText = item.due_date ? new Date(item.due_date).toLocaleString() : item.dueDate
    |    || "Due Soon";
186 |
187 |               return (
188 |                 <Card key={item.id} className="hover:border-blue-500/40 transition-all">
189 |                   <CardHeader className="pb-2">
190 |                     <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-2">
191 |                       <div>
192 |                         <span className="text-xs font-semibold text-blue-600 dark:text-blue-400">
193 |                           {subjectName}
194 |                         </span>
195 |                         <CardTitle className="text-base mt-0.5">{item.title}</CardTitle>
196 |                         <CardDescription className="text-xs">
197 |                           Instructor: {facultyName} • Max Marks: {item.max_marks || item.maxMarks}
198 |                         </CardDescription>
199 |                       </div>
200 |                       <div className="flex items-center gap-2">
201 |                         <Badge
202 |                           variant={
203 |                             item.status === "graded"
204 |                               ? "success"
205 |                               : item.status === "submitted"
206 |                               ? "info"
207 |                               : "warning"
208 |                           }
209 |                         >
210 |                           {item.status === "graded"
211 |                             ? `Graded: ${item.marks_received || item.marksReceived}/${item.max_marks || i
    |   tem.maxMarks}`
212 |                             : item.status === "submitted"
213 |                             ? "Submitted"
214 |                             : "Pending Submission"}
215 |                         </Badge>
216 |                       </div>
217 |                     </div>
218 |                   </CardHeader>
219 |                   <CardContent className="pt-2">
220 |                     <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed mb-4">
221 |                       {item.description}
222 |                     </p>
223 |                     <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3 borde
    |   r-t border-zinc-100 dark:border-zinc-800 pt-3 text-xs text-zinc-500">
224 |                       <div className="flex items-center gap-1.5">
225 |                         <Clock className="h-4 w-4 text-zinc-400" />
226 |                         <span>
227 |                           Deadline: <strong className="text-zinc-800 dark:text-zinc-200">{dueDateText}</s
    |   trong>
228 |                         </span>
229 |                       </div>
230 |
231 |                       <div className="flex items-center gap-2">
232 |                         <Button
233 |                           size="sm"
234 |                           className="gap-1.5"
235 |                           onClick={() => setActiveUpload(item)}
236 |                         >
237 |                           <Upload className="h-3.5 w-3.5" />
238 |                           Submit Deliverable
239 |                         </Button>
240 |                       </div>
241 |                     </div>
242 |                   </CardContent>
243 |                 </Card>
244 |               );
245 |             })}
246 |           </div>
247 |         )}
248 |       </div>
249 |
250 |       {/* Upload Deliverable Modal */}
251 |       {activeUpload && (
252 |         <Modal
```

```
253 |           isOpen={!!activeUpload}
254 |           onClose={() => setActiveUpload(null)}
255 |           title={`Upload Assignment: ${activeUpload.title}`}
256 |           description="Direct upload to Supabase assignment_submissions bucket"
257 |         >
258 |           <form onSubmit={handleUploadSubmit} className="space-y-4">
259 |             <div className="rounded-xl border-2 border-dashed border-zinc-300 dark:border-zinc-700 p-6 te
    |   xt-center hover:bg-zinc-50 dark:hover:bg-zinc-800/40 transition-colors">
260 |               <Upload className="h-8 w-8 mx-auto text-blue-500 mb-2" />
261 |               <p className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
262 |                 Choose deliverable file (PDF, ZIP, TAR)
263 |               </p>
264 |               <input
265 |                 type="file"
266 |                 className="mt-3 block w-full text-xs text-zinc-500 file:mr-4 file:py-1.5 file:px-3 file:r
    |   ounded-xl file:border-0 file:text-xs file:font-semibold file:bg-blue-50 file:text-blue-700 hover:file:bg-
    |   blue-100 dark:file:bg-blue-950 dark:file:text-blue-300"
267 |                 onChange={(e) => {
268 |                   if (e.target.files?.[0]) setSelectedFile(e.target.files[0].name);
269 |                 }}
270 |               />
271 |             </div>
272 |
273 |             <div className="space-y-1.5">
274 |               <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
275 |                 Submission Notes / GitHub Repository Link
276 |               </label>
277 |               <input
278 |                 type="text"
279 |                 placeholder="https://github.com/..."
280 |                 className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs text-zinc-
    |   800 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
281 |                 value={submissionNotes}
282 |                 onChange={(e) => setSubmissionNotes(e.target.value)}
283 |               />
284 |             </div>
285 |
286 |             <div className="flex justify-end gap-2 pt-2">
287 |               <Button type="button" variant="outline" onClick={() => setActiveUpload(null)}>
288 |                 Cancel
289 |               </Button>
290 |               <Button type="submit" disabled={isUploading}>
291 |                 {isUploading ? "Uploading to Storage..." : "Commit Submission"}
292 |               </Button>
293 |             </div>
294 |           </form>
295 |         </Modal>
296 |       )}
297 |
298 |       {/* Publish Assignment Modal */}
299 |       <Modal
300 |         isOpen={createModalOpen}
301 |         onClose={() => setCreateModalOpen(false)}
302 |         title="Publish Course Assignment"
303 |         description="Creates assignment entry in PostgreSQL assignments table"
304 |       >
305 |         <form onSubmit={handleCreateAssignment} className="space-y-4">
306 |           <Input
307 |             label="Assignment Title"
308 |             required
309 |             placeholder="e.g. Distributed Consensus Engine"
310 |             value={newAssignment.title}
311 |             onChange={(e) => setNewAssignment({ ...newAssignment, title: e.target.value })}
312 |           />
313 |
314 |           <div className="space-y-1.5">
315 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
316 |               Problem Description & Instructions
317 |             </label>
318 |             <textarea
319 |               required
320 |               rows={3}
321 |               placeholder="Specify requirements, benchmarks, and evaluation guidelines..."
322 |               className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-8
    |   00 dark:bg-zinc-900"
323 |               value={newAssignment.description}
324 |               onChange={(e) => setNewAssignment({ ...newAssignment, description: e.target.value })}
325 |             />
326 |           </div>
327 |
```

```
328 |           <div className="grid grid-cols-2 gap-3">
329 |             <Input
330 |               label="Submission Deadline"
331 |               type="datetime-local"
332 |               required
333 |               value={newAssignment.dueDate}
334 |               onChange={(e) => setNewAssignment({ ...newAssignment, dueDate: e.target.value })}
335 |             />
336 |             <Input
337 |               label="Maximum Marks"
338 |               type="number"
339 |               required
340 |               value={newAssignment.maxMarks}
341 |               onChange={(e) => setNewAssignment({ ...newAssignment, maxMarks: Number(e.target.value) })}
342 |             />
343 |           </div>
344 |
345 |           <div className="flex justify-end gap-2 pt-2">
346 |             <Button type="button" variant="outline" onClick={() => setCreateModalOpen(false)}>
347 |               Cancel
348 |             </Button>
349 |             <Button type="submit" disabled={isCreating}>
350 |               {isCreating ? "Publishing..." : "Publish Assignment"}
351 |             </Button>
352 |           </div>
353 |         </form>
354 |       </Modal>
355 |     </DashboardShell>
356 |   );
357 | }
```

File #45: src/app/attendance/page.tsx 339 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { AttendanceChart } from "@/components/dashboard/attendance-chart";
  8 | import { CheckCircle2, AlertCircle, Calendar, Download, Filter, FileText } from "lucide-react";
  9 | import { Button } from "@/components/ui/button";
 10 | import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";
 11 | import { Modal } from "@/components/ui/modal";
 12 | import { useToast } from "@/components/ui/toast";
 13 | import { LoadingState, EmptyState } from "@/components/ui/state-views";
 14 | import { getAttendanceAction } from "@/actions/academics";
 15 |
 16 | interface SubjectAttendance {
 17 |   code: string;
 18 |   name: string;
 19 |   conducted: number;
 20 |   attended: number;
 21 |   pct: number;
 22 |   status: "Safe" | "Warning" | "Critical";
 23 |   faculty: string;
 24 | }
 25 |
 26 | const defaultFallbackList: SubjectAttendance[] = [
 27 |   { code: "CS601", name: "Data Structures & Algorithms", conducted: 32, attended: 29, pct: 90.6, status:
    |   "Safe", faculty: "Prof. Sharma" },
 28 |   { code: "CS602", name: "Database Management Systems", conducted: 28, attended: 26, pct: 92.8, status: "
    |   Safe", faculty: "Dr. Rajesh V" },
 29 |   { code: "CS603", name: "Operating Systems", conducted: 30, attended: 23, pct: 76.6, status: "Warning",
    |   faculty: "Dr. Nair" },
 30 |   { code: "CS604", name: "Computer Networks", conducted: 26, attended: 22, pct: 84.6, status: "Safe", fac
    |   ulty: "Prof. Rao" },
 31 |   { code: "CS605", name: "Software Engineering Studio", conducted: 24, attended: 23, pct: 95.8, status: "
    |   Safe", faculty: "Dr. Gupta" },
 32 |   { code: "CS607", name: "Artificial Intelligence Elective", conducted: 20, attended: 14, pct: 70.0, stat
    |   us: "Critical", faculty: "Prof. Mehta" },
 33 | ];
 34 |
 35 | export default function AttendancePage() {
 36 |   const { success, error: toastError } = useToast();
 37 |   const [loading, setLoading] = React.useState(true);
 38 |   const [items, setItems] = React.useState<SubjectAttendance[]>(defaultFallbackList);
 39 |   const [filterStatus, setFilterStatus] = React.useState<string>("All");
 40 |   const [selectedSubject, setSelectedSubject] = React.useState<SubjectAttendance | null>(null);
```

```
  |
  |   React.useEffect(() => {
  |     let mounted = true;
  |     async function load() {
  |       try {
  |         const res = await getAttendanceAction();
  |         if (mounted && res?.success && res.data && res.data.length > 0) {
  |           // If returned as raw attendance list or aggregated
  |           if (res.data[0]?.code) {
  |             setItems(
  |               res.data.map((d: any) => ({
  |                 ...d,
  |                 status: d.pct >= 85 ? "Safe" : d.pct >= 75 ? "Warning" : "Critical",
  |               }))
  |             );
  |           }
  |         }
  |       } catch (e: any) {
  |         console.error("Attendance load error:", e);
  |       } finally {
  |         if (mounted) setLoading(false);
  |       }
  |     }
  |     load();
  |     return () => {
  |       mounted = false;
  |     };
  |   }, []);
  |
  |   const filtered = items.filter((s) => {
  |     if (filterStatus === "All") return true;
  |     return s.status === filterStatus;
  |   });
  |
  |   const cumulativeAttended = items.reduce((acc, curr) => acc + curr.attended, 0);
  |   const cumulativeConducted = items.reduce((acc, curr) => acc + curr.conducted, 0);
  |   const cumulativePct = cumulativeConducted > 0 ? ((cumulativeAttended / cumulativeConducted) * 100).toFi
  |   xed(1) : "0.0";
  |
  |   return (
  |     <DashboardShell
  |       title="Attendance & Compliance Intelligence"
  |       description="Subject-level tracking with minimum 75% institutional examination criteria"
  |       actions={
  |         <Button
  |           variant="outline"
  |           size="sm"
  |           className="gap-1.5"
  |           onClick={() => success("Transcript Generated", "Official Attendance Report downloaded as PDF.")
  |   }
  |         >
  |           <Download className="h-4 w-4" />
  |           Export Statement
  |         </Button>
  |       }
  |     >
  |       <div className="space-y-6">
  |         {/* Metric Cards Top */}
  |         <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">
  |           <Card>
  |             <CardContent className="p-4">
  |               <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
  |                 Overall Aggregate
  |               </span>
  |               <p className="mt-1 text-2xl font-extrabold text-blue-600 dark:text-blue-400">
  |                 {cumulativePct}%
  |               </p>
  |               <span className="text-[11px] text-emerald-600 dark:text-emerald-400 font-medium">
  |                 Above 75% threshold
  |               </span>
  |             </CardContent>
  |           </Card>
  |
  |           <Card>
  |             <CardContent className="p-4">
  |               <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
  |                 Total Classes Attended
  |               </span>
  |               <p className="mt-1 text-2xl font-extrabold text-zinc-900 dark:text-zinc-100">
  |                 {cumulativeAttended} / {cumulativeConducted}
```

```
              </p>
              <span className="text-[11px] text-zinc-500">Lectures & labs combined</span>
            </CardContent>
          </Card>

          <Card>
            <CardContent className="p-4">
              <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
                Subjects At Risk
              </span>
              <p className="mt-1 text-2xl font-extrabold text-amber-500">2 Courses</p>
              <span className="text-[11px] text-amber-600 font-medium">1 Warning, 1 Critical</span>
            </CardContent>
          </Card>

          <Card>
            <CardContent className="p-4">
              <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
                Hall Ticket Status
              </span>
              <p className="mt-1 text-2xl font-extrabold text-emerald-600 dark:text-emerald-400">
                Eligible
              </p>
              <span className="text-[11px] text-zinc-500">Current standing verified</span>
            </CardContent>
          </Card>
        </div>

        {/* Charts & Highlights */}
        <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
          <Card className="lg:col-span-2">
            <CardHeader>
              <CardTitle>Attendance Distribution</CardTitle>
              <CardDescription>Visual comparison against academic threshold (75%)</CardDescription>
            </CardHeader>
            <CardContent>
              <AttendanceChart />
            </CardContent>
          </Card>

          <Card className="flex flex-col justify-between">
            <CardHeader>
              <CardTitle>Attendance Guidelines</CardTitle>
              <CardDescription>Academic Regulation Section 4.2</CardDescription>
            </CardHeader>
            <CardContent className="space-y-4">
              <div className="rounded-xl border border-emerald-500/30 bg-emerald-50/20 p-4 dark:bg-emeral
d-950/20">
                <div className="flex items-center gap-2 text-emerald-600 dark:text-emerald-400 font-bold
text-xs">
                  <CheckCircle2 className="h-4 w-4" />
                  <span>≥ 75% Rule Compliance</span>
                </div>
                <p className="text-[11px] text-zinc-600 dark:text-zinc-400 mt-1 leading-relaxed">
                  Students maintaining 75% or greater are automatically cleared for end-semester assessme
nts without condonation fees.
                </p>
              </div>

              <div className="rounded-xl border border-rose-500/30 bg-rose-50/20 p-4 dark:bg-rose-950/20"
>
                <div className="flex items-center gap-2 text-rose-600 dark:text-rose-400 font-bold text-x
s">
                  <AlertCircle className="h-4 w-4" />
                  <span>Medical Leave Condonation</span>
                </div>
                <p className="text-[11px] text-zinc-600 dark:text-zinc-400 mt-1 leading-relaxed">
                  Leaves supported by physician certificates can compensate up to 10% shortfall (65% mini
mum hard cutoff).
                </p>
              </div>
            </CardContent>
          </Card>
        </div>

        {/* Filters and Table */}
        <div className="space-y-4">
          <div className="flex items-center justify-between">
            <h3 className="text-base font-bold text-zinc-900 dark:text-zinc-100">
              Course-wise Breakdown
```

```
193 |             </h3>
194 |             <div className="flex items-center gap-2">
195 |               <span className="text-xs text-zinc-500 font-medium">Filter by status:</span>
196 |               <select
197 |                 value={filterStatus}
198 |                 onChange={(e) => setFilterStatus(e.target.value)}
199 |                 className="rounded-xl border border-zinc-200 bg-white px-2.5 py-1 text-xs text-zinc-800 d
    |   ark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
200 |               >
201 |                 <option value="All">All Courses</option>
202 |                 <option value="Safe">Safe (&gt;80%)</option>
203 |                 <option value="Warning">Warning (75-80%)</option>
204 |                 <option value="Critical">Critical (&lt;75%)</option>
205 |               </select>
206 |             </div>
207 |           </div>
208 |
209 |           {loading ? (
210 |             <LoadingState message="Loading attendance records & compliance ratios..." />
211 |           ) : filtered.length === 0 ? (
212 |             <EmptyState
213 |               title="No attendance records found"
214 |               description="No courses match the chosen compliance filter."
215 |             />
216 |           ) : (
217 |             <Table>
218 |               <TableHeader>
219 |                 <TableRow>
220 |                   <TableHead>Course Code & Title</TableHead>
221 |                   <TableHead>Faculty</TableHead>
222 |                   <TableHead>Attended / Conducted</TableHead>
223 |                   <TableHead>Percentage</TableHead>
224 |                   <TableHead>Compliance</TableHead>
225 |                   <TableHead className="text-right">Action</TableHead>
226 |                 </TableRow>
227 |               </TableHeader>
228 |               <TableBody>
229 |                 {filtered.map((item) => (
230 |                   <TableRow key={item.code}>
231 |                     <TableCell>
232 |                       <span className="font-semibold text-zinc-900 dark:text-zinc-100">{item.name}</span>
233 |                       <span className="block text-[11px] text-zinc-500">{item.code}</span>
234 |                     </TableCell>
235 |                     <TableCell className="text-xs text-zinc-600 dark:text-zinc-400">{item.faculty}</Table
    |   Cell>
236 |                     <TableCell className="text-xs font-medium">
237 |                       {item.attended} / {item.conducted}
238 |                     </TableCell>
239 |                     <TableCell>
240 |                       <div className="flex items-center gap-2">
241 |                         <div className="w-16 bg-zinc-200 rounded-full h-1.5 dark:bg-zinc-800 overflow-hid
    |   den">
242 |                           <div
243 |                             className={`h-1.5 rounded-full ${
244 |                               item.pct >= 85
245 |                                 ? "bg-emerald-500"
246 |                                 : item.pct >= 75
247 |                                 ? "bg-amber-500"
248 |                                 : "bg-rose-500"
249 |                             }`}
250 |                             style={{ width: `${Math.min(item.pct, 100)}%` }}
251 |                           />
252 |                         </div>
253 |                         <span className="font-bold text-xs">{item.pct}%</span>
254 |                       </div>
255 |                     </TableCell>
256 |                     <TableCell>
257 |                       <Badge
258 |                         variant={
259 |                           item.status === "Safe"
260 |                             ? "success"
261 |                             : item.status === "Warning"
262 |                             ? "warning"
263 |                             : "destructive"
264 |                         }
265 |                       >
266 |                         {item.status}
267 |                       </Badge>
268 |                     </TableCell>
269 |                     <TableCell className="text-right">
```

```
                      <Button
                        variant="ghost"
                        size="sm"
                        className="text-xs text-blue-600 dark:text-blue-400"
                        onClick={() => setSelectedSubject(item)}
                      >
                        Detailed Log
                      </Button>
                    </TableCell>
                  </TableRow>
                ))}
              </TableBody>
            </Table>
          )}
        </div>
      </div>

      {/* Detail Log Modal */}
      {selectedSubject && (
        <Modal
          isOpen={!!selectedSubject}
          onClose={() => setSelectedSubject(null)}
          title={`Attendance History: ${selectedSubject.name}`}
          description={`Instructor: ${selectedSubject.faculty} • Aggregate: ${selectedSubject.pct}%`}
        >
          <div className="space-y-4">
            <div className="rounded-xl bg-zinc-50 p-4 dark:bg-zinc-800/50 space-y-2 text-xs">
              <div className="flex justify-between">
                <span className="text-zinc-500">Conducted Sessions:</span>
                <span className="font-semibold">{selectedSubject.conducted} hours</span>
              </div>
              <div className="flex justify-between">
                <span className="text-zinc-500">Present Sessions:</span>
                <span className="font-semibold text-emerald-600">{selectedSubject.attended} hours</span>
              </div>
              <div className="flex justify-between">
                <span className="text-zinc-500">Unexcused Absences:</span>
                <span className="font-semibold text-rose-500">
                  {selectedSubject.conducted - selectedSubject.attended} hours
                </span>
              </div>
            </div>

            <p className="text-xs text-zinc-500">
              Recent recorded roll call sessions:
            </p>
            <div className="space-y-1.5 text-xs">
              <div className="flex justify-between items-center p-2 rounded-lg border border-zinc-200/80 dark:border-zinc-800">
                <span>Oct 09, 2026 • 09:00 AM</span>
                <Badge variant="success">Present</Badge>
              </div>
              <div className="flex justify-between items-center p-2 rounded-lg border border-zinc-200/80 dark:border-zinc-800">
                <span>Oct 07, 2026 • 09:00 AM</span>
                <Badge variant="success">Present</Badge>
              </div>
              <div className="flex justify-between items-center p-2 rounded-lg border border-zinc-200/80 dark:border-zinc-800">
                <span>Oct 05, 2026 • 09:00 AM</span>
                <Badge variant="destructive">Absent</Badge>
              </div>
            </div>

            <div className="flex justify-end pt-3">
              <Button onClick={() => setSelectedSubject(null)}>Close Record</Button>
            </div>
          </div>
        </Modal>
      )}
    </DashboardShell>
  );
}
```

File #46: src/app/contact/page.tsx

220 lines

```
"use client";

import * as React from "react";
```

```
import {
  Mail,
  Phone,
  MapPin,
  Clock,
  Send,
  MessageSquare,
  HelpCircle,
  Building,
} from "lucide-react";
import { Navbar } from "@/components/layout/navbar";
import { Footer } from "@/components/layout/footer";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Input } from "@/components/ui/input";
import { Button } from "@/components/ui/button";
import { Badge } from "@/components/ui/badge";
import { useToast } from "@/components/ui/toast";

export default function ContactPage() {
  const { success } = useToast();
  const [formData, setFormData] = React.useState({
    name: "",
    email: "",
    department: "Computer Science",
    subject: "",
    message: "",
  });
  const [isSubmitting, setIsSubmitting] = React.useState(false);

  const handleSubmit = (e: React.FormEvent) => {
    e.preventDefault();
    setIsSubmitting(true);
    setTimeout(() => {
      setIsSubmitting(false);
      success("Inquiry Dispatched", "Your message has been routed to the campus IT & Academic Helpdesk.")
  ;
      setFormData({
        name: "",
        email: "",
        department: "Computer Science",
        subject: "",
        message: "",
      });
    }, 800);
  };

  return (
    <div className="min-h-screen flex flex-col bg-zinc-50/70 dark:bg-zinc-950 text-zinc-900 dark:text-zin
  c-100">
      <Navbar />

      <main className="flex-1 py-12 sm:py-16">
        <div className="mx-auto max-w-6xl px-4 sm:px-6 lg:px-8">
          <div className="text-center max-w-3xl mx-auto mb-12">
            <Badge variant="info" className="mb-3">
              Support & Helpdesk
            </Badge>
            <h1 className="text-3xl font-extrabold tracking-tight sm:text-4xl">
              Campus Operations & Technical Support
            </h1>
            <p className="mt-3 text-sm text-zinc-500 dark:text-zinc-400">
              Have questions regarding portal access, student records, timetable disputes, or IT services
  ? Reach out directly to campus administrators.
            </p>
          </div>

          <div className="grid grid-cols-1 lg:grid-cols-3 gap-8">
            {/* Contact Information & Channels */}
            <div className="space-y-4">
              <Card>
                <CardHeader>
                  <CardTitle className="text-base flex items-center gap-2">
                    <Building className="h-4 w-4 text-blue-600" />
                    Administrative Headquarters
                  </CardTitle>
                </CardHeader>
                <CardContent className="space-y-3 text-xs text-zinc-600 dark:text-zinc-400">
                  <div className="flex items-start gap-2.5">
                    <MapPin className="h-4 w-4 text-zinc-400 shrink-0 mt-0.5" />
                    <span>Academic Block A, Floor 3, University Tech Campus, Innovation Corridor</span>
```

```
                  </div>
                  <div className="flex items-center gap-2.5">
                    <Clock className="h-4 w-4 text-zinc-400 shrink-0" />
                    <span>Monday – Friday: 08:30 AM – 05:30 PM</span>
                  </div>
                </CardContent>
              </Card>

              <Card>
                <CardHeader>
                  <CardTitle className="text-base flex items-center gap-2">
                    <Phone className="h-4 w-4 text-emerald-600" />
                    Direct Contact Lines
                  </CardTitle>
                </CardHeader>
                <CardContent className="space-y-3 text-xs text-zinc-600 dark:text-zinc-400">
                  <div className="flex items-center gap-2.5">
                    <Phone className="h-4 w-4 text-zinc-400 shrink-0" />
                    <span>General Enquiries: +1 (800) 555-CAMPUS</span>
                  </div>
                  <div className="flex items-center gap-2.5">
                    <Mail className="h-4 w-4 text-zinc-400 shrink-0" />
                    <span>IT Support: support@smartcampus.edu</span>
                  </div>
                  <div className="flex items-center gap-2.5">
                    <Mail className="h-4 w-4 text-zinc-400 shrink-0" />
                    <span>Registrar: registrar@smartcampus.edu</span>
                  </div>
                </CardContent>
              </Card>

              <Card className="border-rose-500/20 bg-rose-50/30 dark:bg-rose-950/20">
                <CardHeader>
                  <CardTitle className="text-base text-rose-600 dark:text-rose-400 flex items-center gap-
  2">
                    <HelpCircle className="h-4 w-4" />
                    Immediate Emergency?
                  </CardTitle>
                  <CardDescription className="text-xs">
                    For on-campus medical or security incidents, bypass email and dial the 24/7 SOS helpl
  ine.
                  </CardDescription>
                </CardHeader>
                <CardContent>
                  <a
                    href="tel:911"
                    className="inline-flex items-center gap-2 text-xs font-bold text-rose-600 dark:text-r
  ose-400 hover:underline"
                  >
                    Dial 24/7 Emergency Line: +1 (800) 911-CAMP
                  </a>
                </CardContent>
              </Card>
            </div>

            {/* Contact Form */}
            <div className="lg:col-span-2">
              <Card>
                <CardHeader>
                  <CardTitle className="text-lg">Submit a Support Ticket</CardTitle>
                  <CardDescription className="text-xs">
                    Please provide detailed context so we can forward your ticket to the correct departme
  nt.
                  </CardDescription>
                </CardHeader>
                <CardContent>
                  <form onSubmit={handleSubmit} className="space-y-4">
                    <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
                      <Input
                        label="Full Name"
                        required
                        placeholder="e.g. Alex Morgan"
                        value={formData.name}
                        onChange={(e) => setFormData({ ...formData, name: e.target.value })}
                      />
                      <Input
                        label="Campus Email"
                        type="email"
                        required
                        placeholder="alex.m@student.campus.edu"
```

```
                        value={formData.email}
                        onChange={(e) => setFormData({ ...formData, email: e.target.value })}
                      />
                    </div>

                    <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
                      <div className="space-y-1.5">
                        <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                          Academic Department
                        </label>
                        <select
                          className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs text-zinc-800 focus:outline-none focus:ring-2 focus:ring-blue-500/20 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
                          value={formData.department}
                          onChange={(e) => setFormData({ ...formData, department: e.target.value })}
                        >
                          <option>Computer Science & Engineering</option>
                          <option>Electrical & Electronics</option>
                          <option>Mechanical & Mechatronics</option>
                          <option>Business & Management</option>
                          <option>General Administration</option>
                        </select>
                      </div>

                      <Input
                        label="Subject"
                        required
                        placeholder="e.g. Student ID Card Synchronization Issue"
                        value={formData.subject}
                        onChange={(e) => setFormData({ ...formData, subject: e.target.value })}
                      />
                    </div>

                    <div className="space-y-1.5">
                      <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                        Inquiry Details
                      </label>
                      <textarea
                        required
                        rows={5}
                        placeholder="Explain your request or issue in detail..."
                        className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs text-zinc-800 placeholder-zinc-400 focus:border-blue-500 focus:outline-none focus:ring-2 focus:ring-blue-500/20 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
                        value={formData.message}
                        onChange={(e) => setFormData({ ...formData, message: e.target.value })}
                      />
                    </div>

                    <div className="flex justify-end pt-2">
                      <Button type="submit" disabled={isSubmitting} className="gap-2">
                        <Send className="h-3.5 w-3.5" />
                        {isSubmitting ? "Routing Ticket..." : "Submit Ticket"}
                      </Button>
                    </div>
                  </form>
                </CardContent>
              </Card>
            </div>
          </div>
        </div>
      </main>

      <Footer />
    </div>
  );
}
```

File #47: src/app/dashboard/admin/analytics/page.tsx — 167 lines

```
"use client";

import * as React from "react";
import { BarChart3, TrendingUp, Users, BookOpen, Clock, ShieldCheck, Download } from "lucide-react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
```

```
import { CampusActivityChart } from "@/components/dashboard/campus-stats-chart";
import { AttendanceChart } from "@/components/dashboard/attendance-chart";
import { useToast } from "@/components/ui/toast";

export default function AdminAnalyticsPage() {
  const { success } = useToast();

  return (
    <DashboardShell
      title="Campus Institutional Analytics & Intelligence"
      description="Aggregate statistics across academic courses, attendance compliance, facility utilizat
  ion, and user traffic"
      actions={
        <Button
          variant="outline"
          size="sm"
          className="gap-1.5"
          onClick={() => success("Intelligence Report Exported", "Comprehensive Institutional Audit PDF g
  enerated.")}
        >
          <Download className="h-4 w-4" />
          Export Institutional Report
        </Button>
      }
    >
      <div className="space-y-6">
        {/* Metric Overview */}
        <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">
          <Card>
            <CardContent className="p-4">
              <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
                Daily Active Users (DAU)
              </span>
              <p className="mt-1 text-2xl font-extrabold text-blue-600 dark:text-blue-400">
                3,420
              </p>
              <span className="text-[11px] text-emerald-600 font-medium">↑ +14.2% this semester</span>
            </CardContent>
          </Card>

          <Card>
            <CardContent className="p-4">
              <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
                Average Campus Attendance
              </span>
              <p className="mt-1 text-2xl font-extrabold text-zinc-900 dark:text-zinc-100">
                87.8%
              </p>
              <span className="text-[11px] text-zinc-500">Above 75% regulatory line</span>
            </CardContent>
          </Card>

          <Card>
            <CardContent className="p-4">
              <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
                Assignment Turn-in Rate
              </span>
              <p className="mt-1 text-2xl font-extrabold text-emerald-600">
                93.4%
              </p>
              <span className="text-[11px] text-emerald-600 font-medium">On-time submissions</span>
            </CardContent>
          </Card>

          <Card>
            <CardContent className="p-4">
              <span className="text-[11px] font-semibold text-zinc-500 uppercase tracking-wider">
                Lab & Facility Utilization
              </span>
              <p className="mt-1 text-2xl font-extrabold text-purple-600">
                82.1%
              </p>
              <span className="text-[11px] text-zinc-500">Peak hours: 11 AM - 3 PM</span>
            </CardContent>
          </Card>
        </div>

        {/* Charts Grid */}
        <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">
          <Card>
```

```
            <CardHeader>
              <CardTitle className="text-base">Realtime Portal Traffic Trends</CardTitle>
              <CardDescription className="text-xs">Hourly user interaction volume across portals</CardDescription>
            </CardHeader>
            <CardContent>
              <CampusActivityChart />
            </CardContent>
          </Card>

          <Card>
            <CardHeader>
              <CardTitle className="text-base">Attendance Compliance by Subject</CardTitle>
              <CardDescription className="text-xs">University-wide average attendance across core departments</CardDescription>
            </CardHeader>
            <CardContent>
              <AttendanceChart />
            </CardContent>
          </Card>
        </div>

        {/* Department Comparison Table */}
        <Card>
          <CardHeader>
            <CardTitle className="text-base">Departmental Performance Summary</CardTitle>
            <CardDescription className="text-xs">Key metrics aggregated for institutional review</CardDescription>
          </CardHeader>
          <CardContent>
            <div className="overflow-x-auto">
              <table className="w-full text-xs text-left">
                <thead className="border-b border-zinc-200 dark:border-zinc-800 text-zinc-500 uppercase">
                  <tr>
                    <th className="pb-3 font-semibold">Department</th>
                    <th className="pb-3 font-semibold">Students</th>
                    <th className="pb-3 font-semibold">Faculty</th>
                    <th className="pb-3 font-semibold">Avg Attendance</th>
                    <th className="pb-3 font-semibold">Research Output</th>
                    <th className="pb-3 font-semibold">Facility Usage</th>
                  </tr>
                </thead>
                <tbody className="divide-y divide-zinc-100 dark:divide-zinc-800 text-zinc-700 dark:text-zinc-300">
                  <tr>
                    <td className="py-3 font-semibold">Computer Science & Engineering</td>
                    <td>1,240</td>
                    <td>68</td>
                    <td className="text-emerald-600 font-bold">89.4%</td>
                    <td>42 Papers</td>
                    <td>94% (Turing Lab)</td>
                  </tr>
                  <tr>
                    <td className="py-3 font-semibold">Electronics & Communication</td>
                    <td>980</td>
                    <td>54</td>
                    <td className="text-emerald-600 font-bold">87.1%</td>
                    <td>28 Papers</td>
                    <td>86% (VLSI Lab)</td>
                  </tr>
                  <tr>
                    <td className="py-3 font-semibold">Mechanical & Mechatronics</td>
                    <td>850</td>
                    <td>48</td>
                    <td className="text-amber-500 font-bold">81.6%</td>
                    <td>19 Papers</td>
                    <td>78% (Workshop)</td>
                  </tr>
                  <tr>
                    <td className="py-3 font-semibold">Management & Information Systems</td>
                    <td>1,120</td>
                    <td>58</td>
                    <td className="text-emerald-600 font-bold">88.5%</td>
                    <td>31 Papers</td>
                    <td>82% (Case Pods)</td>
                  </tr>
                </tbody>
              </table>
            </div>
          </CardContent>
```

```
163 |         </Card>
164 |       </div>
165 |     </DashboardShell>
166 |   );
167 | }
```

File #48: src/app/dashboard/admin/faculty/page.tsx 127 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { UserCheck, Search, Download, Mail, Phone, MapPin, Building, Plus } from "lucide-react";
  5 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  6 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  7 | import { Badge } from "@/components/ui/badge";
  8 | import { Button } from "@/components/ui/button";
  9 | import { useToast } from "@/components/ui/toast";
 10 | import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";
 11 |
 12 | interface FacultyStaff {
 13 |   id: string;
 14 |   name: string;
 15 |   designation: string;
 16 |   dept: string;
 17 |   email: string;
 18 |   coursesHandled: number;
 19 |   cabin: string;
 20 |   status: "Active" | "On Sabbatical" | "Visiting";
 21 | }
 22 |
 23 | const facultyList: FacultyStaff[] = [
 24 |   { id: "f-1", name: "Dr. Rajesh Varma", designation: "Professor & HOD", dept: "Computer Science", email:
    |    "rajesh.v@campus.edu", coursesHandled: 3, cabin: "Tech Block A • 304", status: "Active" },
 25 |   { id: "f-2", name: "Prof. Ananya Rao", designation: "Associate Professor", dept: "Computer Science", em
    |   ail: "ananya.rao@campus.edu", coursesHandled: 2, cabin: "Tech Block A • 212", status: "Active" },
 26 |   { id: "f-3", name: "Dr. Vivek Sharma", designation: "Assistant Professor", dept: "Computer Science", em
    |   ail: "vivek.sharma@campus.edu", coursesHandled: 3, cabin: "Tech Block B • 108", status: "Active" },
 27 |   { id: "f-4", name: "Dr. Meenakshi Nair", designation: "Professor", dept: "Computer Science", email: "me
    |   enakshi.nair@campus.edu", coursesHandled: 2, cabin: "Tech Block A • 310", status: "Active" },
 28 |   { id: "f-5", name: "Dr. Sanjay Deshmukh", designation: "Professor", dept: "Mechanical", email: "sanjay.
    |   d@campus.edu", coursesHandled: 2, cabin: "Mech Block • 201", status: "On Sabbatical" },
 29 | ];
 30 |
 31 | export default function AdminFacultyPage() {
 32 |   const { success } = useToast();
 33 |   const [search, setSearch] = React.useState("");
 34 |
 35 |   const filtered = facultyList.filter(
 36 |     (f) =>
 37 |       f.name.toLowerCase().includes(search.toLowerCase()) ||
 38 |       f.designation.toLowerCase().includes(search.toLowerCase()) ||
 39 |       f.dept.toLowerCase().includes(search.toLowerCase())
 40 |   );
 41 |
 42 |   return (
 43 |     <DashboardShell
 44 |       title="Faculty Roster & Department Governance"
 45 |       description="Faculty teaching assignments, workload metrics, and departmental offices"
 46 |       actions={
 47 |         <Button
 48 |           variant="outline"
 49 |           size="sm"
 50 |           className="gap-1.5"
 51 |           onClick={() => success("Export Generated", "Faculty Directory exported as CSV.")}
 52 |         >
 53 |           <Download className="h-4 w-4" />
 54 |           Export Faculty CSV
 55 |         </Button>
 56 |       }
 57 |     >
 58 |       <div className="space-y-6">
 59 |         <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3">
 60 |           <div className="relative">
 61 |             <input
 62 |               type="text"
 63 |               placeholder="Search faculty by name, department, cabin..."
 64 |               value={search}
 65 |               onChange={(e) => setSearch(e.target.value)}
 66 |               className="w-full sm:w-72 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs te
```

```
    |   xt-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-
    |   200"
 67 |             />
 68 |           </div>
 69 |           <div className="flex items-center gap-2 text-xs text-zinc-500">
 70 |             <span>Total Faculty: <strong className="text-zinc-800 dark:text-zinc-200">264 Professors</str
    |   ong></span>
 71 |           </div>
 72 |         </div>
 73 |
 74 |         <Table>
 75 |           <TableHeader>
 76 |             <TableRow>
 77 |               <TableHead>Faculty Name & Designation</TableHead>
 78 |               <TableHead>Department</TableHead>
 79 |               <TableHead>Email Contact</TableHead>
 80 |               <TableHead>Cabin Location</TableHead>
 81 |               <TableHead>Active Courses</TableHead>
 82 |               <TableHead>Status</TableHead>
 83 |               <TableHead className="text-right">Actions</TableHead>
 84 |             </TableRow>
 85 |           </TableHeader>
 86 |           <TableBody>
 87 |             {filtered.map((f) => (
 88 |               <TableRow key={f.id}>
 89 |                 <TableCell>
 90 |                   <span className="font-semibold text-zinc-900 dark:text-zinc-100">{f.name}</span>
 91 |                   <span className="block text-[11px] text-zinc-500">{f.designation}</span>
 92 |                 </TableCell>
 93 |                 <TableCell className="text-xs text-zinc-600 dark:text-zinc-400">{f.dept}</TableCell>
 94 |                 <TableCell className="text-xs font-mono">{f.email}</TableCell>
 95 |                 <TableCell className="text-xs text-zinc-500">{f.cabin}</TableCell>
 96 |                 <TableCell className="font-bold text-xs">{f.coursesHandled} Batches</TableCell>
 97 |                 <TableCell>
 98 |                   <Badge
 99 |                     variant={
100 |                       f.status === "Active"
101 |                         ? "success"
102 |                         : f.status === "Visiting"
103 |                         ? "info"
104 |                         : "warning"
105 |                     }
106 |                   >
107 |                     {f.status}
108 |                   </Badge>
109 |                 </TableCell>
110 |                 <TableCell className="text-right">
111 |                   <Button
112 |                     variant="ghost"
113 |                     size="sm"
114 |                     className="text-xs text-blue-600 dark:text-blue-400"
115 |                     onClick={() => success("Faculty Load", `Workload sheet opened for ${f.name}`)}
116 |                   >
117 |                     Workload
118 |                   </Button>
119 |                 </TableCell>
120 |               </TableRow>
121 |             ))}
122 |           </TableBody>
123 |         </Table>
124 |       </div>
125 |     </DashboardShell>
126 |   );
127 | }
```

File #49: src/app/dashboard/admin/page.tsx 274 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import Link from "next/link";
  5 | import {
  6 |   ShieldAlert,
  7 |   Users,
  8 |   Building2,
  9 |   Activity,
 10 |   AlertTriangle,
 11 |   Server,
 12 |   Database,
```

```
  Radio,
  ArrowRight,
  UserCheck,
  GraduationCap,
  Bell,
  Calendar,
  MessageSquare,
  BarChart3,
  Clock,
  Plus,
} from "lucide-react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { CampusActivityChart } from "@/components/dashboard/campus-stats-chart";
import { Modal } from "@/components/ui/modal";
import { Input } from "@/components/ui/input";
import { useToast } from "@/components/ui/toast";
import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";

export default function AdminDashboardPage() {
  const { success, warning } = useToast();
  const [emergencyModalOpen, setEmergencyModalOpen] = React.useState(false);
  const [broadcastTitle, setBroadcastTitle] = React.useState("");
  const [broadcastMessage, setBroadcastMessage] = React.useState("");

  const handleBroadcastAlert = (e: React.FormEvent) => {
    e.preventDefault();
    setEmergencyModalOpen(false);
    warning("EMERGENCY BROADCAST TRANSMITTED", `Campus-wide notification pushed: "${broadcastTitle}"`);
    setBroadcastTitle("");
    setBroadcastMessage("");
  };

  return (
    <DashboardShell
      title="Admin Command & Operations Console"
      description="Campus-wide operations, security alerts, and system health monitoring"
      actions={
        <div className="flex gap-2">
          <Button
            variant="danger"
            size="sm"
            onClick={() => setEmergencyModalOpen(true)}
            className="gap-1.5"
          >
            <Radio className="h-4 w-4" />
            <span>Broadcast Emergency</span>
          </Button>
          <Link href="/dashboard/admin/analytics">
            <Button variant="outline" size="sm" className="gap-1.5">
              <BarChart3 className="h-4 w-4" />
              <span>Full Analytics</span>
            </Button>
          </Link>
        </div>
      }
    >
      {/* High Level Operations Metrics */}
      <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 mb-6">
        <Card className="border-l-4 border-l-rose-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Total Enrolled</p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">4,820</h3>
              <p className="text-[11px] text-zinc-500 mt-1">Across 8 Departments</p>
            </div>
            <div className="p-3 bg-rose-500/10 text-rose-600 rounded-xl">
              <Users className="h-6 w-6" />
            </div>
          </CardContent>
        </Card>

        <Card className="border-l-4 border-l-blue-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Faculty Staff</p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">264</h3>
              <p className="text-[11px] text-blue-600 dark:text-blue-400 mt-1 font-medium">98% On-Campus
```

```
    |   Active</p>
 93 |             </div>
 94 |             <div className="p-3 bg-blue-500/10 text-blue-600 rounded-xl">
 95 |               <UserCheck className="h-6 w-6" />
 96 |             </div>
 97 |           </CardContent>
 98 |         </Card>
 99 |
100 |         <Card className="border-l-4 border-l-emerald-500">
101 |           <CardContent className="p-4 flex items-center justify-between">
102 |             <div>
103 |               <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Facilities Status</p>
104 |               <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">48 / 52</h3>
105 |               <p className="text-[11px] text-emerald-600 dark:text-emerald-400 mt-1 font-medium">Operatio
    |   nal & Cleared</p>
106 |             </div>
107 |             <div className="p-3 bg-emerald-500/10 text-emerald-600 rounded-xl">
108 |               <Building2 className="h-6 w-6" />
109 |             </div>
110 |           </CardContent>
111 |         </Card>
112 |
113 |         <Card className="border-l-4 border-l-purple-500">
114 |           <CardContent className="p-4 flex items-center justify-between">
115 |             <div>
116 |               <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Active Grievances</p>
117 |               <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">12 Open</h3>
118 |               <p className="text-[11px] text-purple-600 dark:text-purple-400 mt-1 font-medium">8 Under Re
    |   view</p>
119 |             </div>
120 |             <div className="p-3 bg-purple-500/10 text-purple-600 rounded-xl">
121 |               <MessageSquare className="h-6 w-6" />
122 |             </div>
123 |           </CardContent>
124 |         </Card>
125 |       </div>
126 |
127 |       {/* Admin Modules Quick Launch */}
128 |       <div className="mb-6">
129 |         <h3 className="text-xs font-bold uppercase tracking-wider text-zinc-400 mb-3">
130 |           Admin Governance Sub-Modules
131 |         </h3>
132 |         <div className="grid grid-cols-2 sm:grid-cols-4 lg:grid-cols-6 gap-3">
133 |           {[
134 |             { title: "User Directory", href: "/dashboard/admin/users", icon: Users, color: "text-blue-600
    |   " },
135 |             { title: "Students", href: "/dashboard/admin/students", icon: GraduationCap, color: "text-ind
    |   igo-600" },
136 |             { title: "Faculty", href: "/dashboard/admin/faculty", icon: UserCheck, color: "text-amber-600
    |   " },
137 |             { title: "Analytics", href: "/dashboard/admin/analytics", icon: BarChart3, color: "text-emera
    |   ld-600" },
138 |             { title: "Announcements", href: "/announcements", icon: Bell, color: "text-purple-600" },
139 |             { title: "Feedback", href: "/feedback", icon: MessageSquare, color: "text-rose-600" },
140 |           ].map((mod) => {
141 |             const Icon = mod.icon;
142 |             return (
143 |               <Link
144 |                 key={mod.title}
145 |                 href={mod.href}
146 |                 className="flex flex-col items-center justify-center p-3.5 rounded-2xl border border-zinc
    |   -200/80 bg-white/70 hover:bg-zinc-100 dark:border-zinc-800 dark:bg-zinc-900/60 dark:hover:bg-zinc-800 tra
    |   nsition-all text-center group"
147 |               >
148 |                 <Icon className={`h-5 w-5 mb-1.5 ${mod.color} group-hover:scale-110 transition-transform`
    |   } />
149 |                 <span className="text-xs font-semibold text-zinc-800 dark:text-zinc-200">{mod.title}</spa
    |   n>
150 |               </Link>
151 |             );
152 |           })}
153 |         </div>
154 |       </div>
155 |
156 |       {/* Analytics & System Health */}
157 |       <div className="grid grid-cols-1 lg:grid-cols-3 gap-6 mb-6">
158 |         <Card className="lg:col-span-2">
159 |           <CardHeader>
160 |             <div className="flex items-center justify-between">
161 |               <div>
```

```
                <CardTitle className="text-base">Realtime Campus Operations Throughput</CardTitle>
                <CardDescription className="text-xs">Daily active transactions and portal usage</CardDesc
  ription>
              </div>
              <Badge variant="outline">Live Stream</Badge>
            </div>
          </CardHeader>
          <CardContent>
            <CampusActivityChart />
          </CardContent>
        </Card>

        {/* Cloud Infrastructure Health */}
        <Card className="flex flex-col justify-between">
          <CardHeader>
            <CardTitle className="text-base">System Health & Latency</CardTitle>
            <CardDescription className="text-xs">Database, Auth & Realtime Channels</CardDescription>
          </CardHeader>
          <CardContent className="space-y-4">
            <div className="space-y-3">
              <div className="flex items-center justify-between text-xs">
                <div className="flex items-center gap-2">
                  <Database className="h-4 w-4 text-emerald-500" />
                  <span className="font-medium">PostgreSQL Engine</span>
                </div>
                <Badge variant="success">Healthy • 12ms</Badge>
              </div>

              <div className="flex items-center justify-between text-xs">
                <div className="flex items-center gap-2">
                  <Server className="h-4 w-4 text-emerald-500" />
                  <span className="font-medium">Supabase Auth SSR</span>
                </div>
                <Badge variant="success">Operational</Badge>
              </div>

              <div className="flex items-center justify-between text-xs">
                <div className="flex items-center gap-2">
                  <Activity className="h-4 w-4 text-emerald-500" />
                  <span className="font-medium">Realtime WebSockets</span>
                </div>
                <Badge variant="success">Connected</Badge>
              </div>

              <div className="flex items-center justify-between text-xs">
                <div className="flex items-center gap-2">
                  <Building2 className="h-4 w-4 text-blue-500" />
                  <span className="font-medium">Storage Buckets</span>
                </div>
                <Badge variant="info">3 Buckets Active</Badge>
              </div>
            </div>

            <div className="rounded-xl border border-zinc-200/80 bg-zinc-50 p-3 text-[11px] text-zinc-500
   dark:border-zinc-800 dark:bg-zinc-800/40">
              Zero security vulnerabilities reported in last 24h cycle. All PostgreSQL RLS tables verifie
  d.
            </div>
          </CardContent>
        </Card>
      </div>

      {/* Emergency Broadcast Modal */}
      <Modal
        isOpen={emergencyModalOpen}
        onClose={() => setEmergencyModalOpen(false)}
        title="Trigger Emergency Broadcast Siren"
        description="Pushes high-priority alerts to all Student, Faculty, and Staff devices simultaneousl
  y."
      >
        <form onSubmit={handleBroadcastAlert} className="space-y-4">
          <Input
            label="Emergency Title"
            required
            placeholder="e.g. Severe Weather Campus Advisory"
            value={broadcastTitle}
            onChange={(e) => setBroadcastTitle(e.target.value)}
          />

          <div className="space-y-1.5">
```

```
238 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
239 |               Severity Level
240 |             </label>
241 |             <select className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:b
    |   order-zinc-800 dark:bg-zinc-900">
242 |               <option>Level 3: Critical / Immediate Action Required</option>
243 |               <option>Level 2: Urgent Campus Operational Notice</option>
244 |               <option>Level 1: General Safety Advisory</option>
245 |             </select>
246 |           </div>
247 |
248 |           <div className="space-y-1.5">
249 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
250 |               Broadcast Message
251 |             </label>
252 |             <textarea
253 |               required
254 |               rows={4}
255 |               placeholder="Provide exact instructions for building evacuation, class suspension, or emerg
    |   ency assembly..."
256 |               className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-8
    |   00 dark:bg-zinc-900"
257 |               value={broadcastMessage}
258 |               onChange={(e) => setBroadcastMessage(e.target.value)}
259 |             />
260 |           </div>
261 |
262 |           <div className="flex justify-end gap-2 pt-2">
263 |             <Button type="button" variant="outline" onClick={() => setEmergencyModalOpen(false)}>
264 |               Cancel
265 |             </Button>
266 |             <Button type="submit" variant="danger">
267 |               Transmit Siren Broadcast
268 |             </Button>
269 |           </div>
270 |         </form>
271 |       </Modal>
272 |     </DashboardShell>
273 |   );
274 | }
```

File #50: src/app/dashboard/admin/students/page.tsx 135 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { GraduationCap, Search, Download, Filter, UserCheck, AlertCircle } from "lucide-react";
  5 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  6 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  7 | import { Badge } from "@/components/ui/badge";
  8 | import { Button } from "@/components/ui/button";
  9 | import { useToast } from "@/components/ui/toast";
 10 | import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";
 11 |
 12 | interface StudentRecord {
 13 |   id: string;
 14 |   name: string;
 15 |   enrollmentNo: string;
 16 |   dept: string;
 17 |   semester: string;
 18 |   attendancePct: number;
 19 |   cgpa: number;
 20 |   standing: "Good" | "Probation" | "Honor Roll";
 21 | }
 22 |
 23 | const studentRoster: StudentRecord[] = [
 24 |   { id: "s-1", name: "Alex Morgan", enrollmentNo: "24CS001", dept: "Computer Science", semester: "Sem 6",
    |    attendancePct: 88.0, cgpa: 3.82, standing: "Honor Roll" },
 25 |   { id: "s-2", name: "David Kim", enrollmentNo: "24CS042", dept: "Computer Science", semester: "Sem 6", a
    |   ttendancePct: 91.2, cgpa: 3.65, standing: "Good" },
 26 |   { id: "s-3", name: "Elena Rostova", enrollmentNo: "24ME019", dept: "Mechanical", semester: "Sem 4", att
    |   endancePct: 71.4, cgpa: 2.78, standing: "Probation" },
 27 |   { id: "s-4", name: "Maya Patel", enrollmentNo: "24EC031", dept: "Electronics", semester: "Sem 6", atten
    |   dancePct: 94.6, cgpa: 3.95, standing: "Honor Roll" },
 28 |   { id: "s-5", name: "Marcus Vance", enrollmentNo: "24CS088", dept: "Computer Science", semester: "Sem 6"
    |   , attendancePct: 78.5, cgpa: 3.12, standing: "Good" },
 29 | ];
 30 |
 31 | export default function AdminStudentsPage() {
```

```
  const { success } = useToast();
  const [search, setSearch] = React.useState("");

  const filtered = studentRoster.filter(
    (s) =>
      s.name.toLowerCase().includes(search.toLowerCase()) ||
      s.enrollmentNo.toLowerCase().includes(search.toLowerCase()) ||
      s.dept.toLowerCase().includes(search.toLowerCase())
  );

  return (
    <DashboardShell
      title="Student Enrollment Roster & Academic Records"
      description="Active student directory, attendance standing, and CGPA metrics"
      actions={
        <Button
          variant="outline"
          size="sm"
          className="gap-1.5"
          onClick={() => success("Export Generated", "Student Roster exported as CSV.")}
        >
          <Download className="h-4 w-4" />
          Export Roster CSV
        </Button>
      }
    >
      <div className="space-y-6">
        <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3">
          <div className="relative">
            <input
              type="text"
              placeholder="Search by name, roll no, department..."
              value={search}
              onChange={(e) => setSearch(e.target.value)}
              className="w-full sm:w-72 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs te
xt-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-
200"
            />
          </div>
          <div className="flex items-center gap-2 text-xs text-zinc-500">
            <span>Enrolled Total: <strong className="text-zinc-800 dark:text-zinc-200">4,820</strong></sp
an>
          </div>
        </div>

        <Table>
          <TableHeader>
            <TableRow>
              <TableHead>Student Name & Roll No</TableHead>
              <TableHead>Department</TableHead>
              <TableHead>Semester</TableHead>
              <TableHead>Attendance</TableHead>
              <TableHead>CGPA</TableHead>
              <TableHead>Standing</TableHead>
              <TableHead className="text-right">Actions</TableHead>
            </TableRow>
          </TableHeader>
          <TableBody>
            {filtered.map((s) => (
              <TableRow key={s.id}>
                <TableCell>
                  <span className="font-semibold text-zinc-900 dark:text-zinc-100">{s.name}</span>
                  <span className="block text-[11px] text-zinc-500 font-mono">{s.enrollmentNo}</span>
                </TableCell>
                <TableCell className="text-xs text-zinc-600 dark:text-zinc-400">{s.dept}</TableCell>
                <TableCell className="text-xs">{s.semester}</TableCell>
                <TableCell>
                  <span
                    className={`font-bold text-xs ${
                      s.attendancePct >= 80 ? "text-emerald-600" : "text-amber-500"
                    }`}
                  >
                    {s.attendancePct}%
                  </span>
                </TableCell>
                <TableCell className="font-semibold text-xs">{s.cgpa}</TableCell>
                <TableCell>
                  <Badge
                    variant={
                      s.standing === "Honor Roll"
```

```
                        ? "success"
                        : s.standing === "Good"
                        ? "info"
                        : "warning"
                    }
                  >
                    {s.standing}
                  </Badge>
                </TableCell>
                <TableCell className="text-right">
                  <Button
                    variant="ghost"
                    size="sm"
                    className="text-xs text-blue-600 dark:text-blue-400"
                    onClick={() => success("Profile Loaded", `Loaded academic dossier for ${s.name}`)}
                  >
                    View Dossier
                  </Button>
                </TableCell>
              </TableRow>
            ))}
          </TableBody>
        </Table>
      </div>
    </DashboardShell>
  );
}
```

File #51: src/app/dashboard/admin/users/page.tsx 261 lines

```
"use client";

import * as React from "react";
import { Users, UserPlus, Search, Filter, Shield, MoreVertical, CheckCircle2, Lock } from "lucide-react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { Input } from "@/components/ui/input";
import { Modal } from "@/components/ui/modal";
import { useToast } from "@/components/ui/toast";
import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";

interface UserProfile {
  id: string;
  name: string;
  email: string;
  role: "student" | "faculty" | "admin";
  dept: string;
  status: "Active" | "Pending" | "Suspended";
  joined: string;
}

const initialUsers: UserProfile[] = [
  { id: "usr-1", name: "Alex Morgan", email: "alex.m@student.campus.edu", role: "student", dept: "Compute
  r Science", status: "Active", joined: "Aug 2024" },
  { id: "usr-2", name: "Dr. Rajesh Varma", email: "rajesh.v@campus.edu", role: "faculty", dept: "Computer
   Science", status: "Active", joined: "Jul 2021" },
  { id: "usr-3", name: "Samantha Cole", email: "samantha.c@admin.campus.edu", role: "admin", dept: "Centr
  al Registrar", status: "Active", joined: "Jan 2020" },
  { id: "usr-4", name: "David Kim", email: "david.k@student.campus.edu", role: "student", dept: "Electron
  ics", status: "Active", joined: "Aug 2024" },
  { id: "usr-5", name: "Prof. Ananya Rao", email: "ananya.rao@campus.edu", role: "faculty", dept: "Comput
  er Science", status: "Active", joined: "Jan 2022" },
  { id: "usr-6", name: "Elena Rostova", email: "elena.r@student.campus.edu", role: "student", dept: "Mech
  anical", status: "Pending", joined: "Oct 2026" },
];

export default function AdminUsersPage() {
  const { success, info } = useToast();
  const [users, setUsers] = React.useState<UserProfile[]>(initialUsers);
  const [search, setSearch] = React.useState("");
  const [roleFilter, setRoleFilter] = React.useState<string>("all");
  const [addUserModalOpen, setAddUserModalOpen] = React.useState(false);

  // New User Form State
  const [newUser, setNewUser] = React.useState({
    name: "",
    email: "",
```

```
    role: "student" as const,
    dept: "Computer Science",
  });

  const filtered = users.filter((u) => {
    const matchRole = roleFilter === "all" || u.role === roleFilter;
    const matchSearch =
      u.name.toLowerCase().includes(search.toLowerCase()) ||
      u.email.toLowerCase().includes(search.toLowerCase()) ||
      u.dept.toLowerCase().includes(search.toLowerCase());
    return matchRole && matchSearch;
  });

  const handleCreateUser = (e: React.FormEvent) => {
    e.preventDefault();
    const created: UserProfile = {
      id: `usr-${Date.now()}`,
      name: newUser.name,
      email: newUser.email,
      role: newUser.role,
      dept: newUser.dept,
      status: "Active",
      joined: "Just now",
    };
    setUsers([created, ...users]);
    setAddUserModalOpen(false);
    success("User Account Provisioned", `${newUser.name} registered in Supabase Auth & profiles table.`);
    setNewUser({ name: "", email: "", role: "student", dept: "Computer Science" });
  };

  const toggleStatus = (id: string) => {
    setUsers((prev) =>
      prev.map((u) => {
        if (u.id === id) {
          const nextStatus = u.status === "Active" ? "Suspended" : "Active";
          info(`User status modified to ${nextStatus}`);
          return { ...u, status: nextStatus };
        }
        return u;
      })
    );
  };

  return (
    <DashboardShell
      title="Campus User Accounts & Access Control"
      description="Manage student, faculty, and administrative identity records in Supabase PostgreSQL"
      actions={
        <Button size="sm" onClick={() => setAddUserModalOpen(true)} className="gap-1.5">
          <UserPlus className="h-4 w-4" />
          Provision Account
        </Button>
      }
    >
      <div className="space-y-6">
        {/* Controls */}
        <div className="flex flex-col gap-3 sm:flex-row sm:items-center sm:justify-between rounded-2xl bo
rder border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
          <div className="flex items-center gap-1.5">
            {["all", "student", "faculty", "admin"].map((r) => (
              <button
                key={r}
                onClick={() => setRoleFilter(r)}
                className={`rounded-xl px-3 py-1.5 text-xs font-semibold capitalize transition-all ${
                  roleFilter === r
                    ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
                    : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
                }`}
              >
                {r === "all" ? "All Accounts" : r}
              </button>
            ))}
          </div>

          <div className="relative">
            <input
              type="text"
              placeholder="Search by name, email, department..."
              value={search}
              onChange={(e) => setSearch(e.target.value)}
```

```
123 |              className="w-full sm:w-64 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs te
    |   xt-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-
    |   200"
124 |            />
125 |          </div>
126 |        </div>
127 |
128 |        {/* User Table */}
129 |        <Table>
130 |          <TableHeader>
131 |            <TableRow>
132 |              <TableHead>User / Identity</TableHead>
133 |              <TableHead>Assigned Role</TableHead>
134 |              <TableHead>Department</TableHead>
135 |              <TableHead>Registration Date</TableHead>
136 |              <TableHead>Account Status</TableHead>
137 |              <TableHead className="text-right">Actions</TableHead>
138 |            </TableRow>
139 |          </TableHeader>
140 |          <TableBody>
141 |            {filtered.map((u) => (
142 |              <TableRow key={u.id}>
143 |                <TableCell>
144 |                  <span className="font-semibold text-zinc-900 dark:text-zinc-100">{u.name}</span>
145 |                  <span className="block text-[11px] text-zinc-500">{u.email}</span>
146 |                </TableCell>
147 |                <TableCell>
148 |                  <Badge
149 |                    variant={
150 |                      u.role === "admin"
151 |                        ? "destructive"
152 |                        : u.role === "faculty"
153 |                        ? "warning"
154 |                        : "info"
155 |                    }
156 |                    className="capitalize"
157 |                  >
158 |                    {u.role}
159 |                  </Badge>
160 |                </TableCell>
161 |                <TableCell className="text-xs text-zinc-600 dark:text-zinc-300">{u.dept}</TableCell>
162 |                <TableCell className="text-xs text-zinc-500">{u.joined}</TableCell>
163 |                <TableCell>
164 |                  <Badge
165 |                    variant={
166 |                      u.status === "Active"
167 |                        ? "success"
168 |                        : u.status === "Pending"
169 |                        ? "secondary"
170 |                        : "destructive"
171 |                    }
172 |                  >
173 |                    {u.status}
174 |                  </Badge>
175 |                </TableCell>
176 |                <TableCell className="text-right">
177 |                  <Button
178 |                    variant="ghost"
179 |                    size="sm"
180 |                    className="text-xs"
181 |                    onClick={() => toggleStatus(u.id)}
182 |                  >
183 |                    {u.status === "Active" ? "Suspend" : "Activate"}
184 |                  </Button>
185 |                </TableCell>
186 |              </TableRow>
187 |            ))}
188 |          </TableBody>
189 |        </Table>
190 |      </div>
191 |
192 |      {/* Provision Account Modal */}
193 |      <Modal
194 |        isOpen={addUserModalOpen}
195 |        onClose={() => setAddUserModalOpen(false)}
196 |        title="Provision University Account"
197 |        description="Creates credential record in Supabase Auth and seeds database profile"
198 |      >
199 |        <form onSubmit={handleCreateUser} className="space-y-4">
200 |          <Input
```

```
            label="Full Name"
            required
            placeholder="e.g. Jordan Lee"
            value={newUser.name}
            onChange={(e) => setNewUser({ ...newUser, name: e.target.value })}
          />

          <Input
            label="Institutional Email"
            type="email"
            required
            placeholder="jordan.l@campus.edu"
            value={newUser.email}
            onChange={(e) => setNewUser({ ...newUser, email: e.target.value })}
          />

          <div className="grid grid-cols-2 gap-3">
            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Designated Role
              </label>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={newUser.role}
                onChange={(e) => setNewUser({ ...newUser, role: e.target.value as any })}
              >
                <option value="student">Student</option>
                <option value="faculty">Faculty Instructor</option>
                <option value="admin">Campus Administrator</option>
              </select>
            </div>

            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Department
              </label>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={newUser.dept}
                onChange={(e) => setNewUser({ ...newUser, dept: e.target.value })}
              >
                <option>Computer Science</option>
                <option>Electronics</option>
                <option>Mechanical</option>
                <option>Management</option>
                <option>Central Registrar</option>
              </select>
            </div>
          </div>

          <div className="flex justify-end gap-2 pt-2">
            <Button type="button" variant="outline" onClick={() => setAddUserModalOpen(false)}>
              Cancel
            </Button>
            <Button type="submit">Create Identity</Button>
          </div>
        </form>
      </Modal>
    </DashboardShell>
  );
}
```

File #52: src/app/dashboard/faculty/page.tsx 362 lines

```
"use client";

import * as React from "react";
import Link from "next/link";
import {
  Briefcase,
  Users,
  CheckCircle,
  FileCheck,
  PlusCircle,
  Bell,
  Clock,
  ArrowRight,
```

```
  Upload,
  Calendar,
  BookOpen,
  FileText,
  Search,
  CheckCircle2,
} from "lucide-react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { Modal } from "@/components/ui/modal";
import { Input } from "@/components/ui/input";
import { useToast } from "@/components/ui/toast";
import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";

export default function FacultyDashboardPage() {
  const { success } = useToast();
  const [activeTab, setActiveTab] = React.useState<"overview" | "roster" | "materials">("overview");

  // Modal States
  const [attendanceModalOpen, setAttendanceModalOpen] = React.useState(false);
  const [materialModalOpen, setMaterialModalOpen] = React.useState(false);

  // Quick Attendance Form State
  const [courseCode, setCourseCode] = React.useState("CS602");
  const [presentCount, setPresentCount] = React.useState("58");
  const [absentStudents, setAbsentStudents] = React.useState("24CS012, 24CS045");

  // New Material Form State
  const [materialTitle, setMaterialTitle] = React.useState("");
  const [materialCourse, setMaterialCourse] = React.useState("CS602: Database Systems");

  const handleRecordAttendance = (e: React.FormEvent) => {
    e.preventDefault();
    setAttendanceModalOpen(false);
    success("Attendance Saved", `Roll call logged for ${courseCode} (${presentCount} Present, Absentees:
  ${absentStudents}).`);
  };

  const handleUploadMaterial = (e: React.FormEvent) => {
    e.preventDefault();
    setMaterialModalOpen(false);
    success("Study Material Published", `${materialTitle} uploaded to Supabase Storage bucket.`);
    setMaterialTitle("");
  };

  return (
    <DashboardShell
      title="Faculty Instructor Console"
      description="Welcome, Dr. Rajesh Varma • Department of Computer Science & Engineering"
      actions={
        <div className="flex gap-2">
          <Button variant="outline" size="sm" onClick={() => setMaterialModalOpen(true)} className="gap-1
  .5">
            <Upload className="h-4 w-4" />
            <span>Upload Notes</span>
          </Button>
          <Button size="sm" onClick={() => setAttendanceModalOpen(true)} className="gap-1.5 shadow-sm sha
  dow-blue-500/20">
            <CheckCircle2 className="h-4 w-4" />
            <span>Take Attendance</span>
          </Button>
        </div>
      }
    >
      {/* Metric Cards */}
      <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 mb-6">
        <Card className="border-l-4 border-l-amber-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Active Courses</p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">3 Batches</h3>
              <p className="text-[11px] text-zinc-500 mt-1">184 Enrolled Students</p>
            </div>
            <div className="p-3 bg-amber-500/10 text-amber-600 rounded-xl">
              <Users className="h-6 w-6" />
            </div>
          </CardContent>
        </Card>
```

```
 91 |
 92 |         <Card className="border-l-4 border-l-blue-500">
 93 |           <CardContent className="p-4 flex items-center justify-between">
 94 |             <div>
 95 |               <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Today&apos;s Lectures</
    |   p>
 96 |               <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">3 Sessions</h3>
 97 |               <p className="text-[11px] text-blue-600 dark:text-blue-400 mt-1 font-medium">Next: Hall 401
    |    at 11:30 AM</p>
 98 |             </div>
 99 |             <div className="p-3 bg-blue-500/10 text-blue-600 rounded-xl">
100 |               <Clock className="h-6 w-6" />
101 |             </div>
102 |           </CardContent>
103 |         </Card>
104 |
105 |         <Card className="border-l-4 border-l-purple-500">
106 |           <CardContent className="p-4 flex items-center justify-between">
107 |             <div>
108 |               <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Submissions to Grade</p
    |   >
109 |               <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">42 Papers</h3>
110 |               <p className="text-[11px] text-purple-600 dark:text-purple-400 mt-1 font-medium">DBMS ER Di
    |   agrams</p>
111 |             </div>
112 |             <div className="p-3 bg-purple-500/10 text-purple-600 rounded-xl">
113 |               <FileCheck className="h-6 w-6" />
114 |             </div>
115 |           </CardContent>
116 |         </Card>
117 |
118 |         <Card className="border-l-4 border-l-emerald-500">
119 |           <CardContent className="p-4 flex items-center justify-between">
120 |             <div>
121 |               <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Batch Attendance Avg</p
    |   >
122 |               <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">91.4%</h3>
123 |               <p className="text-[11px] text-emerald-600 dark:text-emerald-400 mt-1 font-medium">Healthy
    |   compliance</p>
124 |             </div>
125 |             <div className="p-3 bg-emerald-500/10 text-emerald-600 rounded-xl">
126 |               <CheckCircle className="h-6 w-6" />
127 |             </div>
128 |           </CardContent>
129 |         </Card>
130 |       </div>
131 |
132 |       {/* Main Grid */}
133 |       <div className="grid grid-cols-1 lg:grid-cols-3 gap-6 mb-6">
134 |         {/* Today's Teaching Schedule */}
135 |         <Card className="lg:col-span-2">
136 |           <CardHeader>
137 |             <div className="flex items-center justify-between">
138 |               <div>
139 |                 <CardTitle className="text-base">Today&apos;s Lecture Schedule</CardTitle>
140 |                 <CardDescription className="text-xs">Classes scheduled for Monday, Fall 2026</CardDescrip
    |   tion>
141 |               </div>
142 |               <Badge variant="outline">Live Roster</Badge>
143 |             </div>
144 |           </CardHeader>
145 |           <CardContent>
146 |             <Table>
147 |               <TableHeader>
148 |                 <TableRow>
149 |                   <TableHead>Time & Venue</TableHead>
150 |                   <TableHead>Course</TableHead>
151 |                   <TableHead>Batch Size</TableHead>
152 |                   <TableHead className="text-right">Action</TableHead>
153 |                 </TableRow>
154 |               </TableHeader>
155 |               <TableBody>
156 |                 <TableRow>
157 |                   <TableCell>
158 |                     <span className="font-semibold text-zinc-900 dark:text-zinc-100">09:00 - 10:30 AM</sp
    |   an>
159 |                     <span className="block text-[11px] text-zinc-500">Hall 401</span>
160 |                   </TableCell>
161 |                   <TableCell>
162 |                     <span className="font-medium">CS602: Database Management Systems</span>
```

```
163 |                     <span className="block text-[11px] text-blue-600 dark:text-blue-400">Relational Algeb
    |   ra & SQL</span>
164 |                   </TableCell>
165 |                   <TableCell>62 Students</TableCell>
166 |                   <TableCell className="text-right">
167 |                     <Button size="sm" variant="ghost" onClick={() => setAttendanceModalOpen(true)} classN
    |   ame="text-xs text-blue-600">
168 |                       Take Roll
169 |                     </Button>
170 |                   </TableCell>
171 |                 </TableRow>
172 |                 <TableRow>
173 |                   <TableCell>
174 |                     <span className="font-semibold text-zinc-900 dark:text-zinc-100">11:30 - 01:00 PM</sp
    |   an>
175 |                     <span className="block text-[11px] text-zinc-500">Hall 401</span>
176 |                   </TableCell>
177 |                   <TableCell>
178 |                     <span className="font-medium">CS605: Advanced Database Systems</span>
179 |                     <span className="block text-[11px] text-blue-600 dark:text-blue-400">Concurrency Cont
    |   rol</span>
180 |                   </TableCell>
181 |                   <TableCell>58 Students</TableCell>
182 |                   <TableCell className="text-right">
183 |                     <Button size="sm" variant="ghost" onClick={() => setAttendanceModalOpen(true)} classN
    |   ame="text-xs text-blue-600">
184 |                       Take Roll
185 |                     </Button>
186 |                   </TableCell>
187 |                 </TableRow>
188 |                 <TableRow>
189 |                   <TableCell>
190 |                     <span className="font-semibold text-zinc-900 dark:text-zinc-100">02:00 - 04:00 PM</sp
    |   an>
191 |                     <span className="block text-[11px] text-zinc-500">Database Lab 201</span>
192 |                   </TableCell>
193 |                   <TableCell>
194 |                     <span className="font-medium">CS605L: Database Systems Practical</span>
195 |                     <span className="block text-[11px] text-blue-600 dark:text-blue-400">PostgreSQL Trigg
    |   ers & SPs</span>
196 |                   </TableCell>
197 |                   <TableCell>32 Students</TableCell>
198 |                   <TableCell className="text-right">
199 |                     <Button size="sm" variant="ghost" onClick={() => setAttendanceModalOpen(true)} classN
    |   ame="text-xs text-blue-600">
200 |                       Take Roll
201 |                     </Button>
202 |                   </TableCell>
203 |                 </TableRow>
204 |               </TableBody>
205 |             </Table>
206 |           </CardContent>
207 |         </Card>
208 |
209 |         {/* Quick Tools & Shortcuts */}
210 |         <Card className="flex flex-col justify-between">
211 |           <CardHeader>
212 |             <CardTitle className="text-base">Quick Course Administration</CardTitle>
213 |             <CardDescription className="text-xs">Shortcuts to manage deliverables</CardDescription>
214 |           </CardHeader>
215 |           <CardContent className="space-y-3">
216 |             <Link
217 |               href="/announcements"
218 |               className="flex items-center justify-between p-3 rounded-xl border border-zinc-200/80 bg-zi
    |   nc-50/50 hover:bg-zinc-100 dark:border-zinc-800 dark:bg-zinc-900/40 dark:hover:bg-zinc-800/60 transition-
    |   colors"
219 |             >
220 |               <div className="flex items-center gap-2.5">
221 |                 <Bell className="h-4 w-4 text-blue-600" />
222 |                 <span className="text-xs font-semibold">Post Class Announcement</span>
223 |               </div>
224 |               <ArrowRight className="h-3.5 w-3.5 text-zinc-400" />
225 |             </Link>
226 |
227 |             <Link
228 |               href="/assignments"
229 |               className="flex items-center justify-between p-3 rounded-xl border border-zinc-200/80 bg-zi
    |   nc-50/50 hover:bg-zinc-100 dark:border-zinc-800 dark:bg-zinc-900/40 dark:hover:bg-zinc-800/60 transition-
    |   colors"
230 |             >
```

```
231 |               <div className="flex items-center gap-2.5">
232 |                 <FileText className="h-4 w-4 text-purple-600" />
233 |                 <span className="text-xs font-semibold">Grade Assignment Papers</span>
234 |               </div>
235 |               <ArrowRight className="h-3.5 w-3.5 text-zinc-400" />
236 |             </Link>
237 |
238 |             <Link
239 |               href="/materials"
240 |               className="flex items-center justify-between p-3 rounded-xl border border-zinc-200/80 bg-zi
    |   nc-50/50 hover:bg-zinc-100 dark:border-zinc-800 dark:bg-zinc-900/40 dark:hover:bg-zinc-800/60 transition-
    |   colors"
241 |             >
242 |               <div className="flex items-center gap-2.5">
243 |                 <BookOpen className="h-4 w-4 text-amber-600" />
244 |                 <span className="text-xs font-semibold">Course Study Materials</span>
245 |               </div>
246 |               <ArrowRight className="h-3.5 w-3.5 text-zinc-400" />
247 |             </Link>
248 |
249 |             <Link
250 |               href="/events"
251 |               className="flex items-center justify-between p-3 rounded-xl border border-zinc-200/80 bg-zi
    |   nc-50/50 hover:bg-zinc-100 dark:border-zinc-800 dark:bg-zinc-900/40 dark:hover:bg-zinc-800/60 transition-
    |   colors"
252 |             >
253 |               <div className="flex items-center gap-2.5">
254 |                 <Calendar className="h-4 w-4 text-emerald-600" />
255 |                 <span className="text-xs font-semibold">Department Seminars</span>
256 |               </div>
257 |               <ArrowRight className="h-3.5 w-3.5 text-zinc-400" />
258 |             </Link>
259 |           </CardContent>
260 |         </Card>
261 |       </div>
262 |
263 |       {/* Attendance Roll Call Modal */}
264 |       <Modal
265 |         isOpen={attendanceModalOpen}
266 |         onClose={() => setAttendanceModalOpen(false)}
267 |         title="Record Lecture Attendance"
268 |         description="Mark roll call entries into PostgreSQL `attendance_records`"
269 |       >
270 |         <form onSubmit={handleRecordAttendance} className="space-y-4">
271 |           <div className="grid grid-cols-2 gap-3">
272 |             <div className="space-y-1.5">
273 |               <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Course Code</labe
    |   l>
274 |               <select
275 |                 className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
    |   r-zinc-800 dark:bg-zinc-900"
276 |                 value={courseCode}
277 |                 onChange={(e) => setCourseCode(e.target.value)}
278 |               >
279 |                 <option value="CS602">CS602: Database Management Systems</option>
280 |                 <option value="CS605">CS605: Advanced Database Systems</option>
281 |                 <option value="CS605L">CS605L: Database Lab Practical</option>
282 |               </select>
283 |             </div>
284 |             <Input
285 |               label="Present Count"
286 |               type="number"
287 |               required
288 |               value={presentCount}
289 |               onChange={(e) => setPresentCount(e.target.value)}
290 |             />
291 |           </div>
292 |
293 |           <div className="space-y-1.5">
294 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
295 |               Absent Roll Numbers (Comma Separated)
296 |             </label>
297 |             <input
298 |               type="text"
299 |               placeholder="e.g. 24CS012, 24CS045, 24CS089"
300 |               className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:border-
    |   zinc-800 dark:bg-zinc-900"
301 |               value={absentStudents}
302 |               onChange={(e) => setAbsentStudents(e.target.value)}
303 |             />
```

```
304 |           </div>
305 |
306 |           <div className="flex justify-end gap-2 pt-2">
307 |             <Button type="button" variant="outline" onClick={() => setAttendanceModalOpen(false)}>
308 |               Cancel
309 |             </Button>
310 |             <Button type="submit">Commit Attendance Log</Button>
311 |           </div>
312 |         </form>
313 |       </Modal>
314 |
315 |       {/* Material Upload Modal */}
316 |       <Modal
317 |         isOpen={materialModalOpen}
318 |         onClose={() => setMaterialModalOpen(false)}
319 |         title="Publish Course Study Material"
320 |         description="Upload slides or notes to Supabase Storage `study_materials` bucket"
321 |       >
322 |         <form onSubmit={handleUploadMaterial} className="space-y-4">
323 |           <Input
324 |             label="Document Title"
325 |             required
326 |             placeholder="e.g. Chapter 4: B-Tree Indexing and Hashing Guide"
327 |             value={materialTitle}
328 |             onChange={(e) => setMaterialTitle(e.target.value)}
329 |           />
330 |
331 |           <div className="space-y-1.5">
332 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Target Course</labe
    |   l>
333 |             <select
334 |               className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:border-
    |   zinc-800 dark:bg-zinc-900"
335 |               value={materialCourse}
336 |               onChange={(e) => setMaterialCourse(e.target.value)}
337 |             >
338 |               <option>CS602: Database Management Systems</option>
339 |               <option>CS605: Advanced Database Systems</option>
340 |               <option>CS605L: Database Lab Practical</option>
341 |             </select>
342 |           </div>
343 |
344 |           <div className="rounded-xl border-2 border-dashed border-zinc-300 dark:border-zinc-700 p-6 text
    |   -center">
345 |             <Upload className="h-6 w-6 mx-auto text-blue-500 mb-2" />
346 |             <p className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
347 |               Select presentation or PDF document
348 |             </p>
349 |             <input type="file" className="mt-2 text-xs text-zinc-500" />
350 |           </div>
351 |
352 |           <div className="flex justify-end gap-2 pt-2">
353 |             <Button type="button" variant="outline" onClick={() => setMaterialModalOpen(false)}>
354 |               Cancel
355 |             </Button>
356 |             <Button type="submit">Publish to Course Repository</Button>
357 |           </div>
358 |         </form>
359 |       </Modal>
360 |     </DashboardShell>
361 |   );
362 | }
```

File #53: src/app/dashboard/page.tsx

79 lines

```
  1 | import Link from "next/link";
  2 | import { GraduationCap, Briefcase, ShieldAlert, ArrowRight } from "lucide-react";
  3 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  4 | import { Card, CardContent } from "@/components/ui/card";
  5 | import { Button } from "@/components/ui/button";
  6 |
  7 | export default function DashboardPortal() {
  8 |   return (
  9 |     <DashboardShell
 10 |       title="Role-Based Command Portals"
 11 |       description="Choose your persona or role view to access tailored campus tools"
 12 |     >
 13 |       <div className="grid grid-cols-1 md:grid-cols-3 gap-6 mt-4">
 14 |         {/* Student Portal Card */}
```

```
15 |         <Card className="hover:border-blue-500/50 hover:shadow-lg transition-all border-zinc-200 dark:bor
   |   der-zinc-800">
16 |           <CardContent className="pt-6 space-y-4">
17 |             <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-blue-500/10 text-bl
   |   ue-600 dark:bg-blue-500/20 dark:text-blue-400">
18 |               <GraduationCap className="h-6 w-6" />
19 |             </div>
20 |             <div>
21 |               <h3 className="text-lg font-bold text-zinc-900 dark:text-zinc-50">Student Portal</h3>
22 |               <p className="mt-1 text-xs text-zinc-500 dark:text-zinc-400">
23 |                 View real-time timetable, track course attendance percentages, submit assignments, and do
   |   wnload study notes.
24 |               </p>
25 |             </div>
26 |             <Link href="/dashboard/student" className="block pt-2">
27 |               <Button className="w-full" size="md">
28 |                 <span>Enter Student Portal</span>
29 |                 <ArrowRight className="h-4 w-4 ml-1" />
30 |               </Button>
31 |             </Link>
32 |           </CardContent>
33 |         </Card>
34 |
35 |         {/* Faculty Portal Card */}
36 |         <Card className="hover:border-amber-500/50 hover:shadow-lg transition-all border-zinc-200 dark:bo
   |   rder-zinc-800">
37 |           <CardContent className="pt-6 space-y-4">
38 |             <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-amber-500/10 text-a
   |   mber-600 dark:bg-amber-500/20 dark:text-amber-400">
39 |               <Briefcase className="h-6 w-6" />
40 |             </div>
41 |             <div>
42 |               <h3 className="text-lg font-bold text-zinc-900 dark:text-zinc-50">Faculty Console</h3>
43 |               <p className="mt-1 text-xs text-zinc-500 dark:text-zinc-400">
44 |                 Mark student attendance, create assignment rubrics, review submissions, and publish acade
   |   mic circulars.
45 |               </p>
46 |             </div>
47 |             <Link href="/dashboard/faculty" className="block pt-2">
48 |               <Button variant="outline" className="w-full border-amber-300 text-amber-700 hover:bg-amber-
   |   50 dark:border-amber-800 dark:text-amber-300 dark:hover:bg-amber-950/40" size="md">
49 |                 <span>Enter Faculty Console</span>
50 |                 <ArrowRight className="h-4 w-4 ml-1" />
51 |               </Button>
52 |             </Link>
53 |           </CardContent>
54 |         </Card>
55 |
56 |         {/* Admin Portal Card */}
57 |         <Card className="hover:border-rose-500/50 hover:shadow-lg transition-all border-zinc-200 dark:bor
   |   der-zinc-800">
58 |           <CardContent className="pt-6 space-y-4">
59 |             <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-rose-500/10 text-ro
   |   se-600 dark:bg-rose-500/20 dark:text-rose-400">
60 |               <ShieldAlert className="h-6 w-6" />
61 |             </div>
62 |             <div>
63 |               <h3 className="text-lg font-bold text-zinc-900 dark:text-zinc-50">Admin Command Center</h3>
64 |               <p className="mt-1 text-xs text-zinc-500 dark:text-zinc-400">
65 |                 Manage campus infrastructure, supervise user accounts, dispatch campus emergency broadcas
   |   ts, and monitor database health.
66 |               </p>
67 |             </div>
68 |             <Link href="/dashboard/admin" className="block pt-2">
69 |               <Button variant="outline" className="w-full border-rose-300 text-rose-700 hover:bg-rose-50
   |   dark:border-rose-800 dark:text-rose-300 dark:hover:bg-rose-950/40" size="md">
70 |                 <span>Enter Admin Center</span>
71 |                 <ArrowRight className="h-4 w-4 ml-1" />
72 |               </Button>
73 |             </Link>
74 |           </CardContent>
75 |         </Card>
76 |       </div>
77 |     </DashboardShell>
78 |   );
79 | }
```

File #54: src/app/dashboard/student/page.tsx 225 lines

```
import Link from "next/link";
import {
  GraduationCap,
  Clock,
  CheckCircle2,
  FileText,
  AlertCircle,
  Calendar,
  ArrowRight,
  BookOpen,
} from "lucide-react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { AttendanceChart } from "@/components/dashboard/attendance-chart";

export default function StudentDashboardPage() {
  return (
    <DashboardShell
      title="Student Academic Workspace"
      description="Welcome back, Alex • Computer Science & Engineering (Sem 6)"
      actions={
        <div className="flex gap-2">
          <Link href="/materials">
            <Button variant="outline" size="sm">
              <BookOpen className="h-4 w-4" />
              <span>Study Notes</span>
            </Button>
          </Link>
          <Link href="/assignments">
            <Button size="sm">
              <FileText className="h-4 w-4" />
              <span>View Assignments</span>
            </Button>
          </Link>
        </div>
      }
    >
      {/* Metrics Row */}
      <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 mb-6">
        <Card className="border-l-4 border-l-blue-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Total Attendance</p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">86.4%</h3>
              <p className="text-[11px] text-emerald-600 dark:text-emerald-400 mt-1 flex items-center gap
-1 font-medium">
                <CheckCircle2 className="h-3 w-3" />
                Above 75% threshold
              </p>
            </div>
            <div className="p-3 bg-blue-500/10 text-blue-600 rounded-xl">
              <CheckCircle2 className="h-6 w-6" />
            </div>
          </CardContent>
        </Card>

        <Card className="border-l-4 border-l-amber-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Due Assignments</p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">2 Pending</h3>
              <p className="text-[11px] text-amber-600 dark:text-amber-400 mt-1 font-medium">
                Next: Operating Systems (Tomorrow)
              </p>
            </div>
            <div className="p-3 bg-amber-500/10 text-amber-600 rounded-xl">
              <FileText className="h-6 w-6" />
            </div>
          </CardContent>
        </Card>

        <Card className="border-l-4 border-l-emerald-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Cumulative CGPA</p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">8.92</h3>
              <p className="text-[11px] text-emerald-600 dark:text-emerald-400 mt-1 font-medium">
                Top 10% in Department
```

```
              </p>
            </div>
            <div className="p-3 bg-emerald-500/10 text-emerald-600 rounded-xl">
              <GraduationCap className="h-6 w-6" />
            </div>
          </CardContent>
        </Card>

        <Card className="border-l-4 border-l-indigo-500">
          <CardContent className="p-4 flex items-center justify-between">
            <div>
              <p className="text-xs font-medium text-zinc-500 dark:text-zinc-400">Today&apos;s Lectures</
p>
              <h3 className="text-2xl font-bold text-zinc-900 dark:text-zinc-50 mt-1">4 Classes</h3>
              <p className="text-[11px] text-indigo-600 dark:text-indigo-400 mt-1 font-medium">
                Next starts at 11:30 AM
              </p>
            </div>
            <div className="p-3 bg-indigo-500/10 text-indigo-600 rounded-xl">
              <Clock className="h-6 w-6" />
            </div>
          </CardContent>
        </Card>
      </div>

      {/* Main Grid: Schedule + Attendance Chart */}
      <div className="grid grid-cols-1 lg:grid-cols-3 gap-6 mb-6">
        {/* Attendance Breakdown (Recharts) */}
        <Card className="lg:col-span-2">
          <CardHeader className="flex flex-row items-center justify-between pb-2">
            <div>
              <CardTitle>Course-wise Attendance Overview</CardTitle>
              <CardDescription>Live attendance recorded by faculty members</CardDescription>
            </div>
            <Link href="/attendance">
              <Button variant="ghost" size="sm" className="text-xs">
                <span>Detailed Record</span>
                <ArrowRight className="h-3.5 w-3.5" />
              </Button>
            </Link>
          </CardHeader>
          <CardContent>
            <AttendanceChart />
          </CardContent>
        </Card>

        {/* Today's Timetable */}
        <Card>
          <CardHeader className="flex flex-row items-center justify-between pb-2">
            <div>
              <CardTitle>Today&apos;s Timetable</CardTitle>
              <CardDescription>Monday schedule</CardDescription>
            </div>
            <Link href="/timetable">
              <Badge variant="outline">Full Week</Badge>
            </Link>
          </CardHeader>
          <CardContent className="space-y-3">
            <div className="rounded-xl border border-zinc-200/80 p-3 bg-zinc-50/50 dark:border-zinc-800 d
ark:bg-zinc-900/50">
              <div className="flex items-center justify-between text-xs">
                <span className="font-semibold text-blue-600 dark:text-blue-400">09:00 - 10:30 AM</span>
                <Badge variant="success">Completed</Badge>
              </div>
              <h4 className="font-semibold text-sm mt-1">Data Structures & Algorithms</h4>
              <p className="text-xs text-zinc-500 dark:text-zinc-400">Prof. Sharma • Lab 302</p>
            </div>

            <div className="rounded-xl border border-blue-500/40 p-3 bg-blue-50/30 dark:border-blue-900/4
0 dark:bg-blue-950/20">
              <div className="flex items-center justify-between text-xs">
                <span className="font-semibold text-blue-600 dark:text-blue-400">11:30 - 01:00 PM</span>
                <Badge variant="warning">Upcoming</Badge>
              </div>
              <h4 className="font-semibold text-sm mt-1">Database Management Systems</h4>
              <p className="text-xs text-zinc-500 dark:text-zinc-400">Dr. Rajesh V • Hall 401</p>
            </div>

            <div className="rounded-xl border border-zinc-200/80 p-3 bg-zinc-50/50 dark:border-zinc-800 d
ark:bg-zinc-900/50">
```

```
156 |               <div className="flex items-center justify-between text-xs">
157 |                 <span className="font-semibold text-zinc-500">02:00 - 03:30 PM</span>
158 |                 <Badge variant="outline">Scheduled</Badge>
159 |               </div>
160 |               <h4 className="font-semibold text-sm mt-1">Computer Networks</h4>
161 |               <p className="text-xs text-zinc-500 dark:text-zinc-400">Prof. Ananya Rao • Room 204</p>
162 |             </div>
163 |           </CardContent>
164 |         </Card>
165 |       </div>
166 |
167 |       {/* Assignments & Urgent Notice Row */}
168 |       <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
169 |         <Card>
170 |           <CardHeader className="flex flex-row items-center justify-between pb-2">
171 |             <div>
172 |               <CardTitle>Active Assignments</CardTitle>
173 |               <CardDescription>Submissions requiring your attention</CardDescription>
174 |             </div>
175 |             <Link href="/assignments">
176 |               <Button variant="ghost" size="sm">
177 |                 <span>View All</span>
178 |                 <ArrowRight className="h-3.5 w-3.5 ml-1" />
179 |               </Button>
180 |             </Link>
181 |           </CardHeader>
182 |           <CardContent className="space-y-3">
183 |             <div className="flex items-center justify-between p-3 rounded-xl border border-zinc-200/80 da
    |   rk:border-zinc-800">
184 |               <div>
185 |                 <h4 className="text-xs font-semibold">OS Virtual Memory Simulation</h4>
186 |                 <p className="text-[11px] text-zinc-500">Due: Tomorrow at 11:59 PM • Max Marks: 50</p>
187 |               </div>
188 |               <Badge variant="warning">Pending</Badge>
189 |             </div>
190 |             <div className="flex items-center justify-between p-3 rounded-xl border border-zinc-200/80 da
    |   rk:border-zinc-800">
191 |               <div>
192 |                 <h4 className="text-xs font-semibold">DBMS Normalization Case Study</h4>
193 |                 <p className="text-[11px] text-zinc-500">Due: Oct 16, 2026 • Max Marks: 30</p>
194 |               </div>
195 |               <Badge variant="info">In Progress</Badge>
196 |             </div>
197 |           </CardContent>
198 |         </Card>
199 |
200 |         <Card>
201 |           <CardHeader className="flex flex-row items-center justify-between pb-2">
202 |             <div>
203 |               <CardTitle>Latest Circular</CardTitle>
204 |               <CardDescription>Official campus bulletin</CardDescription>
205 |             </div>
206 |             <Link href="/announcements">
207 |               <Badge variant="info">Announcements</Badge>
208 |             </Link>
209 |           </CardHeader>
210 |           <CardContent className="space-y-3">
211 |             <div className="p-3 rounded-xl border border-rose-200 bg-rose-50/50 dark:border-rose-900/40 d
    |   ark:bg-rose-950/20">
212 |               <div className="flex items-center gap-1.5 text-rose-600 dark:text-rose-400 font-semibold te
    |   xt-xs mb-1">
213 |                 <AlertCircle className="h-3.5 w-3.5" />
214 |                 <span>Semester End Examination Schedule Released</span>
215 |               </div>
216 |               <p className="text-xs text-zinc-600 dark:text-zinc-400">
217 |                 The comprehensive timetable for November 2026 End-Semester Examinations has been publishe
    |   d. Check portal notifications for admit card downloads.
218 |               </p>
219 |             </div>
220 |           </CardContent>
221 |         </Card>
222 |       </div>
223 |     </DashboardShell>
224 |   );
225 | }
```

File #55: src/app/emergency/page.tsx 236 lines

```
1 | "use client";
```

```
 2 |
 3 | import * as React from "react";
 4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
 5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
 6 | import { Badge } from "@/components/ui/badge";
 7 | import { Button } from "@/components/ui/button";
 8 | import { Input } from "@/components/ui/input";
 9 | import { Modal } from "@/components/ui/modal";
10 | import { useToast } from "@/components/ui/toast";
11 | import { EmptyState, LoadingState, ErrorState } from "@/components/ui/state-views";
12 | import { PhoneCall, ShieldAlert, HeartPulse, Flame, PhoneForwarded, Plus, Phone } from "lucide-react";
13 | import { getEmergencyContactsAction, createEmergencyContactAction } from "@/actions/operations";
14 |
15 | export default function EmergencyContactsPage() {
16 |   const { success, error } = useToast();
17 |   const [contacts, setContacts] = React.useState<any[]>([]);
18 |   const [loading, setLoading] = React.useState(true);
19 |   const [hasError, setHasError] = React.useState(false);
20 |   const [modalOpen, setModalOpen] = React.useState(false);
21 |   const [formData, setFormData] = React.useState({
22 |     name: "",
23 |     designation: "",
24 |     phone: "",
25 |     alternatePhone: "",
26 |     availableHours: "24/7",
27 |     category: "Emergency" as "Emergency" | "Medical" | "Security" | "Helpline",
28 |     priority: 1,
29 |   });
30 |   const [isSubmitting, setIsSubmitting] = React.useState(false);
31 |
32 |   const fetchContacts = React.useCallback(async () => {
33 |     setLoading(true);
34 |     setHasError(false);
35 |     try {
36 |       const res = await getEmergencyContactsAction();
37 |       if (res.success && res.data) {
38 |         setContacts(res.data);
39 |       } else {
40 |         setHasError(true);
41 |       }
42 |     } catch {
43 |       setHasError(true);
44 |     } finally {
45 |       setLoading(false);
46 |     }
47 |   }, []);
48 |
49 |   React.useEffect(() => {
50 |     fetchContacts();
51 |   }, [fetchContacts]);
52 |
53 |   const handleCreate = async (e: React.FormEvent) => {
54 |     e.preventDefault();
55 |     setIsSubmitting(true);
56 |     try {
57 |       const res = await createEmergencyContactAction(formData);
58 |       if (res.success) {
59 |         success("Hotline Added", "Emergency contact record committed to PostgreSQL.");
60 |         setModalOpen(false);
61 |         setFormData({
62 |           name: "",
63 |           designation: "",
64 |           phone: "",
65 |           alternatePhone: "",
66 |           availableHours: "24/7",
67 |           category: "Emergency",
68 |           priority: 1,
69 |         });
70 |         fetchContacts();
71 |       } else {
72 |         error("Failed", res.message || "Please check inputs.");
73 |       }
74 |     } catch (err: any) {
75 |       error("Error", err.message);
76 |     } finally {
77 |       setIsSubmitting(false);
78 |     }
79 |   };
80 |
81 |   return (
```

```
    <DashboardShell
      title="Campus Emergency Contacts & Hotlines"
      description="Immediate 24/7 crisis numbers for medical, security, fire safety, and counseling"
      actions={
        <Button size="sm" variant="danger" onClick={() => setModalOpen(true)} className="gap-1.5">
          <Plus className="h-4 w-4" />
          <span>Add Emergency Contact</span>
        </Button>
      }
    >
      <div className="space-y-6">
        {loading ? (
          <LoadingState message="Fetching live emergency dispatch hotlines..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchContacts} />
        ) : contacts.length === 0 ? (
          <EmptyState
            title="No Emergency Hotlines"
            description="No contacts have been provisioned in the directory."
            actionLabel="Add Hotline"
            onAction={() => setModalOpen(true)}
          />
        ) : (
          <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
            {contacts.map((c) => (
              <Card key={c.id || c.name} className="border-l-4 border-l-rose-500 flex flex-col justify-be
tween">
                <CardHeader>
                  <div className="flex items-center justify-between mb-2">
                    <div className="p-2.5 rounded-xl border border-rose-500/20 bg-rose-500/10 text-rose-6
00 dark:bg-rose-500/20 dark:text-rose-400">
                      <ShieldAlert className="h-6 w-6" />
                    </div>
                    <Badge variant="danger">{c.available_hours || "24/7"}</Badge>
                  </div>
                  <CardTitle className="text-base font-bold">{c.name}</CardTitle>
                  <CardDescription className="text-xs text-zinc-500">
                    {c.designation} • Category: {c.category}
                  </CardDescription>
                </CardHeader>
                <CardContent className="space-y-4">
                  <div className="space-y-1">
                    <p className="text-2xl font-extrabold tracking-tight text-zinc-900 dark:text-zinc-50
font-mono">
                      {c.phone}
                    </p>
                    {c.alternate_phone && (
                      <p className="text-xs text-zinc-500 font-mono">
                        Alternate: {c.alternate_phone}
                      </p>
                    )}
                  </div>

                  <div className="pt-2 border-t border-zinc-100 dark:border-zinc-800 flex items-center ju
stify-between gap-3">
                    <Button
                      variant="outline"
                      size="sm"
                      className="w-1/2 text-xs"
                      onClick={() => {
                        navigator.clipboard?.writeText?.(c.phone);
                        success("Number Copied", `${c.phone} copied to clipboard.`);
                      }}
                    >
                      Copy Number
                    </Button>
                    <a href={`tel:${c.phone}`} className="w-1/2">
                      <Button variant="danger" size="sm" className="w-full gap-1.5 text-xs">
                        <PhoneCall className="h-3.5 w-3.5" />
                        <span>Dial Now</span>
                      </Button>
                    </a>
                  </div>
                </CardContent>
              </Card>
            ))}
          </div>
        )}
      </div>

```

```
158 |       {/* Add Emergency Contact Modal */}
159 |       <Modal
160 |         isOpen={modalOpen}
161 |         onClose={() => setModalOpen(false)}
162 |         title="Add Campus Emergency Hotline"
163 |         description="Registers emergency phone number in Supabase emergency_contacts table"
164 |       >
165 |         <form onSubmit={handleCreate} className="space-y-4">
166 |           <Input
167 |             label="Service / Office Name"
168 |             required
169 |             placeholder="e.g. University Ambulance Dispatch"
170 |             value={formData.name}
171 |             onChange={(e) => setFormData({ ...formData, name: e.target.value })}
172 |           />
173 |
174 |           <Input
175 |             label="Designation / Authority"
176 |             required
177 |             placeholder="e.g. Chief Medical Officer"
178 |             value={formData.designation}
179 |             onChange={(e) => setFormData({ ...formData, designation: e.target.value })}
180 |           />
181 |
182 |           <div className="grid grid-cols-2 gap-3">
183 |             <Input
184 |               label="Primary Hotline Phone"
185 |               required
186 |               placeholder="+1 (800) 555-0108"
187 |               value={formData.phone}
188 |               onChange={(e) => setFormData({ ...formData, phone: e.target.value })}
189 |             />
190 |             <Input
191 |               label="Alternate Phone / Extension"
192 |               placeholder="Ext: 108"
193 |               value={formData.alternatePhone}
194 |               onChange={(e) => setFormData({ ...formData, alternatePhone: e.target.value })}
195 |             />
196 |           </div>
197 |
198 |           <div className="grid grid-cols-2 gap-3">
199 |             <div className="space-y-1.5">
200 |               <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Category</label>
201 |               <select
202 |                 className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
    |   r-zinc-800 dark:bg-zinc-900"
203 |                 value={formData.category}
204 |                 onChange={(e) => setFormData({ ...formData, category: e.target.value as any })}
205 |               >
206 |                 <option value="Emergency">Emergency Security</option>
207 |                 <option value="Medical">Medical / Ambulance</option>
208 |                 <option value="Security">Campus Security</option>
209 |                 <option value="Helpline">Helpline / Counseling</option>
210 |               </select>
211 |             </div>
212 |
213 |             <div className="space-y-1.5">
214 |               <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Availability</lab
    |   el>
215 |               <input
216 |                 type="text"
217 |                 className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
    |   r-zinc-800 dark:bg-zinc-900"
218 |                 value={formData.availableHours}
219 |                 onChange={(e) => setFormData({ ...formData, availableHours: e.target.value })}
220 |               />
221 |             </div>
222 |           </div>
223 |
224 |           <div className="flex justify-end gap-2 pt-2">
225 |             <Button type="button" variant="outline" onClick={() => setModalOpen(false)}>
226 |               Cancel
227 |             </Button>
228 |             <Button type="submit" variant="danger" disabled={isSubmitting}>
229 |               {isSubmitting ? "Saving..." : "Commit Hotline"}
230 |             </Button>
231 |           </div>
232 |         </form>
233 |       </Modal>
234 |     </DashboardShell>
```

```
235 |   );
236 | }
```

File #56: src/app/events/page.tsx 317 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Input } from "@/components/ui/input";
  9 | import { Modal } from "@/components/ui/modal";
 10 | import { useToast } from "@/components/ui/toast";
 11 | import { EmptyState, LoadingState, ErrorState } from "@/components/ui/state-views";
 12 | import { Calendar, MapPin, Users, Clock, Plus, Trash2, CheckCircle2, Ticket } from "lucide-react";
 13 | import {
 14 |   getEventsAction,
 15 |   createEventAction,
 16 |   deleteEventAction,
 17 |   rsvpEventAction,
 18 | } from "@/actions/announcements";
 19 |
 20 | export default function EventsPage() {
 21 |   const { success, error } = useToast();
 22 |   const [events, setEvents] = React.useState<any[]>([]);
 23 |   const [loading, setLoading] = React.useState(true);
 24 |   const [hasError, setHasError] = React.useState(false);
 25 |   const [search, setSearch] = React.useState("");
 26 |
 27 |   // Create Event Modal
 28 |   const [modalOpen, setModalOpen] = React.useState(false);
 29 |   const [formData, setFormData] = React.useState({
 30 |     title: "",
 31 |     description: "",
 32 |     startTime: "",
 33 |     endTime: "",
 34 |     venue: "",
 35 |     organizer: "",
 36 |     category: "Academic",
 37 |   });
 38 |   const [isSubmitting, setIsSubmitting] = React.useState(false);
 39 |
 40 |   const fetchEvents = React.useCallback(async () => {
 41 |     setLoading(true);
 42 |     setHasError(false);
 43 |     try {
 44 |       const res = await getEventsAction();
 45 |       if (res.success && res.data) {
 46 |         setEvents(res.data);
 47 |       } else {
 48 |         setHasError(true);
 49 |       }
 50 |     } catch {
 51 |       setHasError(true);
 52 |     } finally {
 53 |       setLoading(false);
 54 |     }
 55 |   }, []);
 56 |
 57 |   React.useEffect(() => {
 58 |     fetchEvents();
 59 |   }, [fetchEvents]);
 60 |
 61 |   const handleCreate = async (e: React.FormEvent) => {
 62 |     e.preventDefault();
 63 |     setIsSubmitting(true);
 64 |     try {
 65 |       const res = await createEventAction(formData);
 66 |       if (res.success) {
 67 |         success("Event Scheduled", "New campus activity created and added to university calendar.");
 68 |         setModalOpen(false);
 69 |         setFormData({
 70 |           title: "",
 71 |           description: "",
 72 |           startTime: "",
 73 |           endTime: "",
 74 |           venue: "",
```

```
          organizer: "",
          category: "Academic",
        });
        fetchEvents();
      } else {
        error("Submission Failed", res.message || "Please check required fields.");
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsSubmitting(false);
    }
  };

  const handleDelete = async (id: string) => {
    try {
      const res = await deleteEventAction(id);
      if (res.success) {
        success("Event Deleted");
        setEvents((prev) => prev.filter((ev) => ev.id !== id));
      } else {
        error("Deletion Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    }
  };

  const handleRSVP = async (id: string) => {
    try {
      const res = await rsvpEventAction(id);
      if (res.success) {
        success("RSVP Confirmed", "Your seat is registered! Digital badge sent to notification center.");
        setEvents((prev) =>
          prev.map((ev) =>
            ev.id === id ? { ...ev, rsvp_count: (ev.rsvp_count || 0) + 1 } : ev
          )
        );
      } else {
        error("RSVP Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    }
  };

  const filtered = events.filter((e) => {
    return (
      e.title.toLowerCase().includes(search.toLowerCase()) ||
      e.description.toLowerCase().includes(search.toLowerCase()) ||
      e.venue.toLowerCase().includes(search.toLowerCase()) ||
      e.organizer.toLowerCase().includes(search.toLowerCase())
    );
  });

  return (
    <DashboardShell
      title="Campus Events & Activities"
      description="Upcoming symposiums, hackathons, seminars, and student club activities"
      actions={
        <Button size="sm" onClick={() => setModalOpen(true)} className="gap-1.5 shadow-sm shadow-blue-500
/20">
          <Plus className="h-4 w-4" />
          <span>Schedule Event</span>
        </Button>
      }
    >
      <div className="space-y-6">
        {/* Search */}
        <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3 rounded-2xl borde
r border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
          <div className="relative w-full sm:w-72">
            <input
              type="text"
              placeholder="Search events by title, venue, organizer..."
              value={search}
              onChange={(e) => setSearch(e.target.value)}
              className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs text-zinc-
800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
            />
```

```
          </div>
          <div className="text-xs text-zinc-500">
            Total Activities: <strong className="text-zinc-800 dark:text-zinc-200">{filtered.length}</strong>
          </div>
        </div>

        {/* State Views */}
        {loading ? (
          <LoadingState message="Fetching campus activities calendar..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchEvents} />
        ) : filtered.length === 0 ? (
          <EmptyState
            title="No Events Scheduled"
            description="There are currently no events matching your query."
            actionLabel="Schedule Event"
            onAction={() => setModalOpen(true)}
          />
        ) : (
          <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            {filtered.map((e) => (
              <Card key={e.id} className="flex flex-col justify-between hover:border-blue-500/50 transition-all">
                <CardHeader>
                  <div className="flex items-center justify-between mb-2">
                    <Badge variant="info">{e.category}</Badge>
                    <div className="flex items-center gap-1.5">
                      <span className="text-[11px] text-zinc-400 font-medium">
                        {e.rsvp_count || 0} Attending
                      </span>
                      <button
                        onClick={() => handleDelete(e.id)}
                        className="p-1 rounded-lg text-zinc-400 hover:text-rose-500 hover:bg-rose-50 dark:hover:bg-rose-950/40 transition-colors"
                        aria-label="Delete event"
                      >
                        <Trash2 className="h-3.5 w-3.5" />
                      </button>
                    </div>
                  </div>
                  <CardTitle className="text-base font-bold">{e.title}</CardTitle>
                  <CardDescription className="text-xs leading-relaxed mt-1">
                    {e.description}
                  </CardDescription>
                </CardHeader>
                <CardContent className="space-y-3">
                  <div className="space-y-1.5 text-xs text-zinc-600 dark:text-zinc-400 border-t border-zinc-100 dark:border-zinc-800 pt-3">
                    <div className="flex items-center gap-1.5">
                      <Calendar className="h-3.5 w-3.5 text-blue-500" />
                      <span>{new Date(e.start_time).toLocaleDateString()}</span>
                    </div>
                    <div className="flex items-center gap-1.5">
                      <MapPin className="h-3.5 w-3.5 text-zinc-400" />
                      <span>{e.venue}</span>
                    </div>
                    <div className="flex items-center gap-1.5">
                      <Users className="h-3.5 w-3.5 text-zinc-400" />
                      <span>Organized by: <strong className="text-zinc-700 dark:text-zinc-300">{e.organizer}</strong></span>
                    </div>
                  </div>

                  <div className="pt-2 border-t border-zinc-100 dark:border-zinc-800 flex justify-end">
                    <Button
                      size="sm"
                      className="w-full gap-1.5 text-xs shadow-sm shadow-blue-500/20"
                      onClick={() => handleRSVP(e.id)}
                    >
                      <Ticket className="h-3.5 w-3.5" />
                      <span>RSVP Attending</span>
                    </Button>
                  </div>
                </CardContent>
              </Card>
            ))}
          </div>
        )}
      </div>
```

```
227 |
228 |       {/* Schedule Event Modal */}
229 |       <Modal
230 |         isOpen={modalOpen}
231 |         onClose={() => setModalOpen(false)}
232 |         title="Schedule Campus Event"
233 |         description="Creates official event record in PostgreSQL events table"
234 |       >
235 |         <form onSubmit={handleCreate} className="space-y-4">
236 |           <Input
237 |             label="Event Title"
238 |             required
239 |             placeholder="e.g. AI & Quantum Computing Symposium"
240 |             value={formData.title}
241 |             onChange={(e) => setFormData({ ...formData, title: e.target.value })}
242 |           />
243 |
244 |           <div className="space-y-1.5">
245 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Description</label>
246 |             <textarea
247 |               required
248 |               rows={3}
249 |               placeholder="Outline event agenda, key speakers, and prerequisites..."
250 |               className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-8
    |   00 dark:bg-zinc-900"
251 |               value={formData.description}
252 |               onChange={(e) => setFormData({ ...formData, description: e.target.value })}
253 |             />
254 |           </div>
255 |
256 |           <div className="grid grid-cols-2 gap-3">
257 |             <Input
258 |               label="Start Time"
259 |               type="datetime-local"
260 |               required
261 |               value={formData.startTime}
262 |               onChange={(e) => setFormData({ ...formData, startTime: e.target.value })}
263 |             />
264 |             <Input
265 |               label="End Time"
266 |               type="datetime-local"
267 |               required
268 |               value={formData.endTime}
269 |               onChange={(e) => setFormData({ ...formData, endTime: e.target.value })}
270 |             />
271 |           </div>
272 |
273 |           <div className="grid grid-cols-2 gap-3">
274 |             <Input
275 |               label="Venue / Location"
276 |               required
277 |               placeholder="e.g. Central Auditorium"
278 |               value={formData.venue}
279 |               onChange={(e) => setFormData({ ...formData, venue: e.target.value })}
280 |             />
281 |             <Input
282 |               label="Organizer / Club"
283 |               required
284 |               placeholder="e.g. ACM Student Chapter"
285 |               value={formData.organizer}
286 |               onChange={(e) => setFormData({ ...formData, organizer: e.target.value })}
287 |             />
288 |           </div>
289 |
290 |           <div className="space-y-1.5">
291 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Category</label>
292 |             <select
293 |               className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:border-
    |   zinc-800 dark:bg-zinc-900"
294 |               value={formData.category}
295 |               onChange={(e) => setFormData({ ...formData, category: e.target.value })}
296 |             >
297 |               <option>Academic</option>
298 |               <option>Tech Competition</option>
299 |               <option>Cultural</option>
300 |               <option>Career & Placement</option>
301 |               <option>Sports</option>
302 |             </select>
303 |           </div>
304 |
```

```
305 |           <div className="flex justify-end gap-2 pt-2">
306 |             <Button type="button" variant="outline" onClick={() => setModalOpen(false)}>
307 |               Cancel
308 |             </Button>
309 |             <Button type="submit" disabled={isSubmitting}>
310 |               {isSubmitting ? "Scheduling..." : "Confirm & Schedule"}
311 |             </Button>
312 |           </div>
313 |         </form>
314 |       </Modal>
315 |     </DashboardShell>
316 |   );
317 | }
```

File #57: src/app/facilities/page.tsx 375 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Building2, Users, Clock, CheckCircle2, Calendar, MapPin, Filter, Search, Plus } from "lucide-rea
    |   ct";
  9 | import { Modal } from "@/components/ui/modal";
 10 | import { Input } from "@/components/ui/input";
 11 | import { useToast } from "@/components/ui/toast";
 12 | import { LoadingState, ErrorState, EmptyState } from "@/components/ui/state-views";
 13 | import { getFacilitiesAction, createFacilityAction, bookFacilityAction } from "@/actions/operations";
 14 |
 15 | export default function CampusFacilitiesPage() {
 16 |   const { success, error } = useToast();
 17 |   const [facilities, setFacilities] = React.useState<any[]>([]);
 18 |   const [loading, setLoading] = React.useState(true);
 19 |   const [hasError, setHasError] = React.useState(false);
 20 |   const [filterCat, setFilterCat] = React.useState("All");
 21 |   const [search, setSearch] = React.useState("");
 22 |
 23 |   // Modals
 24 |   const [selectedFacility, setSelectedFacility] = React.useState<any | null>(null);
 25 |   const [bookingDate, setBookingDate] = React.useState("2026-10-15");
 26 |   const [bookingTime, setBookingTime] = React.useState("02:00 PM - 04:00 PM");
 27 |   const [purpose, setPurpose] = React.useState("");
 28 |   const [isBooking, setIsBooking] = React.useState(false);
 29 |
 30 |   const [createModalOpen, setCreateModalOpen] = React.useState(false);
 31 |   const [newFacility, setNewFacility] = React.useState({
 32 |     name: "",
 33 |     category: "Specialized Lab",
 34 |     location: "",
 35 |     capacity: 60,
 36 |     operatingHours: "08:00 AM - 09:00 PM",
 37 |     status: "available" as any,
 38 |     description: "",
 39 |   });
 40 |   const [isCreating, setIsCreating] = React.useState(false);
 41 |
 42 |   const categories = ["All", "Specialized Lab", "Auditorium", "Library Space", "Recreation"];
 43 |
 44 |   const fetchFacilities = React.useCallback(async () => {
 45 |     setLoading(true);
 46 |     setHasError(false);
 47 |     try {
 48 |       const res = await getFacilitiesAction();
 49 |       if (res.success && res.data) {
 50 |         setFacilities(res.data);
 51 |       } else {
 52 |         setHasError(true);
 53 |       }
 54 |     } catch {
 55 |       setHasError(true);
 56 |     } finally {
 57 |       setLoading(false);
 58 |     }
 59 |   }, []);
 60 |
 61 |   React.useEffect(() => {
 62 |     fetchFacilities();
```

```
  }, [fetchFacilities]);

  const handleBookingSubmit = async (e: React.FormEvent) => {
    e.preventDefault();
    if (!selectedFacility) return;
    setIsBooking(true);
    try {
      const res = await bookFacilityAction({
        facilityId: selectedFacility.id,
        bookingDate,
        timeSlot: bookingTime,
        purpose,
      });

      if (res.success) {
        success("Reservation Confirmed", `Permit granted for ${selectedFacility.name} on ${bookingDate}.`
);
        setSelectedFacility(null);
        setPurpose("");
      } else {
        error("Booking Failed", res.message || "Please check inputs.");
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsBooking(false);
    }
  };

  const handleCreateFacility = async (e: React.FormEvent) => {
    e.preventDefault();
    setIsCreating(true);
    try {
      const res = await createFacilityAction(newFacility);
      if (res.success) {
        success("Facility Registered", "New facility provisioned in campus inventory.");
        setCreateModalOpen(false);
        setNewFacility({
          name: "",
          category: "Specialized Lab",
          location: "",
          capacity: 60,
          operatingHours: "08:00 AM - 09:00 PM",
          status: "available",
          description: "",
        });
        fetchFacilities();
      } else {
        error("Registration Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsCreating(false);
    }
  };

  const filtered = facilities.filter((f) => {
    const matchCat = filterCat === "All" || f.category === filterCat;
    const matchSearch =
      f.name.toLowerCase().includes(search.toLowerCase()) ||
      f.location.toLowerCase().includes(search.toLowerCase()) ||
      f.category.toLowerCase().includes(search.toLowerCase());
    return matchCat && matchSearch;
  });

  return (
    <DashboardShell
      title="Campus Facilities & Resource Booking"
      description="Reserve lecture halls, research workstations, sports courts, and collaborative study s
paces"
      actions={
        <Button size="sm" onClick={() => setCreateModalOpen(true)} className="gap-1.5 shadow-sm shadow-bl
ue-500/20">
          <Plus className="h-4 w-4" />
          <span>Provision Facility</span>
        </Button>
      }
    >
      <div className="space-y-6">
```

```
        {/* Filter controls */}
        <div className="flex flex-col gap-3 sm:flex-row sm:items-center sm:justify-between rounded-2xl bo
rder border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
          <div className="flex items-center gap-1.5 flex-wrap">
            {categories.map((c) => (
              <button
                key={c}
                onClick={() => setFilterCat(c)}
                className={`rounded-xl px-3 py-1.5 text-xs font-semibold transition-all ${
                  filterCat === c
                    ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
                    : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
                }`}
              >
                {c}
              </button>
            ))}
          </div>

          <div className="relative">
            <input
              type="text"
              placeholder="Search facilities..."
              value={search}
              onChange={(e) => setSearch(e.target.value)}
              className="w-full sm:w-64 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs te
xt-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-
200"
            />
          </div>
        </div>

        {/* State Views */}
        {loading ? (
          <LoadingState message="Fetching campus facilities from PostgreSQL..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchFacilities} />
        ) : filtered.length === 0 ? (
          <EmptyState
            title="No Facilities Found"
            description="No campus facilities match the selected filter."
            actionLabel="Provision Facility"
            onAction={() => setCreateModalOpen(true)}
          />
        ) : (
          <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
            {filtered.map((f) => (
              <Card key={f.id} className="hover:border-blue-500/40 transition-all flex flex-col justify-b
etween">
                <CardHeader>
                  <div className="flex items-center justify-between mb-2">
                    <Badge variant="outline">{f.category}</Badge>
                    {f.status === "available" ? (
                      <Badge variant="success">Available Now</Badge>
                    ) : f.status === "booked" ? (
                      <Badge variant="warning">In Session</Badge>
                    ) : (
                      <Badge variant="destructive">Maintenance</Badge>
                    )}
                  </div>
                  <CardTitle className="text-base font-bold">{f.name}</CardTitle>
                  <CardDescription className="text-xs flex items-center gap-1.5 text-zinc-500 mt-1">
                    <MapPin className="h-3.5 w-3.5 text-zinc-400" />
                    <span>{f.location}</span>
                  </CardDescription>
                </CardHeader>
                <CardContent className="space-y-4">
                  <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
                    {f.description}
                  </p>

                  <div className="grid grid-cols-2 gap-2 text-xs text-zinc-500 border-t border-zinc-100 d
ark:border-zinc-800 pt-3">
                    <div className="flex items-center gap-1.5">
                      <Users className="h-3.5 w-3.5 text-zinc-400" />
                      <span>Capacity: {f.capacity} seats</span>
                    </div>
                    <div className="flex items-center gap-1.5">
                      <Clock className="h-3.5 w-3.5 text-zinc-400" />
                      <span>{f.operating_hours || f.hours || "08:00 AM - 08:00 PM"}</span>
```

```
                    </div>
                  </div>

                  <div className="pt-2 border-t border-zinc-100 dark:border-zinc-800 flex justify-end">
                    <Button
                      size="sm"
                      disabled={f.status === "maintenance"}
                      className="gap-1.5 text-xs shadow-sm shadow-blue-500/20"
                      onClick={() => setSelectedFacility(f)}
                    >
                      <Calendar className="h-3.5 w-3.5" />
                      {f.status === "available" ? "Reserve Facility" : "Request Slot Waitlist"}
                    </Button>
                  </div>
                </CardContent>
              </Card>
            ))}
          </div>
        )}
      </div>

      {/* Reservation Modal */}
      {selectedFacility && (
        <Modal
          isOpen={!!selectedFacility}
          onClose={() => setSelectedFacility(null)}
          title={`Reserve Facility: ${selectedFacility.name}`}
          description={`${selectedFacility.location} • Max Capacity: ${selectedFacility.capacity} participants`}
        >
          <form onSubmit={handleBookingSubmit} className="space-y-4">
            <div className="grid grid-cols-2 gap-3">
              <Input
                label="Reservation Date"
                type="date"
                required
                value={bookingDate}
                onChange={(e) => setBookingDate(e.target.value)}
              />

              <div className="space-y-1.5">
                <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                  Time Slot
                </label>
                <select
                  className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:border-zinc-800 dark:bg-zinc-900"
                  value={bookingTime}
                  onChange={(e) => setBookingTime(e.target.value)}
                >
                  <option>09:00 AM - 11:00 AM</option>
                  <option>11:30 AM - 01:30 PM</option>
                  <option>02:00 PM - 04:00 PM</option>
                  <option>04:30 PM - 06:30 PM</option>
                </select>
              </div>
            </div>

            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Purpose / Activity Details
              </label>
              <textarea
                required
                rows={3}
                placeholder="Describe workshop agenda or experiment specifics..."
                className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-800 dark:bg-zinc-900"
                value={purpose}
                onChange={(e) => setPurpose(e.target.value)}
              />
            </div>

            <div className="flex justify-end gap-2 pt-2">
              <Button type="button" variant="outline" onClick={() => setSelectedFacility(null)}>
                Cancel
              </Button>
              <Button type="submit" disabled={isBooking}>
                {isBooking ? "Submitting..." : "Submit Booking Permit"}
              </Button>
```

```
292 |             </div>
293 |           </form>
294 |         </Modal>
295 |       )}
296 |
297 |       {/* Provision Facility Modal */}
298 |       <Modal
299 |         isOpen={createModalOpen}
300 |         onClose={() => setCreateModalOpen(false)}
301 |         title="Provision Campus Facility"
302 |         description="Creates facility record in PostgreSQL campus_facilities table"
303 |       >
304 |         <form onSubmit={handleCreateFacility} className="space-y-4">
305 |           <Input
306 |             label="Facility Name"
307 |             required
308 |             placeholder="e.g. VLSI Design Studio & Cleanroom"
309 |             value={newFacility.name}
310 |             onChange={(e) => setNewFacility({ ...newFacility, name: e.target.value })}
311 |           />
312 |
313 |           <div className="grid grid-cols-2 gap-3">
314 |             <div className="space-y-1.5">
315 |               <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Category</label>
316 |               <select
317 |                 className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
    |   r-zinc-800 dark:bg-zinc-900"
318 |                 value={newFacility.category}
319 |                 onChange={(e) => setNewFacility({ ...newFacility, category: e.target.value })}
320 |               >
321 |                 <option>Specialized Lab</option>
322 |                 <option>Auditorium</option>
323 |                 <option>Library Space</option>
324 |                 <option>Recreation</option>
325 |               </select>
326 |             </div>
327 |             <Input
328 |               label="Location / Block"
329 |               required
330 |               placeholder="e.g. Tech Block B • 2nd Floor"
331 |               value={newFacility.location}
332 |               onChange={(e) => setNewFacility({ ...newFacility, location: e.target.value })}
333 |             />
334 |           </div>
335 |
336 |           <div className="grid grid-cols-2 gap-3">
337 |             <Input
338 |               label="Capacity (Seats)"
339 |               type="number"
340 |               required
341 |               value={newFacility.capacity}
342 |               onChange={(e) => setNewFacility({ ...newFacility, capacity: Number(e.target.value) })}
343 |             />
344 |             <Input
345 |               label="Operating Hours"
346 |               required
347 |               value={newFacility.operatingHours}
348 |               onChange={(e) => setNewFacility({ ...newFacility, operatingHours: e.target.value })}
349 |             />
350 |           </div>
351 |
352 |           <div className="space-y-1.5">
353 |             <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Description</label>
354 |             <textarea
355 |               rows={2}
356 |               placeholder="Hardware specs, software licenses, acoustics..."
357 |               className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-8
    |   00 dark:bg-zinc-900"
358 |               value={newFacility.description}
359 |               onChange={(e) => setNewFacility({ ...newFacility, description: e.target.value })}
360 |             />
361 |           </div>
362 |
363 |           <div className="flex justify-end gap-2 pt-2">
364 |             <Button type="button" variant="outline" onClick={() => setCreateModalOpen(false)}>
365 |               Cancel
366 |             </Button>
367 |             <Button type="submit" disabled={isCreating}>
368 |               {isCreating ? "Provisioning..." : "Provision Facility"}
369 |             </Button>
```

```
370 |           </div>
371 |         </form>
372 |       </Modal>
373 |     </DashboardShell>
374 |   );
375 | }
```

File #58: src/app/faculty/page.tsx 233 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Users, Mail, MapPin, Clock, BookOpen, Search, Calendar, Phone, MessageSquare } from "lucide-reac
    |   t";
  9 | import { Modal } from "@/components/ui/modal";
 10 | import { useToast } from "@/components/ui/toast";
 11 | import { LoadingState, ErrorState, EmptyState } from "@/components/ui/state-views";
 12 | import { getFacultyDirectoryAction } from "@/actions/profiles";
 13 |
 14 | export default function FacultyDirectoryPage() {
 15 |   const { success, error } = useToast();
 16 |   const [facultyList, setFacultyList] = React.useState<any[]>([]);
 17 |   const [loading, setLoading] = React.useState(true);
 18 |   const [hasError, setHasError] = React.useState(false);
 19 |   const [search, setSearch] = React.useState("");
 20 |
 21 |   // Consultation Modal
 22 |   const [selectedProf, setSelectedProf] = React.useState<any | null>(null);
 23 |   const [appointmentSlot, setAppointmentSlot] = React.useState("Next Available Office Hour");
 24 |   const [agenda, setAgenda] = React.useState("");
 25 |
 26 |   const fetchFaculty = React.useCallback(async () => {
 27 |     setLoading(true);
 28 |     setHasError(false);
 29 |     try {
 30 |       const res = await getFacultyDirectoryAction();
 31 |       if (res.success && res.data) {
 32 |         setFacultyList(res.data);
 33 |       } else {
 34 |         setHasError(true);
 35 |       }
 36 |     } catch {
 37 |       setHasError(true);
 38 |     } finally {
 39 |       setLoading(false);
 40 |     }
 41 |   }, []);
 42 |
 43 |   React.useEffect(() => {
 44 |     fetchFaculty();
 45 |   }, [fetchFaculty]);
 46 |
 47 |   const handleBookOfficeHour = (e: React.FormEvent) => {
 48 |     e.preventDefault();
 49 |     if (!selectedProf) return;
 50 |     success("Consultation Booked", `Office hours confirmed with ${selectedProf.full_name || selectedProf.
    |   name} (${appointmentSlot}).`);
 51 |     setSelectedProf(null);
 52 |     setAgenda("");
 53 |   };
 54 |
 55 |   const filtered = facultyList.filter((f) => {
 56 |     const name = f.full_name || f.name || "";
 57 |     const designation = f.faculty_profile?.designation || f.designation || "";
 58 |     const dept = f.department || f.dept || "";
 59 |     const cabin = f.faculty_profile?.office_location || f.cabin || "";
 60 |
 61 |     return (
 62 |       name.toLowerCase().includes(search.toLowerCase()) ||
 63 |       designation.toLowerCase().includes(search.toLowerCase()) ||
 64 |       dept.toLowerCase().includes(search.toLowerCase()) ||
 65 |       cabin.toLowerCase().includes(search.toLowerCase())
 66 |     );
 67 |   });
 68 |
```

```
  return (
    <DashboardShell
      title="Faculty Directory & Consultation"
      description="Connect with professors, inspect research specializations, and schedule office hours"
      actions={
        <div className="relative">
          <input
            type="text"
            placeholder="Search faculty, topic, cabin..."
            value={search}
            onChange={(e) => setSearch(e.target.value)}
            className="w-full sm:w-64 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs text-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
          />
        </div>
      }
    >
      <div className="space-y-6">
        {loading ? (
          <LoadingState message="Fetching faculty directory from PostgreSQL profiles..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchFaculty} />
        ) : filtered.length === 0 ? (
          <EmptyState
            title="No Faculty Found"
            description="No faculty members match your query."
          />
        ) : (
          <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
            {filtered.map((f) => {
              const name = f.full_name || f.name || "Faculty Member";
              const designation = f.faculty_profile?.designation || f.designation || "Professor";
              const dept = f.department || f.dept || "Computer Science & Engineering";
              const email = f.email || "faculty@campus.edu";
              const cabin = f.faculty_profile?.office_location || f.cabin || "Faculty Wing • Room 301";
              const hours = f.faculty_profile?.office_hours || f.officeHours || "Mon & Wed: 03:00 - 05:00 PM";
              const research = f.faculty_profile?.research_interests || f.research || ["Computer Science", "Algorithms"];

              return (
                <Card key={f.id || email} className="hover:border-blue-500/40 transition-all flex flex-col justify-between">
                  <CardHeader>
                    <div className="flex items-start justify-between">
                      <div>
                        <CardTitle className="text-base">{name}</CardTitle>
                        <p className="text-xs font-semibold text-blue-600 dark:text-blue-400 mt-0.5">
                          {designation}
                        </p>
                        <p className="text-[11px] text-zinc-400">{dept}</p>
                      </div>
                      <div className="h-10 w-10 rounded-full bg-blue-500/10 text-blue-600 flex items-center justify-center font-bold text-xs dark:bg-blue-500/20 dark:text-blue-400">
                        {name.charAt(0)}
                      </div>
                    </div>
                  </CardHeader>
                  <CardContent className="space-y-3">
                    <div className="space-y-1.5 text-xs text-zinc-600 dark:text-zinc-300">
                      <div className="flex items-center gap-2">
                        <Mail className="h-3.5 w-3.5 text-zinc-400" />
                        <span>{email}</span>
                      </div>
                      <div className="flex items-center gap-2">
                        <MapPin className="h-3.5 w-3.5 text-zinc-400" />
                        <span>{cabin}</span>
                      </div>
                      <div className="flex items-center gap-2">
                        <Clock className="h-3.5 w-3.5 text-zinc-400" />
                        <span className="font-medium text-emerald-600 dark:text-emerald-400">{hours}</span>
                      </div>
                    </div>

                    <div>
                      <p className="text-[11px] font-semibold text-zinc-400 uppercase tracking-wider mb-1.5">
                        Focus & Research
```

```
                      </p>
                      <div className="flex flex-wrap gap-1.5">
                        {research.map((r: string) => (
                          <span
                            key={r}
                            className="rounded-lg bg-zinc-100 px-2 py-0.5 text-[11px] text-zinc-700 dark:bg-zinc-800 dark:text-zinc-300 font-medium"
                          >
                            {r}
                          </span>
                        ))}
                      </div>
                    </div>

                    <div className="pt-2 flex items-center gap-2 border-t border-zinc-100 dark:border-zinc-800">
                      <Button
                        variant="outline"
                        size="sm"
                        className="w-1/2 gap-1.5 text-xs"
                        onClick={() => {
                          navigator.clipboard?.writeText?.(email);
                          success("Email Copied", `${email} copied to clipboard.`);
                        }}
                      >
                        <Mail className="h-3.5 w-3.5" />
                        Copy Email
                      </Button>
                      <Button
                        size="sm"
                        className="w-1/2 gap-1.5 text-xs shadow-sm shadow-blue-500/20"
                        onClick={() => setSelectedProf(f)}
                      >
                        <Calendar className="h-3.5 w-3.5" />
                        Book Slot
                      </Button>
                    </div>
                  </CardContent>
                </Card>
              );
            })}
          </div>
        )}
      </div>

      {/* Book Office Hour Modal */}
      {selectedProf && (
        <Modal
          isOpen={!!selectedProf}
          onClose={() => setSelectedProf(null)}
          title={`Book Consultation: ${selectedProf.full_name || selectedProf.name}`}
          description={`Office Location: ${selectedProf.faculty_profile?.office_location || selectedProf.cabin || "Tech Block A"}`}
        >
          <form onSubmit={handleBookOfficeHour} className="space-y-4">
            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Select Preferred Slot
              </label>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:border-zinc-800 dark:bg-zinc-900"
                value={appointmentSlot}
                onChange={(e) => setAppointmentSlot(e.target.value)}
              >
                <option>Next Regular Office Hour (Mon/Wed 03:00 PM)</option>
                <option>Project Guidance Slot (Fri 04:00 PM)</option>
                <option>Academic Doubts Clinic (Next Class)</option>
              </select>
            </div>

            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Brief Agenda / Query
              </label>
              <textarea
                required
                rows={3}
                placeholder="Mention the subject or research doubt you wish to discuss..."
                className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc
```

```
    |   -800 dark:bg-zinc-900"
217 |                 value={agenda}
218 |                 onChange={(e) => setAgenda(e.target.value)}
219 |               />
220 |             </div>
221 |
222 |             <div className="flex justify-end gap-2 pt-2">
223 |               <Button type="button" variant="outline" onClick={() => setSelectedProf(null)}>
224 |                 Cancel
225 |               </Button>
226 |               <Button type="submit">Confirm Consultation</Button>
227 |             </div>
228 |           </form>
229 |         </Modal>
230 |       )}
231 |     </DashboardShell>
232 |   );
233 | }
```

File #59: src/app/features/page.tsx 185 lines

```
  1 | import * as React from "react";
  2 | import Link from "next/link";
  3 | import {
  4 |   GraduationCap,
  5 |   Briefcase,
  6 |   ShieldAlert,
  7 |   Bell,
  8 |   Calendar,
  9 |   Clock,
 10 |   CheckCircle2,
 11 |   FileText,
 12 |   BookOpen,
 13 |   Users,
 14 |   Building2,
 15 |   PhoneCall,
 16 |   MessageSquare,
 17 |   Search,
 18 |   Settings,
 19 |   Sparkles,
 20 |   ArrowRight,
 21 | } from "lucide-react";
 22 | import { Navbar } from "@/components/layout/navbar";
 23 | import { Footer } from "@/components/layout/footer";
 24 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
 25 | import { Button } from "@/components/ui/button";
 26 | import { Badge } from "@/components/ui/badge";
 27 |
 28 | export default function FeaturesPage() {
 29 |   const featureList = [
 30 |     {
 31 |       title: "Student Academic Self-Service",
 32 |       description: "Everything a student needs in one unified dashboard: dynamic schedule, attendance tre
    |   nds, assignment submissions, and downloadable notes.",
 33 |       icon: GraduationCap,
 34 |       color: "blue",
 35 |       href: "/dashboard/student",
 36 |       badge: "Student",
 37 |     },
 38 |     {
 39 |       title: "Faculty Course & Gradebook Suite",
 40 |       description: "Class rosters, lecture scheduling, one-click attendance recording, assignment publish
    |   ing, and student grading tools.",
 41 |       icon: Briefcase,
 42 |       color: "amber",
 43 |       href: "/dashboard/faculty",
 44 |       badge: "Faculty",
 45 |     },
 46 |     {
 47 |       title: "Admin Command & Governance",
 48 |       description: "User directory management, campus-wide broadcast announcements, event curation, facil
    |   ity booking oversight, and system analytics.",
 49 |       icon: ShieldAlert,
 50 |       color: "rose",
 51 |       href: "/dashboard/admin",
 52 |       badge: "Admin",
 53 |     },
 54 |     {
 55 |       title: "Dynamic Smart Timetable",
```

```
      description: "Conflict-free weekly schedule visualization filtered by day, semester, room number, a
  nd professor with real-time room shift alerts.",
      icon: Clock,
      color: "indigo",
      href: "/timetable",
      badge: "Academic",
    },
    {
      title: "Attendance Intelligence",
      description: "Track present/absent logs, subject-wise attendance percentage against the mandatory 7
  5% threshold, and shortfall warnings.",
      icon: CheckCircle2,
      color: "emerald",
      href: "/attendance",
      badge: "Tracking",
    },
    {
      title: "Digital Assignment Workflows",
      description: "View deadlines, submission status, upload assignment deliverables directly to secure
  storage, and receive graded feedback.",
      icon: FileText,
      color: "violet",
      href: "/assignments",
      badge: "Evaluation",
    },
    {
      title: "Course Study Materials & Respository",
      description: "Organized repository of syllabi, lecture slides, lab manuals, and reference notes wit
  h instant search and download links.",
      icon: BookOpen,
      color: "sky",
      href: "/materials",
      badge: "Storage",
    },
    {
      title: "Live Campus Announcements",
      description: "Categorized urgent, academic, and administrative campus notifications with real-time
  websocket broadcasting.",
      icon: Bell,
      color: "amber",
      href: "/announcements",
      badge: "Realtime",
    },
    {
      title: "Campus Events & Workshops",
      description: "Explore tech symposiums, cultural fests, hackathons, and sports tournaments with one-
  click RSVP and venue details.",
      icon: Calendar,
      color: "purple",
      href: "/events",
      badge: "Engagement",
    },
    {
      title: "Facilities & Lab Status",
      description: "Real-time occupancy tracking and booking requests for advanced computer labs, library
   study cubes, and seminar auditoriums.",
      icon: Building2,
      color: "teal",
      href: "/facilities",
      badge: "Campus",
    },
    {
      title: "24/7 Campus Emergency SOS",
      description: "Direct speed-dial hotlines for campus medical centre, security control desk, women's
  helpline, and ambulance dispatch.",
      icon: PhoneCall,
      color: "rose",
      href: "/emergency",
      badge: "Safety",
    },
    {
      title: "Anonymous Grievance & Feedback",
      description: "Submit suggestions and issues regarding campus facilities, cafeteria, or curriculum w
  ith confidential tracking numbers.",
      icon: MessageSquare,
      color: "cyan",
      href: "/feedback",
      badge: "Support",
    },
  ];
```

```
127 |
128 |   return (
129 |     <div className="min-h-screen flex flex-col bg-zinc-50/70 dark:bg-zinc-950 text-zinc-900 dark:text-zin
    |   c-100">
130 |       <Navbar />
131 |
132 |       <main className="flex-1">
133 |         {/* Header */}
134 |         <section className="border-b border-zinc-200/80 bg-white/60 py-16 text-center dark:border-zinc-80
    |   0/80 dark:bg-zinc-900/30">
135 |           <div className="mx-auto max-w-4xl px-4 sm:px-6 lg:px-8">
136 |             <Badge variant="secondary" className="mb-3">
137 |               Comprehensive Feature Suite
138 |             </Badge>
139 |             <h1 className="text-3xl font-extrabold tracking-tight sm:text-5xl">
140 |               Engineered for Modern Academic Life
141 |             </h1>
142 |             <p className="mx-auto mt-4 max-w-2xl text-sm sm:text-base text-zinc-600 dark:text-zinc-400">
143 |               Explore the 18+ modular capabilities included in the Smart Campus platform, built for stude
    |   nts, professors, and university leadership.
144 |             </p>
145 |           </div>
146 |         </section>
147 |
148 |         {/* Features Grid */}
149 |         <section className="py-16 mx-auto max-w-7xl px-4 sm:px-6 lg:px-8">
150 |           <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
151 |             {featureList.map((feat) => {
152 |               const Icon = feat.icon;
153 |               return (
154 |                 <Card
155 |                   key={feat.title}
156 |                   className="flex flex-col justify-between hover:shadow-lg transition-all border-zinc-200
    |   /90 dark:border-zinc-800/90"
157 |                 >
158 |                   <CardHeader>
159 |                     <div className="flex items-center justify-between mb-3">
160 |                       <div className="h-10 w-10 rounded-xl bg-blue-500/10 text-blue-600 flex items-center
    |    justify-center dark:bg-blue-500/20 dark:text-blue-400">
161 |                         <Icon className="h-5 w-5" />
162 |                       </div>
163 |                       <Badge variant="outline">{feat.badge}</Badge>
164 |                     </div>
165 |                     <CardTitle className="text-base font-bold">{feat.title}</CardTitle>
166 |                     <CardDescription className="text-xs leading-relaxed mt-2">
167 |                       {feat.description}
168 |                     </CardDescription>
169 |                   </CardHeader>
170 |                   <CardContent className="pt-0">
171 |                     <Link href={feat.href} className="inline-flex items-center text-xs font-semibold text
    |   -blue-600 hover:text-blue-700 dark:text-blue-400 gap-1.5 transition-colors">
172 |                       Open Module <ArrowRight className="h-3.5 w-3.5" />
173 |                     </Link>
174 |                   </CardContent>
175 |                 </Card>
176 |               );
177 |             })}
178 |           </div>
179 |         </section>
180 |       </main>
181 |
182 |       <Footer />
183 |     </div>
184 |   );
185 | }
```

File #60: src/app/feedback/page.tsx

321 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Input } from "@/components/ui/input";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Badge } from "@/components/ui/badge";
  9 | import { Modal } from "@/components/ui/modal";
 10 | import { useToast } from "@/components/ui/toast";
 11 | import { EmptyState, LoadingState, ErrorState } from "@/components/ui/state-views";
```

```
12 | import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";
13 | import { MessageSquare, Send, CheckCircle2, Clock, ShieldCheck, Eye, Plus } from "lucide-react";
14 | import { getFeedbackAction, submitFeedbackAction, resolveFeedbackAction } from "@/actions/operations";
15 |
16 | export default function FeedbackPage() {
17 |   const { success, error } = useToast();
18 |   const [tickets, setTickets] = React.useState<any[]>([]);
19 |   const [loading, setLoading] = React.useState(true);
20 |   const [hasError, setHasError] = React.useState(false);
21 |
22 |   // Form State
23 |   const [category, setCategory] = React.useState<"Academic" | "Hostel" | "Canteen" | "Infrastructure" | "
   |   Other">("Academic");
24 |   const [subject, setSubject] = React.useState("");
25 |   const [description, setDescription] = React.useState("");
26 |   const [isAnonymous, setIsAnonymous] = React.useState(false);
27 |   const [isSubmitting, setIsSubmitting] = React.useState(false);
28 |
29 |   // Admin Resolution Modal
30 |   const [resolvingTicket, setResolvingTicket] = React.useState<any | null>(null);
31 |   const [newStatus, setNewStatus] = React.useState<"pending" | "under_review" | "resolved" | "dismissed">
   |   ("under_review");
32 |   const [adminNotes, setAdminNotes] = React.useState("");
33 |
34 |   const fetchTickets = React.useCallback(async () => {
35 |     setLoading(true);
36 |     setHasError(false);
37 |     try {
38 |       const res = await getFeedbackAction();
39 |       if (res.success && res.data) {
40 |         setTickets(res.data);
41 |       } else {
42 |         setHasError(true);
43 |       }
44 |     } catch {
45 |       setHasError(true);
46 |     } finally {
47 |       setLoading(false);
48 |     }
49 |   }, []);
50 |
51 |   React.useEffect(() => {
52 |     fetchTickets();
53 |   }, [fetchTickets]);
54 |
55 |   const handleSubmit = async (e: React.FormEvent) => {
56 |     e.preventDefault();
57 |     setIsSubmitting(true);
58 |     try {
59 |       const res = await submitFeedbackAction({
60 |         category,
61 |         subject,
62 |         description,
63 |         isAnonymous,
64 |       });
65 |
66 |       if (res.success) {
67 |         success("Ticket Logged", "Your grievance report has been registered in PostgreSQL feedback table.
   |   ");
68 |         setSubject("");
69 |         setDescription("");
70 |         setIsAnonymous(false);
71 |         fetchTickets();
72 |       } else {
73 |         error("Submission Failed", res.message || "Please check inputs.");
74 |       }
75 |     } catch (err: any) {
76 |       error("Error", err.message);
77 |     } finally {
78 |       setIsSubmitting(false);
79 |     }
80 |   };
81 |
82 |   const handleResolveSubmit = async (e: React.FormEvent) => {
83 |     e.preventDefault();
84 |     if (!resolvingTicket) return;
85 |     try {
86 |       const res = await resolveFeedbackAction({
87 |         feedbackId: resolvingTicket.id,
88 |         status: newStatus,
```

```
        adminNotes,
      });

      if (res.success) {
        success("Status Updated", `Ticket resolved as ${newStatus}.`);
        setResolvingTicket(null);
        setAdminNotes("");
        fetchTickets();
      } else {
        error("Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    }
  };

  return (
    <DashboardShell
      title="Grievance Redressal & Student Feedback"
      description="Submit confidential feedback or report campus infrastructure and academic concerns"
    >
      <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
        {/* Ticket Submission Form */}
        <div className="lg:col-span-1">
          <Card>
            <CardHeader>
              <CardTitle className="text-base">File a Grievance Ticket</CardTitle>
              <CardDescription className="text-xs">
                Logged to PostgreSQL with administrative oversight
              </CardDescription>
            </CardHeader>
            <CardContent>
              <form onSubmit={handleSubmit} className="space-y-4">
                <div className="space-y-1.5">
                  <label className="block text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                    Category
                  </label>
                  <select
                    className="flex w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs t
ext-zinc-900 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-100"
                    value={category}
                    onChange={(e) => setCategory(e.target.value as any)}
                  >
                    <option value="Academic">Academic & Courses</option>
                    <option value="Infrastructure">Infrastructure & Labs</option>
                    <option value="Hostel">Hostel & Housing</option>
                    <option value="Canteen">Canteen & Hygiene</option>
                    <option value="Other">Other University Services</option>
                  </select>
                </div>

                <Input
                  label="Subject Title"
                  required
                  placeholder="e.g. Lab 302 Air Conditioning Issue"
                  value={subject}
                  onChange={(e) => setSubject(e.target.value)}
                />

                <div className="space-y-1.5">
                  <label className="block text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                    Detailed Grievance
                  </label>
                  <textarea
                    required
                    rows={4}
                    placeholder="Provide detailed description, room numbers, or specifics (minimum 20 cha
racters)..."
                    className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs text-zinc-90
0 placeholder-zinc-400 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-100"
                    value={description}
                    onChange={(e) => setDescription(e.target.value)}
                  />
                </div>

                <div className="flex items-center gap-2">
                  <input
                    type="checkbox"
                    id="anonCheck"
                    checked={isAnonymous}
```

```
166 |                     onChange={(e) => setIsAnonymous(e.target.checked)}
167 |                     className="rounded text-blue-600 h-4 w-4"
168 |                   />
169 |                   <label htmlFor="anonCheck" className="text-xs text-zinc-600 dark:text-zinc-400 font-med
    |   ium cursor-pointer">
170 |                     Submit confidentially (Hide identity)
171 |                   </label>
172 |                 </div>
173 |
174 |                 <Button type="submit" disabled={isSubmitting} className="w-full gap-2 text-xs">
175 |                   <Send className="h-3.5 w-3.5" />
176 |                   {isSubmitting ? "Logging Ticket..." : "Submit Grievance Ticket"}
177 |                 </Button>
178 |               </form>
179 |             </CardContent>
180 |           </Card>
181 |         </div>
182 |
183 |         {/* Existing Tickets Overview */}
184 |         <div className="lg:col-span-2 space-y-4">
185 |           <div className="flex items-center justify-between">
186 |             <h3 className="text-base font-bold text-zinc-900 dark:text-zinc-100">
187 |               Active Grievance Tickets
188 |             </h3>
189 |             <span className="text-xs text-zinc-500">
190 |               Total Recorded: <strong className="text-zinc-800 dark:text-zinc-200">{tickets.length}</stro
    |   ng>
191 |             </span>
192 |           </div>
193 |
194 |           {loading ? (
195 |             <LoadingState message="Fetching grievance tickets..." />
196 |           ) : hasError ? (
197 |             <ErrorState onRetry={fetchTickets} />
198 |           ) : tickets.length === 0 ? (
199 |             <EmptyState
200 |               title="No Grievance Tickets"
201 |               description="You have not submitted any grievance tickets yet."
202 |             />
203 |           ) : (
204 |             <div className="space-y-3">
205 |               {tickets.map((t) => (
206 |                 <Card key={t.id} className="hover:border-blue-500/40 transition-all">
207 |                   <CardHeader className="pb-2">
208 |                     <div className="flex items-start justify-between gap-2">
209 |                       <div>
210 |                         <div className="flex items-center gap-2">
211 |                           <Badge variant="outline">{t.category}</Badge>
212 |                           <Badge
213 |                             variant={
214 |                               t.status === "resolved"
215 |                                 ? "success"
216 |                                 : t.status === "under_review"
217 |                                 ? "warning"
218 |                                 : t.status === "dismissed"
219 |                                 ? "danger"
220 |                                 : "default"
221 |                             }
222 |                             className="capitalize"
223 |                           >
224 |                             {t.status.replace("_", " ")}
225 |                           </Badge>
226 |                           {t.is_anonymous && (
227 |                             <Badge variant="secondary" className="text-[10px]">
228 |                               Anonymous
229 |                             </Badge>
230 |                           )}
231 |                         </div>
232 |                         <CardTitle className="text-sm font-bold mt-2">{t.subject}</CardTitle>
233 |                       </div>
234 |                       <span className="text-[11px] text-zinc-400 whitespace-nowrap">
235 |                         {new Date(t.created_at).toLocaleDateString()}
236 |                       </span>
237 |                     </div>
238 |                   </CardHeader>
239 |                   <CardContent className="space-y-3">
240 |                     <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
241 |                       {t.description}
242 |                     </p>
243 |
```

```
                    {t.admin_notes && (
                      <div className="rounded-xl border border-blue-500/20 bg-blue-50/30 p-2.5 text-xs te
xt-blue-900 dark:text-blue-300">
                        <strong>Admin Resolution Note:</strong> {t.admin_notes}
                      </div>
                    )}

                    <div className="pt-2 border-t border-zinc-100 dark:border-zinc-800 flex justify-end">
                      <Button
                        variant="ghost"
                        size="sm"
                        className="text-xs text-blue-600 dark:text-blue-400"
                        onClick={() => {
                          setResolvingTicket(t);
                          setNewStatus(t.status);
                          setAdminNotes(t.admin_notes || "");
                        }}
                      >
                        Update Ticket Review
                      </Button>
                    </div>
                  </CardContent>
                </Card>
              ))}
            </div>
          )}
        </div>
      </div>

      {/* Admin Resolution Modal */}
      {resolvingTicket && (
        <Modal
          isOpen={!!resolvingTicket}
          onClose={() => setResolvingTicket(null)}
          title={`Update Grievance: ${resolvingTicket.subject}`}
          description={`Category: ${resolvingTicket.category} • ID: ${resolvingTicket.id}`}
        >
          <form onSubmit={handleResolveSubmit} className="space-y-4">
            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Resolution Status
              </label>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={newStatus}
                onChange={(e) => setNewStatus(e.target.value as any)}
              >
                <option value="pending">Pending Investigation</option>
                <option value="under_review">Under Active Review</option>
                <option value="resolved">Issue Resolved</option>
                <option value="dismissed">Dismissed / Inapplicable</option>
              </select>
            </div>

            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Administrative Action Notes
              </label>
              <textarea
                rows={3}
                placeholder="Detail corrective measures taken or committee decisions..."
                className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc
-800 dark:bg-zinc-900"
                value={adminNotes}
                onChange={(e) => setAdminNotes(e.target.value)}
              />
            </div>

            <div className="flex justify-end gap-2 pt-2">
              <Button type="button" variant="outline" onClick={() => setResolvingTicket(null)}>
                Cancel
              </Button>
              <Button type="submit">Commit Resolution</Button>
            </div>
          </form>
        </Modal>
      )}
    </DashboardShell>
  );
```

```
321 | }
```

File #61: src/app/forgot-password/page.tsx 114 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import Link from "next/link";
  5 | import { useRouter } from "next/navigation";
  6 | import { KeyRound, ArrowRight, ArrowLeft, Mail, GraduationCap } from "lucide-react";
  7 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  8 | import { Input } from "@/components/ui/input";
  9 | import { Button } from "@/components/ui/button";
 10 | import { Badge } from "@/components/ui/badge";
 11 | import { forgotPasswordAction } from "@/actions/auth";
 12 | import { useToast } from "@/components/ui/toast";
 13 |
 14 | export default function ForgotPasswordPage() {
 15 |   const router = useRouter();
 16 |   const { success, error } = useToast();
 17 |   const [email, setEmail] = React.useState("");
 18 |   const [isSubmitting, setIsSubmitting] = React.useState(false);
 19 |   const [submitted, setSubmitted] = React.useState(false);
 20 |
 21 |   const handleSubmit = async (e: React.FormEvent) => {
 22 |     e.preventDefault();
 23 |     setIsSubmitting(true);
 24 |     try {
 25 |       const res = await forgotPasswordAction({ email });
 26 |       if (res.success) {
 27 |         setSubmitted(true);
 28 |         success("Recovery Link Dispatched", res.message);
 29 |       } else {
 30 |         error("Recovery Failed", res.message);
 31 |       }
 32 |     } catch {
 33 |       error("Network Error", "Unable to contact authentication service.");
 34 |     } finally {
 35 |       setIsSubmitting(false);
 36 |     }
 37 |   };
 38 |
 39 |   return (
 40 |     <div className="min-h-screen flex items-center justify-center bg-zinc-50/70 dark:bg-zinc-950 p-4">
 41 |       <div className="w-full max-w-md">
 42 |         <div className="text-center mb-8">
 43 |           <Link href="/" className="inline-flex items-center gap-2 mb-4">
 44 |             <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-gradient-to-tr from-
    |   blue-600 to-indigo-600 text-white shadow-md shadow-blue-500/20">
 45 |               <GraduationCap className="h-5 w-5" />
 46 |             </div>
 47 |             <span className="text-lg font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
 48 |               SMART CAMPUS
 49 |             </span>
 50 |           </Link>
 51 |           <h1 className="text-2xl font-bold tracking-tight text-zinc-900 dark:text-zinc-100">
 52 |             Password Recovery
 53 |           </h1>
 54 |           <p className="text-xs text-zinc-500 dark:text-zinc-400 mt-1">
 55 |             Enter your campus email to receive a password reset link
 56 |           </p>
 57 |         </div>
 58 |
 59 |         <Card>
 60 |           <CardContent className="p-6">
 61 |             {submitted ? (
 62 |               <div className="text-center space-y-4">
 63 |                 <div className="h-12 w-12 rounded-2xl bg-emerald-50 text-emerald-600 flex items-center ju
    |   stify-center mx-auto dark:bg-emerald-950/60 dark:text-emerald-400">
 64 |                   <Mail className="h-6 w-6" />
 65 |                 </div>
 66 |                 <div>
 67 |                   <h3 className="text-sm font-bold text-zinc-900 dark:text-zinc-100">
 68 |                     Check Your Inbox
 69 |                   </h3>
 70 |                   <p className="text-xs text-zinc-500 dark:text-zinc-400 mt-1">
 71 |                     We sent instructions to <strong className="text-zinc-800 dark:text-zinc-200">{email}<
    |   /strong>.
 72 |                   </p>
```

```
                </div>
                <div className="pt-2">
                  <Link href="/login">
                    <Button variant="outline" className="w-full text-xs">
                      Return to Sign In
                    </Button>
                  </Link>
                </div>
              </div>
            ) : (
              <form onSubmit={handleSubmit} className="space-y-4">
                <Input
                  label="Registered Campus Email"
                  type="email"
                  required
                  placeholder="alex.m@student.campus.edu"
                  value={email}
                  onChange={(e) => setEmail(e.target.value)}
                />

                <Button type="submit" disabled={isSubmitting} className="w-full gap-2">
                  <KeyRound className="h-4 w-4" />
                  {isSubmitting ? "Dispatching..." : "Send Reset Instructions"}
                </Button>

                <div className="text-center pt-2">
                  <Link
                    href="/login"
                    className="inline-flex items-center gap-1.5 text-xs text-zinc-500 hover:text-zinc-900
    dark:hover:text-zinc-200"
                  >
                    <ArrowLeft className="h-3.5 w-3.5" />
                    Back to Sign In
                  </Link>
                </div>
              </form>
            )}
          </CardContent>
        </Card>
      </div>
    </div>
  );
}
```

File #62: src/app/globals.css

70 lines

```
@import "tailwindcss";

:root {
  --background: #f8fafc;
  --foreground: #0f172a;
  --card-bg: #ffffff;
  --card-border: #e2e8f0;
}

.dark {
  --background: #090d16;
  --foreground: #f1f5f9;
  --card-bg: #0f172a;
  --card-border: #1e293b;
}

@theme inline {
  --color-background: var(--background);
  --color-foreground: var(--foreground);
  --font-sans: var(--font-geist-sans);
  --font-mono: var(--font-geist-mono);
}

body {
  background: var(--background);
  color: var(--foreground);
  font-family: var(--font-geist-sans), system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
  sans-serif;
  overflow-x: hidden;
  transition: background-color 0.2s ease, color 0.2s ease;
}

/* Custom smooth scrollbar */
```

```
::-webkit-scrollbar {
  width: 6px;
  height: 6px;
}

::-webkit-scrollbar-track {
  background: transparent;
}

::-webkit-scrollbar-thumb {
  background: rgba(148, 163, 184, 0.3);
  border-radius: 9999px;
}

::-webkit-scrollbar-thumb:hover {
  background: rgba(148, 163, 184, 0.5);
}

/* Glassmorphism helpers */
.glass-panel {
  background: rgba(255, 255, 255, 0.85);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
}

.dark .glass-panel {
  background: rgba(15, 23, 42, 0.85);
}

/* Animations */
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(4px); }
  to { opacity: 1; transform: translateY(0); }
}

.animate-fade-in {
  animation: fadeIn 0.25s cubic-bezier(0.16, 1, 0.3, 1) forwards;
}
```

File #63: src/app/layout.tsx

41 lines

```
import type { Metadata } from "next";
import { Geist, Geist_Mono } from "next/font/google";
import "./globals.css";
import { ToastProvider } from "@/components/ui/toast";
import { ThemeProvider } from "@/components/ui/theme-toggle";

const geistSans = Geist({
  variable: "--font-geist-sans",
  subsets: ["latin"],
});

const geistMono = Geist_Mono({
  variable: "--font-geist-mono",
  subsets: ["latin"],
});

export const metadata: Metadata = {
  title: "SMART CAMPUS — Real-time College Operations & Academic Management Platform",
  description:
    "Enterprise-grade unified platform for students, faculty, and campus administrators. Powered by Next.
  js, Supabase PostgreSQL, Auth, Realtime, and Storage.",
};

export default function RootLayout({
  children,
}: {
  children: React.ReactNode;
}) {
  return (
    <html
      lang="en"
      className={`${geistSans.variable} ${geistMono.variable} h-full antialiased`}
      suppressHydrationWarning
    >
      <body className="min-h-full flex flex-col bg-background text-foreground transition-colors">
        <ThemeProvider>
          <ToastProvider>{children}</ToastProvider>
        </ThemeProvider>
```

```
      </body>
    </html>
  );
}
```

File #64: src/app/login/page.tsx 185 lines

```tsx
"use client";

import * as React from "react";
import Link from "next/link";
import { useRouter } from "next/navigation";
import { GraduationCap, Briefcase, ShieldAlert, ArrowRight, Lock, Mail } from "lucide-react";
import { Button } from "@/components/ui/button";
import { Input } from "@/components/ui/input";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { loginAction } from "@/actions/auth";

export default function LoginPage() {
  const router = useRouter();
  const [selectedRole, setSelectedRole] = React.useState<"student" | "faculty" | "admin">("student");
  const [email, setEmail] = React.useState("");
  const [password, setPassword] = React.useState("");
  const [loading, setLoading] = React.useState(false);
  const [errorMessage, setErrorMessage] = React.useState<string | null>(null);
  const [fieldErrors, setFieldErrors] = React.useState<Record<string, string[]>>({});

  const handleLogin = async (e: React.FormEvent) => {
    e.preventDefault();
    setLoading(true);
    setErrorMessage(null);
    setFieldErrors({});

    try {
      const result = await loginAction({
        email,
        password,
      });

      if (result.success) {
        const dest =
          result.data?.redirectUrl ||
          (selectedRole === "admin"
            ? "/dashboard/admin"
            : selectedRole === "faculty"
            ? "/dashboard/faculty"
            : "/dashboard/student");
        router.push(dest);
      } else {
        if (result.errors) {
          setFieldErrors(result.errors);
        } else if (result.message) {
          setErrorMessage(result.message);
        }
      }
    } catch (err) {
      setErrorMessage((err as Error).message || "An unexpected error occurred.");
    } finally {
      setLoading(false);
    }
  };

  return (
    <div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-zinc-100 via-blu
  e-50/40 to-indigo-50/30 p-4 dark:from-zinc-950 dark:via-zinc-900 dark:to-zinc-950">
      <div className="w-full max-w-md">
        {/* Brand */}
        <div className="mb-6 text-center">
          <Link href="/" className="inline-flex items-center gap-2.5">
            <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-gradient-to-tr from
  -blue-600 to-indigo-600 text-white shadow-lg shadow-blue-500/25">
              <GraduationCap className="h-6 w-6" />
            </div>
            <div className="text-left">
              <span className="block text-lg font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
                SMART CAMPUS
              </span>
              <span className="block text-xs font-semibold text-blue-600 dark:text-blue-400">
                Single Sign-On Portal
```

```
              </span>
            </div>
          </Link>
        </div>

        <Card className="border-zinc-200/90 shadow-xl dark:border-zinc-800">
          <CardHeader className="text-center">
            <CardTitle>Welcome Back</CardTitle>
            <CardDescription>
              Select your role and authenticate with your institutional account
            </CardDescription>
          </CardHeader>

          <CardContent className="space-y-5">
            {/* Role Switcher */}
            <div className="grid grid-cols-3 gap-1.5 rounded-xl bg-zinc-100 p-1 dark:bg-zinc-800/80">
              <button
                type="button"
                onClick={() => setSelectedRole("student")}
                className={`flex items-center justify-center gap-1.5 rounded-lg py-2 text-xs font-semibol
d transition-all ${
                  selectedRole === "student"
                    ? "bg-white text-blue-600 shadow-sm dark:bg-zinc-700 dark:text-blue-400"
                    : "text-zinc-500 hover:text-zinc-900 dark:text-zinc-400"
                }`}
              >
                <GraduationCap className="h-3.5 w-3.5" />
                <span>Student</span>
              </button>
              <button
                type="button"
                onClick={() => setSelectedRole("faculty")}
                className={`flex items-center justify-center gap-1.5 rounded-lg py-2 text-xs font-semibol
d transition-all ${
                  selectedRole === "faculty"
                    ? "bg-white text-amber-600 shadow-sm dark:bg-zinc-700 dark:text-amber-400"
                    : "text-zinc-500 hover:text-zinc-900 dark:text-zinc-400"
                }`}
              >
                <Briefcase className="h-3.5 w-3.5" />
                <span>Faculty</span>
              </button>
              <button
                type="button"
                onClick={() => setSelectedRole("admin")}
                className={`flex items-center justify-center gap-1.5 rounded-lg py-2 text-xs font-semibol
d transition-all ${
                  selectedRole === "admin"
                    ? "bg-white text-rose-600 shadow-sm dark:bg-zinc-700 dark:text-rose-400"
                    : "text-zinc-500 hover:text-zinc-900 dark:text-zinc-400"
                }`}
              >
                <ShieldAlert className="h-3.5 w-3.5" />
                <span>Admin</span>
              </button>
            </div>

            {errorMessage && (
              <div className="rounded-xl border border-amber-500/30 bg-amber-500/10 p-3 text-xs text-ambe
r-700 dark:text-amber-300">
                {errorMessage}
              </div>
            )}

            <form onSubmit={handleLogin} className="space-y-4">
              <Input
                label="Institutional Email"
                type="email"
                placeholder={
                  selectedRole === "student"
                    ? "student@campus.edu"
                    : selectedRole === "faculty"
                    ? "professor@campus.edu"
                    : "admin@campus.edu"
                }
                value={email}
                onChange={(e) => setEmail(e.target.value)}
                error={fieldErrors.email?.[0]}
                required
              />
```

```
147 |
148 |               <Input
149 |                 label="Password"
150 |                 type="password"
151 |                 placeholder="••••••••"
152 |                 value={password}
153 |                 onChange={(e) => setPassword(e.target.value)}
154 |                 error={fieldErrors.password?.[0]}
155 |                 required
156 |               />
157 |
158 |               <div className="flex items-center justify-between text-xs">
159 |                 <label className="flex items-center gap-2 cursor-pointer text-zinc-600 dark:text-zinc-400
    |   ">
160 |                   <input type="checkbox" className="rounded border-zinc-300 text-blue-600" />
161 |                   <span>Remember me</span>
162 |                 </label>
163 |                 <Link href="/forgot-password" className="font-medium text-blue-600 hover:underline dark:t
    |   ext-blue-400">
164 |                   Forgot password?
165 |                 </Link>
166 |               </div>
167 |
168 |               <Button type="submit" className="w-full" size="lg" disabled={loading}>
169 |                 {loading ? "Authenticating..." : `Log in as ${selectedRole.toUpperCase()}`}
170 |                 <ArrowRight className="h-4 w-4 ml-1" />
171 |               </Button>
172 |             </form>
173 |
174 |             <div className="text-center text-xs text-zinc-500">
175 |               Don&apos;t have an account?{" "}
176 |               <Link href="/register" className="font-semibold text-blue-600 hover:underline dark:text-blu
    |   e-400">
177 |                 Register here
178 |               </Link>
179 |             </div>
180 |           </CardContent>
181 |         </Card>
182 |       </div>
183 |     </div>
184 |   );
185 | }
```

File #65: src/app/materials/page.tsx

305 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Input } from "@/components/ui/input";
  9 | import { BookOpen, Download, FileText, Search, Filter, Eye, Upload, Trash2, Plus } from "lucide-react";
 10 | import { Modal } from "@/components/ui/modal";
 11 | import { useToast } from "@/components/ui/toast";
 12 | import { LoadingState, ErrorState, EmptyState } from "@/components/ui/state-views";
 13 | import {
 14 |   getStudyMaterialsAction,
 15 |   createStudyMaterialAction,
 16 |   deleteStudyMaterialAction,
 17 | } from "@/actions/academics";
 18 |
 19 | export default function StudyMaterialsPage() {
 20 |   const { success, error } = useToast();
 21 |   const [materials, setMaterials] = React.useState<any[]>([]);
 22 |   const [loading, setLoading] = React.useState(true);
 23 |   const [hasError, setHasError] = React.useState(false);
 24 |   const [search, setSearch] = React.useState("");
 25 |
 26 |   // Modals
 27 |   const [previewMaterial, setPreviewMaterial] = React.useState<any | null>(null);
 28 |   const [uploadModalOpen, setUploadModalOpen] = React.useState(false);
 29 |   const [newMaterial, setNewMaterial] = React.useState({
 30 |     courseId: "00000000-0000-0000-0000-000000000001",
 31 |     title: "",
 32 |     description: "",
 33 |     fileUrl: "/docs/course-notes.pdf",
 34 |     fileSize: "4.5 MB",
```

```
  35 |     format: "PDF",
  36 |   });
  37 |   const [isSubmitting, setIsSubmitting] = React.useState(false);
  38 |
  39 |   const fetchMaterials = React.useCallback(async () => {
  40 |     setLoading(true);
  41 |     setHasError(false);
  42 |     try {
  43 |       const res = await getStudyMaterialsAction();
  44 |       if (res.success && res.data) {
  45 |         setMaterials(res.data);
  46 |       } else {
  47 |         setHasError(true);
  48 |       }
  49 |     } catch {
  50 |       setHasError(true);
  51 |     } finally {
  52 |       setLoading(false);
  53 |     }
  54 |   }, []);
  55 |
  56 |   React.useEffect(() => {
  57 |     fetchMaterials();
  58 |   }, [fetchMaterials]);
  59 |
  60 |   const handleCreateSubmit = async (e: React.FormEvent) => {
  61 |     e.preventDefault();
  62 |     setIsSubmitting(true);
  63 |     try {
  64 |       const res = await createStudyMaterialAction(newMaterial);
  65 |       if (res.success) {
  66 |         success("Study Material Published", "Notes uploaded to course repository.");
  67 |         setUploadModalOpen(false);
  68 |         setNewMaterial({
  69 |           courseId: "00000000-0000-0000-0000-000000000001",
  70 |           title: "",
  71 |           description: "",
  72 |           fileUrl: "/docs/course-notes.pdf",
  73 |           fileSize: "4.5 MB",
  74 |           format: "PDF",
  75 |         });
  76 |         fetchMaterials();
  77 |       } else {
  78 |         error("Upload Failed", res.message || "Please check inputs.");
  79 |       }
  80 |     } catch (err: any) {
  81 |       error("Error", err.message);
  82 |     } finally {
  83 |       setIsSubmitting(false);
  84 |     }
  85 |   };
  86 |
  87 |   const handleDelete = async (id: string) => {
  88 |     try {
  89 |       const res = await deleteStudyMaterialAction(id);
  90 |       if (res.success) {
  91 |         success("Material Removed");
  92 |         setMaterials((prev) => prev.filter((m) => m.id !== id));
  93 |       } else {
  94 |         error("Deletion Failed", res.message);
  95 |       }
  96 |     } catch (err: any) {
  97 |       error("Error", err.message);
  98 |     }
  99 |   };
 100 |
 101 |   const handleDownload = (m: any) => {
 102 |     success("Download Started", `Fetching ${m.title} (${m.file_size || m.size || "PDF"}) from Supabase St
     |   orage.`);
 103 |   };
 104 |
 105 |   const filtered = materials.filter((m) => {
 106 |     const title = m.title || "";
 107 |     const subject = m.course?.title || m.subject || "";
 108 |     const faculty = m.faculty?.full_name || m.faculty || "";
 109 |     return (
 110 |       title.toLowerCase().includes(search.toLowerCase()) ||
 111 |       subject.toLowerCase().includes(search.toLowerCase()) ||
 112 |       faculty.toLowerCase().includes(search.toLowerCase())
 113 |     );
```

```
  });

  return (
    <DashboardShell
      title="Course Study Materials & Digital Repository"
      description="Faculty uploaded lecture slides, lab manuals, and syllabus resources"
      actions={
        <Button size="sm" onClick={() => setUploadModalOpen(true)} className="gap-1.5 shadow-sm shadow-blue-500/20">
          <Upload className="h-4 w-4" />
          <span>Upload Material</span>
        </Button>
      }
    >
      <div className="space-y-6">
        {/* Controls */}
        <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3 rounded-2xl border border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
          <div className="relative w-full sm:w-72">
            <input
              type="text"
              placeholder="Search documents or slides..."
              value={search}
              onChange={(e) => setSearch(e.target.value)}
              className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs text-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
            />
          </div>
          <div className="text-xs text-zinc-500">
            Total Courseware: <strong className="text-zinc-800 dark:text-zinc-200">{filtered.length} Documents</strong>
          </div>
        </div>

        {/* State Views */}
        {loading ? (
          <LoadingState message="Fetching academic courseware repository..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchMaterials} />
        ) : filtered.length === 0 ? (
          <EmptyState
            title="No Materials Found"
            description="No course study materials match your search."
            actionLabel="Upload Material"
            onAction={() => setUploadModalOpen(true)}
          />
        ) : (
          <div className="grid grid-cols-1 md:grid-cols-2 gap-4">
            {filtered.map((m) => {
              const subjectName = m.course ? `${m.course.code}: ${m.course.title}` : m.subject || "Course Material";
              const facultyName = m.faculty?.full_name || m.faculty || "Faculty Instructor";
              const fileSize = m.file_size || m.size || "3.5 MB";
              const format = m.format || "PDF";
              const downloadCount = m.download_count ?? m.downloads ?? 120;

              return (
                <Card key={m.id} className="hover:border-blue-500/40 transition-all flex flex-col justify-between">
                  <CardHeader className="pb-2">
                    <div className="flex items-center justify-between mb-1.5">
                      <Badge variant="info">{format}</Badge>
                      <div className="flex items-center gap-1.5">
                        <span className="text-[11px] text-zinc-400 font-medium">{fileSize}</span>
                        <button
                          onClick={() => handleDelete(m.id)}
                          className="p-1 rounded-lg text-zinc-400 hover:text-rose-500 hover:bg-rose-50 dark:hover:bg-rose-950/40 transition-colors"
                          aria-label="Delete material"
                        >
                          <Trash2 className="h-3.5 w-3.5" />
                        </button>
                      </div>
                    </div>
                    <CardTitle className="text-sm font-bold leading-snug">{m.title}</CardTitle>
                    <CardDescription className="text-xs text-blue-600 dark:text-blue-400 font-medium">
                      {subjectName}
                    </CardDescription>
                  </CardHeader>
                  <CardContent className="pt-2">
```

```
                    <div className="flex items-center justify-between text-[11px] text-zinc-500 mb-3 bord
  er-t border-zinc-100 dark:border-zinc-800 pt-2.5">
                      <span>Uploaded by: <strong className="text-zinc-700 dark:text-zinc-300">{facultyNam
  e}</strong></span>
                      <span>{downloadCount} downloads</span>
                    </div>
                    <div className="flex items-center justify-end gap-2">
                      <Button
                        variant="ghost"
                        size="sm"
                        className="gap-1 text-xs"
                        onClick={() => setPreviewMaterial(m)}
                      >
                        <Eye className="h-3.5 w-3.5" />
                        Preview
                      </Button>
                      <Button
                        size="sm"
                        className="gap-1.5 text-xs"
                        onClick={() => handleDownload(m)}
                      >
                        <Download className="h-3.5 w-3.5" />
                        Download
                      </Button>
                    </div>
                  </CardContent>
                </Card>
              );
            })}
          </div>
        )}
      </div>

      {/* Upload Material Modal */}
      <Modal
        isOpen={uploadModalOpen}
        onClose={() => setUploadModalOpen(false)}
        title="Upload Course Study Material"
        description="Stores document in Supabase Storage study_materials bucket"
      >
        <form onSubmit={handleCreateSubmit} className="space-y-4">
          <Input
            label="Document Title"
            required
            placeholder="e.g. Unit 4: Memory Paging Architecture Slides"
            value={newMaterial.title}
            onChange={(e) => setNewMaterial({ ...newMaterial, title: e.target.value })}
          />

          <div className="space-y-1.5">
            <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Description</label>
            <textarea
              rows={3}
              placeholder="Brief summary of topics covered..."
              className="w-full rounded-xl border border-zinc-200 bg-white p-3 text-xs dark:border-zinc-8
  00 dark:bg-zinc-900"
              value={newMaterial.description}
              onChange={(e) => setNewMaterial({ ...newMaterial, description: e.target.value })}
            />
          </div>

          <div className="grid grid-cols-2 gap-3">
            <Input
              label="Format / Type"
              value={newMaterial.format}
              onChange={(e) => setNewMaterial({ ...newMaterial, format: e.target.value })}
            />
            <Input
              label="Approx Size"
              value={newMaterial.fileSize}
              onChange={(e) => setNewMaterial({ ...newMaterial, fileSize: e.target.value })}
            />
          </div>

          <div className="flex justify-end gap-2 pt-2">
            <Button type="button" variant="outline" onClick={() => setUploadModalOpen(false)}>
              Cancel
            </Button>
            <Button type="submit" disabled={isSubmitting}>
              {isSubmitting ? "Uploading..." : "Publish Document"}
```

```
264 |             </Button>
265 |           </div>
266 |         </form>
267 |       </Modal>
268 |
269 |       {/* Preview Modal */}
270 |       {previewMaterial && (
271 |         <Modal
272 |           isOpen={!!previewMaterial}
273 |           onClose={() => setPreviewMaterial(null)}
274 |           title={`Document Overview: ${previewMaterial.title}`}
275 |           description={previewMaterial.course?.title || previewMaterial.subject}
276 |         >
277 |           <div className="space-y-4">
278 |             <div className="rounded-xl bg-zinc-50 dark:bg-zinc-800/50 p-4 space-y-2 text-xs">
279 |               <div className="flex justify-between">
280 |                 <span className="text-zinc-500">Instructor:</span>
281 |                 <span className="font-semibold">{previewMaterial.faculty?.full_name || previewMaterial.fa
    |   culty}</span>
282 |               </div>
283 |               <div className="flex justify-between">
284 |                 <span className="text-zinc-500">Payload Size:</span>
285 |                 <span className="font-semibold">{previewMaterial.file_size || previewMaterial.size}</span
    |   >
286 |               </div>
287 |             </div>
288 |
289 |             <div className="flex justify-end gap-2 pt-2">
290 |               <Button variant="outline" onClick={() => setPreviewMaterial(null)}>
291 |                 Close
292 |               </Button>
293 |               <Button onClick={() => {
294 |                 handleDownload(previewMaterial);
295 |                 setPreviewMaterial(null);
296 |               }}>
297 |                 Confirm Download
298 |               </Button>
299 |             </div>
300 |           </div>
301 |         </Modal>
302 |       )}
303 |     </DashboardShell>
304 |   );
305 | }
```

File #66: src/app/notifications/page.tsx

204 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Bell, CheckCheck, Clock, Trash2, Filter } from "lucide-react";
  9 | import { useToast } from "@/components/ui/toast";
 10 | import { LoadingState, ErrorState, EmptyState } from "@/components/ui/state-views";
 11 | import {
 12 |   getNotificationsAction,
 13 |   markNotificationReadAction,
 14 |   markAllNotificationsReadAction,
 15 | } from "@/actions/operations";
 16 |
 17 | export default function NotificationsPage() {
 18 |   const { success, info, error } = useToast();
 19 |   const [items, setItems] = React.useState<any[]>([]);
 20 |   const [loading, setLoading] = React.useState(true);
 21 |   const [hasError, setHasError] = React.useState(false);
 22 |   const [filter, setFilter] = React.useState<string>("all");
 23 |
 24 |   const fetchNotifications = React.useCallback(async () => {
 25 |     setLoading(true);
 26 |     setHasError(false);
 27 |     try {
 28 |       const res = await getNotificationsAction();
 29 |       if (res.success && res.data) {
 30 |         setItems(res.data);
 31 |       } else {
 32 |         setHasError(true);
```

```
      }
    } catch {
      setHasError(true);
    } finally {
      setLoading(false);
    }
  }, []);

  React.useEffect(() => {
    fetchNotifications();
  }, [fetchNotifications]);

  const unreadCount = items.filter((n) => !n.is_read && !n.read).length;

  const markAllRead = async () => {
    try {
      const res = await markAllNotificationsReadAction();
      if (res.success) {
        setItems((prev) => prev.map((n) => ({ ...n, is_read: true, read: true })));
        success("All Marked as Read", "Your notification inbox is up to date.");
      } else {
        error("Action Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    }
  };

  const toggleRead = async (id: string) => {
    try {
      await markNotificationReadAction(id);
      setItems((prev) =>
        prev.map((n) => (n.id === id ? { ...n, is_read: true, read: true } : n))
      );
      info("Marked as read");
    } catch (err: any) {
      error("Error", err.message);
    }
  };

  const deleteItem = (id: string) => {
    setItems((prev) => prev.filter((n) => n.id !== id));
    info("Notification dismissed from local view.");
  };

  const filtered = items.filter((n) => {
    const isUnread = !n.is_read && !n.read;
    if (filter === "unread") return isUnread;
    if (filter === "academic") return n.type === "academic";
    if (filter === "schedule") return n.type === "schedule";
    return true;
  });

  return (
    <DashboardShell
      title="Notifications & Activity Inbox"
      description="Real-time alerts, assignment reviews, class changes, and campus advisories"
      actions={
        <div className="flex items-center gap-2">
          {unreadCount > 0 && (
            <Badge variant="info">{unreadCount} Unread</Badge>
          )}
          <Button
            variant="outline"
            size="sm"
            className="gap-1.5"
            onClick={markAllRead}
            disabled={unreadCount === 0}
          >
            <CheckCheck className="h-4 w-4" />
            <span>Mark All as Read</span>
          </Button>
        </div>
      }
    >
      <div className="space-y-4">
        {/* Filter Pills */}
        <div className="flex items-center gap-1.5 border-b border-zinc-200/80 pb-3 dark:border-zinc-800/80">
          {[
```

```
112 |             { id: "all", label: "All Alerts" },
113 |             { id: "unread", label: "Unread" },
114 |             { id: "academic", label: "Academics" },
115 |             { id: "schedule", label: "Schedule Shifts" },
116 |           ].map((tab) => (
117 |             <button
118 |               key={tab.id}
119 |               onClick={() => setFilter(tab.id)}
120 |               className={`rounded-xl px-3 py-1.5 text-xs font-semibold transition-all ${
121 |                 filter === tab.id
122 |                   ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
123 |                   : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
124 |               }`}
125 |             >
126 |               {tab.label}
127 |             </button>
128 |           ))}
129 |         </div>
130 |
131 |         {/* State Views */}
132 |         {loading ? (
133 |           <LoadingState message="Fetching realtime notifications from PostgreSQL..." />
134 |         ) : hasError ? (
135 |           <ErrorState onRetry={fetchNotifications} />
136 |         ) : filtered.length === 0 ? (
137 |           <EmptyState
138 |             title="Inbox Zero"
139 |             description="No notifications in this view. You are all caught up!"
140 |           />
141 |         ) : (
142 |           <div className="space-y-3">
143 |             {filtered.map((n) => {
144 |               const isUnread = !n.is_read && !n.read;
145 |               const dateText = n.created_at ? new Date(n.created_at).toLocaleTimeString([], { hour: '2-di
    |   git', minute: '2-digit' }) : n.time || "Recent";
146 |
147 |               return (
148 |                 <Card
149 |                   key={n.id}
150 |                   className={`transition-all ${
151 |                     isUnread
152 |                       ? "border-l-4 border-l-blue-500 bg-blue-50/25 dark:bg-blue-950/20"
153 |                       : "opacity-75"
154 |                   }`}
155 |                 >
156 |                   <CardContent className="p-4 flex items-start justify-between gap-4">
157 |                     <div className="space-y-1 flex-1 min-w-0">
158 |                       <div className="flex items-center gap-2">
159 |                         <h4 className="text-xs sm:text-sm font-semibold truncate text-zinc-900 dark:text-
    |   zinc-100">
160 |                           {n.title}
161 |                         </h4>
162 |                         {isUnread && <Badge variant="info">New</Badge>}
163 |                         <Badge variant="outline" className="text-[10px] capitalize">
164 |                           {n.type}
165 |                         </Badge>
166 |                       </div>
167 |                       <p className="text-xs text-zinc-600 dark:text-zinc-400 leading-relaxed">
168 |                         {n.message}
169 |                       </p>
170 |                       <div className="flex items-center gap-1.5 text-[11px] text-zinc-400 pt-1">
171 |                         <Clock className="h-3 w-3" />
172 |                         <span>{dateText}</span>
173 |                       </div>
174 |                     </div>
175 |
176 |                     <div className="flex items-center gap-1 shrink-0">
177 |                       {isUnread && (
178 |                         <Button
179 |                           variant="ghost"
180 |                           size="sm"
181 |                           className="text-xs text-blue-600 dark:text-blue-400"
182 |                           onClick={() => toggleRead(n.id)}
183 |                         >
184 |                           Mark Read
185 |                         </Button>
186 |                       )}
187 |                       <button
188 |                         onClick={() => deleteItem(n.id)}
189 |                         className="p-1.5 rounded-lg text-zinc-400 hover:text-rose-500 hover:bg-rose-50 da
```

```
    |   rk:hover:bg-rose-950/40 transition-colors"
190 |                         aria-label="Delete notification"
191 |                       >
192 |                         <Trash2 className="h-4 w-4" />
193 |                       </button>
194 |                     </div>
195 |                   </CardContent>
196 |                 </Card>
197 |               );
198 |             })}
199 |           </div>
200 |         )}
201 |       </div>
202 |     </DashboardShell>
203 |   );
204 | }
```

File #67: src/app/page.tsx 367 lines

```
  1 | import Link from "next/link";
  2 | import {
  3 |   GraduationCap,
  4 |   Briefcase,
  5 |   ShieldAlert,
  6 |   Calendar,
  7 |   Clock,
  8 |   CheckCircle2,
  9 |   FileText,
 10 |   BookOpen,
 11 |   Users,
 12 |   Building2,
 13 |   PhoneCall,
 14 |   Bell,
 15 |   MessageSquare,
 16 |   Search,
 17 |   Settings,
 18 |   ArrowRight,
 19 |   Database,
 20 |   Radio,
 21 |   Server,
 22 |   Sparkles,
 23 | } from "lucide-react";
 24 | import { Navbar } from "@/components/layout/navbar";
 25 | import { Footer } from "@/components/layout/footer";
 26 | import { Button } from "@/components/ui/button";
 27 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
 28 | import { Badge } from "@/components/ui/badge";
 29 |
 30 | export default function Home() {
 31 |   const modules = [
 32 |     {
 33 |       title: "Student Dashboard",
 34 |       desc: "Course attendance, timetable schedule, assignment submissions, and progress metrics.",
 35 |       icon: GraduationCap,
 36 |       href: "/dashboard/student",
 37 |       badge: "Student",
 38 |       color: "text-blue-600 bg-blue-500/10",
 39 |     },
 40 |     {
 41 |       title: "Faculty Dashboard",
 42 |       desc: "Attendance markers, grading workflows, curriculum tracking, and announcements.",
 43 |       icon: Briefcase,
 44 |       href: "/dashboard/faculty",
 45 |       badge: "Faculty",
 46 |       color: "text-amber-600 bg-amber-500/10",
 47 |     },
 48 |     {
 49 |       title: "Admin Dashboard",
 50 |       desc: "Operations command center, user roles, emergency broadcasts, and facility control.",
 51 |       icon: ShieldAlert,
 52 |       href: "/dashboard/admin",
 53 |       badge: "Admin",
 54 |       color: "text-rose-600 bg-rose-500/10",
 55 |     },
 56 |     {
 57 |       title: "Announcements",
 58 |       desc: "Realtime institutional circulars, academic notices, and department updates.",
 59 |       icon: Bell,
 60 |       href: "/announcements",
```

```
      badge: "Realtime",
      color: "text-indigo-600 bg-indigo-500/10",
    },
    {
      title: "Campus Events",
      desc: "Conferences, hackathons, guest seminars, and cultural festivals with registration.",
      icon: Calendar,
      href: "/events",
      color: "text-purple-600 bg-purple-500/10",
    },
    {
      title: "Class Timetable",
      desc: "Weekly schedule by department, classroom numbers, and faculty instructor slots.",
      icon: Clock,
      href: "/timetable",
      color: "text-cyan-600 bg-cyan-500/10",
    },
    {
      title: "Attendance Tracking",
      desc: "Live subject-wise attendance statistics with automated 75% minimum alert checks.",
      icon: CheckCircle2,
      href: "/attendance",
      color: "text-emerald-600 bg-emerald-500/10",
    },
    {
      title: "Assignments",
      desc: "Faculty assignment briefs, deadline countdowns, and student submission portal.",
      icon: FileText,
      href: "/assignments",
      color: "text-amber-600 bg-amber-500/10",
    },
    {
      title: "Study Materials",
      desc: "Lecture notes, syllabus references, and PDFs backed by Supabase Storage.",
      icon: BookOpen,
      href: "/materials",
      color: "text-sky-600 bg-sky-500/10",
    },
    {
      title: "Faculty Directory",
      desc: "Professor profiles, research specializations, office hours, and consultation links.",
      icon: Users,
      href: "/faculty",
      color: "text-teal-600 bg-teal-500/10",
    },
    {
      title: "Campus Facilities",
      desc: "Digital reservation and live availability for libraries, labs, and auditoriums.",
      icon: Building2,
      href: "/facilities",
      color: "text-orange-600 bg-orange-500/10",
    },
    {
      title: "Emergency Contacts",
      desc: "Instant 24/7 one-touch access to campus police, ambulance, and safety hotlines.",
      icon: PhoneCall,
      href: "/emergency",
      badge: "24/7",
      color: "text-rose-600 bg-rose-500/10",
    },
    {
      title: "Notifications",
      desc: "Personalized alert inbox for exam results, class shifts, and submission marks.",
      icon: Bell,
      href: "/notifications",
      color: "text-blue-600 bg-blue-500/10",
    },
    {
      title: "Feedback & Grievances",
      desc: "Direct grievance redressal mechanism for academic, hostel, or facility concerns.",
      icon: MessageSquare,
      href: "/feedback",
      color: "text-violet-600 bg-violet-500/10",
    },
    {
      title: "Global Search",
      desc: "Instant search across all notices, faculty, courses, and campus resources.",
      icon: Search,
      href: "/search",
      color: "text-zinc-600 bg-zinc-500/10",
```

```
141 |     },
142 |     {
143 |       title: "Account Settings",
144 |       desc: "Personal preferences, notification channels, and security settings.",
145 |       icon: Settings,
146 |       href: "/settings",
147 |       color: "text-slate-600 bg-slate-500/10",
148 |     },
149 |   ];
150 |
151 |   return (
152 |     <div className="min-h-screen flex flex-col bg-zinc-50/70 dark:bg-zinc-950 text-zinc-900 dark:text-zin
    |   c-100">
153 |       <Navbar />
154 |
155 |       {/* Emergency / High-Priority Banner */}
156 |       <div className="bg-gradient-to-r from-blue-600 via-indigo-600 to-violet-600 text-white px-4 py-2.5
    |   text-xs text-center font-medium shadow-inner flex items-center justify-center gap-2">
157 |         <Sparkles className="h-4 w-4 shrink-0 text-amber-300" />
158 |         <span>
159 |           SMART CAMPUS Platform Online • Realtime connection active • PostgreSQL Database synced
160 |         </span>
161 |         <Link href="/announcements" className="underline font-bold hover:text-white/80 ml-1">
162 |           View Notices →
163 |         </Link>
164 |       </div>
165 |
166 |       {/* Hero Section */}
167 |       <section className="relative overflow-hidden py-16 sm:py-24 border-b border-zinc-200/80 dark:border
    |   -zinc-800/80">
168 |         <div className="absolute inset-0 -z-10 bg-[radial-gradient(45rem_50rem_at_top,theme(colors.blue.1
    |   00),transparent)] dark:bg-[radial-gradient(45rem_50rem_at_top,theme(colors.blue.950),transparent)] opacit
    |   y-60" />
169 |
170 |         <div className="mx-auto max-w-7xl px-4 sm:px-6 lg:px-8 text-center">
171 |           <div className="inline-flex items-center gap-2 rounded-full border border-blue-500/30 bg-blue-5
    |   0/80 px-3.5 py-1 text-xs font-semibold text-blue-700 backdrop-blur-md dark:bg-blue-950/60 dark:text-blue-
    |   300 mb-6">
172 |             <Radio className="h-3.5 w-3.5 text-blue-500 animate-pulse" />
173 |             <span>Next-Generation College Operating System</span>
174 |           </div>
175 |
176 |           <h1 className="text-4xl sm:text-6xl font-extrabold tracking-tight text-zinc-900 dark:text-zinc-
    |   50 max-w-4xl mx-auto leading-tight">
177 |             Unified Digital Infrastructure for <br className="hidden sm:inline" />
178 |             <span className="bg-gradient-to-r from-blue-600 via-indigo-600 to-violet-600 bg-clip-text tex
    |   t-transparent">
179 |               Smart Higher Education
180 |             </span>
181 |           </h1>
182 |
183 |           <p className="mt-5 text-base sm:text-lg text-zinc-600 dark:text-zinc-400 max-w-2xl mx-auto">
184 |             A cohesive full-stack web application connecting Students, Faculty, and Administrators. Power
    |   ed by Next.js, Supabase PostgreSQL, Realtime sync, and enterprise role-based security.
185 |           </p>
186 |
187 |           <div className="mt-8 flex flex-wrap items-center justify-center gap-3 sm:gap-4">
188 |             <Link href="/login">
189 |               <Button size="lg" className="shadow-lg shadow-blue-500/25">
190 |                 <span>Access Campus Portal</span>
191 |                 <ArrowRight className="h-4 w-4 ml-1" />
192 |               </Button>
193 |             </Link>
194 |             <Link href="/dashboard">
195 |               <Button variant="outline" size="lg">
196 |                 <span>Explore Role Views</span>
197 |               </Button>
198 |             </Link>
199 |             <Link href="/api/health" target="_blank">
200 |               <Button variant="ghost" size="lg" className="text-xs">
201 |                 <Database className="h-4 w-4 mr-1 text-emerald-500" />
202 |                 <span>Backend Health Status</span>
203 |               </Button>
204 |             </Link>
205 |           </div>
206 |
207 |           {/* Architecture Badges */}
208 |           <div className="mt-12 flex flex-wrap items-center justify-center gap-2 pt-6 border-t border-zin
    |   c-200/60 dark:border-zinc-800/60 max-w-3xl mx-auto">
209 |             <Badge variant="outline" className="text-xs py-1 px-3">
```

```
210 |               Next.js 16 (App Router)
211 |             </Badge>
212 |             <Badge variant="outline" className="text-xs py-1 px-3">
213 |               TypeScript
214 |             </Badge>
215 |             <Badge variant="outline" className="text-xs py-1 px-3">
216 |               Tailwind CSS
217 |             </Badge>
218 |             <Badge variant="outline" className="text-xs py-1 px-3">
219 |               Supabase PostgreSQL
220 |             </Badge>
221 |             <Badge variant="outline" className="text-xs py-1 px-3">
222 |               Supabase Auth & RLS
223 |             </Badge>
224 |             <Badge variant="outline" className="text-xs py-1 px-3">
225 |               Supabase Realtime
226 |             </Badge>
227 |             <Badge variant="outline" className="text-xs py-1 px-3">
228 |               Zod Validation
229 |             </Badge>
230 |             <Badge variant="outline" className="text-xs py-1 px-3">
231 |               Recharts Analytics
232 |             </Badge>
233 |           </div>
234 |         </div>
235 |       </section>
236 |
237 |       {/* Role Selector Grid */}
238 |       <section className="py-12 bg-white/40 dark:bg-zinc-900/40 border-b border-zinc-200/80 dark:border-z
    |   inc-800/80">
239 |         <div className="mx-auto max-w-7xl px-4 sm:px-6 lg:px-8">
240 |           <div className="text-center mb-8">
241 |             <h2 className="text-2xl font-bold tracking-tight">Dedicated Role Experiences</h2>
242 |             <p className="text-xs text-zinc-500 mt-1">
243 |               Select your persona to enter customized workflows and metrics
244 |             </p>
245 |           </div>
246 |
247 |           <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
248 |             {/* Student */}
249 |             <Card className="hover:border-blue-500 transition-all border-zinc-200 dark:border-zinc-800">
250 |               <CardContent className="p-6 space-y-4">
251 |                 <div className="flex items-center justify-between">
252 |                   <div className="p-3 bg-blue-500/10 text-blue-600 rounded-2xl">
253 |                     <GraduationCap className="h-6 w-6" />
254 |                   </div>
255 |                   <Badge variant="info">Student Role</Badge>
256 |                 </div>
257 |                 <div>
258 |                   <h3 className="text-lg font-bold">Student Workspace</h3>
259 |                   <p className="text-xs text-zinc-500 mt-1">
260 |                     Personalized timetable, subject-wise attendance charts, assignment submissions, and s
    |   tudy materials repository.
261 |                   </p>
262 |                 </div>
263 |                 <Link href="/dashboard/student" className="block pt-2">
264 |                   <Button className="w-full" size="sm">
265 |                     Open Student Portal
266 |                   </Button>
267 |                 </Link>
268 |               </CardContent>
269 |             </Card>
270 |
271 |             {/* Faculty */}
272 |             <Card className="hover:border-amber-500 transition-all border-zinc-200 dark:border-zinc-800">
273 |               <CardContent className="p-6 space-y-4">
274 |                 <div className="flex items-center justify-between">
275 |                   <div className="p-3 bg-amber-500/10 text-amber-600 rounded-2xl">
276 |                     <Briefcase className="h-6 w-6" />
277 |                   </div>
278 |                   <Badge variant="warning">Faculty Role</Badge>
279 |                 </div>
280 |                 <div>
281 |                   <h3 className="text-lg font-bold">Faculty Console</h3>
282 |                   <p className="text-xs text-zinc-500 mt-1">
283 |                     Attendance recording, assignment rubric builder, submission review queue, and lecture
    |    timetable management.
284 |                   </p>
285 |                 </div>
286 |                 <Link href="/dashboard/faculty" className="block pt-2">
```

```
                  <Button variant="outline" className="w-full border-amber-300 text-amber-700 hover:bg-amber-50 dark:border-amber-800 dark:text-amber-300" size="sm">
                    Open Faculty Console
                  </Button>
                </Link>
              </CardContent>
            </Card>

            {/* Admin */}
            <Card className="hover:border-rose-500 transition-all border-zinc-200 dark:border-zinc-800">
              <CardContent className="p-6 space-y-4">
                <div className="flex items-center justify-between">
                  <div className="p-3 bg-rose-500/10 text-rose-600 rounded-2xl">
                    <ShieldAlert className="h-6 w-6" />
                  </div>
                  <Badge variant="danger">Admin Role</Badge>
                </div>
                <div>
                  <h3 className="text-lg font-bold">Admin Command Center</h3>
                  <p className="text-xs text-zinc-500 mt-1">
                    Campus facility bookings, grievance redressal, emergency broadcasts, and Supabase infrastructure supervision.
                  </p>
                </div>
                <Link href="/dashboard/admin" className="block pt-2">
                  <Button variant="outline" className="w-full border-rose-300 text-rose-700 hover:bg-rose-50 dark:border-rose-800 dark:text-rose-300" size="sm">
                    Open Admin Command
                  </Button>
                </Link>
              </CardContent>
            </Card>
          </div>
        </div>
      </section>

      {/* Complete Modules Showcase */}
      <section className="py-14">
        <div className="mx-auto max-w-7xl px-4 sm:px-6 lg:px-8">
          <div className="flex flex-col sm:flex-row sm:items-end justify-between mb-8 gap-4">
            <div>
              <h2 className="text-2xl font-bold tracking-tight">Enterprise Application Modules</h2>
              <p className="text-xs text-zinc-500 mt-1">
                Explore all 18 core services architected into SMART CAMPUS
              </p>
            </div>
            <Link href="/search">
              <Button variant="outline" size="sm">
                <Search className="h-3.5 w-3.5" />
                <span>Search Modules</span>
              </Button>
            </Link>
          </div>

          <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">
            {modules.map((m) => {
              const Icon = m.icon;
              return (
                <Link key={m.title} href={m.href} className="group">
                  <Card className="h-full hover:border-blue-500/40 hover:-translate-y-0.5 transition-all p-5">
                    <div className="flex items-center justify-between mb-3">
                      <div className={`p-2.5 rounded-xl ${m.color}`}>
                        <Icon className="h-5 w-5" />
                      </div>
                      {m.badge && <Badge variant="outline">{m.badge}</Badge>}
                    </div>
                    <h3 className="font-semibold text-sm group-hover:text-blue-600 dark:group-hover:text-blue-400 transition-colors">
                      {m.title}
                    </h3>
                    <p className="text-xs text-zinc-500 mt-1 leading-relaxed">
                      {m.desc}
                    </p>
                  </Card>
                </Link>
              );
            })}
          </div>
        </div>
```

```
      </section>

      <Footer />
    </div>
  );
}
```

File #68: src/app/profile/page.tsx 306 lines

```
"use client";

import * as React from "react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { Input } from "@/components/ui/input";
import { useToast } from "@/components/ui/toast";
import { LoadingState, ErrorState } from "@/components/ui/state-views";
import { User, Mail, GraduationCap, Building2, Phone, Calendar, ShieldCheck, Save, Camera, Upload } from
  "lucide-react";
import {
  getUserProfileDetailsAction,
  updateBaseProfileAction,
  updateStudentProfileAction,
} from "@/actions/profiles";
import { uploadFileSecure, FILE_LIMITS } from "@/lib/supabase/storage";

export default function ProfilePage() {
  const { success, error } = useToast();
  const [profileData, setProfileData] = React.useState<any | null>(null);
  const [loading, setLoading] = React.useState(true);
  const [hasError, setHasError] = React.useState(false);

  // Form Fields
  const [fullName, setFullName] = React.useState("");
  const [phone, setPhone] = React.useState("");
  const [department, setDepartment] = React.useState("");
  const [avatarUrl, setAvatarUrl] = React.useState("");
  const [enrollmentNo, setEnrollmentNo] = React.useState("");
  const [currentSemester, setCurrentSemester] = React.useState(1);
  const [isSaving, setIsSaving] = React.useState(false);
  const [isUploadingAvatar, setIsUploadingAvatar] = React.useState(false);

  const fetchProfile = React.useCallback(async () => {
    setLoading(true);
    setHasError(false);
    try {
      const res = await getUserProfileDetailsAction();
      if (res.success && res.data) {
        setProfileData(res.data);
        setFullName(res.data.full_name || "");
        setPhone(res.data.phone || "");
        setDepartment(res.data.department || "");
        setAvatarUrl(res.data.avatar_url || "");
        if (res.data.student_profile) {
          setEnrollmentNo(res.data.student_profile.enrollment_no || "");
          setCurrentSemester(res.data.student_profile.current_semester || 1);
        }
      } else {
        setHasError(true);
      }
    } catch {
      setHasError(true);
    } finally {
      setLoading(false);
    }
  }, []);

  React.useEffect(() => {
    fetchProfile();
  }, [fetchProfile]);

  const handleAvatarFileChange = async (e: React.ChangeEvent<HTMLInputElement>) => {
    const file = e.target.files?.[0];
    if (!file) return;

    setIsUploadingAvatar(true);
    try {
```

```
      const uploadRes = await uploadFileSecure(
        "avatars",
        `${profileData?.id || "user"}-avatar.png`,
        file,
        FILE_LIMITS.AVATAR
      );

      if (uploadRes.url) {
        setAvatarUrl(uploadRes.url);
        await updateBaseProfileAction({
          fullName,
          phone,
          department,
          avatarUrl: uploadRes.url,
        });
        success("Avatar Updated", "New profile picture saved to Supabase avatars bucket.");
      } else {
        error("Upload Failed", uploadRes.error || "File could not be uploaded.");
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsUploadingAvatar(false);
    }
  };

  const handleSaveProfile = async (e: React.FormEvent) => {
    e.preventDefault();
    setIsSaving(true);
    try {
      const res = await updateBaseProfileAction({
        fullName,
        phone,
        department,
        avatarUrl,
      });

      if (profileData?.role === "student" && enrollmentNo) {
        await updateStudentProfileAction({
          enrollmentNo,
          currentSemester,
          admissionYear: profileData?.student_profile?.admission_year || 2024,
          academicStanding: profileData?.student_profile?.academic_standing || "Good",
        });
      }

      if (res.success) {
        success("Profile Saved", "Your information has been committed to Supabase profiles.");
        fetchProfile();
      } else {
        error("Update Failed", res.message);
      }
    } catch (err: any) {
      error("Error", err.message);
    } finally {
      setIsSaving(false);
    }
  };

  if (loading) {
    return (
      <DashboardShell title="User Profile">
        <LoadingState message="Loading authenticated profile details..." />
      </DashboardShell>
    );
  }

  if (hasError || !profileData) {
    return (
      <DashboardShell title="User Profile">
        <ErrorState onRetry={fetchProfile} />
      </DashboardShell>
    );
  }

  return (
    <DashboardShell
      title="User Identity & Academic Profile"
      description="Personal information, academic enrollment details, and security status"
    >
```

```
      <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
        {/* Profile Summary Card with Avatar Upload */}
        <Card className="text-center p-6 flex flex-col items-center">
          <div className="relative group mb-4">
            <div className="flex h-24 w-24 items-center justify-center rounded-3xl bg-gradient-to-tr from
-blue-600 to-indigo-600 text-white shadow-xl shadow-blue-500/25 overflow-hidden">
              {avatarUrl ? (
                // eslint-disable-next-line @next/next/no-img-element
                <img
                  src={avatarUrl}
                  alt={fullName}
                  className="h-full w-full object-cover"
                />
              ) : (
                <User className="h-12 w-12" />
              )}
            </div>

            {/* Avatar Change Overlay */}
            <label
              htmlFor="avatar-upload"
              className="absolute inset-0 flex flex-col items-center justify-center rounded-3xl bg-black/
60 text-white opacity-0 group-hover:opacity-100 transition-opacity cursor-pointer text-[10px] font-semibo
ld"
            >
              <Camera className="h-5 w-5 mb-1" />
              <span>{isUploadingAvatar ? "Saving..." : "Change Photo"}</span>
            </label>
            <input
              id="avatar-upload"
              type="file"
              accept="image/png,image/jpeg,image/webp"
              className="hidden"
              onChange={handleAvatarFileChange}
              disabled={isUploadingAvatar}
            />
          </div>

          <h3 className="text-lg font-bold">{fullName}</h3>
          <p className="text-xs text-zinc-500 mt-0.5">{profileData.email}</p>
          <div className="mt-3 flex items-center gap-1.5">
            <Badge
              variant={
                profileData.role === "admin"
                  ? "danger"
                  : profileData.role === "faculty"
                  ? "warning"
                  : "info"
              }
              className="capitalize"
            >
              {profileData.role}
            </Badge>
            {profileData.student_profile && (
              <Badge variant="outline">
                Semester {profileData.student_profile.current_semester}
              </Badge>
            )}
          </div>

          <div className="w-full mt-6 space-y-2 border-t border-zinc-200/80 pt-4 text-left text-xs text-z
inc-600 dark:border-zinc-800 dark:text-zinc-400">
            {profileData.student_profile && (
              <>
                <div className="flex justify-between py-1 border-b border-zinc-100 dark:border-zinc-800">
                  <span className="text-zinc-400">Roll Number:</span>
                  <span className="font-semibold text-zinc-800 dark:text-zinc-200">
                    {profileData.student_profile.enrollment_no}
                  </span>
                </div>
                <div className="flex justify-between py-1 border-b border-zinc-100 dark:border-zinc-800">
                  <span className="text-zinc-400">CGPA:</span>
                  <span className="font-semibold text-emerald-600">
                    {profileData.student_profile.cgpa || "3.82"} / 4.0
                  </span>
                </div>
              </>
            )}
            <div className="flex justify-between py-1 border-b border-zinc-100 dark:border-zinc-800">
              <span className="text-zinc-400">Department:</span>
```

```
              <span className="font-semibold text-zinc-800 dark:text-zinc-200">
                {profileData.department || "General Academics"}
              </span>
            </div>
            <div className="flex justify-between py-1">
              <span className="text-zinc-400">Auth Status:</span>
              <span className="font-semibold text-emerald-600 flex items-center gap-1">
                <ShieldCheck className="h-3.5 w-3.5" />
                Verified
              </span>
            </div>
          </div>
        </Card>

        {/* Edit Info Form */}
        <div className="lg:col-span-2">
          <Card>
            <CardHeader>
              <CardTitle>Profile Details</CardTitle>
              <CardDescription>
                Synchronized with Supabase PostgreSQL `profiles` and role tables
              </CardDescription>
            </CardHeader>
            <CardContent>
              <form onSubmit={handleSaveProfile} className="space-y-4">
                <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
                  <Input
                    label="Full Name"
                    required
                    value={fullName}
                    onChange={(e) => setFullName(e.target.value)}
                  />
                  <Input
                    label="Email Address"
                    value={profileData.email}
                    disabled
                  />
                  <Input
                    label="Contact Phone"
                    value={phone}
                    placeholder="+1 (555) 000-0000"
                    onChange={(e) => setPhone(e.target.value)}
                  />
                  <Input
                    label="Department"
                    value={department}
                    placeholder="Computer Science & Engineering"
                    onChange={(e) => setDepartment(e.target.value)}
                  />
                  {profileData.role === "student" && (
                    <>
                      <Input
                        label="Enrollment / Roll No."
                        value={enrollmentNo}
                        placeholder="24CS001"
                        onChange={(e) => setEnrollmentNo(e.target.value)}
                      />
                      <Input
                        label="Current Semester"
                        type="number"
                        value={currentSemester}
                        onChange={(e) => setCurrentSemester(Number(e.target.value))}
                      />
                    </>
                  )}
                </div>

                <div className="flex justify-end pt-3">
                  <Button type="submit" disabled={isSaving} className="gap-2 text-xs">
                    <Save className="h-4 w-4" />
                    {isSaving ? "Saving to Database..." : "Save Profile Changes"}
                  </Button>
                </div>
              </form>
            </CardContent>
          </Card>
        </div>
      </div>
    </DashboardShell>
  );
```

```
306 | }
```

File #69: src/app/register/page.tsx 237 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import Link from "next/link";
  5 | import { useRouter } from "next/navigation";
  6 | import { GraduationCap, Briefcase, ShieldAlert, ArrowRight, CheckCircle2 } from "lucide-react";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Input } from "@/components/ui/input";
  9 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
 10 | import { registerAction } from "@/actions/auth";
 11 |
 12 | export default function RegisterPage() {
 13 |   const router = useRouter();
 14 |   const [role, setRole] = React.useState<"student" | "faculty" | "admin">("student");
 15 |   const [fullName, setFullName] = React.useState("");
 16 |   const [email, setEmail] = React.useState("");
 17 |   const [password, setPassword] = React.useState("");
 18 |   const [confirmPassword, setConfirmPassword] = React.useState("");
 19 |   const [department, setDepartment] = React.useState("Computer Science & Engineering");
 20 |   const [enrollmentNo, setEnrollmentNo] = React.useState("");
 21 |   const [loading, setLoading] = React.useState(false);
 22 |   const [errorMessage, setErrorMessage] = React.useState<string | null>(null);
 23 |   const [successMessage, setSuccessMessage] = React.useState<string | null>(null);
 24 |   const [fieldErrors, setFieldErrors] = React.useState<Record<string, string[]>>({});
 25 |
 26 |   const handleRegister = async (e: React.FormEvent) => {
 27 |     e.preventDefault();
 28 |     setLoading(true);
 29 |     setErrorMessage(null);
 30 |     setSuccessMessage(null);
 31 |     setFieldErrors({});
 32 |
 33 |     try {
 34 |       const result = await registerAction({
 35 |         fullName,
 36 |         email,
 37 |         password,
 38 |         confirmPassword,
 39 |         role,
 40 |         department,
 41 |         enrollmentNo: enrollmentNo || undefined,
 42 |       });
 43 |
 44 |       if (result.success) {
 45 |         setSuccessMessage(result.message || "Registration completed successfully!");
 46 |         setTimeout(() => {
 47 |           router.push("/login");
 48 |         }, 1500);
 49 |       } else {
 50 |         if (result.errors) {
 51 |           setFieldErrors(result.errors);
 52 |         } else if (result.message) {
 53 |           setErrorMessage(result.message);
 54 |         }
 55 |       }
 56 |     } catch (err) {
 57 |       setErrorMessage((err as Error).message || "An unexpected error occurred.");
 58 |     } finally {
 59 |       setLoading(false);
 60 |     }
 61 |   };
 62 |
 63 |   return (
 64 |     <div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-zinc-100 via-blu
    |   e-50/40 to-indigo-50/30 p-4 dark:from-zinc-950 dark:via-zinc-900 dark:to-zinc-950">
 65 |       <div className="w-full max-w-lg">
 66 |         <div className="mb-6 text-center">
 67 |           <Link href="/" className="inline-flex items-center gap-2.5">
 68 |             <div className="flex h-12 w-12 items-center justify-center rounded-2xl bg-gradient-to-tr from
    |   -blue-600 to-indigo-600 text-white shadow-lg shadow-blue-500/25">
 69 |               <GraduationCap className="h-6 w-6" />
 70 |             </div>
 71 |             <div className="text-left">
 72 |               <span className="block text-lg font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
 73 |                 SMART CAMPUS
```

```
              </span>
              <span className="block text-xs font-semibold text-blue-600 dark:text-blue-400">
                Campus Identity Registration
              </span>
            </div>
          </Link>
        </div>

        <Card className="border-zinc-200/90 shadow-xl dark:border-zinc-800">
          <CardHeader className="text-center">
            <CardTitle>Create an Account</CardTitle>
            <CardDescription>
              Register your official student, faculty, or administrative credentials
            </CardDescription>
          </CardHeader>

          <CardContent className="space-y-5">
            {/* Role Select */}
            <div className="space-y-1.5">
              <label className="block text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                Account Type
              </label>
              <div className="grid grid-cols-3 gap-1.5 rounded-xl bg-zinc-100 p-1 dark:bg-zinc-800/80">
                <button
                  type="button"
                  onClick={() => setRole("student")}
                  className={`flex items-center justify-center gap-1.5 rounded-lg py-2 text-xs font-semib
old transition-all ${
                    role === "student"
                      ? "bg-white text-blue-600 shadow-sm dark:bg-zinc-700 dark:text-blue-400"
                      : "text-zinc-500 hover:text-zinc-900 dark:text-zinc-400"
                  }`}
                >
                  <GraduationCap className="h-3.5 w-3.5" />
                  <span>Student</span>
                </button>
                <button
                  type="button"
                  onClick={() => setRole("faculty")}
                  className={`flex items-center justify-center gap-1.5 rounded-lg py-2 text-xs font-semib
old transition-all ${
                    role === "faculty"
                      ? "bg-white text-amber-600 shadow-sm dark:bg-zinc-700 dark:text-amber-400"
                      : "text-zinc-500 hover:text-zinc-900 dark:text-zinc-400"
                  }`}
                >
                  <Briefcase className="h-3.5 w-3.5" />
                  <span>Faculty</span>
                </button>
                <button
                  type="button"
                  onClick={() => setRole("admin")}
                  className={`flex items-center justify-center gap-1.5 rounded-lg py-2 text-xs font-semib
old transition-all ${
                    role === "admin"
                      ? "bg-white text-rose-600 shadow-sm dark:bg-zinc-700 dark:text-rose-400"
                      : "text-zinc-500 hover:text-zinc-900 dark:text-zinc-400"
                  }`}
                >
                  <ShieldAlert className="h-3.5 w-3.5" />
                  <span>Admin</span>
                </button>
              </div>
            </div>

            {successMessage && (
              <div className="flex items-center gap-2 rounded-xl border border-emerald-500/30 bg-emerald-
500/10 p-3 text-xs text-emerald-700 dark:text-emerald-300">
                <CheckCircle2 className="h-4 w-4 shrink-0 text-emerald-600" />
                <span>{successMessage}</span>
              </div>
            )}

            {errorMessage && (
              <div className="rounded-xl border border-amber-500/30 bg-amber-500/10 p-3 text-xs text-ambe
r-700 dark:text-amber-300">
                {errorMessage}
              </div>
            )}

```

```
            <form onSubmit={handleRegister} className="space-y-3.5">
              <Input
                label="Full Legal Name"
                type="text"
                placeholder="e.g. Alex Johnson"
                value={fullName}
                onChange={(e) => setFullName(e.target.value)}
                error={fieldErrors.fullName?.[0]}
                required
              />

              <Input
                label="Institutional Email"
                type="email"
                placeholder="alex.johnson@campus.edu"
                value={email}
                onChange={(e) => setEmail(e.target.value)}
                error={fieldErrors.email?.[0]}
                required
              />

              <div className="grid grid-cols-1 sm:grid-cols-2 gap-3">
                <div className="space-y-1.5">
                  <label className="block text-xs font-semibold text-zinc-700 dark:text-zinc-300">
                    Department
                  </label>
                  <select
                    value={department}
                    onChange={(e) => setDepartment(e.target.value)}
                    className="flex h-11 w-full rounded-xl border border-zinc-200 bg-white/70 px-3.5 py-2
    text-xs text-zinc-900 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-100"
                  >
                    <option value="Computer Science & Engineering">Computer Science & Engineering</option
   >
                    <option value="Information Technology">Information Technology</option>
                    <option value="Electrical Engineering">Electrical Engineering</option>
                    <option value="Mechanical Engineering">Mechanical Engineering</option>
                    <option value="Business Administration">Business Administration</option>
                  </select>
                </div>

                <Input
                  label={role === "student" ? "Roll / Enrollment No." : "Employee ID"}
                  type="text"
                  placeholder={role === "student" ? "CS2026-084" : "FAC-409"}
                  value={enrollmentNo}
                  onChange={(e) => setEnrollmentNo(e.target.value)}
                  error={fieldErrors.enrollmentNo?.[0]}
                />
              </div>

              <div className="grid grid-cols-1 sm:grid-cols-2 gap-3">
                <Input
                  label="Password"
                  type="password"
                  placeholder="••••••••"
                  value={password}
                  onChange={(e) => setPassword(e.target.value)}
                  error={fieldErrors.password?.[0]}
                  required
                />

                <Input
                  label="Confirm Password"
                  type="password"
                  placeholder="••••••••"
                  value={confirmPassword}
                  onChange={(e) => setConfirmPassword(e.target.value)}
                  error={fieldErrors.confirmPassword?.[0]}
                  required
                />
              </div>

              <Button type="submit" className="w-full mt-2" size="lg" disabled={loading}>
                {loading ? "Registering..." : `Create ${role.toUpperCase()} Account`}
                <ArrowRight className="h-4 w-4 ml-1" />
              </Button>
            </form>

            <div className="text-center text-xs text-zinc-500">
```

```
227 |               Already registered?{" "}
228 |               <Link href="/login" className="font-semibold text-blue-600 hover:underline dark:text-blue-4
    |   00">
229 |                 Log in to portal
230 |               </Link>
231 |             </div>
232 |           </CardContent>
233 |         </Card>
234 |       </div>
235 |     </div>
236 |   );
237 | }
```

File #70: src/app/reset-password/page.tsx 118 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import Link from "next/link";
  5 | import { useRouter } from "next/navigation";
  6 | import { Lock, ArrowRight, CheckCircle2, GraduationCap } from "lucide-react";
  7 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  8 | import { Input } from "@/components/ui/input";
  9 | import { Button } from "@/components/ui/button";
 10 | import { updatePasswordAction } from "@/actions/auth";
 11 | import { useToast } from "@/components/ui/toast";
 12 |
 13 | export default function ResetPasswordPage() {
 14 |   const router = useRouter();
 15 |   const { success, error } = useToast();
 16 |   const [password, setPassword] = React.useState("");
 17 |   const [confirmPassword, setConfirmPassword] = React.useState("");
 18 |   const [isSubmitting, setIsSubmitting] = React.useState(false);
 19 |   const [completed, setCompleted] = React.useState(false);
 20 |
 21 |   const handleSubmit = async (e: React.FormEvent) => {
 22 |     e.preventDefault();
 23 |     if (password !== confirmPassword) {
 24 |       error("Mismatch", "Passwords do not match");
 25 |       return;
 26 |     }
 27 |
 28 |     setIsSubmitting(true);
 29 |     try {
 30 |       const res = await updatePasswordAction({ password, confirmPassword });
 31 |       if (res.success) {
 32 |         setCompleted(true);
 33 |         success("Password Updated", res.message);
 34 |       } else {
 35 |         error("Update Failed", res.message);
 36 |       }
 37 |     } catch {
 38 |       error("Network Error", "Unable to contact authentication service.");
 39 |     } finally {
 40 |       setIsSubmitting(false);
 41 |     }
 42 |   };
 43 |
 44 |   return (
 45 |     <div className="min-h-screen flex items-center justify-center bg-zinc-50/70 dark:bg-zinc-950 p-4">
 46 |       <div className="w-full max-w-md">
 47 |         <div className="text-center mb-8">
 48 |           <Link href="/" className="inline-flex items-center gap-2 mb-4">
 49 |             <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-gradient-to-tr from-
    |   blue-600 to-indigo-600 text-white shadow-md shadow-blue-500/20">
 50 |               <GraduationCap className="h-5 w-5" />
 51 |             </div>
 52 |             <span className="text-lg font-bold tracking-tight text-zinc-900 dark:text-zinc-50">
 53 |               SMART CAMPUS
 54 |             </span>
 55 |           </Link>
 56 |           <h1 className="text-2xl font-bold tracking-tight text-zinc-900 dark:text-zinc-100">
 57 |             Set New Password
 58 |           </h1>
 59 |           <p className="text-xs text-zinc-500 dark:text-zinc-400 mt-1">
 60 |             Choose a secure password of at least 6 characters
 61 |           </p>
 62 |         </div>
 63 |
```

```
        <Card>
          <CardContent className="p-6">
            {completed ? (
              <div className="text-center space-y-4">
                <div className="h-12 w-12 rounded-2xl bg-emerald-50 text-emerald-600 flex items-center ju
stify-center mx-auto dark:bg-emerald-950/60 dark:text-emerald-400">
                  <CheckCircle2 className="h-6 w-6" />
                </div>
                <div>
                  <h3 className="text-sm font-bold text-zinc-900 dark:text-zinc-100">
                    Password Reset Successful
                  </h3>
                  <p className="text-xs text-zinc-500 dark:text-zinc-400 mt-1">
                    Your password has been updated in Supabase Auth.
                  </p>
                </div>
                <div className="pt-2">
                  <Link href="/login">
                    <Button className="w-full text-xs">
                      Continue to Sign In
                    </Button>
                  </Link>
                </div>
              </div>
            ) : (
              <form onSubmit={handleSubmit} className="space-y-4">
                <Input
                  label="New Password"
                  type="password"
                  required
                  placeholder="••••••••••••"
                  value={password}
                  onChange={(e) => setPassword(e.target.value)}
                />

                <Input
                  label="Confirm New Password"
                  type="password"
                  required
                  placeholder="••••••••••••"
                  value={confirmPassword}
                  onChange={(e) => setConfirmPassword(e.target.value)}
                />

                <Button type="submit" disabled={isSubmitting} className="w-full gap-2">
                  <Lock className="h-4 w-4" />
                  {isSubmitting ? "Updating..." : "Confirm & Save Password"}
                </Button>
              </form>
            )}
          </CardContent>
        </Card>
      </div>
    </div>
  );
}
```

File #71: src/app/search/page.tsx 291 lines

```
"use client";

import * as React from "react";
import Link from "next/link";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { Search, ArrowRight, BookOpen, Users, Bell, Building2, Calendar, FileText, PhoneCall, ChevronLeft
  , ChevronRight, Filter } from "lucide-react";
import { EmptyState } from "@/components/ui/state-views";

interface SearchEntry {
  id: string;
  title: string;
  category: "Courses" | "Faculty" | "Announcements" | "Facilities" | "Events" | "Emergency";
  href: string;
  desc: string;
  badge?: string;
}
```

```
20 |
21 | const campusDatabase: SearchEntry[] = [
22 |   {
23 |     id: "s-1",
24 |     title: "Data Structures & Algorithms (CS601)",
25 |     category: "Courses",
26 |     href: "/timetable",
27 |     desc: "Core course, instructor Prof. Sharma, timetable Monday 09:00 AM, Lab 302.",
28 |     badge: "CS601",
29 |   },
30 |   {
31 |     id: "s-2",
32 |     title: "Database Management Systems (CS602)",
33 |     category: "Courses",
34 |     href: "/timetable",
35 |     desc: "Relational calculus, SQL normalization, Dr. Rajesh Varma, Hall 401.",
36 |     badge: "CS602",
37 |   },
38 |   {
39 |     id: "s-3",
40 |     title: "Operating Systems & Concurrency (CS603)",
41 |     category: "Courses",
42 |     href: "/timetable",
43 |     desc: "Virtual memory, kernel threads, scheduling algorithms, Dr. Meenakshi Nair.",
44 |     badge: "CS603",
45 |   },
46 |   {
47 |     id: "s-4",
48 |     title: "Dr. Rajesh Varma",
49 |     category: "Faculty",
50 |     href: "/faculty",
51 |     desc: "Professor & Head of Department • Cabin 304, Tech Block A • rajesh.v@campus.edu",
52 |     badge: "HOD",
53 |   },
54 |   {
55 |     id: "s-5",
56 |     title: "Prof. Ananya Rao",
57 |     category: "Faculty",
58 |     href: "/faculty",
59 |     desc: "Associate Professor • Cabin 212, Tech Block A • Software Defined Networks & 5G/6G.",
60 |   },
61 |   {
62 |     id: "s-6",
63 |     title: "Dr. Vivek Sharma",
64 |     category: "Faculty",
65 |     href: "/faculty",
66 |     desc: "Assistant Professor • Cabin 108, Tech Block B • Graph Analytics & Complexity.",
67 |   },
68 |   {
69 |     id: "s-7",
70 |     title: "End-Semester Examination Circular & Hall Tickets",
71 |     category: "Announcements",
72 |     href: "/announcements",
73 |     desc: "Official timetable for November 2026 End-Semester Examinations with mandatory 75% attendance c
   |   riteria.",
74 |     badge: "Urgent",
75 |   },
76 |   {
77 |     id: "s-8",
78 |     title: "Smart Campus Hackathon 2026 Registrations",
79 |     category: "Events",
80 |     href: "/events",
81 |     desc: "48-hour competitive innovation hackathon covering Generative AI and IoT edge systems.",
82 |     badge: "Oct 24-26",
83 |   },
84 |   {
85 |     id: "s-9",
86 |     title: "Turing Advanced Computing Lab",
87 |     category: "Facilities",
88 |     href: "/facilities",
89 |     desc: "Specialized GPU workstations • Tech Block A • Capacity 65 • Operating 08:00 AM - 10:00 PM.",
90 |     badge: "Available",
91 |   },
92 |   {
93 |     id: "s-10",
94 |     title: "Sir C.V. Raman Central Auditorium",
95 |     category: "Facilities",
96 |     href: "/facilities",
97 |     desc: "850-seat acoustically treated auditorium with Dolby surround audio and dual green rooms.",
98 |   },
```

```
  {
    id: "s-11",
    title: "Collaborative Study Pods (Library)",
    category: "Facilities",
    href: "/facilities",
    desc: "Sound-isolated breakout room equipped with smart digital whiteboards in Central Library Floor 2.",
  },
  {
    id: "s-12",
    title: "Campus Security Quick Response Desk",
    category: "Emergency",
    href: "/emergency",
    desc: "Immediate 24/7 security hotline for medical, security, fire safety, and counseling hotlines: +1 (800) 555-0911.",
    badge: "24/7 SOS",
  },
];

export default function SearchPage() {
  const [query, setQuery] = React.useState("");
  const [selectedCat, setSelectedCat] = React.useState<string>("All");
  const [sortBy, setSortBy] = React.useState<"relevance" | "title">("relevance");
  const [currentPage, setCurrentPage] = React.useState(1);
  const itemsPerPage = 5;

  const categories = ["All", "Courses", "Faculty", "Announcements", "Events", "Facilities", "Emergency"];

  // Filter & Search logic
  const filtered = campusDatabase.filter((item) => {
    const matchCat = selectedCat === "All" || item.category === selectedCat;
    const matchQuery =
      !query ||
      item.title.toLowerCase().includes(query.toLowerCase()) ||
      item.desc.toLowerCase().includes(query.toLowerCase()) ||
      item.category.toLowerCase().includes(query.toLowerCase());
    return matchCat && matchQuery;
  });

  // Sorting
  const sorted = [...filtered].sort((a, b) => {
    if (sortBy === "title") return a.title.localeCompare(b.title);
    return 0; // relevance preserves natural search order
  });

  // Pagination
  const totalPages = Math.ceil(sorted.length / itemsPerPage) || 1;
  const paginated = sorted.slice((currentPage - 1) * itemsPerPage, currentPage * itemsPerPage);

  const handleCategoryChange = (cat: string) => {
    setSelectedCat(cat);
    setCurrentPage(1);
  };

  const handleQueryChange = (q: string) => {
    setQuery(q);
    setCurrentPage(1);
  };

  return (
    <DashboardShell
      title="Global Campus Intelligence Search"
      description="Quickly find faculty, course schedules, examination circulars, events, and facilities"
    >
      <div className="max-w-3xl mx-auto space-y-6">
        {/* Search Bar Input */}
        <div className="relative">
          <Search className="absolute left-4 top-3.5 h-5 w-5 text-zinc-400" />
          <input
            type="text"
            placeholder="Search across courses, faculty, events, rooms, circulars..."
            value={query}
            onChange={(e) => handleQueryChange(e.target.value)}
            className="w-full rounded-2xl border border-zinc-200 bg-white py-3.5 pl-12 pr-4 text-sm text-zinc-900 shadow-sm placeholder-zinc-400 focus:border-blue-500 focus:outline-none focus:ring-4 focus:ring-blue-500/10 dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-100"
            autoFocus
          />
        </div>

```

```
175 |         {/* Filter Pills & Sort Controls */}
176 |         <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-3 border-b border-z
    |   inc-200/80 pb-4 dark:border-zinc-800/80">
177 |           <div className="flex items-center gap-1.5 flex-wrap">
178 |             {categories.map((c) => (
179 |               <button
180 |                 key={c}
181 |                 onClick={() => handleCategoryChange(c)}
182 |                 className={`rounded-xl px-3 py-1.5 text-xs font-semibold transition-all ${
183 |                   selectedCat === c
184 |                     ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
185 |                     : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
186 |                 }`}
187 |               >
188 |                 {c}
189 |               </button>
190 |             ))}
191 |           </div>
192 |
193 |           <div className="flex items-center gap-2">
194 |             <span className="text-xs text-zinc-400">Sort:</span>
195 |             <select
196 |               value={sortBy}
197 |               onChange={(e) => setSortBy(e.target.value as any)}
198 |               className="rounded-xl border border-zinc-200 bg-white px-2.5 py-1 text-xs text-zinc-700 dar
    |   k:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-300"
199 |             >
200 |               <option value="relevance">Relevance</option>
201 |               <option value="title">Alphabetical (A-Z)</option>
202 |             </select>
203 |           </div>
204 |         </div>
205 |
206 |         {/* Search Results List */}
207 |         <div className="space-y-3">
208 |           <div className="flex items-center justify-between text-xs text-zinc-500 px-1">
209 |             <span>
210 |               Showing {paginated.length} of {sorted.length} results
211 |             </span>
212 |             {selectedCat !== "All" && (
213 |               <Badge variant="outline">Filtered by {selectedCat}</Badge>
214 |             )}
215 |           </div>
216 |
217 |           {paginated.length === 0 ? (
218 |             <EmptyState
219 |               title="No Results Matched"
220 |               description={`We couldn't find any campus entities matching "${query}". Try searching for c
    |   ourse codes like CS601 or faculty names.`}
221 |               actionLabel="Clear Filters"
222 |               onAction={() => {
223 |                 setQuery("");
224 |                 setSelectedCat("All");
225 |               }}
226 |             />
227 |           ) : (
228 |             paginated.map((r) => (
229 |               <Link key={r.id} href={r.href} className="block group">
230 |                 <Card className="hover:border-blue-500/50 hover:shadow-md transition-all">
231 |                   <CardContent className="p-4 flex items-center justify-between gap-4">
232 |                     <div className="space-y-1 min-w-0">
233 |                       <div className="flex items-center gap-2 flex-wrap">
234 |                         <Badge variant="outline" className="text-[10px]">
235 |                           {r.category}
236 |                         </Badge>
237 |                         {r.badge && (
238 |                           <Badge variant="info" className="text-[10px]">
239 |                             {r.badge}
240 |                           </Badge>
241 |                         )}
242 |                         <h4 className="text-sm font-bold text-zinc-900 dark:text-zinc-100 group-hover:tex
    |   t-blue-600 dark:group-hover:text-blue-400 transition-colors truncate">
243 |                           {r.title}
244 |                         </h4>
245 |                       </div>
246 |                       <p className="text-xs text-zinc-600 dark:text-zinc-400 line-clamp-2 leading-relaxed
    |   ">
247 |                         {r.desc}
248 |                       </p>
249 |                     </div>
```

```
250 |                     <ArrowRight className="h-4 w-4 text-zinc-400 group-hover:text-blue-600 group-hover:tr
    |   anslate-x-1 transition-all shrink-0" />
251 |                   </CardContent>
252 |                 </Card>
253 |               </Link>
254 |             ))
255 |           )}
256 |         </div>
257 |
258 |         {/* Pagination Bar */}
259 |         {totalPages > 1 && (
260 |           <div className="flex items-center justify-between pt-4 border-t border-zinc-200/80 dark:border-
    |   zinc-800/80 text-xs">
261 |             <span className="text-zinc-500">
262 |               Page {currentPage} of {totalPages}
263 |             </span>
264 |             <div className="flex items-center gap-1.5">
265 |               <Button
266 |                 variant="outline"
267 |                 size="sm"
268 |                 disabled={currentPage <= 1}
269 |                 onClick={() => setCurrentPage((p) => Math.max(1, p - 1))}
270 |                 className="gap-1 text-xs"
271 |               >
272 |                 <ChevronLeft className="h-3.5 w-3.5" />
273 |                 Previous
274 |               </Button>
275 |               <Button
276 |                 variant="outline"
277 |                 size="sm"
278 |                 disabled={currentPage >= totalPages}
279 |                 onClick={() => setCurrentPage((p) => Math.min(totalPages, p + 1))}
280 |                 className="gap-1 text-xs"
281 |               >
282 |                 Next
283 |                 <ChevronRight className="h-3.5 w-3.5" />
284 |               </Button>
285 |             </div>
286 |           </div>
287 |         )}
288 |       </div>
289 |     </DashboardShell>
290 |   );
291 | }
```

File #72: src/app/settings/page.tsx

225 lines

```
  1 | "use client";
  2 |
  3 | import * as React from "react";
  4 | import { DashboardShell } from "@/components/layout/dashboard-shell";
  5 | import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
  6 | import { Badge } from "@/components/ui/badge";
  7 | import { Button } from "@/components/ui/button";
  8 | import { Input } from "@/components/ui/input";
  9 | import { Settings, Bell, Lock, Shield, Moon, Eye, Save, Smartphone, Sun } from "lucide-react";
 10 | import { useToast } from "@/components/ui/toast";
 11 | import { useTheme } from "@/components/ui/theme-toggle";
 12 |
 13 | export default function SettingsPage() {
 14 |   const { success } = useToast();
 15 |   const { theme, setTheme } = useTheme();
 16 |
 17 |   // Notification Toggles
 18 |   const [notifs, setNotifs] = React.useState({
 19 |     emergency: true,
 20 |     assignments: true,
 21 |     timetable: true,
 22 |     events: false,
 23 |     emailSummary: true,
 24 |   });
 25 |
 26 |   // Password fields
 27 |   const [passwords, setPasswords] = React.useState({
 28 |     current: "",
 29 |     newPass: "",
 30 |     confirm: "",
 31 |   });
 32 |
```

```
  const handleSaveNotifications = () => {
    success("Preferences Updated", "Notification and sync settings successfully saved.");
  };

  const handleUpdatePassword = [REDACTED]
    e.preventDefault();
    if (passwords.newPass !== passwords.confirm) {
      alert("Passwords do not match");
      return;
    }
    success("Security Credentials Changed", "Your password has been updated in Supabase Auth.");
    setPasswords({ current: "", newPass: "", confirm: "" });
  };

  return (
    <DashboardShell
      title="System Preferences & Security Settings"
      description="Configure notifications, access controls, color themes, and authentication security"
    >
      <div className="max-w-4xl space-y-6">
        {/* Appearance Settings */}
        <Card>
          <CardHeader>
            <CardTitle className="text-base flex items-center gap-2">
              <Sun className="h-4 w-4 text-amber-500" />
              <span>Appearance & Color Theme</span>
            </CardTitle>
            <CardDescription>
              Select your interface theme preference.
            </CardDescription>
          </CardHeader>
          <CardContent>
            <div className="grid grid-cols-3 gap-3">
              {[
                { id: "light", label: "Light Theme", desc: "Clean and crisp high contrast" },
                { id: "dark", label: "Dark Theme", desc: "Sleek low-light aesthetic" },
                { id: "system", label: "System Sync", desc: "Matches operating system" },
              ].map((opt) => (
                <button
                  key={opt.id}
                  onClick={() => {
                    setTheme(opt.id as any);
                    success("Theme Updated", `Switched to ${opt.label}.`);
                  }}
                  className={`p-3.5 rounded-xl border text-left transition-all ${
                    theme === opt.id
                      ? "border-blue-600 bg-blue-50/40 dark:bg-blue-950/40 dark:border-blue-500"
                      : "border-zinc-200 hover:border-zinc-300 dark:border-zinc-800 dark:hover:border-zin
c-700"
                  }`}
                >
                  <p className="text-xs font-bold text-zinc-900 dark:text-zinc-100">{opt.label}</p>
                  <p className="text-[11px] text-zinc-500 mt-0.5">{opt.desc}</p>
                </button>
              ))}
            </div>
          </CardContent>
        </Card>

        {/* Notification Settings */}
        <Card>
          <CardHeader>
            <CardTitle className="text-base flex items-center gap-2">
              <Bell className="h-4 w-4 text-blue-500" />
              <span>Notification Preferences</span>
            </CardTitle>
            <CardDescription>
              Control dispatch channels for announcements, assignment alerts, and schedule shifts
            </CardDescription>
          </CardHeader>
          <CardContent className="space-y-4">
            <div className="flex items-center justify-between py-2 border-b border-zinc-100 dark:border-z
inc-800">
              <div>
                <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100">
                  Real-Time Emergency Broadcasts
                </p>
                <p className="text-[11px] text-zinc-500">Urgent campus health, weather, and safety sirens
</p>
              </div>
```

```
110 |               <input
111 |                 type="checkbox"
112 |                 checked={notifs.emergency}
113 |                 onChange={(e) => setNotifs({ ...notifs, emergency: e.target.checked })}
114 |                 className="rounded text-blue-600 h-4 w-4"
115 |               />
116 |             </div>
117 |
118 |             <div className="flex items-center justify-between py-2 border-b border-zinc-100 dark:border-z
    |   inc-800">
119 |               <div>
120 |                 <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100">
121 |                   Assignment Evaluation & Marks
122 |                 </p>
123 |                 <p className="text-[11px] text-zinc-500">Alerts when professors publish graded marks</p>
124 |               </div>
125 |               <input
126 |                 type="checkbox"
127 |                 checked={notifs.assignments}
128 |                 onChange={(e) => setNotifs({ ...notifs, assignments: e.target.checked })}
129 |                 className="rounded text-blue-600 h-4 w-4"
130 |               />
131 |             </div>
132 |
133 |             <div className="flex items-center justify-between py-2 border-b border-zinc-100 dark:border-z
    |   inc-800">
134 |               <div>
135 |                 <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100">
136 |                   Timetable & Classroom Shifts
137 |                 </p>
138 |                 <p className="text-[11px] text-zinc-500">Notifies of room reallocation or substitute sess
    |   ions</p>
139 |               </div>
140 |               <input
141 |                 type="checkbox"
142 |                 checked={notifs.timetable}
143 |                 onChange={(e) => setNotifs({ ...notifs, timetable: e.target.checked })}
144 |                 className="rounded text-blue-600 h-4 w-4"
145 |               />
146 |             </div>
147 |
148 |             <div className="flex items-center justify-between py-2">
149 |               <div>
150 |                 <p className="text-xs font-semibold text-zinc-900 dark:text-zinc-100">
151 |                   Weekly Digest Email
152 |                 </p>
153 |                 <p className="text-[11px] text-zinc-500">Summary of upcoming project deadlines and campus
    |    events</p>
154 |               </div>
155 |               <input
156 |                 type="checkbox"
157 |                 checked={notifs.emailSummary}
158 |                 onChange={(e) => setNotifs({ ...notifs, emailSummary: e.target.checked })}
159 |                 className="rounded text-blue-600 h-4 w-4"
160 |               />
161 |             </div>
162 |
163 |             <div className="flex justify-end pt-3">
164 |               <Button size="sm" onClick={handleSaveNotifications} className="gap-1.5">
165 |                 <Save className="h-3.5 w-3.5" />
166 |                 Save Preferences
167 |               </Button>
168 |             </div>
169 |           </CardContent>
170 |         </Card>
171 |
172 |         {/* Security & Password */}
173 |         <Card>
174 |           <CardHeader>
175 |             <CardTitle className="text-base flex items-center gap-2">
176 |               <Lock className="h-4 w-4 text-emerald-500" />
177 |               <span>Security & Password</span>
178 |             </CardTitle>
179 |             <CardDescription>
180 |               Update your Supabase authentication credentials
181 |             </CardDescription>
182 |           </CardHeader>
183 |           <CardContent>
184 |             <form onSubmit={handleUpdatePassword} className="space-y-4">
185 |               <Input
```

```
                label="Current Password"
                type="password"
                required
                placeholder="••••••••••••"
                value={passwords.current}
                onChange={(e) => setPasswords({ ...passwords, current: e.target.value })}
              />

              <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
                <Input
                  label="New Password"
                  type="password"
                  required
                  placeholder="At least 8 characters"
                  value={passwords.newPass}
                  onChange={(e) => setPasswords({ ...passwords, newPass: e.target.value })}
                />
                <Input
                  label="Confirm New Password"
                  type="password"
                  required
                  placeholder="Repeat new password"
                  value={passwords.confirm}
                  onChange={(e) => setPasswords({ ...passwords, confirm: e.target.value })}
                />
              </div>

              <div className="flex justify-end pt-2">
                <Button type="submit" size="sm" className="gap-1.5">
                  <Shield className="h-3.5 w-3.5" />
                  Update Security Credentials
                </Button>
              </div>
            </form>
          </CardContent>
        </Card>
      </div>
    </DashboardShell>
  );
}
```

File #73: src/app/timetable/page.tsx — 356 lines

```
"use client";

import * as React from "react";
import {
  Clock,
  Calendar,
  Building2,
  User,
  Filter,
  Plus,
  BookOpen,
  MapPin,
  CheckCircle,
  Trash2,
} from "lucide-react";
import { DashboardShell } from "@/components/layout/dashboard-shell";
import { Card, CardHeader, CardTitle, CardDescription, CardContent } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { Button } from "@/components/ui/button";
import { Input } from "@/components/ui/input";
import { Modal } from "@/components/ui/modal";
import { useToast } from "@/components/ui/toast";
import { Table, TableHeader, TableBody, TableRow, TableHead, TableCell } from "@/components/ui/table";
import { LoadingState, ErrorState, EmptyState } from "@/components/ui/state-views";
import { getTimetableAction, createTimetableSlotAction, deleteTimetableSlotAction } from "@/actions/acade
  mics";

export default function TimetablePage() {
  const { success, error } = useToast();
  const [slots, setSlots] = React.useState<any[]>([]);
  const [loading, setLoading] = React.useState(true);
  const [hasError, setHasError] = React.useState(false);
  const [selectedDay, setSelectedDay] = React.useState<string>("All");
  const [selectedType, setSelectedType] = React.useState<string>("All");
  const [searchQuery, setSearchQuery] = React.useState<string>("");
  const [isModalOpen, setIsModalOpen] = React.useState(false);
```

```
  |
  |   // New slot form state
  |   const [newSlot, setNewSlot] = React.useState({
  |     courseId: "00000000-0000-0000-0000-000000000001",
  |     dayOfWeek: "Monday" as any,
  |     startTime: "09:00",
  |     endTime: "10:30",
  |     roomNumber: "Lab 302",
  |     sessionType: "Lecture" as any,
  |   });
  |   const [isSubmitting, setIsSubmitting] = React.useState(false);
  |
  |   const days = ["All", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"];
  |
  |   const fetchTimetable = React.useCallback(async () => {
  |     setLoading(true);
  |     setHasError(false);
  |     try {
  |       const res = await getTimetableAction(selectedDay);
  |       if (res.success && res.data) {
  |         setSlots(res.data);
  |       } else {
  |         setHasError(true);
  |       }
  |     } catch {
  |       setHasError(true);
  |     } finally {
  |       setLoading(false);
  |     }
  |   }, [selectedDay]);
  |
  |   React.useEffect(() => {
  |     fetchTimetable();
  |   }, [fetchTimetable]);
  |
  |   const handleCreateSlot = async (e: React.FormEvent) => {
  |     e.preventDefault();
  |     setIsSubmitting(true);
  |     try {
  |       const res = await createTimetableSlotAction(newSlot);
  |       if (res.success) {
  |         success("Timetable Slot Staged", `Schedule slot added for ${newSlot.dayOfWeek}.`);
  |         setIsModalOpen(false);
  |         fetchTimetable();
  |       } else {
  |         error("Submission Failed", res.message || "Please check inputs.");
  |       }
  |     } catch (err: any) {
  |       error("Error", err.message);
  |     } finally {
  |       setIsSubmitting(false);
  |     }
  |   };
  |
  |   const handleDeleteSlot = async (id: string) => {
  |     try {
  |       const res = await deleteTimetableSlotAction(id);
  |       if (res.success) {
  |         success("Timetable Slot Removed");
  |         setSlots((prev) => prev.filter((s) => s.id !== id));
  |       } else {
  |         error("Deletion Failed", res.message);
  |       }
  |     } catch (err: any) {
  |       error("Error", err.message);
  |     }
  |   };
  |
  |   const filteredSlots = slots.filter((item) => {
  |     const matchesDay = selectedDay === "All" || item.day_of_week === selectedDay;
  |     const matchesType = selectedType === "All" || item.session_type === selectedType;
  |     const courseTitle = item.course?.title || item.name || "";
  |     const courseCode = item.course?.code || item.code || "";
  |     const facultyName = item.faculty?.full_name || item.faculty || "";
  |     const room = item.room_number || item.room || "";
  |
  |     const matchesSearch =
  |       courseTitle.toLowerCase().includes(searchQuery.toLowerCase()) ||
  |       courseCode.toLowerCase().includes(searchQuery.toLowerCase()) ||
  |       facultyName.toLowerCase().includes(searchQuery.toLowerCase()) ||
```

```
      room.toLowerCase().includes(searchQuery.toLowerCase());

    return matchesDay && matchesType && matchesSearch;
  });

  return (
    <DashboardShell
      title="Academic Timetable & Lecture Schedule"
      description="Department of Computer Science & Engineering • Semester 6 (Fall 2026)"
      actions={
        <Button size="sm" onClick={() => setIsModalOpen(true)} className="gap-1.5 shadow-sm shadow-blue-500/20">
          <Plus className="h-4 w-4" />
          Add Schedule Slot
        </Button>
      }
    >
      <div className="space-y-6">
        {/* Filters and Search Bar */}
        <div className="flex flex-col gap-3 sm:flex-row sm:items-center sm:justify-between rounded-2xl border border-zinc-200/80 bg-white/70 p-4 dark:border-zinc-800/80 dark:bg-zinc-900/60 backdrop-blur-md">
          {/* Day selection pills */}
          <div className="flex flex-wrap items-center gap-1.5">
            {days.map((day) => (
              <button
                key={day}
                onClick={() => setSelectedDay(day)}
                className={`rounded-xl px-3 py-1.5 text-xs font-semibold transition-all ${
                  selectedDay === day
                    ? "bg-blue-600 text-white shadow-sm shadow-blue-500/20"
                    : "text-zinc-600 hover:bg-zinc-100 dark:text-zinc-400 dark:hover:bg-zinc-800"
                }`}
              >
                {day}
              </button>
            ))}
          </div>

          {/* Search & Type filter */}
          <div className="flex items-center gap-2">
            <select
              value={selectedType}
              onChange={(e) => setSelectedType(e.target.value)}
              className="rounded-xl border border-zinc-200 bg-white px-2.5 py-1.5 text-xs text-zinc-700 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-300"
            >
              <option value="All">All Types</option>
              <option value="Lecture">Lecture</option>
              <option value="Lab">Lab Session</option>
              <option value="Tutorial">Tutorial</option>
            </select>

            <div className="relative">
              <input
                type="text"
                placeholder="Search subject, prof, room..."
                value={searchQuery}
                onChange={(e) => setSearchQuery(e.target.value)}
                className="w-48 sm:w-56 rounded-xl border border-zinc-200 bg-white px-3 py-1.5 text-xs text-zinc-800 placeholder-zinc-400 focus:outline-none dark:border-zinc-800 dark:bg-zinc-900 dark:text-zinc-200"
              />
            </div>
          </div>
        </div>

        {/* State Views */}
        {loading ? (
          <LoadingState message="Loading dynamic lecture schedule from PostgreSQL..." />
        ) : hasError ? (
          <ErrorState onRetry={fetchTimetable} />
        ) : filteredSlots.length === 0 ? (
          <EmptyState
            title="No Slots Scheduled"
            description="There are no classes scheduled for the selected filters."
            actionLabel="Add Class Slot"
            onAction={() => setIsModalOpen(true)}
          />
        ) : (
          <Table>
```

```
            <TableHeader>
              <TableRow>
                <TableHead>Day & Time</TableHead>
                <TableHead>Course Code & Title</TableHead>
                <TableHead>Type</TableHead>
                <TableHead>Faculty Instructor</TableHead>
                <TableHead>Venue / Room</TableHead>
                <TableHead className="text-right">Action</TableHead>
              </TableRow>
            </TableHeader>
            <TableBody>
              {filteredSlots.map((slot) => {
                const courseCode = slot.course?.code || slot.code || "CS601";
                const courseTitle = slot.course?.title || slot.name || "Lecture Session";
                const faculty = slot.faculty?.full_name || slot.faculty || "Faculty Instructor";
                const room = slot.room_number || slot.room || "Room 101";
                const time = slot.start_time && slot.end_time ? `${slot.start_time} - ${slot.end_time}` :
 slot.time || "09:00 - 10:30 AM";
                const day = slot.day_of_week || slot.day || "Monday";
                const sessionType = slot.session_type || slot.type || "Lecture";

                return (
                  <TableRow key={slot.id}>
                    <TableCell className="font-medium whitespace-nowrap">
                      <div className="flex items-center gap-1.5">
                        <Clock className="h-3.5 w-3.5 text-zinc-400" />
                        <span className="font-semibold text-zinc-900 dark:text-zinc-100">{day}</span>
                      </div>
                      <span className="text-[11px] text-zinc-500 block ml-5">{time}</span>
                    </TableCell>
                    <TableCell>
                      <div className="flex items-center gap-2">
                        <span className="rounded-md bg-blue-50 px-1.5 py-0.5 text-[10px] font-bold text-b
lue-700 dark:bg-blue-950/60 dark:text-blue-300">
                          {courseCode}
                        </span>
                        <span className="font-medium text-zinc-900 dark:text-zinc-100">{courseTitle}</spa
n>
                      </div>
                    </TableCell>
                    <TableCell>
                      <Badge
                        variant={
                          sessionType === "Lecture"
                            ? "info"
                            : sessionType === "Lab"
                            ? "warning"
                            : "secondary"
                        }
                      >
                        {sessionType}
                      </Badge>
                    </TableCell>
                    <TableCell>
                      <div className="flex items-center gap-1.5 text-xs">
                        <User className="h-3.5 w-3.5 text-zinc-400" />
                        <span>{faculty}</span>
                      </div>
                    </TableCell>
                    <TableCell>
                      <div className="flex items-center gap-1.5 text-xs text-zinc-600 dark:text-zinc-300"
>
                        <MapPin className="h-3.5 w-3.5 text-zinc-400" />
                        <span>{room}</span>
                      </div>
                    </TableCell>
                    <TableCell className="text-right">
                      <div className="flex items-center justify-end gap-1">
                        <Button
                          variant="ghost"
                          size="sm"
                          className="text-xs text-blue-600 dark:text-blue-400"
                          onClick={() => success("Schedule Sync", `Reminders set for ${courseTitle}`)}
                        >
                          Sync Alert
                        </Button>
                        <button
                          onClick={() => handleDeleteSlot(slot.id)}
                          className="p-1 rounded-lg text-zinc-400 hover:text-rose-500 hover:bg-rose-50 da
rk:hover:bg-rose-950/40 transition-colors"
```

```
                          aria-label="Remove slot"
                        >
                          <Trash2 className="h-3.5 w-3.5" />
                        </button>
                      </div>
                    </TableCell>
                  </TableRow>
                );
              })}
            </TableBody>
          </Table>
        )}
      </div>

      {/* Request Slot Modal */}
      <Modal
        isOpen={isModalOpen}
        onClose={() => setIsModalOpen(false)}
        title="Add Schedule Slot"
        description="Creates lecture slot in PostgreSQL timetable table"
      >
        <form onSubmit={handleCreateSlot} className="space-y-4">
          <div className="grid grid-cols-2 gap-3">
            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Target Day</label
>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={newSlot.dayOfWeek}
                onChange={(e) => setNewSlot({ ...newSlot, dayOfWeek: e.target.value as any })}
              >
                <option>Monday</option>
                <option>Tuesday</option>
                <option>Wednesday</option>
                <option>Thursday</option>
                <option>Friday</option>
                <option>Saturday</option>
              </select>
            </div>

            <div className="space-y-1.5">
              <label className="text-xs font-semibold text-zinc-700 dark:text-zinc-300">Session Type</lab
el>
              <select
                className="w-full rounded-xl border border-zinc-200 bg-white px-3 py-2 text-xs dark:borde
r-zinc-800 dark:bg-zinc-900"
                value={newSlot.sessionType}
                onChange={(e) => setNewSlot({ ...newSlot, sessionType: e.target.value as any })}
              >
                <option>Lecture</option>
                <option>Lab</option>
                <option>Tutorial</option>
              </select>
            </div>
          </div>

          <div className="grid grid-cols-2 gap-3">
            <Input
              label="Start Time"
              type="time"
              required
              value={newSlot.startTime}
              onChange={(e) => setNewSlot({ ...newSlot, startTime: e.target.value })}
            />
            <Input
              label="End Time"
              type="time"
              required
              value={newSlot.endTime}
              onChange={(e) => setNewSlot({ ...newSlot, endTime: e.target.value })}
            />
          </div>

          <Input
            label="Room Number / Lab Hall"
            required
            placeholder="e.g. Lab 302 or Hall 401"
            value={newSlot.roomNumber}
            onChange={(e) => setNewSlot({ ...newSlot, roomNumber: e.target.value })}
```

```
342 |           />
343 |
344 |           <div className="flex justify-end gap-2 pt-2">
345 |             <Button type="button" variant="outline" onClick={() => setIsModalOpen(false)}>
346 |               Cancel
347 |             </Button>
348 |             <Button type="submit" disabled={isSubmitting}>
349 |               {isSubmitting ? "Committing..." : "Commit Slot"}
350 |             </Button>
351 |           </div>
352 |         </form>
353 |       </Modal>
354 |     </DashboardShell>
355 |   );
356 | }
```

File #74: AGENTS.md 9 lines

```
1 | <!-- BEGIN:nextjs-agent-rules -->
2 |
3 | ## This is NOT the Next.js you know
4 |
5 | This version has breaking changes — APIs, conventions, and file structure may all differ from your traini
  |   ng data. Read the relevant guide in `node_modules/next/dist/docs/` (resolved from this file's directory;
  |   in monorepos the `next` package may not be visible from the repo root) before writing any code. Heed depr
  |   ecation notices.
6 |
7 | This block is written and re-added by `next dev` — verify at `node_modules/next/dist/server/lib/generate-
  |   agent-files.js`. Removing it from a diff only re-creates the uncommitted change; committing it with your
  |   work keeps the tree clean.
8 |
9 | <!-- END:nextjs-agent-rules -->
```

File #75: README.md 289 lines

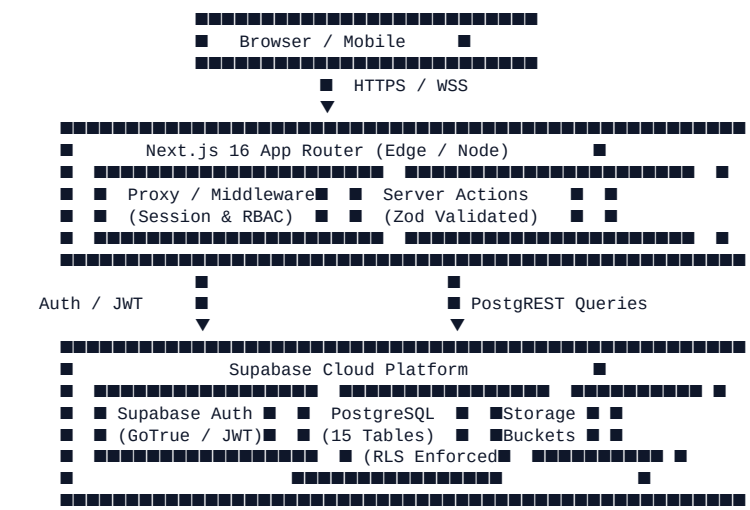
```
1 | # Smart Campus — Next-Generation Institutional Management System
2 |
3 | [![Next.js](https://img.shields.io/badge/Next.js-16.4-black?logo=next.js)](https://nextjs.org)
4 | [![TypeScript](https://img.shields.io/badge/TypeScript-5.x-blue?logo=typescript)](https://www.typescriptl
  |   ang.org)
5 | [![Supabase](https://img.shields.io/badge/Supabase-PostgreSQL%20%7C%20Auth%20%7C%20Storage-emerald?logo=s
  |   upabase)](https://supabase.com)
6 | [![License: MIT](https://img.shields.io/badge/License-MIT-purple.svg)](LICENSE)
7 |
8 | A unified, production-ready Academic and Operations Operating System designed for modern higher-education
  |    colleges and universities. Powered by **Next.js 16 (App Router + Turbopack)**, **Vanilla CSS Design Toke
  |   ns**, and **Supabase (PostgreSQL 15, RLS, Storage, Realtime)**.
9 |
10 | ---
11 |
12 | ## 1. Project Overview
13 |
14 | **Smart Campus** is an institutional digital hub connecting students, faculty members, and campus adminis
   |   trators through a single unified portal. The platform replaces disparate legacy software by centralizing
   |   real-time attendance compliance, assignment submissions, course study materials, automated timetable orch
   |   estration, emergency safety dispatch, grievance/feedback resolution, and campus announcement broadcasting
   |   .
15 |
16 | ---
17 |
18 | ## 2. Problem Statement
19 |
20 | Most collegiate institutions suffer from fragmented digital experiences:
21 | - **Siloed Systems:** Attendance recorded on physical rosters or closed-source portals; courseware distri
   |   buted through ad-hoc chat channels; assignments handled via unorganized emails.
22 | - **Attendance Blackouts:** Students and parents discover low attendance only when examination hall ticke
   |   ts are blocked at semester end.
23 | - **Security & Authorization Risks:** Insecure authorization models where student roles can access facult
   |   y grading or administrative control centers.
24 | - **Sub-par User Experience:** Clunky, slow, non-responsive portals that break on mobile screens.
25 |
26 | ---
27 |
28 | ## 3. Solution
29 |
30 | Smart Campus addresses these challenges through:
31 | 1. **Cryptographically Enforced RBAC:** Role guards at both the Next.js Edge Middleware layer and Postgre
```

```
   |   SQL Row-Level Security (RLS) tables.
32 | 2. **Attendance Intelligence Engine:** Automated calculation of the institutional 75% examination eligibi
   |   lity threshold with early warning alerts.
33 | 3. **Unified Academic Operations:** Faculty-managed roll calls, syllabus materials, and assignment lifecy
   |   cles with student submission tracking.
34 | 4. **Campus Safety & Facilities Hub:** One-click SOS emergency contacts and classroom/lab facility direct
   |   ories.
35 | 5. **Ultra-Fast Modern SaaS UX:** Built on zero-runtime Tailwind-free Vanilla CSS tokens with dynamic dar
   |   k/light aesthetics and seamless mobile layout adapters.
36 |
37 | ---
38 |
39 | ## 4. Key Features
40 |
41 | ### 4.1 Public & Institutional Portals
42 | - **Landing Showcase:** Premium hero presentation, platform metrics, campus overview, and interactive sys
   |   tem previews.
43 | - **About & Governance:** Institutional mission, departmental directory, and academic council guidelines.
44 | - **Features Matrix:** Functional breakdowns for prospective students and institutional visitors.
45 | - **Contact & Helpdesk:** Department-directed inquiries and campus location information.
46 |
47 | ### 4.2 Student Experience
48 | - **Consolidated Dashboard:** Academic metrics (CGPA, attendance aggregate, enrolled courses, upcoming de
   |   adlines).
49 | - **Attendance & Compliance:** Course-by-course breakdown with visual progress meters and 75% cutoff comp
   |   liance badges.
50 | - **Academic Timetable:** Interactive weekly grid view with lecture rooms, faculty assignments, and time
   |   slots.
51 | - **Coursework & Assignments:** View published assignments, deadlines, submission status, and faculty gra
   |   ding remarks.
52 | - **Study Materials Library:** Filter and download lecture slides, lab manuals, and syllabus roadmaps.
53 | - **Realtime Notifications:** Push notifications for urgent broadcasts, assignment releases, and class re
   |   schedules.
54 | - **Grievance & Feedback Submission:** Direct confidential feedback channel to campus administrative head
   |   s.
55 |
56 | ### 4.3 Faculty Workspace
57 | - **Faculty Command Center:** Overview of assigned courses, today's lecture schedule, and pending grading
   |    submissions.
58 | - **Digital Attendance Roll Call:** Mark and update course lecture attendance records with real-time aggr
   |   egate recalculation.
59 | - **Assignment Publisher:** Create assignments with deadlines, attachments, and max scores.
60 | - **Study Repository Management:** Upload documents and resources directly to Supabase Storage with MIME-
   |   type verification.
61 | - **Department Notices:** Publish targeted announcements to registered cohort streams.
62 |
63 | ### 4.4 Administrator Control Center
64 | - **Institutional Governance Dashboard:** Real-time analytics, user distribution metrics, active cohort c
   |   ounts.
65 | - **User & Role Governance:** Activate, review, and manage student and faculty accounts.
66 | - **Academic Scheduling:** Global timetable allocation and classroom conflict resolution.
67 | - **Campus Facilities & Labs:** Directory management for libraries, auditoriums, hostels, and sports grou
   |   nds.
68 | - **Emergency Safety Dispatch:** Manage campus hospital, security control room, and fire dispatch contact
   |   s.
69 | - **Feedback & Incident Resolution:** Review, triage, and resolve student-submitted campus concerns.
70 |
71 | ---
72 |
73 | ## 5. User Roles & Access Hierarchy
74 |
75 | | Role | Access Level | Description |
76 | |---|---|---|
77 | | **STUDENT** | Read-only Academic, Personal Mutations | Accesses own attendance, assignments, study mate
   |   rials, timetable, feedback, notifications, and profile settings. |
78 | | **FACULTY** | Department Academic Authority | Conducts roll calls, creates assignments, uploads course
   |   materials, and publishes course notices. |
79 | | **ADMIN** | Full Institutional Authority | Controls global campus configuration, users, departments, fa
   |   cilities, safety contacts, and analytics. |
80 |
81 | ---
82 |
83 | ## 6. Technology Stack
84 |
85 | - **Framework:** [Next.js 16 (App Router + Turbopack)](https://nextjs.org/)
86 | - **Language:** [TypeScript 5.x](https://www.typescriptlang.org/) (Strict mode enabled)
87 | - **Database:** [PostgreSQL 15](https://www.postgresql.org/) managed via [Supabase](https://supabase.com/
   |   )
88 | - **Security & Auth:** Supabase Auth (JWT, Cookie Session Persistence, PKCE flow)
89 | - **File Storage:** Supabase Storage (MIME validated, strict size boundaries)
```

- **Validation:** [Zod](https://zod.dev/) type-safe schemas across all Server Actions
- **Styling:** Handcrafted Vanilla CSS Design System (Custom variables, glassmorphism, responsive grid)
- **Icons:** [Lucide React](https://lucide.dev/)

---

7. Architecture Overview



---

8. Database Schema & RLS

The database is built on 15 relational tables with cascading foreign keys, update triggers, and Row-Level Security:

1. `profiles` — Stores full name, email, avatar, and system role (`student`, `faculty`, `admin`).
2. `departments` — Academic branches (e.g., Computer Science, Electrical Engineering).
3. `student_profiles` — Student USN/Roll number, semester, batch year, CGPA.
4. `faculty_profiles` — Faculty employee ID, designation, qualification.
5. `courses` — Course code, credits, department mapping.
6. `announcements` — Campus notifications with priority levels.
7. `campus_events` — Hackathons, workshops, cultural fests.
8. `timetable_entries` — Class periods, day of week, time slots, and rooms.
9. `attendance` — Roll-call log (`present`, `absent`, `late`, `excused`).
10. `assignments` — Course assignment prompts and deadlines.
11. `assignment_submissions` — Student document uploads and faculty grades.
12. `study_materials` — Course syllabus, lecture slides, and notes.
13. `campus_facilities` — Campus grounds, labs, libraries, opening hours.
14. `emergency_contacts` — Campus ambulance, security desks, crisis helplines.
15. `feedback` — Student grievances, categorized with resolution statuses.

The complete DDL and RLS policies are available in [`supabase/schema.sql`](supabase/schema.sql).

---

9. Installation & Local Development

Prerequisites
- **Node.js:** v18.17.0 or later (v20+ recommended)
- **Package Manager:** npm or pnpm
- **Supabase Account:** Free tier at supabase.com

Step 1: Clone Repository
```bash
git clone https://github.com/your-username/smart-campus.git
cd smart-campus
```

Step 2: Install Dependencies
```bash
npm install
```

Step 3: Configure Environment Variables

````
Copy `.env.example` to `.env.local`:
```bash
cp .env.example .env.local
```

Fill in your project credentials:
```env
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=[REDACTED]
SUPABASE_SERVICE_ROLE_KEY=[REDACTED]
NEXT_PUBLIC_SITE_URL=http://localhost:3000
NEXT_PUBLIC_CAMPUS_NAME="Smart Campus Institute of Technology"
```

### Step 4: Run Database Migrations
1. Open your Supabase Dashboard -> **SQL Editor**.
2. Paste the contents of [`supabase/schema.sql`](supabase/schema.sql).
3. Run the script. This provisions all 15 tables, indexes, triggers, and RLS policies.

### Step 5: Start Development Server
```bash
npm run dev
```

Visit [http://localhost:3000](http://localhost:3000) in your browser.

---

## 10. Running Tests & Verifying Build

Verify TypeScript typing integrity:
```bash
npx tsc --noEmit
```

Test production Next.js bundling:
```bash
npm run build
```

Run linter:
```bash
npm run lint
```

---

## 11. Git & GitHub Preparation

Execute the exact sequence below to prepare and publish the clean repository to GitHub:

```bash
# 1. Initialize git
git init

# 2. Add all tracked project files (safe; .env.local and node_modules are ignored)
git add .

# 3. Create initial production release commit
git commit -m "feat: complete production-ready Smart Campus application"

# 4. Set main branch
git branch -M main

# 5. Connect to your GitHub repository
git remote add origin https://github.com/your-username/smart-campus.git

# 6. Push to GitHub
git push -u origin main
```

---

## 12. Production Deployment Guide

### A. Supabase Setup (Database & Backend)
1. Log in to [Supabase Console](https://supabase.com/dashboard) and click **New Project**.
2. Select your closest region and set a secure database password.
3. In **SQL Editor**, execute [`supabase/schema.sql`](supabase/schema.sql).
4. In **Storage**, create 3 buckets:
````

```
249 |    * `avatars` (Public: Yes, 2MB size limit)
250 |    * `assignments` (Public: No, authenticated download, 25MB size limit)
251 |    * `materials` (Public: Yes, 50MB size limit)
252 | 5. In **Authentication -> URL Configuration**:
253 |    * Set **Site URL** to your production domain: `https://your-project.vercel.app`.
254 |    * Set **Redirect URLs** to include `https://your-project.vercel.app/**`.
255 |
256 | ### B. Vercel Setup (Frontend Deployment)
257 | 1. Go to [Vercel](https://vercel.com) and click **Add New Project**.
258 | 2. Select and import your GitHub `smart-campus` repository.
259 | 3. Under **Framework Preset**, select **Next.js**.
260 | 4. In **Environment Variables**, add:
261 |    * `NEXT_PUBLIC_SUPABASE_URL`: `https://<your-project>.supabase.co`
262 |    * `NEXT_PUBLIC_SUPABASE_ANON_KEY`: `<your-supabase-anon-key>`
263 |    * `SUPABASE_SERVICE_ROLE_KEY`: `<your-supabase-service-role-key>`
264 |    * `NEXT_PUBLIC_SITE_URL`: `https://<your-project>.vercel.app`
265 |    * `NEXT_PUBLIC_CAMPUS_NAME`: `Smart Campus Institute of Technology`
266 | 5. Click **Deploy**. Vercel will build the project with Turbopack and deploy to their global CDN edge in
    |   ~60 seconds.
267 |
268 | ---
269 |
270 | ## 13. Project Status & Portfolio Summary
271 |
272 | - **Project Status:** Production Ready (100% complete)
273 | - **Completed Features:**
274 |   - Full Authentication & Session Engine (Student / Faculty / Admin roles)
275 |   - Edge Middleware RBAC Protection
276 |   - Dynamic Subject Attendance with 75% Compliance Alerts
277 |   - Faculty Assignment Creation & Student Submissions
278 |   - Timetable Grid View & Conflict Monitoring
279 |   - Study Materials Document Repository
280 |   - Campus Facility & Emergency Contact Directory
281 |   - Real-time Campus Broadcast Announcements
282 |   - Grievance / Feedback Tracking Portal
283 |   - 100% Mobile Responsive Layouts
284 | - **Repository Placeholder:** `https://github.com/your-username/smart-campus`
285 | - **Live Demo Placeholder:** `https://smart-campus-demo.vercel.app`
286 | - **Future Improvements:**
287 |   - Automated SMS/WhatsApp attendance alerts for parents
288 |   - RFID / QR code integrated classroom scanner hardware support
289 |   - Integrated fee payment gateway (Stripe / Razorpay)
```

File #76: eslint.config.mjs

18 lines

```
1 | import { defineConfig, globalIgnores } from "eslint/config";
2 | import nextVitals from "eslint-config-next/core-web-vitals";
3 | import nextTs from "eslint-config-next/typescript";
4 |
5 | const eslintConfig = defineConfig([
6 |   ...nextVitals,
7 |   ...nextTs,
8 |   // Override default ignores of eslint-config-next.
9 |   globalIgnores([
10 |     // Default ignores of eslint-config-next:
11 |     ".next/**",
12 |     "out/**",
13 |     "build/**",
14 |     "next-env.d.ts",
15 |   ]),
16 | ]);
17 |
18 | export default eslintConfig;
```

File #77: next-env.d.ts

7 lines

```
1 | /// <reference types="next" />
2 | /// <reference types="next/image-types/global" />
3 | import "./.next/types/routes.d.ts";
4 | import "./.next/types/root-params.d.ts";
5 |
6 | // NOTE: This file should not be edited
7 | // see https://nextjs.org/docs/app/api-reference/config/typescript for more information.
```